

WORLD OF OIL DERIVATIVES

A guide to financial oil
trading in a modern age

GREG NEWMAN



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First edition

For everyone at Onyx, as we strive towards our collective mission

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FOREWORD

JOEL COUSE

The intimate interplay between energy and price volatility has driven increasingly sophisticated hedging and trading strategies. In the early days of the industry, volatility was contained and its financial impact limited through vertical integration, inventory control, and long-term fixed-price contracts. However, the development of energy derivative markets has introduced flexible risk management solutions that stabilise revenue streams through hedging. Speculators, traders, and market makers provide contract liquidity. The use of financial derivatives, both traded and over-the-counter, has rapidly expanded in value and volume as the market environment has become ever more favourable.

I have personally witnessed much of this transformation in oil trading; initially from 1992 through 2016 in my role as head of market analysis for TotalEnergies (formerly Elf Trading during the 1990s) and subsequently as Chief Energy Economist for the company and later its advisor to the International Energy Agency. Over those 30 years, paper oil trading volumes have increased over seventy-fold for futures and options, while volumes in the over-the-counter (OTC) oil market have gone from almost non-existent to a deep and nearly ubiquitous market across underlying physical indices. This significant transformation surpasses the volume of growth in physical trading and reflects the broader development of financial markets, computer power, information availability, and the constant drive for gains.

“Creative” perhaps best describes this period. Traders used new tools to generate better information and better instruments for securing gains. The interface with financial markets expanded both in breadth and depth. Price Reporting Agencies developed more granular and robust methodologies that used futures and OTC contracts to enrich the information flow for assessments. The Platts e-Window, supported by Intercontinental Exchange, increased OTC price transparency among buyers, sellers, and brokers. The physical and financial sides of the markets became increasingly intertwined, gradually moving away from the shadows of bilateral trading.

However, the most significant transformative events came as much from internal as external drivers. Financial and geopolitical crises caused major

shake-ups – losses in both trading and hedging. In some cases, substantive regulatory interventions were implemented. Following the Great Financial Crisis new EU and US laws forced bilateral settlement of OTC energy trading to clearing on exchanges. At the same time, US regulators put an end to proprietary trading by US banks in commodity markets.

Central clearing party services, already existing for other OTC markets, quickly adapted to the growing variety and volume of energy market contracts. This adaptation brought about transparency and competitiveness, leading to a rapid growth in liquidity and attracting new players as OTC trading definitively shifted to the screen. The collapse of oil prices in 2015 expedited the withdrawal of banks, paving the way for proprietary traders.

While the surge in OTC trading volumes is substantial in energy market terms, it still represents only a fraction of the larger OTC financial markets today. Nonetheless, this exposure has allowed for the development of cross-market relative value trading, targeting specific oil market segments against industrial sectors such as petrochemicals, aviation, or shipping.

Enter Greg Newman. Fresh from University College London with a biochemical engineering degree in 2012, he joined Mandara Capital, a global commodity derivative trading company. He discovered a market in full transformation and eventually became the head of the crude and light distillate derivatives desk. It was during this time that he met Omar Kayaam, his future fellow founding partner in Onyx Capital, which they established in 2016. Onyx Capital's streamlined approach to derivatives trading and hedging has enabled it to capture a significant market share.

Greg Newman, and Onyx, thrive on their rich accumulation of OTC derivative market trade data. Their modelling demonstrates the degree to which trading drivers have extended over the past decade from pure physical supply & demand indicators to the internal balances and flows within the paper markets. Statistical regularity in behaviour arises as the number of trades and traders increases with the financialization of these markets. In a mid-2023 Onyx “World of oil derivatives” podcast, Greg likened the market to a spider’s web of spread trades that connects all the OTC and futures contracts so that a move on one reverberates across all the others. The illustration is remarkable.

Some would prefer to isolate oil trading to purely physical transactions, eliminating any undue influence and volatility driven by financial actors. However, the presence of the latter has substantially improved market

liquidity, price discovery and hedging possibilities over the past decade. Today, the OTC markets help increase physical market transparency as an input to the work of price reporting agencies. Trading derivatives via the e-window and use of clearing now delivers pricing as consistent as that for futures. Companies like Onyx winnow out inefficiencies, narrowing the bid-ask spread and ensuring an orderly pricing of contracts.

Greg envisages widening the scope of players in the OTC market, boosting liquidity by bringing in retail hedgers and retail investors. But this requires deepening their knowledge and understanding of oil markets. This book aims to play that role. As such, I recommend this exhaustive review of the market and its mechanisms to industry professionals, particularly those not trading, policy analysts seeking to broaden their understanding, as well as young graduates aiming to become traders and confirmed financial traders that have yet to delve into the oil market. Greg's simple, precise and straightforward style makes this detailed, technical text eminently readable.

The book provides a thorough overview to understand the functioning of the oil market and particularly the financial side focussing OTC derivatives. It covers historical developments on the physical side of the oil industry as well as contemporary detail on its financial sphere. It provides the background needed in vocabulary as well as each of the different product markets. It explains how trades are done and undone, strategies for holding a position in the market, clearing mechanisms, settlement mechanics, a discussion of regulation, etc. But most importantly, Greg shares a wealth of his personal experience that illustrates and animates the text.

I hope you find the book as useful as I do and enjoy the work as much, or more.

PREFACE

This book is about the mysterious, exciting and complicated virtual world of trading oil derivatives.

I've been in this world for over ten years and have been fascinated by how it works, the people who inhabit it and its explosive growth. For a long time, I've wanted to tell the story of how it developed and how it operates.

At Onyx, the trading and services business I started with my partner Omar, we have a constant stream of new colleagues keen to make a career at the trading desks. We introduce them to this complicated world, explain new concepts and processes, and train them in the skills and disciplines necessary to become competent but reliable practitioners of our dark arts. Of course, not all succeed: it's a demanding business. But most do, and I firmly believe there are two reasons for this: we recruit thoughtfully, selecting the candidates with the best mix of personal skills, and we train methodically to ensure they thoroughly understand how the business works. That's a major reason for this book: to teach those wanting to join the world of oil derivatives.

But many people familiar with trading are also interested in examining how this market works more closely and are keen to participate. So I hope I can explain what they need to know before they launch into trading – it has its own style and many pitfalls for the unguarded.

And finally, I hope that what I say will be interesting and useful to journalists and financial authors, who may understand how the market operates but want clarity on some of the more esoteric details – even seasoned traders in the physical market might find it helpful to see their activities through the eyes of the traders in 'paper barrels'.

I think that anyone interested in commodities trading should also be interested in trading in oil. As with the commodity markets, a few huge companies motivated to keep the oil market to themselves have historically dominated it. For a long time, they have been successful, but I passionately believe that information technology, new processes and a widening interest are loosening their hold. The market is becoming more reactive, responsive and transparent – in a word, democratic. I want to help.

ACKNOWLEDGEMENTS

As I embarked on the journey of writing this book, I struggled to decide the direction it should take. Should this be a recollection of my trading experience, or should it be a rigorous textbook? It was never about how many words I could get onto a page, but rather how to collate my thoughts into a coherent narrative that would empower the reader as they went on to trade, analyse or merely observe the oil derivative market.

When I first approached a professional copywriter for this, the initial response was that it was far too complicated and they wouldn't feel comfortable guiding me. In stepped my father, Michael, who, in his mid-seventies, is still bailing his son out of trouble. Dad helped me, as an outsider to the industry, really get to grips with what I was trying to achieve. He dug deep into this world with me and helped create a structure to allow things to flow. He was very helpful with the initial copy before it was passed to Proofed.com, who excelled in professionalism and efficiency.

Having mentioned my Dad, I should also mention my brother, Chris. His experience in physical oil and previously as a salesman at an investment bank was my introduction to the industry and shone a light into areas that weren't open to me. Undoubtedly, his experiences helped shape this book.

Beyond my immediate family, I want to acknowledge my second family, the team at Onyx, and of course, no one more than my business partner Omar Kayaam. Omar has always given me the freedom to experiment and has never held me back. Even when he hasn't fully agreed with or grasped what I've been trying to achieve, he has always backed me. Without him, this book would certainly not exist.

Next, I acknowledge the traders at Onyx, who helped me double-check things or verify points for accuracy. Manpreet Sandhu, who took over the crude desk from me as I moved out of trading, has been on hand to check passages that refer to the North Sea and speculative trading. Benny Williams looked over various points to help me flesh out the logic, tackling the more conceptual topics, like his observations of the Power market through the lens of a market maker. Also helpful have been Bart Pratt on the fuel oil desk, James Todd on Gasoline and Dylan Prod'Homme Bennett, who has a particular knack for directional trading so early in his career.

Outside of trading, Adam Butler, who runs the data team, created a visual representation of the network that holds the financial oil market together, which I used in this book. He has been pivotal in forming the Onyx Commitment of Traders dataset, which has allowed the research at Onyx to blossom, quantifies and visualises the philosophy of how we view markets at Onyx and underpins important concepts in this book.

In the same vein, Ehsan Massah has been influential in developing research at Onyx, building the team and relentlessly driving improvements. At the same time, he has been the leader in introducing some of the newer concepts tackled in this book to the industry. Consequently, we are going from strength to strength, which presents the ideal opportunity to write and release this book. Ehsan also assisted directly, reviewing my writing to ensure accuracy and identify gaps.

Martha Dowding, a new member of the Onyx research team, has also been highly valuable. Martha tackled much of the work on references, an arduous task. Martha went through the book with me in its entirety, like Ehsan, pointing out areas for improvement, helping me put the finishing touches to areas and helping form graphics. I will always remember the 5 am starts with Martha to put in this time before the working day. These times were vital for me to shape this book into the desired form. The other team members, Audrey Byrne and Vincent Wu, have also influenced the book, particularly by providing examples of positioning and flow analysis.

I would also like to thank Miles Hemsall, who has been the bedrock of the creation of the Onyx research business and, like my brother, has taken me through areas in which I have no experience. A crucial area is supply and demand analysis, which is firmly rooted in the oil market and in which Miles has long had cutting-edge involvement, having built multiple commercial divisions in the largest and best-known research houses from practically the ground up. Miles has been a mentor and the partner I needed to go out on a limb and create a new research concept while facing the inevitable backlash and uncertainty that came with it. This book has been able to flow in part due to the insight he imparted to me over the years. But it is also important to acknowledge how he backed me to continue pushing forward against the tide of the industry status quo.

The Onyx media team has been of utmost importance. Ijeoma Ogiemwonyi was responsible for all of the book's design elements from the front cover to the last page. She took all my crass drawings and turned them

into legible, clean graphics and for that I'm grateful. As media lead, Mhairi Beveridge oversaw the project management of the book, from the draft to the copy editors all the way through to the book's release. This mammoth task involved liaising with the copyeditors, including a lot of back and forth; coordinating the team; identifying proofreaders and getting it over the line. Being a self-published book, this part of the journey involved the highest risk and was no doubt daunting, so I am hugely grateful for the work done.

I sincerely hope that those in the Onyx team who assisted directly and even those who didn't but were nonetheless vital to the Onyx journey and, therefore, an influence on this book can in some way take some pride in the development and release of the World of Oil Derivatives.

Outside of Onyx, others have proofread the book. Of particular support has been Joel Couse, who graciously agreed to write the foreword. Joel's enthusiasm for the subject gave me the confidence I needed to push through, and I am grateful for the discussions we had and the beginnings of a longer-term friendship – I hope.

Kurt Chapman, the godfather of market-making in the Dated Brent market (and the market I spent most of my trading career in), influenced the book indirectly, as someone I've learnt from, but also directly, having kindly agreed to review the book. After a time in competition with Kurt, it has been enjoyable to get together and discuss the market to help solidify some concepts in this book.

A close friend of mine, and brilliant systematic trader, Robert Wolstenholme, has been a great sparring partner over the last few years over what really drives markets in oil and beyond. I very much appreciate the conversations we had, which undoubtedly supported the creation of this book. Rob also helped directly by rephrasing a few areas, and his review of the book was of great use.

Nathan Ohayon was also kind enough to review the book – with his promise to be honest if it wasn't any good! Nathan is another one who has been a great sparring partner over the years concerning understanding the drivers of oil markets, and his view, as ever, has been important to me.

Paul Chapman leads a fantastic industry podcast and has tackled endless complexities in both the financial and physical markets. Our podcast together a year before starting on the journey of writing this book contributed significantly to its timing. Paul excelled at drawing out the themes of financialisation, and getting his views on how this book panned out was

important to me.

Harry Tchlingurian was my first mentor in the industry, having worked as an intern in his research team at BNP Paribas. Harry guided my early understanding of oil and its dynamics and, alongside my brother, provided the basis for my knowledge of how banks in the oil derivative market function. Like Joel, Harry embraced the topic of financialisation, and his personal support for my voice in this area has been a great confidence booster.

So many people have influenced my time in the oil markets, and many deserve mention who aren't included here. My sincere thanks to all who contributed. This book would not have been possible without you.

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1

INTRODUCTION

History of Oil and Oil Trading

The story of oil begins in prehistory; indigenous people in many regions of the world were aware of this black stuff which seeped out of the ground. For much of its life, it has been treated as a nuisance – nasty viscous stuff which sticks to your feet. It was useful as an early antibiotic and disinfectant; it was used to waterproof boats.

Oil grew to be the vitally important commodity it is today in the early 1800s in the United States of America (US) when it was discovered that although it was a poor substitute for whale oil for lighting – it is too smoky and smelly – it could be distilled into a lighter product which burnt cleanly and didn't smell too bad: kerosene. Kerosene lamps became the primary light source after dark – before coal gas was piped into cities and towns.

Of course, the invention of the internal combustion engine catapulted oil to stardom. When Henry Ford adopted the internal combustion engine for his mass-produced vehicles, a ready supply of distilled oil (gasoline) was essential. Interestingly, Ford could have chosen to power his Model T with electricity, but lead-acid batteries were far too heavy. Ladies didn't like internal combustion engines: they were too noisy and smelly, and you had to start them with a cranking handle, but gasoline's unbeatable energy density won – and does to this day.

After that, oil-driven engines were everywhere: in cars, lorries and ships – Churchill had the British Navy convert its warships from coal to oil-burning – and oil was used for electricity generation. So there was a growing demand for oil and its products, first in the US and quickly spreading through the rest of the world.

Because Rockefeller pioneered the vertical integration of oil production – oilfield drilling, transporting by rail, refining and distribution – it became widely available throughout the country and, therefore, the energy provider of choice. Demand soared. New oilfields were discovered, and everyone rushed to exploit them. An entire industry was created, the giants of which are around today: Standard Oil, Chevron, Texaco and others; the British

joined in to exploit oilfields in Russia, Iran, the Middle East and Asia (with the help of US oil companies).

You can read this fascinating story in books like *The Prize*^{[\[1\]](#)} and other books I will reference later. But, for now, the main point is that around 1850, oil went from not being even a footnote in human history to the most important commodity in the world. It wasn't just useful – to industry, to transport and later, increasingly to pharmaceuticals and plastics – it was vital. It became a source of great wealth and embedded in politics. Buying and selling oil became a sophisticated and growing market, which I have more to say about in the next chapter.

Politics dominated the oil business in all its forms as demand grew meteorically, spurred first by the world wars and then by consumer demand in the years of reconstruction and plenty that followed. Throughout this period, big US oil companies monopolised the technology, pricing and, to a great extent, availability. The market for buying and selling was the domain of these oil majors and specialist brokers who traded on long-term bespoke contracts.

Then, countries outside the US that supplied oil to the industrial world began to want more control over their major national asset. When the Organization of the Petroleum Exporting Countries (OPEC) refused to sell oil to the US in the wake of the Arab–Israeli war of 1967, old certainties began to weaken. The world wanted ever more oil, and as a result of these weakening certainties, independent traders saw the opportunity for a 'spot' market to supply oil without the traditional suppliers and their contracts. This market made it possible to trade oil for instant delivery between any two parties. But to develop fully, this market had to provide several assurances: the creditworthiness of the participants, the grade and quality of the oil product and the logistics of delivery and payment. And that, for us, is where the oil market became interesting.

The Physical Oil Market

What is Oil?

Crude oil is a dark, viscous, foul-smelling liquid found in underground deposits in many parts of the world, though it was in western Pennsylvania in the US that it was first commercially exploited. In its natural form, it is not

very useful: it has to be processed or refined – to produce commercially valuable products.

Every refinery publishes a ‘slate’ of the distillates it will produce and in what quantities based on the kind of crude oil it distils – or ‘cracks’. Oils from different parts of the world have different characteristics and are classified broadly as light or heavy depending on density, and sweet or sour depending on sulphur content. A light, sweet crude, like West Texas Intermediate (WTI) or North Sea Brent, generally commands a premium price over a heavy sulphurous crude, like Saudi Arabian Arab Heavy.

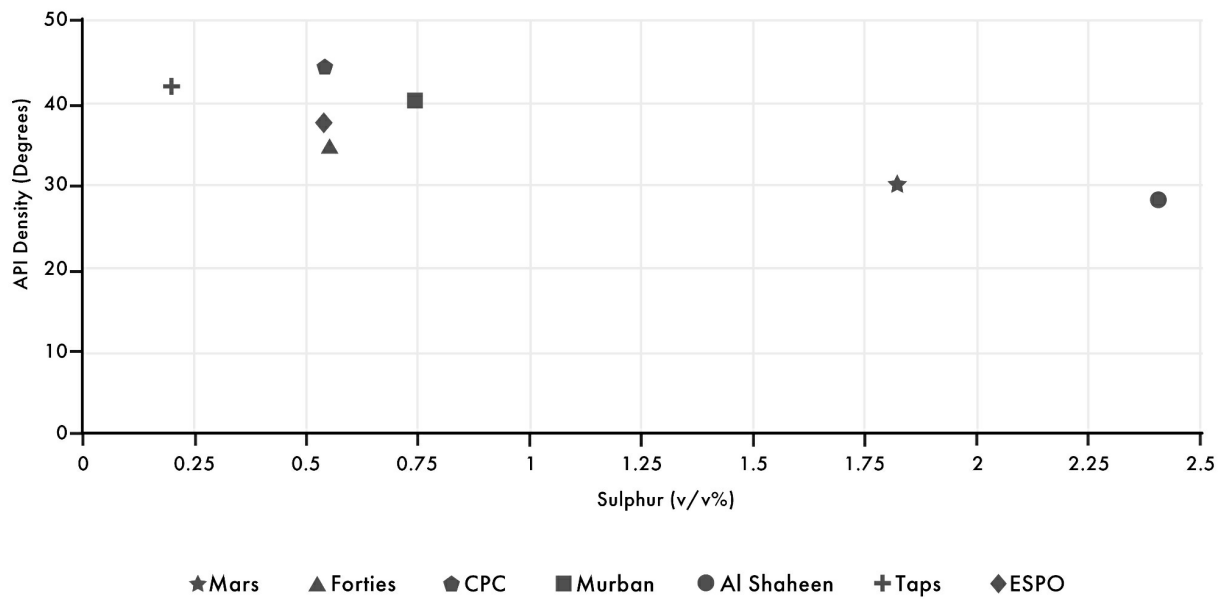


Figure 1.1 Sulphur content and density of various Crude Grades

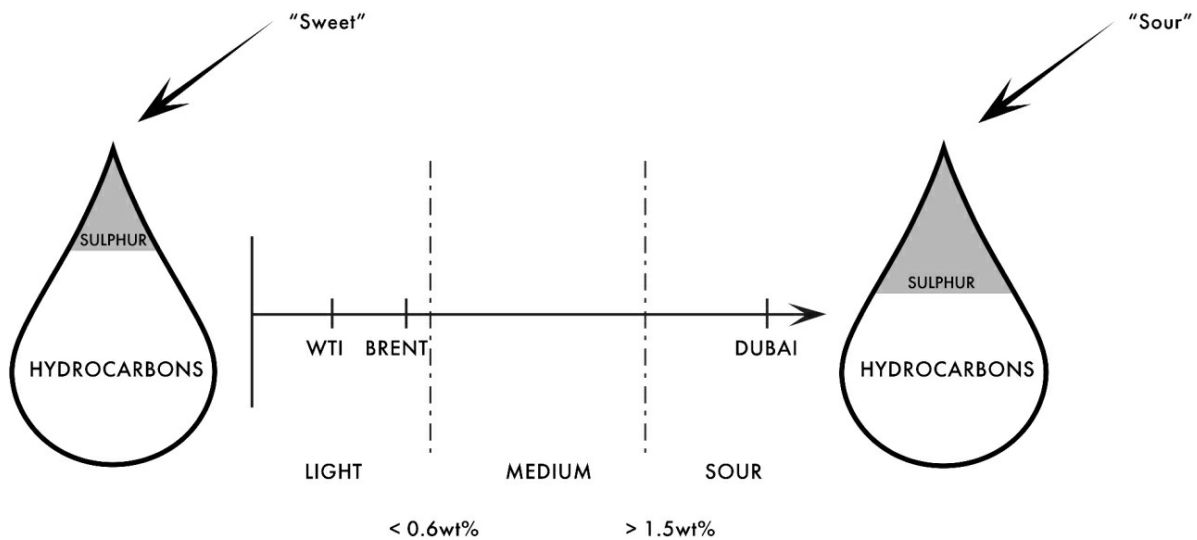


Figure 1.2 The Sweet and Sour Scale of Crude Oil

Crude oil from an oilfield is transported to a refinery and, from there, is boiled in a closed system (that is, the oil is sealed from the outside air). During the boiling process, the crude evaporates and flows through a network of pipes. As it cools, most of it (not the gases like butane) condenses back into a liquid, first in a light, almost colourless form (gasoline and naphtha),

then into heavier and denser liquids (kerosene, jet fuel and diesel) and finally thick residues ('bunker fuel' used in large ships and asphalt). All these distillates have commercial uses.

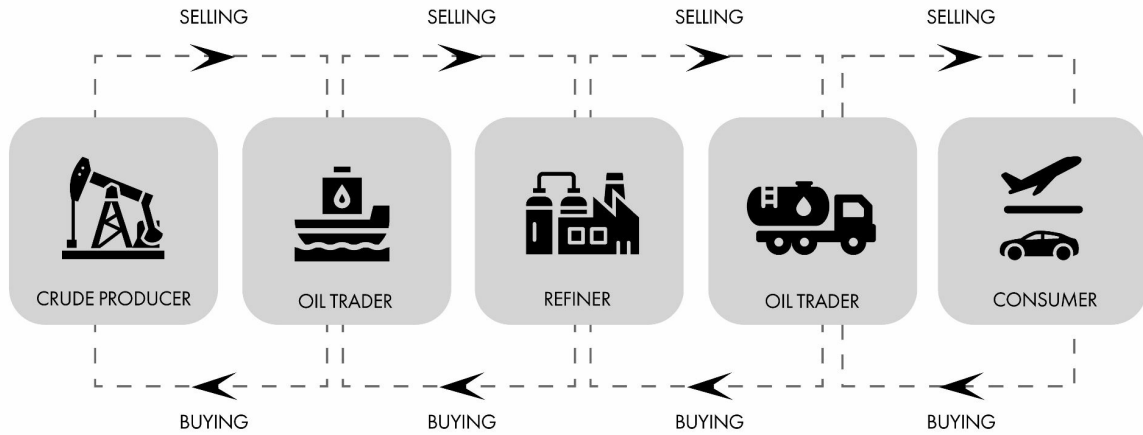


Figure 1.3 Producer-to-consumer Chain

Oil Market Sectors

The oil market is divided into three main sectors:

- The upstream sector encompasses exploring, drilling and producing crude oil and natural gas.
- The midstream sector covers transporting and storing crude oil and natural gas from the production site to refineries or processing plants.
- The downstream sector includes refining, processing and distributing crude oil and natural gas into various products like gasoline, diesel and jet fuel.

These end products are then sold to end users, such as consumers or industrial customers. Each sector plays a crucial role in the functioning of the oil market and energy supply to global consumers and businesses. But for the purposes of this book, we focus on the activities of downstream players and their participation in the financial oil markets.

Buying Oil

Let's travel back into the world of the 1950s and imagine that you are a paraffin (kerosene) retailer for domestic heaters and lamps. You need a regular supply of this fuel oil to be shipped to a depot, from where you can move it in trucks to your retailers nationwide.

How do you buy the oil? It's impractical to pick up the phone and call your local refinery, even if you know where it is. So instead, you call the marketing and trading arm of a physical oil company, like BP, or an independent broker, and you ask for 1,000 barrels to be delivered to your storage tanks. Simple enough in principle, but there are already many factors at play here: the delivery date, the payment method, the product quality and, of course, the price.

All these and more will need to go into the contract, assuming BP trusts your creditworthiness, so a bank and a credit team will need to be involved. And the contract you sign with BP to get your kerosene will be unique to your relationship with BP – if you want to change supplier to Texaco, you'll need another agreement.

Now, let's turn this scenario around. Think of yourself as an employee of the refinery. You want to find customers for your oil products, and the

process is much the same as we imagined before. This method of buying oil will work and, in fact, was the standard way of trading from the 1900s to the 1960s.

But what happened then?

- Politics
- Demand

A Brief History

In the early years of oil trading in the mid-19th century, the US first supplied kerosene domestically and then to Europe in the 1860s. Then, around 1873, the entrepreneurial Nobel brothers began the industrial-scale exploitation of the oilfield on the western shore of the Caspian Sea in Azerbaijan, which had been known for centuries but by 1902 was producing more oil than the US^[2]. After that, oil was found all over the Middle East and Asia. These oilfields were developed by corporations like the so-called Seven Sisters, the giants of US oil, and British and French companies in partnership with their governments.

By the 1960s, there was a growing movement in oil-producing countries to take back control of what was often their primary economic asset. OPEC was formed in 1960, and its embargo on oil sales to the US following the Arab–Israeli war in 1973 sent shock waves through the market. By then, many oilfields were pumping many grades of crude, many refineries were producing a widening range of petroleum products of varying quality, there were tankers of increasing size and there was a growing network of pipelines to connect oilfields to refineries. Powering it all was an ever-increasing demand for this liquid gold. So the market for buying and selling oil had to become more responsive.



1.4 Timeline of Oil Discovery and Development

An essential step along the way was the emergence of independent traders who would buy oil from refineries and sell it to customers. The major oil companies which so dominated the physical supply of oil were (and are) vertically integrated oil companies (IOCs) with upstream, midstream and downstream operations, and they wanted to control the whole supply chain – Chevron simultaneously pumps oil in Indonesia and sells petrol through its Texaco forecourts in Ireland. So the independent trader would, at least notionally, source gasoline from any refinery at a price set to make an attractive margin. The market evolved quickly to where oil companies could buy or sell physical oil under a spot or a future-delivery contract. In the beginning, this was all traded bilaterally, but with the formation of the commodity exchange, the efficiency and sophistication of trade would be much improved.

Commodities Exchanges

A commodities exchange is an old idea: the first one we'd recognise as

such might have been the Bourse in Antwerp in 1531, which was set up to enable shares trades to finance voyages to the East Indies for silk and spices. The Royal Exchange in London was modelled on the Bourse in Antwerp, as were many others, with traders and brokers using an ‘open outcry’ system of buying and selling face to face across a trading floor or ‘pit’. Almost all of today’s exchanges have long since migrated to electronic dealing, although the London Metal Exchange still (as of 2023) uses its ‘ring’, as it has for nearly 150 years.

An exchange performs several functions:

- It **provides a location** for buyers and sellers to trade standard contracts (physically or virtually).
- It **assures the trustworthiness** of the participants (who must be accepted by the exchange as authorised traders).
- It **provides the means for ‘price discovery’** by providing benchmark prices for the contracts it supports and monitoring and displaying the prices for all deals.
- It **provides access to liquidity**, which is the certainty that there are always contracts for purchase or sale.
- It **provides a clearing service** by matching buyers and sellers for all contracts traded.

For an exchange system to work, there must be standard **contracts**. These specify the commodity and its quality, the quantity to be traded, the settlement date, delivery details and, of course, the price. But it’s important to realise that the contract doesn’t set the actual price – the participants do this through their trades – but it does set the price increments or ‘ticks’. For example, on the Chicago Mercantile Exchange (CME), the crude contract (referred to by its ‘ticker’ code ‘CL’) is shown in Figure 1.5^[3].

CONTRACT UNIT	1,000 barrels
PRICE QUOTATION	U.S. dollars and cents per barrel
TRADING HOURS	CME Globex: Sunday - Friday 5:00 p.m. - 4:00 p.m. CT with a 60-minute break each day beginning at 4:00 p.m. CT TAS: Sunday - Friday 5:00 p.m. - 1:30 p.m. CT CME ClearPort: Sunday 5:00 p.m. - Friday 4:00 p.m. CT with no reporting Monday - Thursday from 4:00 p.m. - 5:00 p.m. CT
MINIMUM PRICE FLUCTUATION	0.01 per barrel = \$10.00 TAS: Zero or +/- 10 ticks in the minimum tick increment of the outright
PRODUCT CODE	CME Globex: CLCME ClearPort: CLClearing: CLTAS: CLTTAM: "CL1", "CL2", "CLL"
LISTED CONTRACTS	Monthly contracts listed for the current year and the next 10 calendar years and 2 additional contract months. List monthly contracts for a new calendar year and 2 additional contract months following the termination of trading in the December contract of the current year.
SETTLEMENT METHODS	Deliverable
TERMINATION OF TRADING	Trading terminates 3 business day before the 25th calendar day of the month prior to the contract month. If the 25th calendar day is not a business day, trading terminates 4 business days before the 25th calendar day of the month prior to the contract month.
TAM OR TAS RULES	Trading at Settlement (TAS) is subject to the requirements of Rule 524.A. TAS trades off a "Base Price" of zero (equal to the daily settlement price) to create a differential versus the daily settlement price in the underlying futures contract month. The TAS clearing price equals the daily settlement price of the underlying futures contract month plus or minus the TAS transaction price. Trading at Marker (TAM) is analogous to Trading at Settlement (TAS) wherein parties are permitted to trade at a differential to a not-yet-known price. TAM uses a marker price, whereas TAS uses the Exchange-determined daily settlement price for the underlying futures contract month.

Figure 1.5 The Chicago Mercantile Exchange Crude Oil Future Contract Specifications

But how can price discovery begin? How do traders know where to start negotiating a price? The commodities exchange provides access to benchmark prices. Brent for international trades, WTI in the US domestic market and Dubai in the Middle East and Asian markets. The CME establishes the price of the WTI futures contract, while the Intercontinental

Exchange (ICE) sets the price for Brent futures.

A trader on the CME can look up the price quoted for this contract and buy or sell it with no intention of actually taking (or making) delivery of those 1,000 barrels of oil. Instead, they might simply speculate that the oil price will rise, so they can buy the contract now and sell it in the future to make a margin on the transaction.

Figure 1.5 shows a contract in WTI crude futures on the CME. If you're an authorised trader on the CME, you could elect to buy 1,000 barrels (the minimum contract size) of a contract month in the future. For example, you could buy WTI crude futures for December 2023 delivery, which will be available to purchase at the prevailing market price. Let's say you buy 1,000 barrels at a price of \$77.08 per barrel (or \$77.08/bbl using the common abbreviation); you will have a notional value of WTI crude futures of \$77,080. You could hold this contract until the 'first notice day' (20 November 2023 in this example), when you must take delivery of that oil and find something to do with it or make a cash settlement for the contract value. Before that, you can sell it at any time at the price prevailing when you make the sale. You could win or lose. Quite often, you'll roll the contract – close your position by paying the difference between the price you bought at and the current price and buy the contract again with a later maturity date.

You will notice that I described a futures contract in this example because spot contracts – delivering physical products within days – are of little interest to speculative traders when nearing the contract expiry (and physical delivery). So the exchange is mostly concerned with futures and other derivatives, as we'll see when we explore that market later. And don't forget that the exchange is a business – it makes money by taking a percentage of every trade. So it wants to maximise activity, which is provided not by trades in physical oil (though those do occur) but by the huge volumes of speculative trades in derivative contracts.

An efficient exchange provides the ability to find contracts to buy and the knowledge that you will always be able to sell in a liquid market.

Derivatives

Most trades on commodities exchanges are not directly for the purchase or sale of the underlying commodity. In fact, less than 3% of futures contracts settle as physical delivery^[4]. Most trades are made for hedging or are simply

speculative. This is true for all derivatives in commodities exchanges, especially oil, which is the most commonly traded commodity^[5].

The first oil contract listed was on the NYMEX (the New York Mercantile Exchange, which was taken over by the CME in 2008^[6]). This was a Heating oil futures contract, in 1978. A **futures contract** is a prime example of a derivative: it is useful to producers and users to secure the price of an oil product, but because it can be bought and sold without ever taking physical delivery, the physical commodity is irrelevant to the speculative trader who is buying from or selling to the exchange. In the oil market, futures are settled monthly on a specific day related to the loading times of the type of oil to which they refer.

Other derivatives from the wider world of commodities trading are also used in oil trading:

- A **swap (or Block Future as it is called when cleared on the exchange)** in oil derivatives is a contract that settles daily to a price reporting agency (PRA) assessment.
- An **option** is a contract which one party buys for a fee (the premium) for the opportunity to buy or sell a contract at a specific price in the future.

At their core, derivatives are mechanisms for trading risk, time, cashflows and interest rates.

A market is defined by the activities of its participants. Market dynamics are influenced by the following:

- The volumes traded
- The contract types
- The contract prices
- The individual 'flows' – the trading pattern – can significantly affect price discovery and market conditions through the trading day and over the long term.

The main participants are the following:

- **Producers** prospect for, drill and pump oil.
- **Refiners** distil oil into products like gasoline and, increasingly via hydrocracking, into petrochemical precursors like propylene.
- **Oil traders** match buyers and sellers of physical products and arrange delivery logistics.
- **End users** like airlines, merchant shipping companies and chemical manufacturers consume the end products.

- **Market makers** buy and sell contracts and ensure their liquidity (availability).
- **Hedge funds** speculatively trade in derivatives to profit from price fluctuations.
- **Institutions** like pension funds use oil derivatives to balance risk and profit.
- **Retail traders and individuals** generally trade for themselves, speculating on this dynamic and well-governed marketplace for profit.

The Financialisation of the Oil Market

The major theme of this book is the dramatic growth of the financial trading of oil contracts and its effect on the physical market where oil is pumped, refined and used. For much of its history, oil was traded through bilateral contracts between two parties at a negotiated price. As a result, the oil market was dominated by the major producers and traders and the large banks which financed them – like JP Morgan Chase and Goldman Sachs. To a large extent, that's still true today – you still need much capital to trade in the physical market.

The oil price shocks of the 1970s and the financial crisis of 2008–2009 set in motion huge changes in the way oil is traded due to increasing demand, uncertain supply and banks' decreasing capital – not least because US banks were forbidden to trade on their own behalf. As a result, independent traders stepped in, and standard contracts evolved to allow trading through exchanges. Following the lead of the commodities and equities exchanges, the oil exchanges began to list increasing numbers of standard contracts for traders to buy and sell – without demanding large capital accounts. So interest in these contracts rocketed as producers, speculators and, increasingly, individuals used them to manage risk and as opportunities to make money.

The world of oil derivatives hardly existed ten years ago. Now the volume of trades in this market is many times larger than in the physical market, with the volume traded in Brent futures alone being fourteen times the daily global crude production^[7]. I argue that this rise in financial volumes has become not just a factor but the dominant influence on the consumer oil price. The *democratisation* of the market, which has taken the power of

pricing away from the major oil producers and introduced transparency, governance and data-driven financial instruments, has brought remarkable efficiencies in how this precious commodity is bought and sold. It's a journey that is not finished yet.

The rest of this book deals with trading in derivatives and, specifically, oil-based derivatives. I will explain how the trading process works and try to give some guidance, based on my experience at Onyx Capital Group and earlier companies, about how you can participate in this market successfully. It's a market in which the prepared investor can do well. It's also one of the best ways for a naïve participant to lose money – a lot of money.

2

FOUNDATION

In this section, I want to describe all the components of the oil derivatives trading world. If you're new to oil trading, this will give you the background you need to understand the more detailed sections to come. If you're an experienced trader, this should all be familiar, but confirming your understanding of my viewpoint will still be useful.

Commodities Trading

Oil is a relatively late entry into the long list of commodities humans have traded since the earliest times: soft commodities like soybeans, corn, coffee and pork bellies and hard commodities like metals. The key characteristic of commodities is that they are standardised, so they can be bought and sold sight unseen (unlike a racehorse). The benchmark contract makes it possible to trade oil with varying characteristics and its distilled products.

The main participants in the oil commodities market are the producers who find, drill and pump oil and ship it to refineries; the traders who buy the various distillates the refinery produces and sell them to users in industry; and the network of facilitators who enable the whole process – the banks, regulators, exchanges and, just as important, traders who help match buyers with sellers and trade in risk.

It's easy to see why producers and users want to secure firm pricing for the oil they sell or buy. The concept of hedging against volatile price movements in the market is as ancient as the first futures contract. But as soon as a commodity – oil – can be represented by a tradable contract, a whole world of possibilities opens for buying and selling contracts (and further contracts defined by and derived from them) rather than the underlying physical product.

This accelerating interest in trading derivative contracts is what I call the financialisation of the oil market. It is becoming increasingly more significant to the pricing of this commodity as well as its supply and demand because the

people who expose themselves to price risk in the market and take positions in a commodity they have no interest in possessing – the speculators – are becoming the main drivers of the oil markets and, therefore, the price of its products like gasoline at the petrol pump.

Accessing Oil Derivative Trading

If you want to buy and sell oil in the physical market, you will need to deal through a marketing and trading arm of an IOC like BP or an oil trade house like Vitol. It is practically impossible for an individual to buy or sell oil. These institutions govern the whole market and only deal with entities with plenty of capital. An organisation like an airline or a major transport company will buy oil through an oil trader, and investment banks will provide finance. This is bilateral trading: the parties know each other, and a bank guarantees a line of credit.

An individual wanting to speculatively trade derivatives will need to open an account with a retail brokerage, where the broker will take the credit risk on the individual, much as a bank would with a corporate entity (after due diligence, KYC checks and confirmation that the client understands the trading process and has the funds). People who want exposure to the oil market as part of a general investment strategy will typically do this through a hedge fund or an investment fund which will have a credit agreement with an authorised participant on an exchange who will provide a bilateral price for the oil derivative contracts you want to trade secured against the fund's liquidity. Brokers make money by charging a fee on each transaction or, more commonly, through the bid/offer prices they quote on each trade.

Hedge funds and specialist traders like Onyx deal mainly in derivatives by accessing exchanges through an exchange member who will (in concert with other exchange members) guarantee any default by the trader. For example, the exchange member may be a bank granting credit to trade bilaterally through them and will tightly monitor the trader's performance against the account and the risk exposure.

You can access the market directly or through a bilateral line with an investment bank-type institution. You need to consider the terms for doing one or the other as you define your business strategy, which is discussed in more detail in the Hedging section of this book.

It is difficult to enter the oil derivative market, and when you look at

how difficult it was for traders to get access, it is easy to see why. For example, the exchange-traded futures WTI, Brent, Gasoil, Heating oil and RBOB have been available on exchanges for over 25 years, but to trade them, you need enough cash to cover the deemed trading risk (known as the margining requirement), which can be very restrictive when you start to trade in typical oil market volumes.

Oil swaps pre-2012 were even less accessible: you could only trade if you had an ISDA (pronounced *is-dah*; International Swaps and Derivatives Association) agreement with a bank or counterparty. These OTC-traded oil swap contracts are much more closely linked to the physical world than the exchange-traded futures and provide reference prices for the world's delivered oil and end-user products. There is a scene in the film *The Big Short* (2015^[8]) about the financial crash of 2008, where two owners of the hedge fund Brownfield Capital are looking for an ISDA with JP Morgan Chase, which they describe as a way to access the 'big boy table' and 'make high-level trades not available to stupid amateurs'. They are turned away for 'only' having capital (or assets under management in the hedge fund world) of \$30 million. In fact, the JP Morgan Chase representative says they need to have at least \$1.5 billion – so there is simply no chance they will gain access.

Just like the Brownfield Capital traders, small corporates or wannabe traders have sat on the sidelines of exchanges wanting to trade oil swaps for a long time – until exchanges started listing swaps and exchange-traded futures and barriers to financing were brought down from 2012 onwards.

In the oil world, the strength of the balance sheet is used more commonly than precious cash reserves, especially for securing finance for hedging transactions. Physical assets or company stock equity can be used as collateral for the finance needed to trade oil derivatives. Many factors go into evaluating the risk of providing a line of credit – a major one is knowing the person making the decisions on the other side and whether they are trustworthy. Credit providers, like banks, want to know your strategy and what you are trying to achieve, as they are well aware of the risk of ruin in trading oil derivatives. Just like in the scene from *The Big Short*, banks prefer to deal with customers with a brand and evidence of stability, which means more than impressive trading performance with a small balance sheet. But banks need to take risks to make a profit, and they began to see the opportunity in providing access to the oil market for smaller participants – at a cost.

Investment Banks

Before 2012, investment banks were heavy traders in oil markets and made money from providing finance to traders in return for off-market numbers (outside of the best bid/offer spread in the market) that the bank's trading team could use in its open market trades. When the market was opaque – difficult to get pricing and trading information – investment banks thrived and dominated through their perceived solidity and inside knowledge. This started to change in 2012 when trading activity moved from being dominated by the bilateral market to the exchange.

Investment banks have trading desks that trade freely in the oil market, using the banks' sophisticated systems and information technology to provide access to exchange prices and efficient, well-managed messaging systems and multiple phone lines. These banks are extremely difficult to compete with because they have dedicated lines to the largest trade houses and major producers and are in constant communication with brokers worldwide, giving their traders visibility of market prices. In addition, investment banks have armies of sales professionals to procure hedging trades, known as 'flow', after their administration teams have vetted and assessed the creditworthiness of investors requiring financing. Each trader will try to quote their best price against the market, and salespeople must gauge the highest (if the counterpart is buying) or lowest (if the counterpart is selling) price they can offer that will still allow them to complete the transaction while generating the highest possible profit. The trading team is happy because they can take that trade at a price which gives them a good margin and back the risk in the market for a better price. A salesperson records the difference between the trader's price and the price quoted to the counterparty as the sales margin, known as the 'CC' (client contribution).

In the early days, the salesman's job was simply to charge as much as possible and maximise the flow of trades. Unfortunately, this tended to encourage reckless trading behaviour, which caused tension with compliance teams in the bank trying to minimise exposure to defaults or fraud.

An anecdote from those days: *A bank salesman executed a jet fuel hedge through which an airline bought jet fuel swaps on an outright basis. When the 2008 crisis reduced travel demand, the airline quickly lost revenue. Nevertheless, it needed to finance the market-to-market loss on the swaps with the investment bank, which was calling in the variation margin. This*

loss became so large that the upfront cash required to maintain the hedge threatened to bankrupt the business. The airline CEO made this abundantly clear in an email to the salesman in which he wrote, 'If this is a serious request to call in \$80m in 3 weeks, the bank may as well take the keys'.

Investment banks swarmed in as the derivative world became more established, and many participants routinely traded. At one point, at least twenty investment banks worldwide were involved in the derivative market. At the height of the investment banks' dominance in the 2000s, there was a battle between the banks centred on attracting the best traders – or at least the most well-known – and trying to develop a name as the best in the business using those well-known traders as their brand representatives. Expensively developed research was given out free to banks' clientele – nowadays, this research is in the public domain. Investment banks and their bilateral business line struggled from 2012 onwards, and while some have continued to do very well, the market consists of far fewer banks. These days, there are also independent financing houses that offer services in competition with investment banks.

In trades where pricing is visible to both parties, the spread between the market price and that traded with the counterparty is referred to as 'credit spread' and is taken as fair payment for banks' costs and risk of default. When it is a fixed spread, each party should be satisfied as long as that amount is reasonable.

With increasing trade price visibility, companies are less willing to accept what the larger trade houses, majors and investment banks offer. Instead, they want to direct their own hedging strategy, which has increased the demand for at-market prices and standardised hedging options.

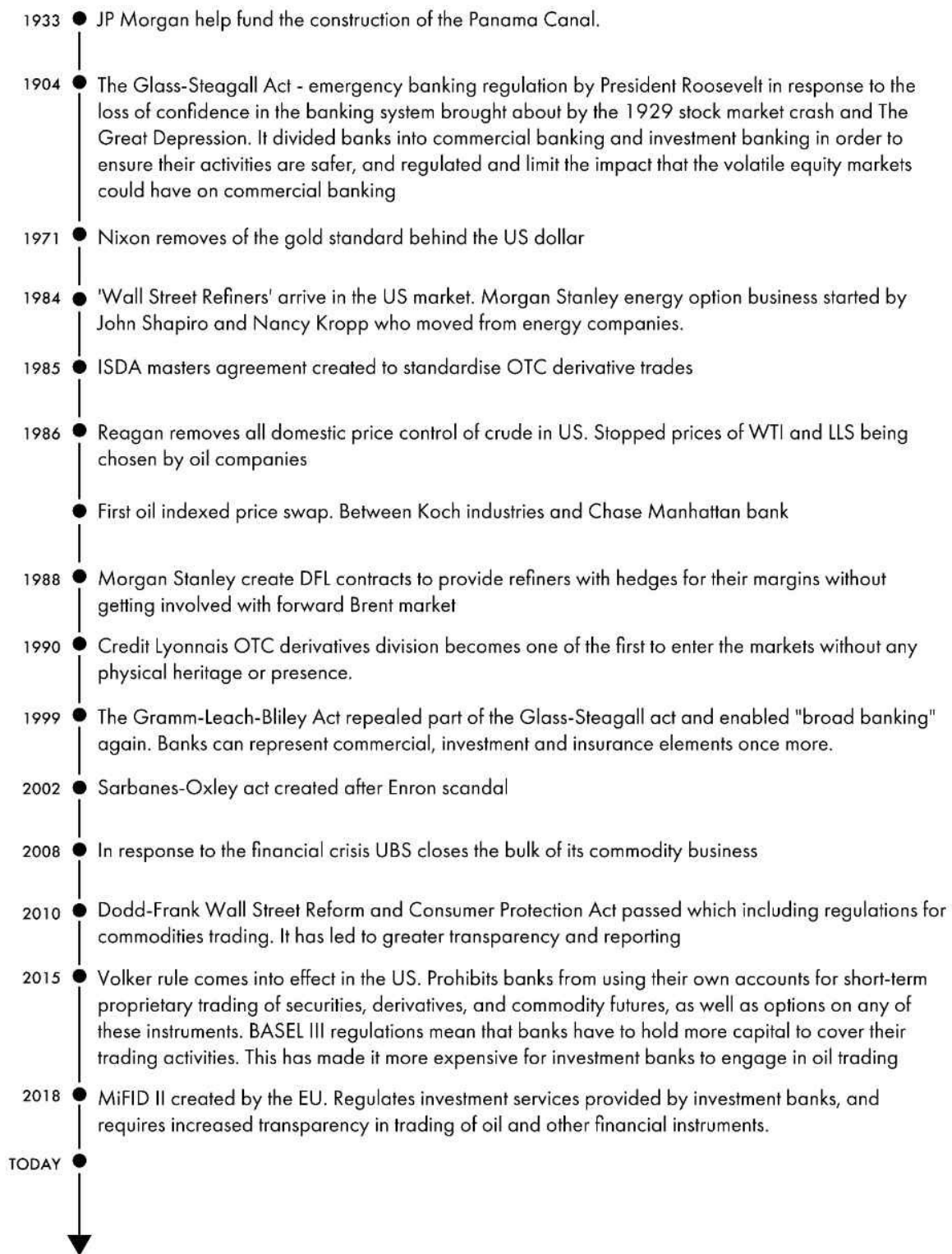


Figure 2.1 Timeline of Key Events For Investment Banks in Oil Markets

Bilateral Lines for Trading with Known Counterparts

Bilateral lines with counterparties is how the physical market functions. Traders in physical oil can trade physically delivered contracts on exchanges or convert existing derivatives contracts into physical contracts via the EFP (exchange futures for physical) contract mechanism.

An EFP contract in the Brent market, for example, is a mechanism that delivers the underlying physical crude (crude from oil fields Brent, Forties, Oseberg, Ekofisk, Troll and WTI Midland, shortened to BFOETM) under the STASCO BFOETM 2022 General Terms and Conditions^[9]. In these contracts, a large balance sheet (at least assets in the hundreds of millions) is deemed essential. For example, I tried to establish a bilateral line with a trade house to trade EFPs with Onyx some time ago, and I was essentially laughed out of the room when they checked our cash reserves. The EFP cargo was 600 KB (600,000 barrels), which at the notional price at the time meant we would have needed to ringfence assets valued at around \$40 million for one cargo. To trade in the more than ten cargos for the strategy I wanted to implement would have required assets valued at more than the entire business was worth.

Large trade houses and majors still sometimes prefer to trade financial contracts bilaterally, partly because the cost of capital for trading on exchanges has grown with rising interest rates and because the counterparty is known in bilateral trade. Since there is no regulatory obligation to report trades outside the exchange, trading bilaterally keeps the details of the deal private, which can be valuable in situations such as exiting a position.

Exchanges

Which came first, the contract or the exchange?

Of course, the oil business was built on contracts (in the Rockefeller era, not all contracts benefitted both parties). Even in the world of physical oil, these evolved to fairly standard, boilerplate terms. But these contracts represent (for the most part) bilateral deals which sit outside the exchange, where the counterparties face each other, and the default risk is carried by institutions like banks or governments.

Derivatives contracts are standard in almost every particular except price. This characteristic makes them tradeable between counterparties, and they can be offset with trades with other counterparties. The other is that they are predominantly traded through an *exchange*, which finds and matches entities wishing to trade and guarantees the creditworthiness of the counterparties who do not need to know one another. Then, at the expiry of the contract, it settles the accounts of each counterparty, moving funds between them to match.

One of the key features of an exchange is that it offers ready availability – liquidity – in the contracts it defines and offers for trade. It is critical that a participant can look for and buy a contract at any time with the knowledge that they will be able to sell it at any time. Offering a new contract is not a trivial decision. The exchange must be sure that it will be easily tradeable. There have been some notable failures – or at least damp squibs – simply because the market isn't interested.

The First Oil Derivative

The oil exchanges most often used are the CME, ICE and SGX (Singapore Exchange). They are modelled on the exchanges for commodities and financial instruments like currencies, stocks and bonds, which have existed, in some cases, for centuries.

In 1978, the NYMEX introduced the first oil derivative contract, a futures contract for heating oil, to provide a hedging option for this heavily-traded oil. However, the next contract, for WTI futures, didn't appear on the exchange until 1983 because it was necessary first to create an accepted method for establishing a price – a benchmark. Oil traders agreed on WTI crude oil as a pricing benchmark because it was heavily traded and of

consistent quality. Oil price volatility and the geopolitical situation made it a good time to introduce a new oil futures contract. NYMEX merged with the Commodity Exchange, giving it the resources and expertise to develop and launch the WTI futures contract.

Note: NYMEX is now part of the CME.

Exchanges are the centralised marketplace for trading oil derivatives. The main purpose of an exchange is to bring together as many participants as possible and impose order on the execution of trades with a framework of rules backed by government regulations. The largest oil companies and traders use exchanges to monitor pricing and trading activity in the financial oil market to help develop their trading strategies. The increase in the value of financial trades on exchanges worldwide and the information available from these trading venues has improved the visibility of trades and pricing, ensuring better market regulation.

Of course, not all traders celebrated this increased transparency, especially those in dominant positions before financial oil trading moved to exchanges. However, based on the increase in trading volumes and the number of market participants, this transparency has caused growth in the industry. It has also led to substantial improvement in the liquidity of previously sporadically traded contracts, making financial oil trading more efficient. As a result, the oil trading world is simultaneously becoming more open and transparent and increasingly tightly regulated.

There are a few main exchanges for the oil market:

- European products traditionally trade on the ICE.
- US products traditionally trade on the CME.
- Singapore-specific products trade on the SGX (petrochemical contracts and some fuel oil contracts).
- In Japan, a Dubai crude contract priced in yen trades on the TOCOM (Tokyo Commodity Exchange).
- In the Middle East, an Oman crude contract is physically delivered on the DME (Dubai Mercantile Exchange), which is useful for settling and delivering locally traded contracts.
- The Shanghai exchange is growing in influence and lists crude and product futures that are deliverable in China.

To trade on the exchange, you must either be a member or trade through a member. To become a member of an exchange, you need to pay an annual fee and have significant financial resources. A clearing member offering to

clear trades on your behalf will require you to have enough capital to satisfy their interpretation of your risk. In addition, exchanges ask for an initial margin to fund a portion of the notional value of your contracts. The ICE, CME and SGX websites state their initial margin requirements.

In summary, the following are the main features of oil trading exchanges:

- They are a place (exclusively online) to trade standard oil derivative contracts.
- They define contract terms.
- They match contract sellers and buyers.
- On the expiry or settlement of a contract, they manage clearing, which involves transferring funds between the counterparties' accounts. Managing settlement is arguably the most important function of exchanges, and you need to know how it works.
- Only members can trade on an exchange – exchange members can act for third parties (brokers, hedge funds or retail traders).

Clearing Function

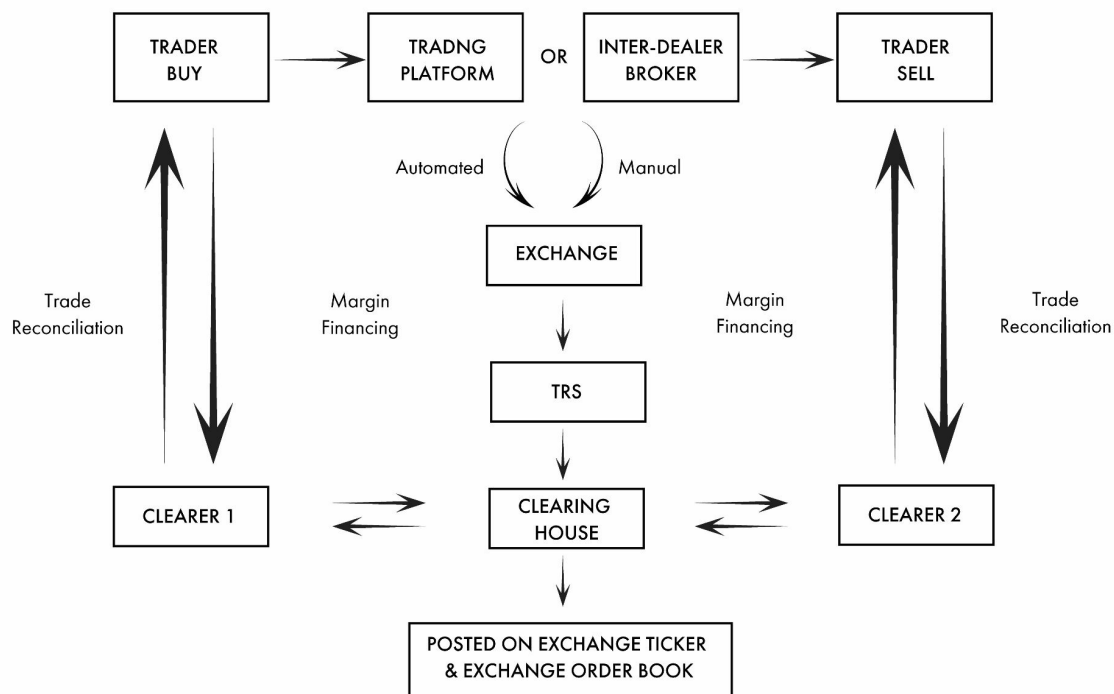


Figure 2.2 The Clearing Function of a Derivative Trade

When you interact with an exchange, you do so through the user interface of their computer system. At the back end of the system is the clearing function. There are two options when brokering a deal:

- It can be traded directly on the exchange via a trading venue using trade execution software.
- It can be confirmed through an interdealer broker who posts the trade details to the exchange on your behalf.

If you are not an exchange member, your trade is accepted by the exchange member you deal through, who checks your history and ensures you have the financial resources needed for the deal. If your bid (buy) or offer (sell) request is acceptable, it is submitted electronically to the clearing function of the exchange, which finds a counterparty that goes through the same process.

Clearing houses have a risk management framework comprising procedures and processes, with oversight and governance through committees. The members are responsible for the trading companies accessing the exchange, so the cash flow from every position as the market moves is transferred from clearing member to clearing member.

Once your request has been submitted, exchange members who form the

clearing house assume the risk of default on the trade. Therefore, if the clearing house notices any issues with the details of your trade or those of your counterparty, they halt the deal and give the parties to the deal a chance to sort out any issues. These interactions take place through the TRS (trade registration system).

Once the trade is accepted, it will be registered on the TRS and will go to the clearing members. When the clearing members confirm the trade, the transaction details will appear on the exchange ticker.

Since they will bear any losses, the first step is ensuring that the clearing members are sufficiently capitalised and have the necessary expertise and operational infrastructure. Participants can apply to be a clearing member, and the clearing house will vet each applicant to ensure they meet these requirements.

Exchange Operating Margin

To fund its operations, a clearing house sets the deposit amount – an *operating* or an *initial margin* – for its members. Members agree on this amount for every contract offered by the exchange. This cash is sought from each clearing member, who will, in turn, source these funds from the clients to whom they are providing access to the exchange. This operating margin allows each trading participant just to open the contract, and it is based on a SPAN (Standard Portfolio Analysis of Risk) algorithm, which estimates the worst intra-day move. The main component of this analysis is volatility, based on the estimated percentage move of that contract multiplied by the volume of contracts traded, and this is what will determine the margin. For example

Brent is trading today at \$80/bbl

Worst intra-day move is 15% of today's Brent price = \$12/bbl

The minimum lot size of Brent = 1 KB

Therefore, the initial margin = \$12,000

Once the initial margin is in place, market movement is considered to calculate a *variation margin*. Each contract has a defined settlement price that the exchange will post, and the difference between a trader's entry price and this settlement price is used to determine the variation margin. If it is a loss, the clearing member will post the necessary cash, which will be transferred to the counterparty trader via their clearing member. For example:

Entry price (or yesterday's settled price) = \$72/bbl

Net direction = Short

Today's settled price = \$73/bbl

Net volume of the contract held (for example) = 150 KB

Therefore, the variation margin = $150 \text{ KB} \times \$1/\text{bbl} = \$150,000$

The clearing member must pay this amount and will, in turn, take that margin from their client. The calculations can become even more complicated if the exchange allows for a variation margin rate.

The variation margin = (market value of the derivative contract – initial value of the derivative contract) $\times$ variation margin rate

Using the above example, if the agreed variation margin rate is 10%, the variation margin would be 10% of the dollar difference.

Variation margin = $\$150,000 \times 10\% = \$15,000$

Each exchange has its own calculation for the variation rate.

Because of the vast numbers of transactions, these calculations are done by computer systems using algorithms. The settlement price is important and can vary greatly for infrequently traded products. The variation margin can result in losses, with a trader losing cash from their account. This can be especially frustrating for market participants when exchange settlements are way off-market, which is common for products outside the core exchange-traded futures. In order to create contracts for these off-market trades, an exchange needs to source data, which sometimes means buying data for oil swaps. If an exchange cannot source data, it must rely on what has been traded.

Cross Exchange Margin

In some instances, the exchange margin does not reflect the actual risk of your trading strategy. For example, you might be holding risk by being long on one exchange through a specific future but hedging it with a contract on another exchange. In this case, your position would be delta neutral, but you would be required to fund the margin on both exchanges. This makes it difficult to be active on multiple exchanges simultaneously.

PRODUCT	ICE POSITION	CME POSITION	NET POSITION
WTI	+1000	-900	+100
BRENT	+1500	-1600	-100

Figure 2.3 Cross-exchange Margin Example

Trading Operations

Trading operations are essential functions of the middle- and back-office administration of trading companies that are not part of the exchange but are necessary for risk management. These operations involve various processes, one of which is the reconciliation process. Reconciliation ensures that the trading company's assessment of its position aligns with the clearing member's assessment and the profit and loss calculation.

To reconcile the positions, a clearing member will send trade activity and position files to the trading company daily, usually in the early morning before the trading day begins. The file sent by the clearing member will include all open buy/sell positions with the entry price of those trades, alongside the trade activity from a particular day.

The position determined internally must match the clearing house's position exactly, but there may be a difference in the profit and loss calculation because the clearing bank calculates the profit and loss by comparing the entry prices of all trades with the settlement values of those traded contracts sourced from the exchange. However, traders may have their own assessment of settlement prices, which can differ from the clearing bank's assessment.

This discrepancy can be contentious, as traders are marking their own books. Therefore, the middle- and back-office operations use third-party sources to verify the variance between the exchange and trader assessments. Naturally, traders and risk managers may argue about this.

To reconcile the positions and calculate the daily profit or loss, the middle- and back-office operational staff will compare the files received from the clearing member with the trading activity from the day before. If the positions match, they will calculate the profit or loss by valuing the positions against current market prices using marked-to-market assessments.

Derivative Contracts

The heart of derivatives trading is the contract or, rather, the multiplicity of contracts you can trade. If you want a non-standard deal, or if you want to buy or sell a physical product, then it has to be traded directly with the counterparty or through a broker as a bilateral or *OTC* contract.

Derivative contracts define precise details on what physical oil is being referenced on settlement and by what PRA, including how it is cleared and margined and the minimum tradeable quantity.

The contracts that make up the oil derivative market are known as **futures**, **swaps** and **options**.

Exchange-traded Futures

Futures are exchange-traded instruments that are settled according to a methodology defined by the exchange that lists it, and the contract is specific to that exchange. A future can require the delivery of physical oil on the expiry of the contract (which rarely happens) or a cash payment settled to an *index* or *benchmark* price.

Exchange-traded futures contracts are settled on one day, known as *expiry*, or, for Brent futures, the *Brent Index*. Brent crude futures contracts expire on the last business day of the month, two months prior to the delivery month of the physical oil cargo used in the contract assessment. Up to that time, trading is allowed during periods known as *tenors*, which correspond with the loading period of the underlying oil grade. For example, the November 2023 Brent futures reflect the November 2023 loading of crude at the North Sea oilfields that underpin the contract and settle on 29 September 2023 (the last business day of the month).

Figure 2.4 shows a Brent futures contract from ICE. The main features of the contract are the quantity traded (multiples of 1 KB), the minimum trading price increment of 0.01¢ and the settlement (expiration) details. The price is not specified, of course; that's what the market decides through trading – the price discovery mechanism.

The second page of the contract details how specific the exchange is on how this contract can be traded.

Trading Screen Product Name	Brent Crude Futures
Trading Screen Hub Name	North Sea
Contract Symbol	B
Contract Size	1,000 barrels
Unit of Trading	Any multiple of 1,000 barrels
Currency	US Dollars and cents
Trading Price	One cent (\$0.01) per barrel
Minimum Price Fluctuation	One cent (\$0.01) per barrel
Expiration Date	Trading shall cease at the end of the designated settlement period on the last Business Day of the second month preceding the relevant contract month (e.g. the March contract month will expire on the last Business Day of January).
If the day on which trading is due to cease would be wither:	
(i)	the Business Day preceding Christmas Day, or
(ii)	the Business Day preceding New Year's Day, then trading shall cease on the next preceding Business Day.

Figure 2.4 Brent Futures Contract (source: ICE)[\[10\]](#)

Exchanges list many different tradeable contracts to the same level of detail as described above – more than 650 on ICE – and you need to know the details of those you intend to trade.

Swaps

An oil swap, also referred to as a *block future* if cleared on the

exchange, is a financial oil contract with a daily settlement price. The contract details are specified by a PRA that calculates the price daily using a publicly advertised methodology. This methodology will combine financial and physical prices to determine the final settlement price.

Oil swaps differ from futures in that the contract month, also known as the tenor, reflects the period it will settle in, not the loading period of physical oil. For example, a November Brent first line future contract's final price is determined by the average settlement prices on all the trading days in the PRA's defined calendar for that month (usually the working day calendar of the region the contract is for).

There are also *futures swaps*, or the *1st line futures* as defined by the exchange, for which the exchange provides a pricing calendar similar to a PRA-indexed contract. The difference is explained further in the next section.

The daily settlement price for a traditional oil swap is closer to the spot physical price than what is used for exchange-traded futures, as it uses physical oil prices that are loading in a nearer time period (more 'prompt'). For example, Brent futures reflect oil loading two months forward from the day it expires to the Brent Index, whereas Dated Brent uses the same physical price but for transactions 10 to 30 days forward from the day the contract is settled.

Options

Options are contracts that give purchasers the right, but not the obligation, to buy or sell the underlying asset at a defined price, known as the 'strike' price. These can be created on any contract and can be a standard or 'vanilla' option readily available on an exchange or a more complex contract available through a broker. There are also options on exchange futures time spreads, known as *calendar spread options* (CSOs), which settle the day before the futures' front month (the next expiry date for the future) expires.

Oil Derivative Contracts Network

For this reason, they are the ‘go-to’ contracts to gain exposure to the outright oil price, known as the ‘flat price’ in the industry. It is also for this reason that all other contracts are in some way derived from these two core contracts.

Every other contract is calculated based on one of these futures plus one or more differentials. Therefore, we look at their differentials when trading every other contract to determine its independent market movement. In other words, we use differentials against Brent or WTI to remove the correlation to the flat price.

Currently, the market holds most of its risk (open interest) in Brent and WTI prompt-month contracts. It is common for traders who want to gain outright exposure to a new contract first to trade these contracts and then, as a second step, trade the differentials to the new contract. If a trader wants to trade a contract directly, then most likely, the price they will receive will be a function of Brent or WTI plus the differential, which is how market makers provide flat price liquidity on contracts outside of Brent and WTI.

For example, *a fuel oil trader wanting to trade a 3.5% Mediterranean residual fuel oil contract may trade that contract directly or build up the exposure themselves.*

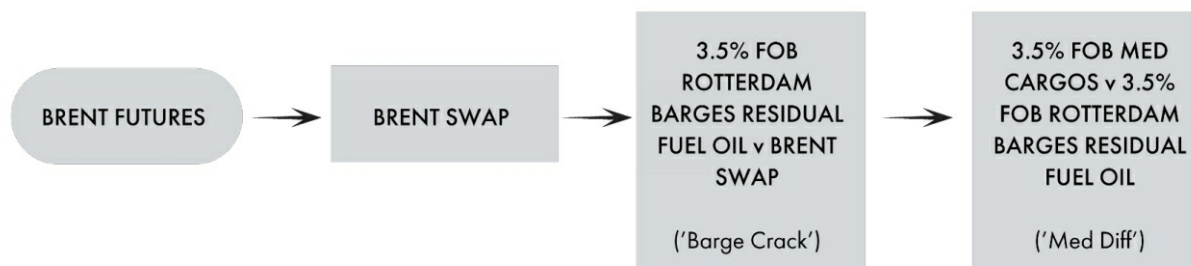


Figure 2.6 Process of Transferring Risk Using Different Contracts

Why Would a Trader Do This?

Within every market, contracts go from the most to least liquid. Therefore, it is common for traders to trade with the most liquid contracts first and trade into more niche contracts if desired. The entire oil derivative complex is organised by contract liquidity. As mentioned, the most liquid contracts are Brent and WTI crude futures. These are traded directly on exchanges, and there is a constant price while the markets are open, making it straightforward to trade at will.

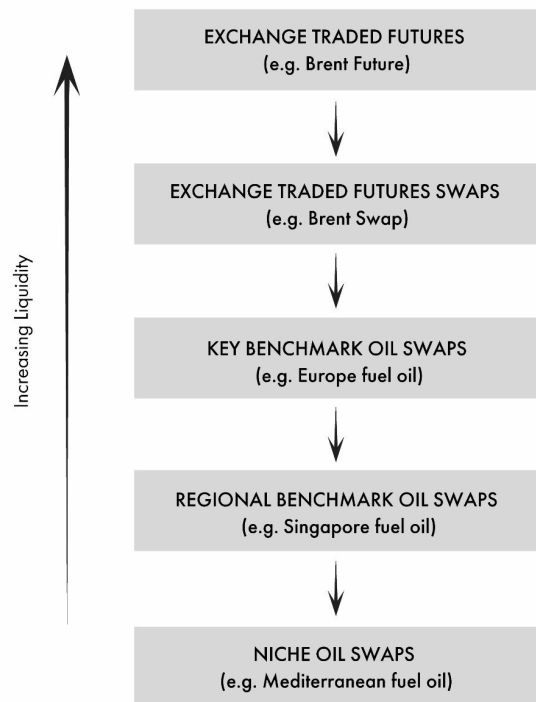


Figure 2.7 Liquidity Chain of Contracts

As shown in Figure 2.7, we can develop the remaining oil derivative contracts from Brent and WTI crude futures. The refined product exchange-traded futures are the most liquid after Brent and WTI. Three refined products also trade this way but are slightly less liquid in their outright price: Heating Oil (US diesel), Gasoil (European diesel) and RBOB (US gasoline) contracts.

These refined products trade most commonly as differentials to Brent, known as exchange-traded futures ‘cracks’. They are named cracks in reference to the cracking process refiners use to convert crude as feedstock into refined products. The final refined product price versus the crude feedstock is, therefore, the cracking margin, i.e. the value a refiner gets from refining the crude.

From these exchange-traded futures, we derive the exchange-traded futures swaps, called *1st line futures* by exchanges.

It is useful to convert exchange-traded futures to exchange-traded futures swaps to gain exposure to daily settlements, usually to align with physical contracts, or to create differentials in the oil swap market, which settle daily. Details of how to convert Brent futures to Brent swaps (and other

exchange-traded futures follow the same logic) are covered in the settlement chapter.

Interestingly, while the sum of these contracts dwarfs oil swap contracts in terms of volume, they are increasingly similar in terms of daily open interest (risk in the market).

Geography of Benchmarks

To build out the global oil swap network, there are varying physical crude grades, refined product specifications and specific markets for similar physical products in different regions. Therefore, the geography, as well as the physical product, is benchmarked by the swap network. The key regions are linked to the main physical flows in the US, Europe and East of Suez as follows:

- The Gulf Coast and New York Harbour in the US
- Rotterdam, Amsterdam and Antwerp for Northwest Europe
- Mediterranean for wider Europe
- Singapore
- Abu Dhabi and Dubai in the Middle East

Oil Swap Markets

Crude Oil

Physical market

Physical market uses

Crude oil is extracted from the ground and undergoes refining processes to produce gasoline, diesel fuel, and other transportation fuels. It is also refined to create essential components for manufacturing products like plastics, fertilisers, and chemicals. Additionally, crude oil serves as an energy source for heating, electricity generation, and industrial processes after refining.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
BENCHMARK SWAPS	WTI Houston	"MEH"	Dated Brent		Dubai	
KEY DIFFERENTIAL	WTI Houston vs Trade month	"HTT"	Dated Brent vs Brent swap	"DFL"	Brent swap vs Dubai swap	"Brent Dubai"
ARBITRAGE DIFFERENTIAL	WTI Houston vs Dated Brent	"MEH/Dated"	Dated Brent vs Dubai	"Dated Dubai"	Dated Brent vs Dubai	"Dated Dubai"
CROSS PRODUCT DIFFERENTIAL	LLS vs Mars	"E5/YV"	Urals vs Dtd		Murban vs Dubai	"Murban Dubai"
NICHE CONTRACTS	Mars vs WTI, WTI Midland vs WTI	"YV"	CPC vs Dtd		Oman vs Brent Murban vs Brent	

Figure 2.9 Crude Oil Contracts

Dated Brent Ecosystem

Dated Brent swaps trade like all the other swaps: they trade in contract months and settle daily to a PRA settlement. However, in addition, they can be traded in *daily* and *weekly* Dated Brent contracts, which can be cleared against BFOETM swaps to form a Contract for Difference contract known as *CFDs*. The *CFDs* are cleared oil swap contracts like any other in the oil derivatives market and do not resemble the *CFD* mechanism used in retail trading.

The Dated Brent market is the most complex and granular swap contract in the oil derivatives market, and the mathematical link between the Brent futures and the Dated Brent market is of great importance. Brent futures and Dated Brent swap contracts are used in physical crude negotiations for as much as 70% of the world's physical oil trades^[11].

The contracts that make up the Brent complex are reflected in Figure 2.10.

CONTRACT NAME	COMPONENTS	DEFINITION	HOW IT'S TRADED
Dated Brent	Dated Brent	Oil swaps priced by Platts settlement values	Traded in the oil swaps market by contract month
Daily Dated Brent	Dated Brent	Oil swaps priced by Platts settlement values	Traded in the oil swaps market by contract day
Brent Futures / Brent swaps	Brent Futures / Brent swaps	Exchange traded futures	Directly on Exchange
BFOETM Partial	BFOETM Partial	Cash settled financial contract reflecting six physical grades that underpin the settlement of Brent. Trades in 100KB clips with same counterpart trades, partial becomes EFP	Traded bilaterally or financial exposure gained from CFDs in oil swaps market
Exchange Futures for Physical (EFP)	BFOETM Partial vs Brent Futures	A contract which obligates 700KB cargo of BFOETM crude to be exchanged with Brent Futures of same loading period	Bilateral only
DFL	Dated Brent vs Brent swap	Month assessment of Dated Brent versus Brent Swap of same calendar	Oil swaps market
CFD	Weekly Dated Brent vs Month one BFOETM Partial	Weekly assessment of Dated Brent versus BFOETM partial of the same calendar	Oil swaps market

Figure 2.10 The Brent Complex Components

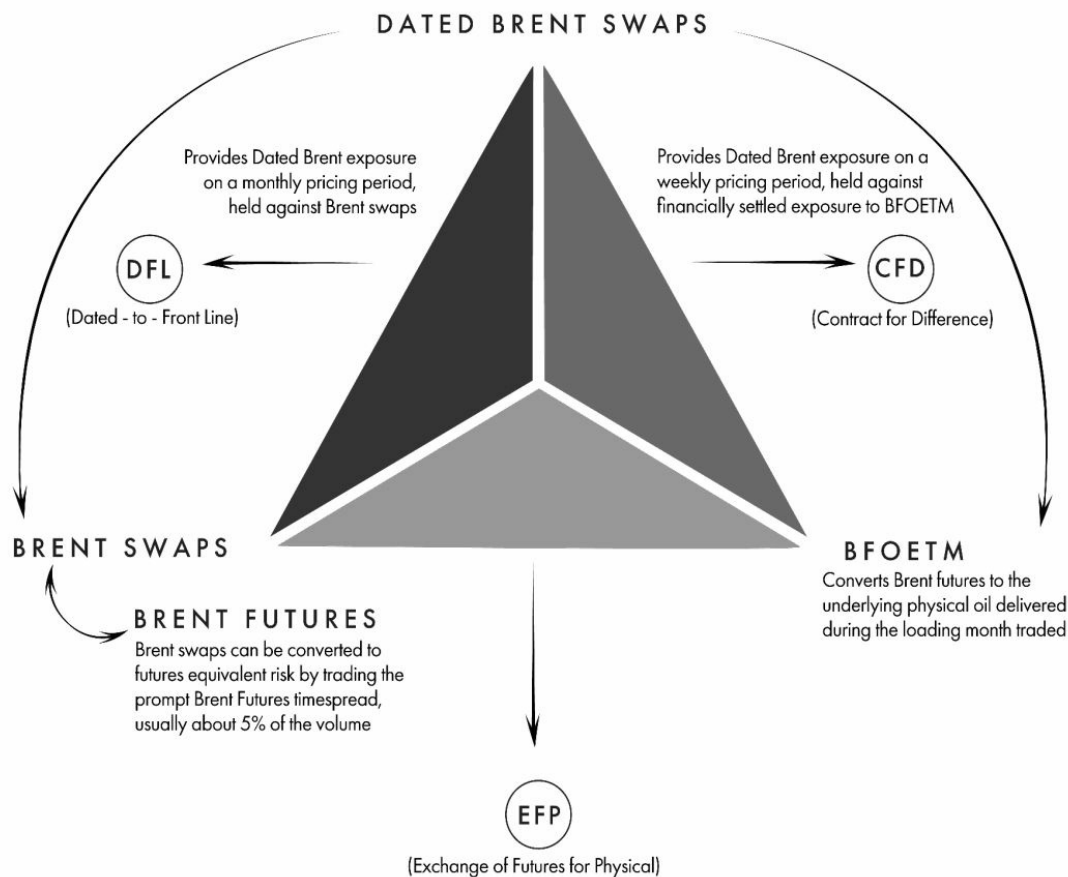


Figure 2.11 The Intra-Brent Complex Interactions

Refined Products

Refiners take crude oil into their system and process it into refined products, which are then sold to end users. The difference in price between the end product and the crude feed processed by the refinery is known as the gross refinery margin, while the net margin includes fees and the cost of running the refinery.

The heat required to yield each refined product in a simple crude distillation unit in a refinery creates a natural hierarchy. The higher up this hierarchy, the smaller the carbon chain of the refined product. In the oil market, traders refer to this spread of refined products as ‘across the barrel’. So when people refer to the ‘light end of the barrel’, for example, they usually mean gasoline, naphtha, NGLs (natural gas liquids) and up, and the ‘heavy end of the barrel’ refers to distillates and residual fuel oil.

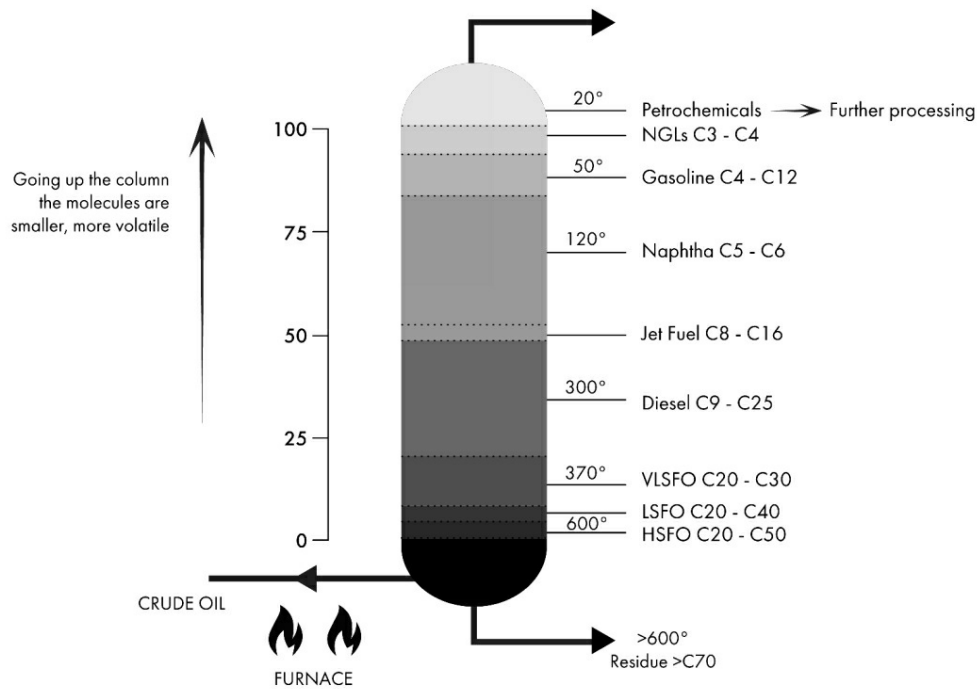


Figure 2.12 The Refinery Column

The oil derivative market has expanded over the years from benchmarks for distillates and gasoline to all the major groups of refined products.

DERIVATIVE MARKET	PRIMARY USE IN PSYICAL MARKET
Petrochemicals	Plastica, detergents, drugs, fertilizers, explosives etc
NGLs	Heating fuel, feedstock for producing Petchems
Naphtha	Blending Component for gasoline, feedstock for producing petrochemicals
Gasoline	Driving fuel
Jet Fuel/Kerosene	Avaition fuel, heating fuel, cooking fuel
Diesel/Heating Oil/Gasoil	Driving fuel, heating fuel
Very Low Sulphur FueOil	IMO 2020 compliant shipping fuel
Residual Fuel Oil/Low Sulphur Fuel Oil	Shipping fuel, power generation, feedstock for further processing in refiner's secondary units

Figure 2.13 Refined Products and their Applications

The major oil products without a traded derivative contract are vacuum Gasoil, a secondary feedstock; bitumen; and asphalt, used to form roads.

Residual High Sulphur Fuel Oil

Physical Product

Physical Product Uses

As the name suggests, residual fuel oil is a by-product of a refinery's crude oil processing. The product has a high-density chemical makeup that requires a huge amount of energy to process and break down the oil into a high-value product. The high density of the product and its low price made it the fuel of choice for ships around the world until IMO 2020, a globally mandated regulation that capped emissions at 0.5%^[12]. Residual fuel is still used in ships with scrubbing technology to remove sulphur or those that do not comply with IMO 2020. The fuel is also used in direct burning for electricity generation and even as feedstock for complex refineries, either directly or blended with other crudes. Residual fuel also acts as the derivative market proxy for physical asphalt and tarmac.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	USGC HSFO	"Gulf Coast Fuel"	3.5% FOB Rotterdam Barges	"Barges"	Singapore 380	"Sing 380"
KEY DIFFERENTIAL	USCG HSFO vs 3.5% FOB Rotterdam Barges	"Gulf Coast arb"	3.5% FOB Rotterdam Bares vs Brent swap	"Barge crack"	Fuel Oil 380 CST Singapore vs Brent	"380 Crack"
ARBITRAGE DIFFERENTIAL	USCG HSFO vs 380 Singapore	"Cross-arb"	3.5% Fuel Oil FOB MED vs 3.5% FOB Rotterdam Barges	"Med North"	Singapore 380 vs 3.5% FOB Rotterdam Barges	"Fuel East West"
CROSS PRODUCT DIFFERENTIAL	USGC Marine Fuel 0.5% Barges vs. Gulf Coast HSFO	"US Hi5"	0.5% FOB Rotterdam Barges vs Fuel Oil 3.5% FOB Rotterdam Barges Swap	"Hi5"	Marine Fuel 0.5% FOB Singapore (Platts) vs 380 CST Singapore	"Sing Hi5"
NICHE CONTRACTS	New York 1% Fuel Oil vs USGC HSFO	"US 1% Hi5"	Fuel Oil 1% FOB NWE Cargoes vs 3.5% FOB Rotterdam Barges	"HILO"	Fuel Oil 180 CST Singapore vs 380 CST Singapore	"Visco"

Figure 2.14 High Sulphur Fuel Oil Contracts

Contract Setup

As with all oil swap contracts, the market dynamics for the derivative

market for residual fuel oil are influenced by how the contracts are linked to each other.

- The most liquid and important contract is the 3.5% Barges FOB Rotterdam vs Brent swap contract, known as the 'Barge crack'. From here, the rest of the contracts are linked, so the activity in this contract drives much of the fuel oil market.
- The outright prices of the benchmark fuel oil contracts in each region are derived from the Barge crack and Brent 1st line futures. The outright 3.5% Barges price is determined by the Barge crack plus the Brent crude 1st line future, while the 380 Singapore outright contract is derived from the outright price on the 3.5% Barges contract plus the 380 Singapore vs 3.5% Barges differential, known as the 'East West'.
- The same process is applied for the main US fuel oil benchmark differential, known as the 'arb' (3.5% Gulf Coast vs 3.5% Barges). Adding the two geographical differentials together results in what is known as the 'cross arb', or 3.5% Gulf Coast vs 380 Singapore.
- The Visco contract, used to manage risk between the viscosity differences in residual fuel, is calculated using the 180 Singapore against the 380 Singapore contract.

Contract Hedging

- Flat price contracts: Utilities and shipping end-user hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- Visco: managing blending economies
- E/W: managing cargo voyages and structural flow typically moving from west to east
- Arb: Hedging voyages from US Gulf Coast to European markets
- Cross Arb: Hedging voyages from US Gulf Coast to Singapore and Eastern markets.

Very Low Sulphur Fuel Oil/0.5% Marine Fuel

Physical Product

Physical Product Uses

Very low sulphur fuel oil (VLSFO) contracts benchmark IMO 2020-compliant fuel for the shipping industry. The sulphur content must be below 0.5%, and this fuel is typically a blend of fuels, such as ultra-low sulphur fuel oil (0.1% sulphur^[13]) blends with Gasoil and high- and low-sulphur residual fuel mixes.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	Marine Fuel 0.5% FOB USGC	"0.5 Gulf Coast"	Marine Fuel 0.5% FOB Rotterdam Barges	"0.5% Barges"	Marine Fuel 0.5% FOB Singapore	"Sing 0.5"
KEY DIFFERENTIAL	USGC Marine Fuel 0.5% vs Brent Futures	"Fibres"	Marine Fuel 0.5% FOB Rotterdam Barges vs Brent	"Euro 0.5 Crack"	Marine Fuel 0.5% FOB Singapore vs Brent	"Sing 0.5 Crack"
ARBITRAGE DIFFERENTIAL	USGC Marine Fuel 0.5% vs 0.5% FOB Rotterdam Barges	"0.5% Arb"	Marine Fuel 0.5% FOB Med Cargoes (Platts) vs Marine Fuel 0.5% FOB Rotterdam Barges	"0.5% Med Arb"	0.5% FOB Singapore vs 0.5% Rotterdam Barges	"0.5 East /West"
CROSS PRODUCT DIFFERENTIAL	0.5% FOB USGC Barges vs USGC HSFO	"US H15"	Marine Fuel 0.5% FOB Rotterdam Barges vs 3.5% FOB Rotterdam Barges	"H15"	Marine Fuel 0.5% FOB Singapore vs Singapore Gasoil	"Asian FOGO"
NICHE CONTRACTS	Marine Fuel 0.5% FOB USGC Barges (Platts) vs 3.5% FOB Rotterdam Barges	-	Marine Fuel 0.5% FOB Rotterdam Barges vs Fuel Oil 1% FOB NWE Cargoes (Platts)	-	Marine Fuel 0.5% FOB Singapore (Platts) vs 380 CST Singapore	"Asian H15"

Figure 2.15 Very Low Sulphur Oil Contracts

Contract Setup

- Until 2022, liquidity was mainly in FOGO or 0.5% vs Gasoil. However, during the peak volatility surrounding the Russia–Ukraine war, Gasoil levels far superseded normal historical levels and were an independent focus for many speculators, exceeding the range on the FOGO. This caused an inverse Gasoil crack and is not a reliable indication of VLSFO markets. Therefore, attention turned to the VLSFO vs Brent swap crack as the main liquidity pool. This is a good example

of how liquidity can drive the main contract in a given market. As a result, VLSFO moved to the crack, and differentials were priced off this contract.

- The outright price of VLSFO is determined by the Euro crack. In much the same way as the residual fuel market outlined above, the rest of the contracts are derived from this outright price and differentials.

Contract Hedging

- Flat price contracts: shipping and utilities end-user hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- E/W: hedging voyages and structural flow, typically moving from west to east
- Hi5: managing the blending economics of 3.5% fuel oil into the marine fuel pool
- Niche contract differentials: managing the basis risk from core contracts

Gasoil/Diesel/Heating Oil

Physical Product

Physical Product Uses

Physical Gasoil, diesel and heating oil are closely linked as products, with specifications differing regionally.

Diesel is primarily used as a fuel for diesel engines in cars, trucks and other vehicles. It has a lower viscosity and density than heating oil, which makes it easier to pump and burn. Gasoil contains additives that improve its performance in cold weather conditions, such as anti-gelling agents. On the other hand, heating oil is a heavier fuel oil primarily used for heating buildings and homes. It has a higher viscosity and density than Gasoil, which makes it more difficult to pump and burn. Heating oil is typically dyed red to differentiate it from other types of fuel oils, and it may also contain additives to improve its performance in cold weather conditions.

While Gasoil and heating oil have different properties, they are derived

from the same crude oil feedstock and are refined similarly. The main difference between the two is their use and specific requirements for each application.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME		EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	NY Harbour ULSD	"HO"	PIE low Sulphur Gasoil 0.1%	"IPE GO"	Singapore 10ppm Sulphur Gasoil	"S10"
	10ppm Gulf coast ULSD	"USG 10ppm ULSD"	ARA 0.1% Barge Gasoil Gasoil 0.1% CIF MED	"BARGE GO" "Gasoil 0.1 MED"	Singapore 0.05% Gasoil	"Sing LSGO"
KEY DIFFERENTIAL	Heating oil vs Brent	"Hobbit"	ULSD 10ppm CIF NWE Cargoes vs Brent	"Diesel NWE crack"	Singapore Gasoil vs Brent	"Eastern Gasoil crack"
ARBITRAGE DIFFERENTIAL	Heating oil vs Low Sulphur Gasoil	"HOGO"	Singapore 10ppm Gasoil vs ICE Low Sulphur Gasoil	"GO E/W"	Singapore 10ppm Gasoil vs ICE Low Sulphur Gasoil	"GO E/W"
CROSS PRODUCT DIFFERENTIAL	NY Harbour ULSD vs Brent	"Heat crack"	Diesel 10PPM ARA Barges vs Low Sulphur Gasoil	"Diesel Diff"	Singapore Jet Kerosene Cargoes vs Singapore Gasoil 10 ppm	"Regrade" "AG gasoil"
NICHE CONTRACTS	Gulf coast ULSD vs NYH ULSHO	"Up/Down"	Low Sulphur Gasoil vs Marine Fuel 0.5% FOB Rotterdam Barges	"GOFO"	Middle East Gasoil FOB Arab Gulf	"AG gasoil"

Figure 2.16 Diesel/Heating Oil/Gasoil Contracts

Contract Setup

In the distillates derivatives market, diesel and heating oil are grouped, as the physical products that price these contracts have very similar specifications. For example, the physical product that underpins the exchange-traded futures for Gasoil used to be 0.05% Gasoil but changed to 10 parts per million in 2015^[14]. Gasoil and heating oil futures in the US are interchangeable with ultra-low sulphur diesel. However, there is still a 0.1% sulphur Gasoil swap in Europe.

- In the diesel/heating oil/Gasoil market, liquidity is centred around exchange-traded futures cracks and swap cracks. The cracks plus the Brent 1st line future implies a Gasoil/diesel/heating oil outright contract

- The swap contracts trade at a differential to these outright prices.

Contract Hedging

- Flat price contracts: utilities, commercial and shipping end-user hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- E/W: managing cargo voyages, typically moving from east to west
- Gasoil differentials: managing the basis risk from exchange-traded futures
- Niche contract differentials: managing the basis risk from core contracts

Jet Fuel/Kerosene

Physical Product

Physical Product Uses

Physical jet fuel is a type of aviation fuel specifically designed to power the jet engines used in most commercial and military aircraft. Jet fuel is typically made from kerosene and has a higher flash point and a lower freezing point than other fuel oils, making it less volatile and more stable at high altitudes. Jet fuel also has a higher energy density than other aviation fuels, which allows aircraft to fly long distances without refuelling.

On the other hand, kerosene is a fuel oil used for various purposes, including heating and lighting. Kerosene is similar to jet fuel because it is made from refined petroleum and has high energy density, but it has a lower flash point and higher freezing point than jet fuel. Kerosene is often used as a substitute for jet fuel in emergencies or when jet fuel is unavailable.

Both jet fuel and kerosene are highly flammable and require careful handling and storage to ensure safety. They are also subject to strict regulations and quality standards to ensure that they meet the specific requirements of the aviation industry.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	Gulf coast Jet Fuel	"Jet USGC"	European Jet Kerosene cargoes NWE	"Jet NWE"	Singapore Jet Kerosene	"Sing kero"
	NY Buckeye Jet Fuel	"NYH jet fuel"	Jet Rotterdam barges	"barge jet"		
KEY DIFFERENTIAL	NY Buckeye Jet Fuel vs. NY Harbor ULSD		Jet CIP NEW Cargoes vs Brent	"jet fuel crack"	Singapore 10ppm Gasoil vs ICE Low Sulphur Gasoil	"Regrade"
ARBITRAGE DIFFERENTIAL	NY Buckeye Jet Fuel vs. Jet Rotterdam (Legs)		Jet FOB Rotterdam Barges vs Jet CIF NWE Cargoes	"Jet Euro Arb"	Singapore Jet Kerosene vs NWE Jet	"Jet EW"
CROSS PRODUCT DIFFERENTIAL	Gulf coast Jet Fuel vs NYH ULSHO	"NYH jet/HO"	Jet CIF NWE Cargoes vs Low Sulphur Gasoil	"European jet diff"	Singapore Jet Kerosene vs Dubai swap	"Asian/ Eastern jet fuel crack"
NICHE CONTRACTS	-	-	Jet FOB MED cargoes	"Jet MED"	Middle East Jet Kero Arab Gulf	"AG"

Figure 2.17 Jet Fuel/Kerosene Market Contracts

Contract Setup

Like the Gasoil/diesel/heating oil complex, the jet fuel derivatives market primarily sources its liquidity from exchange-traded futures cracks and trades as differentials to the flat price implied by Gasoil or heating oil contracts. Within the differentials, the Jet North West Europe (NWE) differential to Gasoil exchange-traded futures swaps is the core contract in Europe, whereas the regrade is the main liquidity pool for jet fuel/kerosene in Eastern markets.

Contract Hedging

- Flat price contracts: airline hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- Gasoil differentials: managing the basis risk from exchange-traded futures
- Niche contract differentials: managing the basis risk from core swap contracts

Naphtha

Physical Product

Physical Product Uses

Physical naphtha is a term used to describe a group of highly volatile, flammable and colourless liquid hydrocarbons commonly used as solvents, diluents and feedstocks in various industrial processes. It is obtained by refining petroleum and is typically composed of alkanes, cycloalkanes and aromatics with carbon numbers ranging from five to twelve.

Naphtha is used in various industries, including the production of gasoline and petrochemicals and as a solvent in manufacturing paints, coatings and cleaning agents. It is also used as a feedstock in producing olefins, which are important building blocks for many chemicals and plastics. Additionally, naphtha is used as a fuel in industrial applications, such as gas turbines, and as a blendstock for diesel and gasoline.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	Natural Gasoline, OPIS Mt. Belvieu Non-TET	"US Natural Gasoline"	Naphtha CIF NWE Cargoes	"NWE Naphtha"	Naphtha C+F Japan Cargo	"MOPJ"
KEY DIFFERENTIAL	Natural Gasoline, OPIS Mt. Belvieu Non-TET vs WTI	"US Natural Gasoline crack"	Naphtha CIF NWE Cargoes vs Brent	"Naphtha Crack"	Naphtha C+F Japan (Platts) vs Brent	"MOPJ Crack"
ARBITRAGE DIFFERENTIAL	-	-	Europe arb - NWE and MED	"Europe Arb"	Naphtha C+F Japan vs Naphtha CIF NWE Cargoes	"Naphtha East/West"
CROSS PRODUCT DIFFERENTIAL	-	-	Naphtha NWE vs EBOB	"Gasnaph"	Singapore Morgas 92 Unleaded (Platts) vs Singapore Naphtha	"92 MOPJ"
NICHE CONTRACTS	-	-	Propane, Argus CIF ARA vs Naphtha CIF NWE Cargoes	"Pronap"	Argus Propane Far East Index vs. Japan C&F Naphtha	"Asian Pronap"

Figure 2.18 Naphtha Contracts

Contract Setup

- The naphtha derivatives market relies heavily on the naphtha crack differential for liquidity. This differential serves as the basis for determining the outright price of naphtha, which is derived from the liquidity of Brent futures.
- Additionally, the Japanese naphtha prices (Mean of Platts Japan or MOPJ) are inferred from the naphtha crack and E/W differentials.
- The GasNap differential is an important factor in driving the naphtha crack, as it is the most liquid differential in Europe that links the naphtha and gasoline markets.

Contract Hedging

- Flat price contracts: petrochemical producer hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- E/W: managing voyages that usually travel from west to east
- GasNap: managing the blending economics for naphtha into gasoline finished products
- Niche contract differentials: managing the basis risks from core swap contracts

Gasoline

Physical Product

Physical Product Uses

Gasoline, also known as petrol, is primarily used as a fuel for internal combustion engines in various modes of transportation, such as cars, motorcycles, boats, aeroplanes and trucks. It is the most commonly used fuel for personal transportation vehicles worldwide^[15].

Gasoline is also used as a solvent and a raw material in manufacturing various chemicals and plastics.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	NYH RBOB Gasoline	"RBOB"	Argus Eurobob OXY FOB Rotterdam Barge	"EBOB"	Singapore Mogas 92 Unleaded	"Sing 92"
KEY DIFFERENTIAL	RBOB Gasoline vs Brent	"RBBR"	Argus Eurobob OXY FOB Rotterdam Barge vs Brent	"EBOB Crack"	Singapore Mogas 92 Unleaded vs Brent	"92 Crack"
ARBITRAGE DIFFERENTIAL	Singapore Mogas 92 Unleaded vs Eurobob Oxy FOB Rotterdam Barges	"X-Arb"	NYH RBOB Gasoline vs Argus Eurobob Oxy FOB Rotterdam Barge	"Arb"	Singapore Mogas 92 Unleaded vs Eurobob FOB Rotterdam Barge	"Sing 92 East/West"
CROSS PRODUCT DIFFERENTIAL	RBOB Gasoline vs Eurobob Oxy FOB Rotterdam Barge	"RB nap"	Eurobob OXY FOB Rotterdam Barges vs Naphtha CIF NWE Cargoes	"Gasnaph"	Singapore Morgas 92 Unleaded vs Singapore Naphtha	"92 MOPI"
NICHE CONTRACTS	Gulf Coast CBOB A	"Calbob"	Premium Unleaded Gasoline 10ppm FOB Med Cargoes vs Eurobob Oxy FOB Rotterdam Barges	"Med/North Gasoline"	Singapore Mogas 95 Unleaded vs Singapore Mogas 92 Unleaded	"Octane spread"

Figure 2.19 Gasoline Contracts

Contract Setup

- The primary liquidity source for gasoline is RBOB vs Brent (RBBR) exchange-traded futures contracts, which serve as the core liquidity pool.
- The RBBR contract significantly impacts the European gasoline market and is connected to the EBOB vs Brent differential via the 'arb' contract. As long as the 'arb' (RBOB vs EBOB) contract remains stable, any fluctuations in the RBBR differential will be mirrored in the EBOB vs Brent (EBOB crack). This interconnectedness between contracts and differentials is critical to ensuring the gasoline market's liquidity and stability.

Contract Hedging

- Flat price contracts: end-user hedging
- Time spreads: managing storage and voyage time
- Cracks: managing refinery component margins
- E/W: managing voyages that usually travel from east to west

- (Europe)
- Arb: managing voyages that usually travel from Europe to the US (EBOB to RBOB)
- Niche contract differentials: managing the basis risks from core swap contracts

Natural Gas Liquids

Physical Product

Physical Product Uses

NGLs are a group of hydrocarbons, including ethane, propane, butane, isobutane and pentane, typically found in natural gas streams. These liquids are used in the following ways across many industries:

- Fuel: NGLs are used as a fuel source in various applications, such as home heating and cooking, and in industrial processes and transportation.
- Petrochemicals: NGLs are used as a feedstock to produce petrochemicals, which are used in manufacturing various products, including plastics, synthetic rubber and fibres.
- Refining: NGLs are used in the refining process to help produce gasoline and other transportation fuels.
- Blending: NGLs are often blended with other products, such as crude oil and gasoline, to improve their properties and enhance performance.
- Extraction: NGLs are used to extract oil and natural gas from underground reservoirs and can help increase the productivity of these operations.
- Other uses: NGLs are used to produce fertilisers, solvents and aerosol propellants.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	Mont Belvieu LDH Propane	"LST"	European Propane CIF ARA	"European propane"	Saudi CP	"CP"
	Conway Propane	"Conway"			Far East Index	"FEI"
KEY DIFFERENTIAL	Conway vs LST	"North South"	European Propane CIF ARA vs Naphtha Cargoes CIF NEW	"Pronap"	FEI vs Japan C&F Naphtha	"Eastern Pronap"
ARBITRAGE DIFFERENTIAL	Propane, OPIS Mt. Belvieu TET vs Propane, Far East Index	"Eastern Arb"	Mont Belvieu vs Propane CIF NWE	"European Arb"	FEI vs Propane CIF NWE	"East/West"
CROSS PRODUCT DIFFERENTIAL					CP vs Mont Belvieu LDH Propane	"CP vs LST"
	C4 ENT vs C3 LST	"C4/C3"	CIF ARA vs Naphtha CIF NWE Cargoes	"Pronap"	Far East Index vs Naphtha C+F Japan Cargoes	"Asian GasNap"
NICHE CONTRACTS	Mont Belvieu Enterprise	"ENT"	European Butane CIF ARA	"Euro Butane"	Butane, Far East Index	"C4 FEI"

Figure 2.20 Natural Gas Liquid Contracts

Contract Setup

- The derivatives of NGLs include propane, butane and ethane. Unlike other refined product derivatives, NGL contracts are driven by independently priced flat contracts. Although there is some correlation between NGL contracts and Brent and WTI futures, this correlation varies significantly depending on the buying or selling flow of the NGL contract relative to Brent futures.
- The primary sources of liquidity in the NGL market are the LST, CP, NWE and FEI propane contracts, along with the arbitrage differentials.
- Butane and ethane prices are determined based on differentials to propane.
- Additionally, some flow from naphtha can impact the NGL market through the propane vs naphtha differential, which is commonly referred to as 'ProNap'.

Contract Hedging

- Flat price: producer and utilities hedging
- ProNap: managing switching economics from naphtha to propane for producing petrochemicals

- E/W: managing voyages usually travelling from west to east
- Arb: managing exports from the US
- C4/C3: moving propane liquidity to butane risk
- Niche contracts: managing the basis risks from core swap contracts

Petrochemicals

Physical Product

Physical Product Uses

Petrochemicals are a group of chemical compounds derived from petroleum or natural gas. They are used as raw materials for producing various products, including plastics, synthetic fibres, rubber, detergents, solvents and many other industrial and consumer goods. Some common uses of petrochemicals include:

- **Plastics:** petrochemicals are used to produce various types of plastics, which are used in a wide range of applications, from packaging and construction materials to medical devices and electronic components.
- **Synthetic fibres:** petrochemicals are also used to produce synthetic fibres, such as polyester and nylon, which are used in clothing, upholstery and other textiles.
- **Rubber:** petrochemicals are used to produce synthetic rubber, which is used in various products, including tires, hoses and gaskets.
- **Detergents and solvents:** petrochemicals are used to produce detergents and solvents, which are used in cleaning products, paints and other industrial applications.
- **Adhesives and sealants:** petrochemicals are used to produce adhesives and sealants, which are used in a wide range of industrial and consumer applications, including construction, automotive and aerospace.
- **Pharmaceuticals:** petrochemicals are used as raw materials to produce some pharmaceuticals, such as pain relievers and antibiotics.

Derivative Market

BENCHMARK REGION	AMERICAS		EUROPE		EAST OF SUEZ	
	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO	EXCHANGE NAME	MARKET LINGO
FLAT PRICE	Benzene, IHS MARKIT US Contract Price	-	Ethylene, ICIS FD NWE	-	SGX Platts PX CFR Taiwan/China	"PX"
	Ethylene, OPIS PCW Choctaw				SGX Platts BZ FOB Korea	"BZ"
	Ethylene, OPIS PCW Mont Belvieu Enterprise				Ethylene, ICIS CFR SE Asia	
CROSS PRODUCT DIFFERENTIAL	-	-	-	-	Ethylene, ICIS CFR NE Asia	
					SGX PX-Naphtha Spread	"PX/MOPJ" or "PXN"
NICHE CONTRACTS	Polymer Grade Propylene (PGP), OPIS PCW Mont Belvieu	-	-	-	SGX BZ-Naphtha Spread	"BZ/MOPJ" or "BZN"
					SGX Platts Methanol CFR China	"Methanol"
					Propylene, I.C.I.S. CFR NE Asia	
					Propylene, I.C.I.S. CFR SE Asia	
					SGX ICIS SM CFR China	"SM"
					SGX ICIS MEG CFR China	"MEG"

Figure 2.21 Petrochemicals' Contracts

Contract Setup

- In the derivatives market, the main source of liquidity is the petrochemical vs naphtha differential, with Eastern contracts as the main activity.
- There are a variety of petrochemical-related contracts, such as PX (paraxylene), BZ (benzene), MT (methanol) and SM (styrene monomer). Although the liquidity is yet to pick up in these contracts, their average daily volume is 20–70 kt/day^[16], significantly smaller than naphtha contracts (2,000 kt/day).

Petrochemical derivatives are in the early stages of development; therefore, there is very little open interest and volume, and the market dynamics between the contracts have yet to be meaningfully established.

Contract Hedging

- Flat price: Producer and end users hedging their physical exposure
- PX-MOPJ: Hedging naphtha-fed petrochemical cracking margins
- BZ-MOPJ: Hedging naphtha-fed petrochemical cracking margins

Refinery Margin

Physical Product

Physical Product Uses

A refinery margin is a measure of the profitability of a petroleum refinery. It represents the difference between the cost of crude oil and the revenue generated by selling refined petroleum products.

$$\text{GROSS REFINERY MARGIN} = \text{COST OF REFINED PRODUCTS} - \text{COST OF FEEDSTOCK}$$

Figure 2.22 Gross Refinery Margin Calculation

Refinery margins are typically expressed as a dollar amount per barrel of crude oil processed. For example, if a refinery processes a barrel of crude oil that costs \$60 and generates \$80 in revenue from the sale of refined products, the refinery margin would be \$20 per barrel.

In general, higher refinery margins indicate greater profitability, while lower margins can signal financial challenges impacting run rates and production of refined oil products. Consequently, the petroleum industry and investors closely watch refinery margins, as they provide insight into the financial health of individual refineries and the broader petroleum market.

Derivative Market

The forward refinery margin can be estimated by calculating each refined product derivative ‘crack’. You can then assign a weight to each crack according to the expected percentage of the total volume produced per barrel of crude – different refineries produce varying proportions of refined products per barrel of crude due to their unique units and processes. The yields of refined products can also differ based on the chemical composition of the crude used and the complexity of the refinery.

The Onyx Refinery Benchmark was created to establish a benchmark for refiner’s yields by determining the output of European refiners through

various sources and making assumptions about modern-day crude flows.

The refinery margin can be tracked live using monthly product derivative prices, provided there is an ability to assess fair value. The refinery margin is valuable in determining a refinery's profitability and making informed decisions about refining operations.

24 Feb 2023 11:23:00 GMT ☀ Market Open

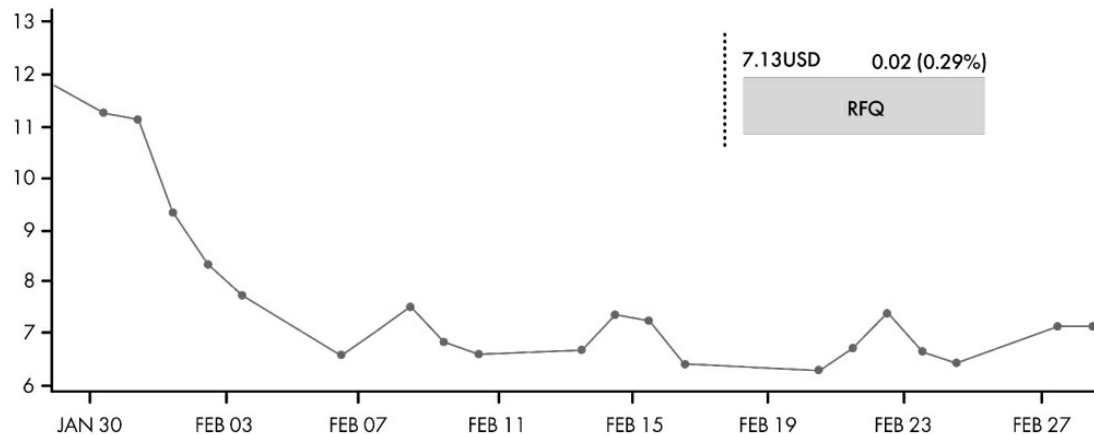


Figure 2.23 European Refinery Margin Price Action^[17]

PRODUCT	EXCHANGE	M1	WEIGHT	CONVERSION
LPG	European Propane CIF ARA (Argus) Futures	3.7456	6.00%	12.40
Naphtha	European Naphtha Cargoes CIF NWE (Platts) Futures	3.901543	10.50%	8.90
Gasoline	Gasoline Euro-bob Oxy NWE Barges (Argus) Futures	19.44003	31.00%	8.33
Jet Fuel	Jet Fuel Cargoes CIF NWE (Platts) Futures	3.062953	4.00%	7.88
Gasoil	ULSD 10PPM Cargoes CIF NWE (Platts) Futures	18.97658	28.00%	7.45
HSFO	European 3.5% Fuel Oil Barges FOB Rdam (Platts) Futures	8.9504	12.50%	6.35
LSFO	European 1% Fuel Cargoes FOB NWE (Platts) Futures	2.863678	4.00%	6.35
VLSFO	Marine Fuel 0.5% FOB Rotterdam Barges (Platts) Futures	2.324409	4.00%	6.35
Finished Product(bbl)	SUM	63.2652	100.00%	
Crude	Dated Brent (Platts) Financial Futures	65.91	100.00%	1.00
Refinery Margin(bbl)		-2.644804		

Figure 2.24 European Refinery Margin Calculation Example^[18]

Contract Expiry and Settlement

Any contract you buy or sell has a notional value during the trading day, as reflected on the computer systems at the exchange. But what is the actual value of the contract, and how is it calculated?

Contract settlements and expiries are when the financial chickens come home to roost. All financial contracts use a price related to a physical product to settle the final price. At this point, the physical and financial markets align. The final settled prices of financial contracts are then used as a reference for the enormous number of contracts worldwide.

First, the financial contracts are held by the derivatives market and need to be settled to determine the definitive cash flow at the end of the trading cycle defined for the contract (daily for futures). The definitive cash flow is the difference between the price at entry into the contract and the settlement price. If it is a physically delivered contract, the settlement price is the price you pay to take delivery of the physical product (if you are the buyer) or the price you get for the physical product (if you are the seller).

The Price Reporting Agencies

PRAs are key service providers in the oil market, providing widely used price assessments that serve as benchmark settlement prices for physical and derivative contracts. PRAs like Platts, Argus and ICIS offer daily settlements based on their proprietary indexes for various grades of crude oil and refined products worldwide.

These prices are used as benchmarks for physical trades and derivative contracts like futures and swaps. Exchanges cannot settle their own contracts because physical prices are used as a basis for the settlement to bring the value in line with the physical market. These physical trades happen almost entirely off-exchange bilaterally between major oil companies and trade houses. PRAs solve this problem by collecting data from market participants and industry contacts to calculate prices for various crude oil products worldwide.

PRAs use an ‘assessment window’, a specific period during which market participants can submit bids, offers and trades for a particular product at a specific location. These data are combined with other sources to arrive at the prevailing market price for that product at that location. The assessment window typically lasts from thirty minutes to a full trading day and is a crucial component of the price discovery process.

But the assessment window can also affect market prices, as traders may make substantial bids or offers that differ significantly from the market trend to change the aggregate price or affect market sentiment.

Focus on Settlement

Oil traders' primary focus is the settlement process because so much depends on these prices. Working days are fixed around settlement times, and preparation for future settlements is a continuous job for traders at trade houses and majors. These individuals have the greatest exposure, as they hold enormous physical contracts linked to settlement prices and take opportunistic delivery of or sell physical oil.

Trade houses and IOCs manage huge financial contract positions that are cash-settled at these settlement times, and they want to participate in and benefit from the settlement periods of contracts worldwide. For example, to make the scarcity of a given oil or oil product known to the PRA, these companies will participate in the public tender, which is used in the contract's assessment. If they can't sell the oil product, offering the physical oil in the public tender can be advantageous to demonstrate to the PRA how much oil is available. If the price on offer gets low enough, the seller may even find a buyer who is not willing to trade in the open physical market.

Know the Contract

A trader in financial contracts must know how contracts will settle. Settlement is the only definite aspect of the financial trading cycle, as price fluctuations up to that point depend on classic trader psychology like in any other financial market. You must know precisely what you are trading or risk ruining all your previous good work at settlement time.

If a hedge fund trader looking for exposure to oil trading in Brent holds their contract to expiry, they will be exposed to the price of physical BFOETM oil. The price difference on expiry using the physical BFOETM prices can be very different to the day before when the contract was simply a function of derivative buyers and sellers. If the trader rolls their position to the next month's contract, they are again exposed to the sentiment of the market participants who trade oil.

Two Physical Markets

There are two physical markets. One is traded in the PRA assessment

windows, and the other in the wider, global spot physical market.

The first difference between the two markets is access. To participate in the tendering window or be included in volume-weighted average calculations of a global benchmark settlement price, the aspiring physical trader needs to be accepted as a counterpart by industry participants, ensuring they can trade. The problem this creates is that it tends towards a select group of companies while discounting transactions from smaller entities that may be willing to trade at more attractive prices. Even if there was a willing buyer or seller for at a given level in a PRA assessment window, they can't do anything about it if they are not an approved participant.

The second difference is timing. The physical market for the PRA assessment window involves trades with loading periods in the future, around 20 days forward on average – but as much as 60 days on some contracts. In the case of Brent futures, for example, the loading period used to settle the final price is for delivery two months forward. This means that the physical that sets the Brent price on settlement has two months to discover a new price before it becomes a spot cargo.

The Physical and Derivative Relationship

The relationship between the physical spot pricing and the next month derivative constitutes the time spread, which holds the two worlds together to create the oil market ecosystem. The time spread is the difference between the contract value in one month and the next. Because of this simplicity, it is easy to think that time spreads are a consequence of independent movements in outright prices. In oil derivatives, at least, this is not the case.

The link from the physical is as follows:

- Physical traders negotiate the physical oil price as a differential to a benchmark.
- The PRA uses the physical differential prices (at least the ones they see from their assessment windows) to settle the price of the benchmark.
- The outright price of the settled financial contract is determined by the next month derivative price plus the physical differential observed by the PRA.
- This logic binds the spot price and month 1 (M1) derivatives together and creates meaning in the time spread value.

The physical differential, and consequently the spot vs M1 price, is the maximum influence physical can have on derivative prices. This influence is critical to the understanding of oil markets. It is why one can't work without the other in the current system and, therefore, why you can't ignore derivative markets.

If the derivative price was not deemed to reflect the underlying physical fundamentals, then because of how derivatives are settled, the physical fundamentals should be reflected by the physical differential, bringing the contract value back in line on settlement. Right? No. The physical relationship between derivatives and physical is through the prompt spread, not the outright prompt price. The value of spot vs M1 has real physical ramifications related to storage. These physical limitations provide a robust range for spot vs M1, which can only be exceeded in physical market extremes.

The time spread is the derivative exposure to the spot vs M1 price. When the prompt month of the time spread starts to settle, the price is set using a physical differential to M1, resulting in $\text{spot vs M1} = \text{time spread value}$. This is the case for most oil derivative contracts; however, some contracts have a slightly different mechanism. For example, the Dated Brent swap price is settled by physical spot values vs 10–30 days forward derivative prices, not simply spot vs M1.

Contango

Time spreads are the difference in price between one contract and the next. If this difference is discounted from the front month to the second month, it is said to be in contango. If the contango is deep enough – a negative enough time spread – it will cover the cost of buying the physical oil, storing it from one month to the next and profiting from the difference. So if there is enough storage capacity, physical traders will make these trades when the contango reaches an economically viable level. The physical oil trader in this scenario would be willing to buy as much physical as there is storage available at a cost that makes the trade workable.

If the price of storage increases, bids for physical by the oil trader will decrease, decreasing the time spread. If all storage capacity on land is full, the time spread will decrease further to a level incentivising storage through chartering tankers.

Only if there is so much oversupply that every bit of land storage and every available tanker in the world is full can the time spread level decrease further and test extremes. Given the storage infrastructure available in the world for oil, this has never happened.

Real-world example: *The WTI contract is priced against crude marketed in Cushing, Oklahoma. This area is inland, with pipeline infrastructure bringing crude in and out, with the only other available modes of transportation being railways and trucks. When there is an oversupply, the pricing mechanism comes under pressure and is vulnerable to bearish distortions.*

In certain scenarios, the pipelines entering Cushing from producing regions are being pumped at maximum capacity and storage in Cushing reaches near capacity and is unable to absorb the incoming buying flows. This happened in 2015 when US domestic production was increasing relentlessly, and supply out of the storage chokepoint to refiners couldn't match the incoming product.

The mechanism by which WTI futures contracts are settled makes it susceptible to bearish price moves in the time spreads. If a trader has plenty of oil storage in Cushing and can direct a lot of oil into Cushing from other areas, that trader can flood the Cushing market with oil as settlement on the prompt WTI futures contract nears. As the trader owns a lot of the storage needed to take delivery of the oil being offered into Cushing, if the market is

weak such that storage is largely full in the Cushing storage facility, what can the market do with incoming oil? The physical differentials can go as low as the physical WTI crude sellers are willing to offer them.

This happened in 2015 when time spreads reached a deep contango for prompt time spreads that some referred to as a 'super contango'.

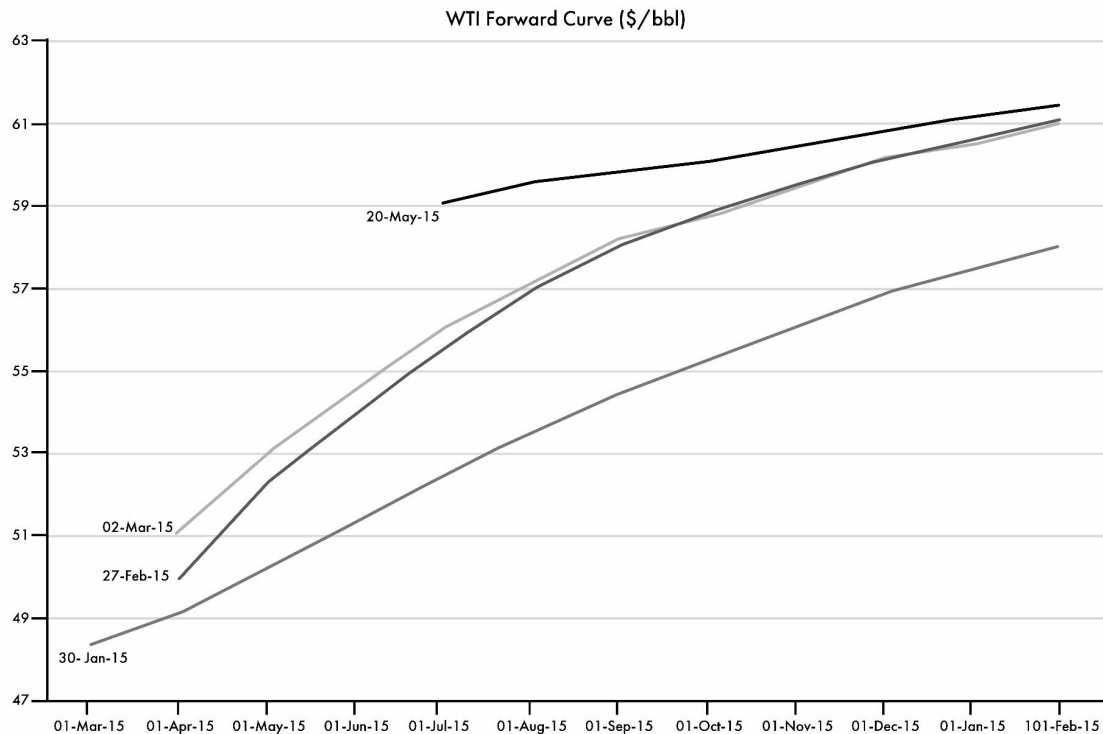


Figure 2.25 Forward Curve Comparison to Show the ‘Super Contango’ in the WTI Futures Contract in 2015

Backwardation

Storage trades provide a floor on the downside of the spot vs M1 in the derivative market. But what about the other way, in other words, when the time spreads are positive – known as backwardated markets? If the time spread gets high enough, those who are holding oil in storage can benefit by selling the oil they have in the open market and buying it back cheaper for next month’s delivery. Generally speaking, it isn’t logical to hold oil in storage in backwardated markets, where the selling price is always better in the first month.

Given the increased supply, this physical selling in the open market weighs on physical differentials. But if there is enough demand, all the world’s oil storage can be drained and sustained buying keeps driving the physical differential higher. If you have no oil in storage, and the demand is still outpacing the supply in prompt markets, then in theory, this could keep going as high as the physical oil traders are willing to bid for the physical.

In the case of extreme backwardation, buyers have other limits, even if there is demand. Physical purchases require financing, and in these extreme times, the notional value of the commodity can get too high relative to the balance sheet of trading companies. So, even if they have the opportunity, oil traders can be hamstrung by this financing restriction, which can affect even the largest of trade houses. In 2022, for example, Trafigura sought a credit facility of an additional \$1.2 billion from a consortium of banks to manage the ‘Unprecedented increase in commodity prices as a result of the war in Ukraine^[19]’. But others couldn’t source such funding, leading to concerns about a lack of participation. At the very least, the cash crunch had a directional impact on oil derivative markets as traders were forced to unwind their positions – this is covered in the World of Oil Derivatives section of this book.

Real World Example: *In 2022, the Brent time spreads reached astronomical levels as the Russia–Ukraine conflict and sparing OPEC supply were met with rapidly growing demand for commodities in the global economy. As a result of the lows Brent reached during the pricing war in April 2020, the physical crude oil market had been steadily drained of the floating storage built up during that time and the storage on land. Even before the war, the market had been starting to test the extremes of a lack of oil supply. So once the war was officially underway and sanctions were imposed on Russia, the oil market was in an unprecedented supply crunch.*

As a result, Brent time spreads reached a high of \$5.20/bbl on the expiry of the September 2022/October 2022 time spread. In the oil swaps market, the spot vs M1 equivalent for the North Sea benchmark Dated Brent (spot vs an average of 10–30 days prices) reached \$5.45/bbl.

It is at the extremes, where storage is either full or there is a supply drought, that the range on the spot vs M1 (or time spread) will extend. But as the above examples show, it is still very unlikely to go much beyond –\$5/bbl to +\$5/bbl. This means that even in the most extreme scenarios, the maximum impact the physical differential can have on the settlement of derivative contracts is this range.

This point needs to be understood as a vital principle in analysing oil markets. The derivative price can be deemed by market participants as overstretched, and the legacy supply and demand philosophy will be that it all comes back to physical in the end. But it is indisputable that the maximum retracement the outright price of a derivative like Brent crude futures will

have once physical is used to settle the price is in the range of $-\$5/\text{bbl}$ to $+\$5/\text{bbl}$.

There is another dynamic at play, which traders call optionality. The time spread can be thought of as a premium that you need to pay to hold storage in backwardated markets and get paid for in contango markets. In contango markets, you can hold the physical in storage and profit outright, with the time spread giving you a cushion to get long. In backwardated markets, it is more tricky, but paying the time spread to hold the physical in storage will allow you to profit from extreme moves in the spot market, which may supersede the time spread value on the benchmark you are pricing to. A trader could also be very technical and look to profit from volatility spikes through delta hedging the time spread to capture the risk more efficiently.

Even if the extremes of these ranges come into play, market prices become regulatory rather than physical market economics driven. During these times, like the recent example in 2022, the exchange and governments will intervene by suspending trading sessions, cancelling trades (as with the London Metals Exchange [LME] and the skyrocketing Nickel prices) or applying price caps (like the European natural gas contract, Title Transfer Facility [TTF]). Such pressures on the financial system are rare but have been put to the test in the last three years, as the above examples demonstrate.

The Time Spread and the Forward Curve

It is crucial to understand the physical vs the derivative market relationship in the context of price discovery in the forward curve. The forward curve does not reflect the market's expectation of forward supply and demand.

It is easy to think that outright prices are all independent markets, and therefore the time spread is the difference between these outright prices. It is better to see the prompt outright contract as one market and the time spreads down the curve each as an independent market. The outright prices are, therefore, the result of the prompt minus the time spreads.

For example, if the nearest contract was August 2023, December 2023 will be priced as:

$$Z_{23} = Q_{23} - (Q_{23}/U_{23} + U_{23}/V_{23} + V_{23}/X_{23} + X_{23}/Z_{23})$$

Where Q = August, U = September, V = October, X = November, Z =

December

Or using example numbers, December 2023 = \$80/bbl – (\$0.50/bbl + \$0.30/bbl + \$0.20/bbl + \$0.10/bbl) = \$78.90/bbl

The time spreads have physical storage implications, which hold them in a range; however, time spreads are used by traders as a general reflection of their view on physical supply and demand. Due to their stability relative to outright prices, oil traders trade the time spreads in higher volumes than the outright price if they reflect a view in the physical market, given the heavy influence of physical prices in how the time spread value is settled.

The liquidity and open interest of time spreads are consequently very high, particularly for prompt time spreads, when compared with back-end outright prices. Given this, the back-end price liquidity is derived from the prompt price and the time spreads. If a trader is looking to buy or sell the back-end outright price, a market maker (automated or voice-brokered) will imply a bid/offer off the sum of the prompt and the time spreads, knowing that the liquidity to exit the risk is via the two.

Therefore, when you read that back-end prices have an independent fundamental justification for being, this is wrong. In fact, understanding the back-end liquidity and price discovery process is vital for speculators and hedgers. I've heard, for example, that treasurers and analysts sometimes use the forward curve as a price prediction tool. To be fair to those who do this, predicting future prices is notoriously difficult, and I doubt this process yields any worse results than listening to the research houses and banks, but the logic is incorrect.

Real World Example: *In 2021, the market was emerging from the price war of April 2020, and COVID restrictions were easing steadily. A significant uptick in deferred demand was expected as the months rolled on. The logic behind this expectation was hard to argue with, but why did the Brent crude futures market, the leading global benchmark for oil, move in backwardation in 2021, causing back-end outright prices to be lower than the prompt even as the world was largely still in lockdown?*

The market moved into backwardation as the health of the physical market improved in the spot market. There was robust demand and a balanced physical market, so the spot vs M1 prices, and consequently the derivative time spread value, needed to reflect that.

In theory, the market could have been forced into contango by extreme buying of back-end outright prices; however, this buying would have needed

to outpace buying in the prompt and overwhelm buying in time spreads, both of which have greater open interest and liquidity. Even if this had happened, when it came to settlement, the time spread would convert to physical spot prices vs M1 derivatives, and physical oil traders would have happily bid for physical oil into backwardated levels, with prompt demand strong enough and far too much competition to store if there was a contango.

Settlement Mechanics

Exchange-traded Futures Settlements

The exchange-traded futures have a final settlement price on the expiry of the contract, but there are also assessments of the price every trading day. The key assessment is the end-of-day settlement, used to price swaps off exchange-traded futures. The end-of-day settlement is usually done at 19.30 London time but can be earlier during bank holidays or US holidays. In addition, there is a published assessment of the exchange-traded futures price during the day, typically at 16.30 Singapore time and 16.30 London time, to coincide with PRA assessment periods. Finally, the exchange can announce an assessment of contracts to accommodate any changes resulting from holidays, or a PRA assessment, such as the five assessments introduced on the day Brent futures expire, called the Brent Index.

Traders use exchange-traded futures for hedging swaps and options contracts with different expiry timings and procedures. Therefore, the exchange offers tools for traders to manage their futures hedges as assessments materialise. As contracts expire, traders use exchange contracts to enter and exit positions at the assessment price.

For a breakdown of how Brent futures settle, refer to the Appendix.

Trade at Settlement

Trade at Settlement (TAS) is a contract that becomes a live exchange-traded future only when the exchange-traded future settles and at the settlement price determined by the exchange. TAS allows a trader to execute a trade at the settlement price of a futures contract rather than at the current market price.

TAS reflects the end-of-day settlement for exchange-traded futures. The exchange determines the TAS price based on the volume-weighted average price by taking all the trades, weighting the prices by trade volume and calculating the average. This process occurs for a two-minute period for most exchange-traded futures, known as the TAS 'window'.

The main purpose of TAS is to provide a way to trade in or out of hedges precisely at the moment of settlement, thereby systematically offsetting price exposure risk at that time. It is therefore used mainly for administrative purposes, and it is common to get junior or operations staff to manage this process.

For example, the bid/offer volume on live Brent futures is around 25 KB on each side at any moment during a busy trading day, but TAS can be as much as 5,000 KB on each side or more. TAS contracts are usually live at 19.30 London time but earlier when there is a US holiday like Thanksgiving. Keeping a record of when TAS is scheduled and executing a trade before the settlement window begins is very important. If you are relying on this contract to manage a hedge, missing the TAS time will leave you exposed until you enter the open market to trade the exchange-traded future yourself. This exposure is to the outright price of futures, the most volatile contract in oil derivatives, so even a small position can be devastating. Therefore, despite being a relatively mundane administrative process, TAS execution is a highly critical period for the market.

TAS does come with risks. If you wait until the last minute of the settlement window, the TAS price becomes very volatile as the algorithms start to calculate TAS prices vs the impending settlement prices. I experienced this in a particular heart-in-mouth moment as my team and I were encouraged by our trading manager to be 'smart with it'. We were market makers, so although we had very low profit and loss swings, we had many swaps rolling off at TAS. Therefore, the \$0.01/bbl at that cumulative volume added up. So in an attempt to get \$0.00/bbl or better, thinking I could be a hero, I held risk into the settlement window only to see the algorithms drive the TAS price to \$0.04/bbl bid and had to lift \$0.05/bbl. This is a worthwhile lesson, as it almost cost me my job. It also serves as a lesson for the trading manager, who should always be aware of the potential risks, even with an administrative trade like this.

Trade at Settlement (TAS)

TAS contracts trade at differentials to today's settlement price

TAS BIDS			TAS OFFERS
		0.03	4976KB
		0.02	4865KB
		0.01	5132KB
4982KB	0.00		
5021KB	-0.01		
5114KB	-0.2		

STABLE PRE-SETTLEMENT PERIOD

SETTLEMENT PERIOD (TAS WINDOW)

2 minute window between
14:28 - 14.30 EST

VOLUME WEIGHTED AVERAGE PRICE (VWAP)

of the trades in the settlement period is assessed

TODAY'S BRENT FUTURES SETTLEMENT PRICE

determined and published

OR SHORT A LIVE FUTURES CONTRACT

with an entry level of settlement price +
differential trade

Figure 2.26 Trade at Settlement (TAS) Mechanism

TAS liquidity helps the market by ensuring buyers and sellers are always available at the settlement price. Ideally, you trade with someone who is hedging their roll-off of swaps or other contracts. However, if hedging volume is not being matched, this volume will find its way into the open market and potentially be used against you, as the short story below demonstrates.

Short Story: Trade at Settlement Involvement in Negative West Texas Intermediate Prices

TAS contracts have sometimes been the centre of controversy, as in 2020 amid the pricing war between Saudi Arabia and Russia. There is a common public perception that oil went negative, but this was fleeting and based on a poor understanding of the TAS contract.

A few canny traders had noticed that the market was thin on the buy side due to the heavy oversupply in physical crude oil markets and the physically

delivered expiry of the prompt (nearest available) WTI crude futures contract coming the next day. They knew that in this state of excessive oversupply, the physical product delivered under the contract would be susceptible to these conditions. For the WTI contract, physical WTI oil is delivered to a landlocked storage facility in Cushing, Oklahoma, with finite space and limited capacity to store the physical oil of the trades flooding the market on expiry. This creates difficulties for exchanges trying to ensure that the futures market remains robust in extreme scenarios like this. The susceptibility of the contract to oversupply was also highlighted in a Commodities Futures Trading Commission (CFTC) investigation in 2011 when a trader said in an email that there was a 's*** load of money to be made shorting'[\[20\]](#) WTI.

The market state in which the prompt WTI contract was found the day before expiry set the stage for a one-way fight for liquidity, since bids in the market would be scarce if those long in the contract were going to exit. Therefore, the traders were confident in selling these contracts short, as they could probably see the evidence of a market under pressure. They exerted considerable leverage by buying the TAS contracts and selling the WTI Futures immediately in the open market. By doing this, they ensured that the longest time they would have a short exposure was until 19.30 London time when the buy order of the TAS contract would convert into a long position and close out the shorts they had entered into earlier. There were huge volumes of TAS contracts, and they were able to trade tens of thousands of lots of these contracts, which, when sold into the open market on a thin liquidity day like the one in this example, can hugely impact the price. That day saw an unprecedented negative move in the price for WTI, as the contracts continued to be sold throughout the TAS window, with traders hitting bids down to lows of -\$37 per barrel.

The speculating traders banked huge sums, which was remarkable when it turned out they were a small trading outfit based in a tiny office in Essex. But it would be a mistake to say that this was a trade with no risk; in fact, it was anything but. By selling in the open market, they were exposed to the market moving against them, which could have been by many dollars per barrel. During that time, the volatility had exploded, and a \$10 per barrel move was not out of the question. If they had sold 5,000 contracts, their exposure would have been at least \$50 million. These traders, I'm sure, knew it was a once-in-a-lifetime trade with the stars aligning like they never had before.

The nature of this trade was contentious and subject to debate, but the ramifications were huge. Oil prices going negative was big news, reported worldwide, and it captured the attention of people with no interest in oil markets. As is almost always the case, commentators explained the phenomenon as an imbalance of supply and demand when it was more a function of derivatives nuance, as the investigation by Kit Chellel at Bloomberg eventually proved.

Minute Markers

Minute markers are similar to TAS but for various times in the day and are calculated over one minute rather than two minutes. Minute markers are provided at important times in the market, usually aligning with roll-off periods in PRA assessment windows or the Brent Index. For example, there is a minute marker at 16.30 Singapore time for Eastern contracts and at 16.30 London time for European-based contracts. For the Brent Index, there are three additional minute markers (10.30, 12.30 and 14.30 London time), the usual minute marker at 16:30 and TAS, the settlement prices of which are used, in part, to calculate the final price for expiring contracts.

Exchange-traded ‘Futures Swaps’ Settlement

Exchange-traded futures have a one-day expiry, and oil swaps settle every day, according to a PRA assessment. As a result, hedging swaps with futures creates a headache when swaps start the settlement process. To help manage this more efficiently, exchanges provide futures swap contracts, which exchanges call 1st line futures.

Futures swaps trade just like PRA-assessed oil swaps, except instead of using a PRA, the exchange settles a price for the futures swap based on the nearest available future. The Brent swap, for instance, settles every day to the Brent futures settlement price determined by the TAS mechanism.

To price a futures swap, you need to link the pricing days with the nearest available future. For example, a January 2023 Brent swap has 21 pricing days, the rest of the month being weekends when the exchange is not open. During January, the nearest available Brent future is the March 2023 Brent future, which expires on 31 January. So, for twenty pricing days, the TAS price of the March 2023 Brent future can be used. However, the March

Brent future will expire to the Brent Index on 31 January (the last working day of the month). So the next available future that will work as an effective hedge on that day will be the next month's futures, in this example, April 2023 Brent futures.

100KB March Brent Futures vs 100KB of Brent Swaps

JANUARY (TRADING DAYS)	MARCH FUTURES	APRIL FUTURES	JANUARY SWAPS
1			-5KB
2			-5KB
21			-5KB
22	+95KB	+5KB	-5KB
TOTAL	+95KB	+5KB	+100KB

Risk is 'delta neutral' as the contracts will move in tandem until settlement

1. BRENT FUTURES HAVE ONE EXPIRING DAY, WHEREAS SWAPS HAVE DAILY.
2. MARCH FUTURES EXPIRES ON LAST WORKING DAY, SO APRIL FUTURES IS USED TO PRICE LAST DAY OF SWAP.
3. A 'FUTURES EQUIVALENT' HEDGE FOR BRENT SWAPS THEREFORE INCLUDES 21 DAYS OF MARCH AND 1 DAY OF APRIL.

JANUARY (TRADING DAYS)	MARCH FUTURES	APRIL FUTURES	JANUARY SWAPS
1			0KB
2			-5KB
21			-5KB
22	+95KB	+5KB	-5KB
TOTAL	+95KB	+5KB	-95KB

Risk is now net long by 5KB as one day of Jan has rolled off, whilst Futures stay live until last day of month.

4. TAS CONTRACT CAN BE USED TO CLOSE OUT FUTURES HEDGE IN THE EXACT VOLUME THAT HAS ROLLED OFF IN THE SWAP, AND AT THE SAME PRICE. GROSS PNL WILL STAY ZERO IF TAS IS TRADED AT 0.00 TO SETTLEMENT.

JANUARY (TRADING DAYS)	MARCH FUTURES	APRIL FUTURES	JANUARY SWAPS
1			0KB
2			-5KB
21			-5KB
22	+90KB	+5KB	-5KB
TOTAL	+90KB	+5KB	-95KB

Selling March TAS in 5KB will reduce position back to 'delta neutral'

Figure 2.7 Using Trade at Settlement to Manage Swap Roll-off

When traders use exchange-traded futures to hedge against an oil swap, they must be aware of the difference. This can be best explained through a real example. A physical trader wants to hedge the outright price of a physical naphtha cargo to be delivered in the future by buying Brent futures. The outright naphtha price correlates well with Brent futures, so this will be a sufficient hedge up to a point. But the oil trader will be left with a short naphtha vs Brent futures exposure until the physical naphtha cargo is priced

and the Brent future is traded.

Naphtha basis risk is reflected in the crack – naphtha vs Brent swaps. But if you want to exit the naphtha vs Brent risk via the crack and you've hedged with Brent futures, then you are short naphtha vs Brent swaps and short Brent futures vs Brent swaps because the Brent swaps price is equivalent to a short Brent futures time spread risk. So when reconciling the hedge performance against the movement of the physical price, this needs to be accounted for.

Getting Into Futures

Naphtha Crack Example

TRADE 1				TRADE 2			TRADE 3		
SELL NWE NAPHTHA				BUY BRENT FUTURE			BUY NAPHTHA CRACK		
							(Naphtha - Brent Swaps)		
'Implied' Naphtha Crack risk									
TRADE 1				TRADE 2			TRADE 3		
TRADING DAY	NWE NAPHTHA	BRENT FUTURES	BRENT SWAPS	NWE NAPHTHA	BRENT FUTURES	BRENT SWAPS	NWE NAPHTHA	BRENT FUTURES	BRENT SWAPS
1	-5KB			-5KB	+5KB			+5KB	-5KB
2	-5KB			-5KB	+5KB			+5KB	-5KB
21	-5KB			-5KB	+5KB			+5KB	-5KB
22	-5KB			-5KB	+5KB			+5KB	-5KB
TOTAL	-100KB	0	0	-100KB	+100KB	0		+100KB	-100KB

See the appendix for an example of how a new swap methodology would be developed, including the important factors to consider.

Common vs Non-common Pricing

When trading Brent swaps in isolation – through the broker market – the pricing days that the Brent swap will follow will be based on the ICE Brent futures calendar (which only has holidays for Easter, Christmas Day and New Year's Day). When trading oil swap differentials with Brent swaps as one of the legs, Brent swap pricing days are defined by the PRA oil swap it is linked to. For example, the DFL contract is Dated Brent vs Brent swaps. Brent swaps will follow the same calendar as Dated Brent, a European calendar with UK bank holidays.

So if you hold an isolated Brent swap against a Brent swap from an oil swap differential like the DFL, you may have a pricing day calendar mismatch, leading to different monthly average prices. It is easy for this to go unnoticed, particularly if you trade many of these contracts regularly. So if you have pooled all your Brent swap risk from multiple contracts and hedged with Brent futures, you must account for these differing calendars.

Platts Market on Close Contracts

PRAs have assessment periods and methodologies to settle their indexes, which set the daily price of financial oil contracts. However, they also offer a Market on Close (MOC) assessment, which works like TAS and minute markers for exchange-traded futures but for oil swaps.

Platts MOC, also known as trade at Platts settlement, is settled through the Platts e-window accessible online via the Platts exchange. It is straightforward, as the PRA takes the last traded price in the assessment period. If there is no trade close to the end of the window, then the PRA uses its discretion to set a price just above the last outstanding bid or offer or a fair value in between.

Compliance and Regulation

You can't have a market without regulation, and the oil derivative market has been regulated since NYMEX listed its first oil derivatives. The CFTC was the first regulator with teeth. It started in the US in 1974 with the remit of overseeing the growth of derivatives and the authority to make rules and demand reporting. Since then, policing of financial oil contracts has come under the regulation of the multi-trillion-dollar derivatives market, of which energy contracts are estimated to represent around 19% as of 2015^[21].

The main purpose of the financial oil market from the regulators' point of view is to provide hedging and price discovery, although policymakers and regulators seem to accept that speculation enhances contract liquidity, making it useful for a robust market to function^[22]. In 1997, the CFTC's commissioner, in a speech to his peers, advocated the need for futures, citing the onion market as an example^[23]. The onion futures market was formed in the 1940s. It was trading on the CME until, in 1955, two traders bought almost all the futures contracts and took physical delivery of around 14,000 tons, warehousing them in Chicago, then going short on onion futures. They drove the price down so low that many growers were forced out of business, and farmers lobbied the US Congress to stop the futures trading. Indeed, in 1958, Congress passed The Onion Futures Act banning onion futures trading – it is still in effect.

The CME was hit hard. Onion futures were 20% of the exchange's business^[24], and many argued that the ban would only increase price volatility. The scenario gave economists the perfect test case. Studies years later found more cash market volatility in onion prices before and after futures trading was permitted. 'In other words,' the CFTC commissioner concluded in his speech, 'futures markets don't cause volatility; they respond to and decrease volatility'.

This was an extraordinary scheme, but as a result, the impact of derivatives, like futures, on the oil market is constantly under scrutiny and subject to debate. Interestingly, the commissioner commented in the same speech that onion growers commenting on the volatility believed that upward prices resulted from supply and demand, but aggressive downward price moves were linked to 'greedy speculators'. This sentiment also exists in oil markets; however, end users and governments lament high prices and volatility and blame this price environment on speculation. Consequently,

governments have (and do) intervened. They see the price at the pump as politically useful at election time – nowhere more so than in the US and other big economies like India. It was recently revealed that in 2002, Tony Blair's UK government was looking to use the 'change of regime' in Iraq to flood the oil market and was encouraging Nigeria and Venezuela to leave OPEC^[25] after a spike in crude oil prices.

Financial oil trading is subject to much of the same regulation as the wider derivative world, with a few exceptions. Oil derivative markets are related to a tangible asset – physical oil – so making a case for non-compliant activity is more difficult than for purely financial securities like bonds or foreign exchange. The separation of physical and derivative transactions and how they are linked makes it complicated for those overseeing the market. What constitutes a hedge? How do you know what is a hedge and what is speculation? It is easier for regulators to create a legal framework and oversee purely financial activity, such as those of investment banks and hedge funds. But this is beginning to change as influence in the market shifts to the oil trader, as recent legal cases have demonstrated, examples of which I go into later in this chapter.

Dodd-Frank

The biggest regulatory interventions, and consequently the most significant effects on the oil derivative market and its operation, came after financial crises, most recently after 2008. A series of high-profile scandals – Societe Generale, where trader Jerome Kerviel lost \$7.2 billion on stock index futures, and Kweku M. Adoboli, who lost \$2.3 billion trading in UBS’s Synthetic Equities team^[26] – put pressure on governments and regulators to step up. Public sentiment against the financial sector was palpable, and the oil derivative market was not immune to regulatory change.

One of the biggest regulatory changes in the last fifteen years is the Volker Rule, named after the chairman of the US Federal Reserve at the time, Paul A. Volker. Initially proposed in 2011 and implemented in 2013, the rule forbade investment banks from trading on their own account. It echoes the Glass–Steagall Act of 1933, which after the financial turmoil of the Great Depression, separated the commercial (retail) and investment activities within banks. It was repealed in 1999 – one of the reasons for the financial crash of 2008–2009. Before the Act was enacted in 1933, investment banks were the dominant providers of stock liquidity, financing and speculative risk. The restriction on proprietary trading took a considerable revenue stream from investment banks; although market-making and hedging client orders were permitted, the restrictions were enough to cause top traders at the banks to head for the exit and move to trade houses, major oil producers and hedge funds. What constituted a hedge or making a market in securities at banks became difficult to define and defend, which made market-making activity seriously restrictive for the trading floors. The restrictions on market-making in banks are one of the reasons why independent market-making firms like Onyx Commodities were able to start and flourish. The combination of oil swaps moving to exchanges, which increased price visibility, and the retrenchment of the large market makers from investment banks left a vacuum for liquidity providing.

Although the Volker Rule was tweaked in 2020 to allow investment in private equity funds or hedge funds, trading by investment banks in oil derivatives is still forbidden.

Basel III

Another important set of financial regulations affecting oil trading financing is the Basel Accords, originally issued in 1988 by the Basel Committee for Banking Supervision. The Accords have undergone three iterations and are due for a fourth in 2023. These regulations are designed to limit the impact on the financial system of banks taking too much risk by strengthening capital requirements and corporate governance. Specifically, the notional value of the trading contracts would now be shown on the balance sheet of investment banks, counting towards the calculation of minimum capital reserves.

Basel III (2018) introduced the leverage ratio, which is calculated by dividing Tier 1 capital (equity capital and declared reserves) by the bank's leverage exposure, and requires banks to keep this in excess of 3%^[27]. The impact on oil derivative markets was similar to other derivative markets, which suffered from how leverage exposure was calculated: the notional value was determined by simply summing the value of contracts on the balance sheet at the prevailing market price. It made no accommodation for how these contracts were offset against each other – it measured gross rather than net risk exposure.

This made it difficult for the banks' margin-financing market makers of derivatives, since the business model is to hold market risk to offset contracts and make small margins between them. In the oil derivatives market, this led to trading 'switches', where market participants offered a price for a group of contracts that offset each other. These were expensive and restrictive, and the Basel regulations were blamed for the collapse of a prominent market maker, Geneva Energy Markets. 'The notional value of our book was in excess of \$50 billion', the company founder, Mark Vonderheide, told Bloomberg News, 'however, the actual risk of the book was always relatively low, with VaR (value-at-risk) at around \$2 million at any given time'^[28]. The effect of the leverage ratio calculation was eventually mitigated by the Standardised Approach for Counterparty Credit Risk (SA-CCR) calculation in 2017, which allowed for net exposure.

The rules will become more stringent with Basel IV, which is reported to be an extensive overhaul of the current system, making it effectively a new framework^[29]. For example, the rules prevent the use of internal risk models to assess the credit risk of large corporates with a turnover above 500 million EUR and standardise how banks determine capital requirements for counterparts.

European Markets Infrastructure Regulation

Introduced in 2012, the EMIR (European markets infrastructure regulation) covered the OTC trading of financial derivatives – trading bilaterally outside exchanges. Under EMIR, any OTC (or bilateral) trade must be sent to the regulator, who analyses the transaction. This, of course, affects banks, the largest provider of financing for these deals, and increases price visibility on the trades that have been generating large sums for banks through the bid/offer process. Now, every deal needs to be analysed with a credit valuation adjustment, which determines the value of the deal net of the credit exposure for the bank to see if the deal is worth doing^[30].

EMIR also forced physical trading firms with derivatives trading exceeding certain thresholds to trade through a central clearing counterparty, which restricts the market from ever going back to OTC, despite the liquidity issues in 2022.

Markets in Financial Instruments Directive II

The MiFID (markets in financial instruments directive) is a European regulation that enhances trading transparency, particularly stocks and shares. It was introduced across the EU in 2007 and updated in 2018 with MiFID II, which broadened its scope to include derivatives and provide protection for different classes of investors: professional, retail and counterparty. For example, oil firms using derivatives to hedge were exempt from MiFID I rules, so the OTC hedge against a physical cargo did not need to be reported. However, MiFID II includes these firms and has moved the onus for reporting trade data from exchanges to trading firms, which must now report directly to regulators.

This move led to sizeable investments in information technology and infrastructure to ensure accurate reporting^[31].

Within the data being analysed by the regulator are position limits, which aim to restrict the ability of one firm to dominate the market. For Brent futures contracts, the maximum position limit on the front month contract is 75,000 lots or 75 Mbbl, which is calculated as around 13.5% of ‘deliverable supply’. Deliverable supply is not based on the BFOETM oil deliverable through the EFP mechanism on Brent expiry, but rather the UK, Norway,

Russia, Libya, Algeria, Angola, Nigeria and the rest of Europe over 2016^[32], as they are substitutable crude oil grades for physical Brent crude. For tenors or contract months outside of the front month, the limit is 300 Mbbl, equating to roughly 10% of the total open interest of Brent futures, using 2016 data. Holding this many contracts outright on Brent futures would be an enormous risk, as the seven-day ATR (average true range) could easily surpass \$10 per barrel, equivalent to a \$750 million loss in just a week – if you were exposed to the peak to trough in 2022, it would have cost you \$70 per barrel, or \$5 billion+!

The reality is that this position size is not restrictive for companies on an outright basis but rather through the Brent futures time spreads, which the majors and the top five trade houses holding the prompt time spread can easily exceed.

Outside of trading, MiFID II regulates the ‘unbundling of research’ – brokerage and market-making services have to be separate from research, and the cost of each should be clearly defined. Banks have long used their research data to attract clientele to the financial and market-making business, but they must now charge for this, as it can’t be seen as a financial promotion or an incentive to trade. The let-out for the bank is that this doesn’t include macroeconomic analysis, and how that is defined can be tricky, allowing the rules to be circumnavigated. Banks have also reduced their costs to a minimum by decreasing their research, so the overall impact has been negligible. However, the use of macroeconomic research has diminished in value given the widespread democratisation of information in this space, which I discuss in the analysis section of this book.

Exchange Rules

As an oil derivative market participant, you will see the most direct impact of regulation from the exchanges, who impose their rules on you for clearing your trades. The rules of exchanges are not the same as those of the regulating bodies like the Financial Conduct Authority (FCA) or CFTC, but exchanges are obliged to create a regulatory framework to prevent market abuse. For example, exchanges monitor trading activity data and can demand evidence of trading communications to justify a trading action.

The exchange will impose several types of limits on your trading, as illustrated in Figure 2.31.

LIMIT TYPE	
Position limits	As determined by regulators; for example, under MiFID II.
Block minimums	Trades directly between counterparts, or via inter-dealer brokerages, are subject to minimum position sizes that are larger than trades executed directly on the exchange. Most oil swap trades are executed this way, as are a proportion of futures.
Block reporting times	Trades not done directly on the exchange are subject to a time limit to get the trade reported to the exchange. The window of time from the deal confirmed to reporting varies from 5 to 15 minutes.
Price and volume reasonability	The price and volume reasonability check is in place to avoid execution errors outside the upper and lower limit on price and threshold for trading volumes
Market manipulation	This is broad by design and is deemed to be committed when a person enters a transaction, places an order to trade or carries out any other behaviour that: Gives, or is likely to give, false or misleading signals as to the supply of, demand for or price of a financial instrument. Secures, or is likely to secure, the price of one or several financial instruments. This includes attempted market manipulation, meaning you do not need to profit or succeed in manipulating markets to be in breach.

Figure 2.30 Trading Limits Imposed by the Exchange

Examples of Market Manipulation

Marking or Banging the Close or Open

This is a pattern of trading activity intended to manipulate the settlement price of an instrument during its settlement period (but can also happen before the open, for example, influencing the opening price of an instrument). The exchange looks for aggressive orders, for example, buys well above the settlement price or sells well below the settlement price within the settlement period.

A real-world example: In 2008, floor broker Daniel Shak, trading on his own account, engaged in a trading strategy deemed manipulative by the CFTC^[33]. The CFTC also found that Shak exceeded the position limit set by NYMEX for this contract, violating the Commodity Exchange Act. Shak's strategy involved using WTI futures against the WTI TAS (see Chapter 4). On two separate occasions, Shak sold TAS contracts and bought WTI futures contracts just before or during the settlement period of WTI futures.

In the two minutes before the settlement window, 22% of all outright buy orders of WTI futures were executed by Shak, which grew to 64% of all purchases during the two-minute TAS window. The buying was observed to be consistently at higher prices during this period and therefore deemed to be 'banging the close'.

Shak agreed to a \$400,000 fine, a permanent ban on trading financial instruments in crude oil and a two-year ban on trading any financial instrument during a closing period.

Spoofing

This involves attempts to deceive the market into thinking that an instrument has more interest, liquidity or depth than it does by placing large orders on one side to cause traders to execute smaller orders on the opposite side. Once the intended orders are filled, the larger orders are deleted. The Commodity Exchange Act defines spoofing as ‘bidding or offering with the intent to cancel the bid or offer before execution’.

A real-world example: *On 30 September 2020, the CFTC ordered Sunoco LP to pay \$450,000 for ‘spoofing in crude oil, gasoline and heating oil futures contracts’^[34]. One of Sunoco’s former traders was found to have placed a ‘genuine order with a small visible quantity’ that he wanted to get filled. On the opposite side of the market, he placed a larger order, which he then cancelled shortly after placing it. He repeated this enough times that it created a pattern the CFTC could prove was the trading activity known as ‘spoofing’.*

Washing or Crossing

This involves simultaneously trading buys and sells in the same instrument to create fictitious activity on an account or in several accounts owned by the same trader.

A real-world example: *At JP Morgan Chase, ‘on ten separate occasions between 1 January 2011 and 30 June 2011, in an effort to manage position limits, traders employed by JPM executed block trades between separate legal entities with the same beneficial owner in WTI or Gasoline’, the notice said, referring to the WTI, the main US crude oil futures contract. The notice went on to state that ‘The [NYMEX Business Conduct Committee] Panel also found that in each of these ten instances, the trader was the sole decision maker for both the buy and sell side of the trade’^[35].*

Inside Information

Insider dealing refers to having an unfair advantage by using inside information to enter into market transactions (whether on your account or for a third party) to which that information relates to the detriment of others who do not have this information.

For commodity derivatives, the definition of inside information consists of three tests. The information

- Must relate directly or indirectly to such commodity derivatives or directly to the related spot commodity contract
- Must be non-public, precise and price sensitive
- Must be reasonably expected or required to be disclosed per legal or regulatory provisions

Market Rumours

Refer to one of the following activities that traders are prohibited from engaging in:

- Fabricating something of market interest and communicating it to another person, whether to affect market prices, levels of trading activity or otherwise
- Broadcasting any information which you either know or strongly suspect is untrue
- Passing on any information which you either know or strongly suspect may stem from inside sources (i.e. sources which are non-public and under a duty of confidentiality)

Further, if the rumour is known or suspected to be based on inside information, you may not trade or encourage others to trade based on that rumour.

Benchmark Manipulation

A more nuanced manipulation that has created a precedent in recent years involves the act of forcing financial benchmarks to extremes to benefit oil traders' physical prices.

Global commodity trading giant Glencore Ltd admitted to engaging in price manipulation in the fuel oil market spanning the eight years from 2011 to 2019^[36]. This admission resulted from a guilty plea to one count of conspiracy, as filed in the District of Connecticut. Glencore employees

reportedly collaborated to manipulate benchmark price assessments for fuel oil products. As a result of this activity, Glencore was ordered to pay a staggering \$1.5 billion in criminal fines and forfeitures, with \$341 million going towards a criminal fine and \$144 million in criminal forfeiture. Additionally, over \$1.1 billion will be paid to the US CFTC in a related, parallel proceeding involving market manipulation and Foreign Corrupt Practices Act violations.

Recording of Trading Communications

The communication of trading activities must be recorded – for example, through a voice recording if you are trading through a brokerage – and the trader must ensure that the transaction adheres to the rules. Inter-dealer brokerages have been implicated in not following this rule over the last few years. If an exchange becomes aware of a breach of this rule, it will hand out fines and report suspicious activity to the regulator. However, exchange rules differ from exchange to exchange.

Anti-competition

The increasing regulation and compliance in the oil derivative industry have undoubtedly helped some companies become monopolistic in their behaviours.

Exchanges have their own clearing services, but the information technology to facilitate trading goes beyond that needed for contract matching and automated clearing. Exchanges sell swathes of data and have research services and trading venues. It is widely known that exchanges in the oil derivatives market are in direct competition with inter-dealer brokerages, which makes the self-regulating nature of the exchange setup an issue for some, particularly in cases where the exchange has been so aggressive.

In 2015 the ICE bought Trayport^[37], a UK trading software venue popular for power, gas and NGL derivatives trading. But the ownership of another sizeable trading venue by an exchange as large as the ICE was too much for the regulator to ignore. In 2017, the ICE was ordered to sell Trayport, which was bought by the Canadian TMXGroup, which was considered not large enough to disrupt competition^[38].

No one should be against competition, and monopolistic activity has the same damaging impact on the financial oil market as on any other market.

Future of Compliance and Regulation and its Impact on the Oil Derivative Market

The regulatory environment in all derivatives markets has developed considerably since the 2008 financial crisis, and many of the proposed rulings have been assimilated successfully. This will continue over the next few years and impact the oil derivatives market. Basel IV will cause consequential change.

Convictions for market abuse have deterred poor behaviour and will continue to do so. The consequent tightening of regulations helps establish trust and democratises markets, which is a primary goal of the regulators.

Transparency will undoubtedly become vital to how traders analyse the markets. The need to record and report trading activity in granular detail will undoubtedly continue. In addition, increasing restrictions on position sizing will increase democratisation in the movement of financial oil prices. Therefore, position data like the Commitment of Traders (COT) from NYMEX, ICE and Onyx will become increasingly important.

In the equity market, hedge funds with assets under management of over \$100 million are required to reveal their stock holdings^[39], which has allowed many amateur commentators on YouTube to analyse the market positions of well-known firms like Warren Buffet's Berkshire Hathaway or Michael Burry's Scion Capital. Could we soon see this type of thing in the oil derivatives market?

Perhaps unsurprisingly, the physical market has been getting much more attention since the Russian war on Ukraine, with the consequent inflation and cost of living crisis. Financial nuances have weakened the outright price of benchmark oil futures. Any price rise was blamed firmly on Putin's war, OPEC and production cuts, so politicians have focused on those issues.

A potential danger to markets is actually to the exchange mechanism itself, such as the recent price controls on natural gas contracts and cancelling nickel trades on the LME in 2022. The direct involvement of the LME in the market mechanism is very worrying and would be hard to get right if tried again.

Exchanges are independent businesses that set the rules and act as self-regulators, which is potentially problematic. Oppressive rules may push traders from one exchange to another, while constant fines by exchanges on

inter-dealer brokers is a worrying trend, as they have the freedom to penalise their competition.

3 TRADING

Oil Market Influencers

Sentiment in the oil market is constantly influenced by people and organisations pushing targeted information into the public domain. The weighting the industry gives each bit of information from a source depends on the market participant.

For example, a statement from the US government of its intention to release crude oil from the US strategic petroleum reserve would generally be considered a bearish headline, as there would be more supply in the market. However, a trader holding a long position in Brent futures might regard it as immaterial to the long-term performance of the contract because they might already subscribe to the view of a research company that there will be a supply crunch from a lack of capital investment in new oil production, which would counter any short-term bad news. So they might elect to hold their position, resulting in no impact on the Brent futures market from this trader.

The point is that it's only when traders react that news has any impact on the market. By the same logic, if all the potential buyers have already bought derivatives contracts and are committed, then no amount of new bullish information will drive markets higher until the settlement date of those contracts, when physical prices are used to determine the final price.

Influencers understand how important sentiment is to market dynamics and try to influence participants' reactions to suit their agendas. The industry refers to this manipulation by influencers as 'talking their book', that is, spinning a narrative that suits the market direction the person is exposed to. Every trader effectively does this when they speak about how they see the market. For example, a trader who doubled down on their short position after a pipeline issue that turned out to be immaterial might explain that they just wanted to express their view, felt the fundamentals were weak and couldn't see how the pipeline would have any influence.

Real-world example: *I remember one time a trader at a bank confessed that he'd fallen for this several times when trading the Brent Futures time spreads. A particularly influential trade house had talked the market up to*

the bank trader and was convincing in its argument, leading the trader to buy time spread contracts. But when he did, he found a wall of selling, and the market started to move aggressively in the other direction. After this scenario played out not once but three times, he suspected it was because the trader from the influential trade house was going in the opposite direction and was in the process of taking off his position.

In the media, the most persuasive voices come from

- OPEC+ and ministers or representatives
- Geopolitical leaders (the US president's administration, EU leaders, China's Communist Party representatives, etc.)
- Major oil or oil trading company representatives
- Independent or government-backed research and analytics companies
- Investment bank analysts

Most traders in the market receive comments and opinions from *Bloomberg* and *Reuters* journalists. The *Financial Times* and *The Wall Street Journal* have dedicated oil reporting less frequently and reliably than *Bloomberg* and *Reuters*, except for large announcements. Oil producers and governments regularly talk to journalists at news companies and make directly quotable announcements or (often more usefully) unofficial statements off the record.

There is a constant battle to control the narrative, and it is increasingly accepted that psychology influences the market as much or more than the underlying physical trading conditions. More often than not, there are several weekly headlines from journalists, which commentators and analysts use to form their predictions and the advice they give to their subscribers.

Right now, if you were to search for oil-related news and information, you would find it very difficult to distinguish between the physical oil market and the derivatives market. Generally speaking, the news media find it difficult to differentiate one from the other. A seemingly significant event in the physical world affecting much physical oil may have very little impact on the derivatives market. Sophisticated oil traders will be cautious unless it is relevant to the underlying physical oil a liquid derivatives contract is designed to reflect. Conversely, a very localised event which impacts a relatively small amount of physical oil can have repercussions for derivatives contracts and knock-on effects on the rest of the derivatives world.

Organization of the Petroleum Exporting Countries

OPEC has been highly influential ever since it was formed by Saudi Arabia, Kuwait, Iraq, Venezuela and Iran in 1960, and even more as OPEC+ (with the addition of Russia, Mexico, Azerbaijan and seven other oil-exporting countries as associates) in the last few years. As the proportion by world volume of its crude exports has diminished, the cartel has evolved its strategy beyond simply price fixing and production control.

Oil Leaders, the book by ex-Saudi oil minister Ibrahim Al Muhanna, gives a remarkable insight into the evolution of this strategy, specifically the deliberate effort to gain influence with words and actions. He writes, ‘Oil market fundamentals are very important, but market psychology is equally important ... At times of crisis and uncertainty, sentiment overshadows fundamentals and influences the behaviour of financial investors in the oil market’^[40]. He details how he, and by inference Saudi Arabia, thinks oil markets are influenced and states clearly that how market fundamentals are *perceived* is more important than the facts themselves. The book confirms that the statements and observations released to the market from OPEC+ are carefully thought through and are a significant aspect of its overall strategy. As you might expect, the tone of the book emphasises Saudi thinking: Al Muhanna portrays his country as the most important OPEC member in both production and strategy. You get the impression that Saudi oil ministers endanger their careers if they do not succeed in managing market prices in line with OPEC’s strategy. The statements made by these ministers indicate that the pressure can get to them: during one period in September 2020 when the outright price was reversing periodically despite a coordinated OPEC+ strategy to see the market rally, Saudi Prince Abdulaziz bin Salman publicly announced, ‘I want the guys on the trading floors to be as jumpy as possible. I’m going to make sure whoever gambles on this market will be ouching like hell’. And you never know when OPEC is testing the market with planted headlines as a diversion from its real intention.

Real-world example: On 21 November 2022, *The Wall Street Journal* published an article reporting that ‘delegates’ were discussing raising OPEC production by 500,000 barrels per day^[41], leading to a severe market reaction and flat price Brent futures selling off from \$85 to \$82/bbl.

Following this reaction, a Saudi representative Tweeted a denial, and the flat price rallied back.

Sometimes, the OPEC media strategy can be transparent, and at other times, it can be very unclear. The ultimate objective is usually well-masked until after the fact. The understanding of how to influence the price of oil has come from decades of experiments and iterations of tactics, and the last four years have been a testament to that. Dramatic changes in production coupled with leading media statements have adroitly driven market sentiment from bull to bear and back. For example, according to Al Muhanna, a coordinated campaign in 2019 to talk down the markets involved private meetings with journalists, analysts, oil trading companies and even hedge funds, followed by a detailed article in the *Financial Times*. If oil derivatives market participants believe that the power is with OPEC, this kind of thing can directly affect traders' decisions. At other times, the collective market sentiment can go against OPEC or OPEC+ and its statements for too often 'crying wolf'. When someone begins to act defensively, they are likely losing control, and you must remember when analysing these organisations that they are still humans making human decisions and are just as susceptible to the vagaries of human psychology. What is key to interpreting OPEC+'s motives is that they don't like volatility; as Al Muhanna confirmed, 'Volatility hurts everyone. It shortens planning horizons'[\[42\]](#).

OPEC+ influences the physical market and changes trading decisions with coordinated changes to its daily crude oil production. This is the core tactic that goes back to the very beginning of OPEC. However, with the financialisation of oil markets, production cuts have evolved alongside targeted media briefings.

For example, over the last three years, OPEC+ has repeatedly intervened in physical production. In 2020, Saudi Arabia flooded the market with crude oil, selling at deep discounts to the benchmarked futures price on the back of a refusal from Russia to agree to a supply cut in OPEC+ crude during the early days of the pandemic. After forty-five days of free fall in oil markets, Russia gave in, but the US had to intervene to stabilise the crude oil supply. This led to the largest cut in history, at 19 Mbbl per day. The cut was later slowly unwound in a coordinated ramp-up in supply.

The Brent crude futures price went on a relentless rally from the lows set by the pricing war between Saudi Arabia and Russia, and OPEC+ firmly held onto the momentum throughout the unwinding period. The message from the

industry as prices rose was that the rushed entry into renewables led to neglect of the oil industry and, consequently, a lack of investment, which had caused the supply shortage.

The lack of investment claim was hard to prove and even hard to believe, given how oversupplied we were just a year prior, but of course, it was believed. The lack of investment explanation was so prevalent in the industry that it was taken as gospel. If you contested it to an industry participant, they said something like, ‘You clearly don’t understand physical oil markets’.

Saudi Aramco reported in 2022 that they could produce 12 Mbbl per day. This amount was produced in 2020 to flood the market. Production was barely 10 Mbbl per day^[43] just a year or so later while they were telling Macron, Boris Johnson and Joe Biden that there was no way they could produce more. Could it really be true that 2 Mbbl per day were taken completely offline from a lack of investment in just a year? Either way, the industry accepted it without question.

Why was there seemingly a lack of critical thinking from the industry? Possibly because a higher flat price environment suits many influential people in the oil market. Higher prices mean more investment, and sure enough, hedge funds rushed to get exposure to financial oil prices in response to demand from investors who had not participated in the rally and had believed the renewable narrative.

Higher notional prices also mean banks can charge more, and analysts aligned with the bullish consensus can say how good their research has always been. Producers make profits hand over fist, and oil traders get better margins on their physical products. Everyone wins except the end user, and the real end user has no bargaining power. They either pay the price or don’t get the product to use in their everyday lives.

The OPEC decision not to ramp up exports despite the Russian war taking so much oil offline was explained as ‘the market reacting against fundamentals’. There were even renewed cuts in late 2022 despite the war and again into the second quarter of 2023, as the macroeconomic environment started to be called into question.

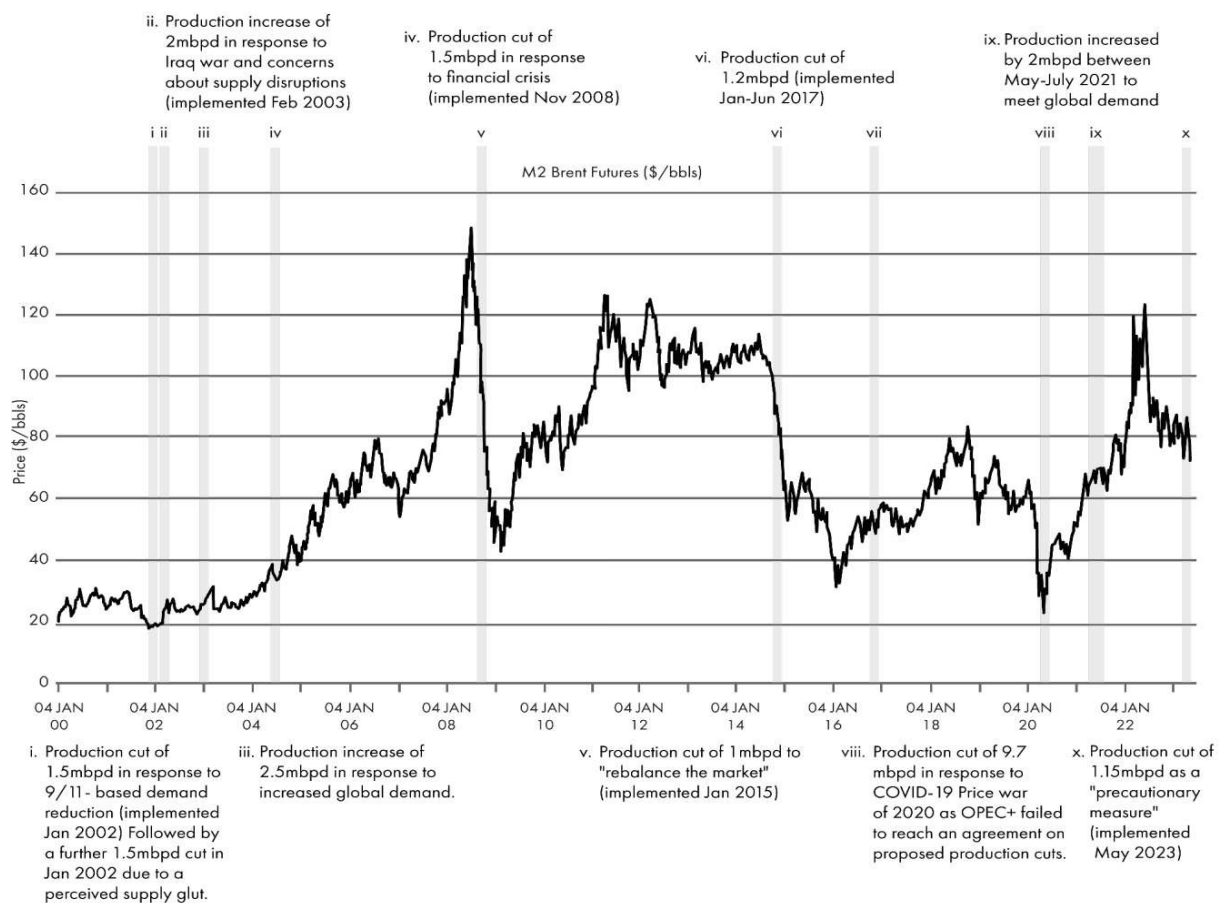


Figure 3.1 OPEC Production Quota Changes Against Rolling Month 2 Brent Futures Price

Geopolitical Leaders

Foreign policy and oil have historically been very tightly linked. Sometimes it can seem that it is a battle between the US for lower prices and Saudi Aramco for higher prices. The US has been much slower than the OPEC+ countries to attempt a coordinated strategy to influence the market, but Trump, whose Tweets heavily influenced traders' actions, kicked this strategy up a gear.

The Biden administration fought back as the Saudis squeezed the oil market in 2022 and did not relent despite the Ukraine crisis, claiming a lack of capacity and attempts to lower volatility and attacking the green movement towards renewables. Government visits to Saudi Arabian oil ministers were considered influential but didn't work with Johnson and Biden.

The US government's best weapon is its strategic petroleum reserve, which was put to effective use in 2022 when Biden ordered the release of 1 Mbbl per day for six months. This was a much more effective signal to the market than the usual lump sum release in the past. The extra oil provided a reliable stream to eat away at the incremental demand and reduce volatility in the physical oil market. During the same time, it was signalled by the International Energy Agency (IEA) that 'partners' would add a further 32.7 Mbbl over the same period. The IEA is an intergovernmental organisation established, among other things, to promote energy security and involves thirty member countries, including European countries, Canada, New Zealand, Japan and Mexico.

Major Oil or Oil Trading Company Representatives

Shale oil's influence on the global oil market was confirmed with a considerable sell-off of crude futures in 2014. Shale oil producers were hedging forward production in considerable volumes as production increased, which created new selling flows into the derivatives market that hadn't been there before. The pressure on the fundamental market was also evident, and eventually, it became clear that shale oil volumes would remain, as would the active hedging of those volumes.

This led to an adjustment in OPEC's strategy. From then, shale oil production became crucial in evaluating the market. US statistics regarding production are deemed reliable, including actual production numbers verified by the government, storage, rig counts and exports to the global market. These, like any supply and demand statistic, are retrospective. So when forecasting forward production, the intentions of the companies themselves, as well as the US administration's policies, are influential.

In the shale oil sector, large trade houses and majors now own many shale oil fields, but many independent producers are still free to make their own plans. The executives of these companies usually release statements to the market and tend to be more forthcoming than the public relations machines of larger oil companies. For example, the CEO of shale oil company Pioneer Natural Resources, Scott Sheffield, announced the reversal of all the company's hedges in 2022^[44] in a clear signal to the market that he

felt the market fundamentals were strong and continued to express his bullish view into the following year, with regular appearances in the media.

Executives at majors like BP and Shell are careful about what they say and rarely make aggressive statements. Instead, you generally only hear from them on quarterly investor calls, in which they give vague summaries of where they see the market going and production targets. These large corporations constantly fight the narrative that they are harming the world. Therefore, they choose not to be seen to be talking the market up. Trading profits is another contentious point for the public, so these companies deliberately avoid publishing these results. The CEO of Total once said, 'Oil trading is a secret'^[45].

Much more aggressive are the executives of the independent trade houses. Companies like Vitol, Trafigura and Mercuria have their CEOs or heads of research call out the market at times, and it is clear they are trying to influence the market. They will talk suggestively at conferences or directly with journalists at *Bloomberg* or *Reuters*. They will also commonly give a flat price target range. Executives like this are aware of their influence in the market, but just like OPEC, their public voice and actions are frequently blurred.

Research Houses

Most independent research houses published analyses and analytics firmly rooted in the evaluation of **supply and demand** data. Traditionally, a supply and demand analyst would provide predictions by observing global trends and comparing them with past trends. The analysis presented a viewpoint, and its integrity was based on how much the market trusted that company or its spokesperson. However, the importance of this style of research has recently been called into question, with supply-and-demand-focused research houses acknowledging the importance of financial markets on the movement of prices.

To understand supply and demand twenty years ago, you needed to have physical assets and gauge interest by operating those assets. It may be a production arm, refiner, storage, end user arm or, in the case of the integrated major oil company, all of the above. As so few companies had this level of sophistication in their physical system, these oil companies had an unbeatable advantage. Therefore, having a strong relationship with a large oil company

or government contacts in producing nations was seen as an informational advantage that justified subscribing to and following an independent research house. As that research company gained credibility, the oil trading community would increasingly listen to what it had to say, and eventually, it would gain enough credibility to influence the markets. At that point, the company's analyses became self-fulfilling concepts because everyone would read the analyses and act on them. After all, that was what everyone else was doing.

Similar to oil executives, spokespeople for research houses regularly voice their views to the market, for example, by reacting to price movements in oil markets. These spokespeople often appear on news channels; this is the public relations machine at work. Each research company will try to support its views to maintain credibility, so it is forced to justify its opinions when they are 'wrong' and go on victory laps when they are 'right'.

While this scenario still exists to some degree, the commentary provided by third-party analysts has become more data-driven as the market has evolved. Buyers of oil research demand justification for supply and demand forecasting, as the data on physical assets has been considerably democratised. New entrants are almost entirely data-focused and label themselves as data analysts, such as those who publish data on storage and ship tracking.

In very recent times, the classic research house has turned its attention to analysing financial markets. The focus is on analysing trader positioning in key contracts worldwide, but this is a nascent research area with much development ahead. Greater detail on this can be found in the Oil Derivatives Market Analysis section.

Investment Banks

In much the same way as independent research houses, investment banks will also offer their views publicly to the market. Investment banks have a legacy in the oil market, and despite low participation in oil derivatives, their opinions are still respected. Typically, investment banks will release their forecasted views on the price of Brent crude quarterly. These forecasts will then be revised based on market events or targets not being reached.

As with independent research houses, banks' analyses are focused on supply and demand evaluations. When investment banks were heavily influential in the oil market, pre-2008, analysts at banks would hold similar sway in the markets as OPEC+ does at present. Banks' influence was bolstered by their heavy involvement in trading oil derivatives and even physical products before regulations ended this activity. The banks had insight into global flows and advised on and executed the hedges of some of the biggest companies in the world, which gave them a perceived powerful insight into the markets. To some of the older oil traders especially, the viewpoints of these banks are still important and represent the status quo.

Oil Derivatives Market Analysis

What *really* drives oil derivatives prices? There isn't a simple answer, and there isn't a unifying equation that links it all. The best we can do is to break down the derivatives market into its constituent parts, understand how traders use it and evaluate the dynamics at play. It is a large industry with many varied forces, making it extremely interesting.

The question of what fundamentally drives the market is probably the most contentious discussion in the oil derivatives world. Some legacy oil professionals insist it is all about the physical market in oil, but some profitable financial traders don't even know what is happening in the physical world. As a result, the discussion has become a search for a philosophy, a constant search for the market 'truth' which constantly evolves and recreates itself. What I try to make clear to new entrants is that when it comes to predicting markets, no one knows anything for sure.

What can be definitively said is that what drives the oil derivatives market is the buyers and sellers who trade in it. Therefore, a thorough market

analysis must identify what will influence contract decision-making. So physical markets are important, but it is very easy to get bogged down in detail that goes way beyond the buying and selling of contracts. This section will show that analysing the physical market is just the first step in defining what drives the oil derivatives market.

The more you know about who the traders are, why they trade and what affects their behaviour, the more you will be able to follow the market and ultimately make sounder predictions.

To have a complete view of oil derivative markets, you have to do your best to evaluate at least the following (and this is not an exhaustive list):

- The future demand for and supply of physical oil
- Derivatives market positioning (long/short and entry levels)
- Derivatives market flows
- Derivatives market behaviour
- Derivatives market conditions (volatility and liquidity)
- Derivatives market historical price action (technical analysis)

Supply and Demand Models

Supply and demand models are fundamental economic models used to understand and predict the dynamics of a market, including the oil market. These models are created by economists, market analysts and researchers within private companies and governmental or intergovernmental organisations, including oil companies, financial institutions, consultancy firms, governmental agencies or international organisations like the IEA or OPEC. The models are tools used to analyse the relationship between producers and end users and demonstrate how prices are influenced by the availability of a product (supply) and the desire of end users to purchase it (demand). The models guide business strategy, investment decisions, policy-making and forecasting future price trends.

While these models provide a fundamental understanding of market dynamics, real-world prices are also affected by other elements not directly incorporated in basic supply and demand models, including market speculation, decisions of major oil producers and strategic reserves policies.

Nevertheless, supply and demand analysis is rooted firmly in professional oil traders' psyches, and you won't find many traders working at trade houses, majors or hedge funds who don't have some kind of modelling

in place.

There is still a strong belief that as oil is a physical commodity, it has a tangible value driven by traditional market forces, and this belief can't and won't change. If you are just joining an oil company, chances are you will be working with this viewpoint. If you're going through a company training programme, expect to become very familiar with the concepts and data that make up this analysis. It is hard to argue that these forces don't influence the physical market, and physical prices have their part in derivatives pricing. But a major theme of this book is that there is a distinction between the forces that govern the physical markets and those that influence derivative markets. How supply and demand correlate with the derivatives price for oil is difficult to determine.

When evaluating supply and demand in the physical oil market, you should be aware that almost every oil company in the world is working on its own models, investing large sums of capital in new data sources and hiring data engineers and scientists to process and evaluate data. If you are new to this field, you will have no obvious advantage when it comes to modelling the world's production and consumption in granular detail. You are up against oil majors who have physical assets worldwide all the way down the supply chain. They receive real-time data on the global production of crude oil, the transport of that oil, the demand from refiners that process the crude oil, the utilisation and capacity of storage around the world and, of course, the core demand from end users. You are also against numerous competing research houses that have been doing this for 20 years or more and have networks worldwide that feed them niche, granular information.

The good news is that this type of analysis has become increasingly democratised in the last ten years. The market for supply and demand data and the analysis of those data is hugely valuable, confirmed by the strong recurring revenues and valuations these businesses command. The value of the oil market is so well recognised that businesses go to extreme lengths to obtain any data that will even slightly improve the timing or accuracy of this analysis.

Some companies fly aircraft equipped with infrared cameras over storage facilities to identify the capacity and usage of tanks to pre-empt publicly released information that moves markets. Telegraph poles are rigged with cameras to overlook oil assets, widgets are placed on buildings and other infrastructure to measure the flow of electricity and consequent activity of

refineries, and satellite imagery is bought. Ship tracking software has been revolutionised by machine learning and the streamlining of publicly available data sets, so any movement of oil worldwide can now be mapped. Expected exports are no longer guesswork but are monitored in real time, and a tanker diversion can be accounted for and the consequent impact assessed by traders.

Such ingenuity has meant that this information is available to all who have the means to acquire it, levelling the playing field in the professional oil trading world. Such is the pace of democratisation of this information that to obtain an edge, you must go back to before the physical movement of oil begins: when ships are chartered weeks before they move, refineries begin major alterations in cracking units and coil trucks are booked. The subsequent hedging of these decisions is the closest thing to knowing the decisions themselves. A refinery's hedging programme will reveal the expected commitment and, therefore, the utilisation rate; the arbitrage hedge will reveal the intention to move physical barrels. This is new, but it will be a commonplace analysis in just a few years, such is the speed of evolution in oil data.

Building a Supply and Demand Model

In summary, to build your supply and demand model, you first need raw, up-to-date and reliable data on the following:

- Physical crude oil production
- Pipeline infrastructure
- Refinery operations
- Inventory levels
- Ports
- Ship tracking

You then need credible sources to forecast the following:

- Forward macroeconomic demand
- Forward crude and condensate supply
- Forward refiner output

To obtain this information as raw and reliable data sets is no mean feat. You will need to sift through government websites, civil servant reporting data, company announcements and consultancy company postings and fill any gaps by talking to the oil community worldwide. You need to test the

accuracy of this data, which can be hugely time-consuming considering the multitude of sources published and the various periods and time zones.

For example, company announcements that need to be studied come from companies that can reveal information on fleet turnarounds, road vehicle supply (cars, light vehicles or heavy vehicles), farming output, heating usage, airline travel (fleet, engine and ticket sales), shipping, etc.

Examples of government reporting data are outlined in Figure 3.2.

NAME	PURPOSE	DESCRIPTION
JODI (Joint Organisation Data Initiative)	Global Database	Cross barrel (crude to NGL) See data from over 100 countries, production to refinery output etc. Data from 2002
IEA	Inter - government funded (under the framework of OECD in 1974), politicised data	"Analysis, data and policy recommendations" Conducts research, collects, and analyses data on energy markets
EIA	Data for US stocks, imports, and exports, across barrel and across PADDs	US government -run, "Independent and impartial energy information"
API	"Represents the Americans oil and gas industry"	A trade association that represents the oil and gas industry in the United States and sets standards for developing and producing oil and natural gas.
Euroilstock	"Rapid European oil information reporting system founded by the oil industry and the Commission of the European Communities"	Euroilstock collects and analyses data on the European downstream oil industry from the 16 member companies.
METI	Japanese government data	Ministry in charge of economic and industrial policies, energy conservation, renewable energy, and resource regulations.
ANP	National Agency of Petroleum, Natural Gas and Biofuels. Brazil government data	Regulates the Brazilian oil and gas industry through licenses, monitoring compliance, and implementing policies for energy security and environmental sustainability.
Insights Global (previously PJK International)	Independent market research company	Provides ARA storage hub data, terminal information, and other data services.
FOIZ	Fujairah Oil Industry Zone, oil inventory data for Fujairah	Oil storage and trading hub located on the eastern coast of the United Arab Emirates. It offers a range of storage and logistics services to oil traders, including blending, bunkering, and refining, and serves as a strategic gateway for oil shipments between the Middle East and other regions
EMA	The energy market authority of Singapore	"Develops a dynamic energy sector in Singapore". Provides Singapore inventory data

Figure 3.2 Supply and Demand Reporting Data

Once you have a strong data set, you can

- Calculate current or 'day one' supply and demand balance figures by crude and products
- Estimate future supply and demand balance figures by crude and products using forecasts

From here, you will have a relatively clear, conclusive number indicating a surplus or deficit for the market you focus on. Then, research houses will assume that a supply deficit or surplus will lead to the associated

derivatives market prices going up or down.

	Q1'23	Q2'23	Q3'23	Q4'23	Q1'24	Q2'24	Q3'24	Q4'24
NORTH AMERICA	1327	559	857	1173	466	446	879	453
DEMAND	5560	5443	5123	5609	5343	5666	5144	5527
SUPPLY	6887	6002	5980	6782	5809	6112	6023	5980
EUROPE	-901	-798	-1190	-1346	-1343	-1574	-1508	-1220
DEMAND	6002	6809	6690	6345	6223	6574	6900	6232
SUPPLY	5101	6011	5500	4999	4880	5000	5392	5012
MIDDLE EAST	1190	1164	969	1007	1363	1109	1122	1178
DEMAND	1800	1809	1867	1923	1900	1890	1978	1922
SUPPLY	2990	2973	2836	2930	3263	2999	3100	3100

Figure 3.3 A Typical Example of Supply and Demand Data

How Supply and Demand Models Are Used by the Wider Market

It isn't just traders that use supply and demand modelling. Many companies are exposed to the oil price and need to assess the long-term trends, which could be used to justify a hedging strategy for a producer or for the valuation of an oil asset.

When trying to predict prices, most traders and analysts know the pitfalls of the supply and demand approach, but you need to start somewhere. It is reasonable to begin by assessing the state of the underlying oil market and the factors that could change it. Market participants justify paying their subscriptions for third-party supply and demand analyses to benchmark their models, which often have much variance. In the end, a consensus is powerful among traders. This consensus makes decision-makers comfortable with their decisions. Knowing that other market participants are also reading particular third-party research (or using the conclusions at least) assures them they aren't exposed. Market participants like to interact and engage with each other's views, and discussing the latest conclusions from third-party research will reaffirm the consensus.

Research houses will publish the following:

- **Price trends** on key benchmarks with the greatest weighting on the outright price (some cracks, time spreads and swaps differentials like Jet and VLSFO vs Brent)
- **Price prediction** on key benchmarks (but cannot provide specific trade ideas without the regulations to do so)

As a result, the general oil audience will know who thinks what based on the information shared by the research houses.

Taking Supply and Demand Modelling to Derivative Execution

Supply and demand analysis yields a binary signal: bullish or bearish. But you must conduct a much more thorough evaluation to convert this view into a trade. You should answer the following questions:

- Which contract should you choose?
- Which location should your trade reflect?
- Which contract best reflects that view?
- How do you determine if the contract is under- or overvalued?
- What are the physical levels?
- What is the market risk (positioning, open interest and entry price)?

The wise trader knows precisely what they are taking a view on and can reflect that view as a hypothesis about how the market will move next. Forming a hypothesis and turning this into a speculative trade is covered in the Speculative Trading section.

What Are the Challenges with Supply and Demand Models?

First, the only accurate, current supply picture available is already retrospective. Supply and demand data lag by at least a few days, so building a picture of last week is difficult enough, let alone today. Given the reliance on estimates and the time data are published, you frequently have an incomplete picture.

Second, everything forward-looking in a supply and demand model is based on assumptions, and the reliability of those assumptions is impossible to confirm. Companies exaggerate, people lie, but, more importantly, things always happen unexpectedly. No one modelled a viral pandemic which left airlines doubly impacted by money hedges with no physical demand to offset the losses with savings on the physical product. With so many factors going into a supply and demand model, any inaccuracies can have knock-on effects that make the final number difficult to trust. For example, gross domestic product projections by central banks or government-backed think tanks are often proved inaccurate by the final figures. If forward demand, even in six months, is based on these projections, it can impact the reliability of those forecasts.

Supply and demand modelling also pays no attention to forward curve economics or hedging. For example, one-year forward refinery margins could be very healthy, and many refiners could have hedged these prices. At the same time, the supply and demand model could be predicting a lack of supply. However, because the refiner has hedged, this guarantees refined products no matter the demand forecast or the supply and demand picture.

After a report has been published with an outlook for a given period, it becomes impossible to know whether this surplus or deficit is already priced in by the market. The trading community can act on supply and demand forecasts on a speculative basis, but once those traders have committed to that view by executing the trade, the forecast will have no further impact on the market. If supply and demand models are heavily democratised information (i.e. many traders and analysts who look at the market this way have come to a consensus), the market is expected to price in this information very quickly.

The results produced using this method present an incomplete analysis. What is the appropriate instrument selection now that you have this supply

and demand-driven view? Should this instrument be based on the outright price, a differential or a time spread? Supply and demand models do not provide the necessary details to evaluate derivatives market conditions, historical pricing, market position and so on.

Supply and demand modelling used solely to predict prices also assumes a causal link between the physical and derivatives markets. This is the biggest pitfall. The financial volumes dwarf the physical volumes. The contracts are partly settled to the forward curve, making the financial influence too significant to limit the explanation to the physical market. It also doesn't consider participant motives in the derivatives market and the consequent distortions that can occur.

The trading on derivatives market settlement periods is mostly about what everyone believes the supply and demand situation is or will be. In a market that still does most of its research in this area, the resulting information will significantly impact trader decision-making, so it is a circular argument I believe we will see a move away from a reliance on supply and demand models for pricing prediction, but it is also not entirely clear how this would relate to 'dislocations^[46]' and the market being 'broken^[47]'.

I believe that in the future, traders who rely on binary supply and demand information will be weeded out of the trading pool as it becomes clear that more expansive analysis based on the positioning and flows of financial oil markets is required. You can then make the case that if the physical market is traded too far below or above derivatives, the benchmark financial contract must correct accordingly. This is exactly what most oil traders with a physical oil approach to derivatives are banking on, but the pricing doesn't back this up well enough for this to be the only conclusive driver of markets. If you take a deeper dive into the mechanics of derivatives, it is clear that this view is incomplete.

Market Positioning

Once you know the market participants and understand what drives their flows, you need to understand how they are positioned. Knowing how the market is positioned and which market participants are making up those positions can be used to your advantage. Much like a poker player makes decisions based on how they've gauged the hand of the player across from

them, so you should look to base your decisions on the actions market participants will likely take next.

It is a good practice to quantify the market's risk, which is achieved through open interest data. Open interest is the term used for derivatives contracts that have been traded but have not yet been closed out. It is available on the exchanges by contract and can be assessed daily against recent changes and historical ranges. For every buyer, there must be a seller, so the larger the open interest, the greater the risk (or live exposure to oil prices) in the market. However, finding information on open interest is only an initial step, as open interest numbers can be misleading or are at least more complicated than a simple number.

How Does Money Flow Through Financial Contracts?

Money in the oil market flows from the producer pumping oil and selling it at a price above the cost of extraction. As soon as the producer hedges, some of that money goes into the derivatives market. Intermediaries trade among themselves and, finally, with the end user, who also hedges. If the producer hedges low and prices rise, they have effectively given the market money.

Example

Let's assume that on 1 March 2023, a producer of crude oil with established operations knows it will produce 1 Mbbl over three months. It knows it will need to sell this crude oil on the physical market, and the product will be subject to price fluctuations until the time of the sale. Therefore, it hedges the production price using the derivative market by selling 1 Mbbl of a June 2023 Dated Brent swap contract at \$60/bbl.

In June, the Dated Brent contract prices out at \$80/bbl, and therefore the producer loses \$20/bbl on the derivative contract – \$20 million.

However, the producer will now sell in the physical market using Dated Brent as the benchmark. Therefore, it makes \$80/bbl on the 1 Mbbl – \$80 million. The producer spent an equivalent of \$50/bbl to produce the crude, generating a \$30 million profit. Although the producer lost \$20 million on the hedge, it made \$30 million on the physical. Therefore, net of all transactions and costs, it generates \$10 million.

Where did this \$80 million come from? The simple answer is the end user, who is always the ultimate buyer of the commodity. However, on the way, intermediaries brought the oil to the end user and took a margin, increasing the final cost to the end user. The oil trader acts like a market maker, buying from the producer at \$80/bbl and selling at a higher price to a refiner who needs crude oil to process into refined products. Taking \$0.50 per barrel as a reasonable price difference, the oil trader will have made \$500,000 on the 1 Mbbl. In effect, the money has come from the producer, who could have sold it to the refiner for \$0.50 more than to the oil trader. However, the oil trader brings expertise, a network and logistics knowledge that allows them to generate this premium, which the producer may not be able or willing to do. An IOC, like BP or Shell, will do this itself. However, it will still separate the transaction from upstream to the marketing and trading departments and audit the prices at each stage.

It is the same for the refinery. The refiner processes the crude oil into products and generates net revenues by selling the refined products at a higher price than the cost of the feedstock – the refinery margin – in this example, \$3.50/bbl. The refiner will sell the products to an oil trader and gauge the margin from the buying price of the oil trader, who will then make a profit when selling to the end user.

The end user then pays in full for the oil products produced by the refiner. Figure 3.4 shows this cash flow.

	BUY PRICE	SELL PRICE	DIFFERENCE	
Producer	\$50 (cost of extraction)	\$80	\$30	\$30m (\$80m less cost of extraction)
Oil Trader	\$80	\$80.50	50c	\$500k
Refiner	\$80.50	\$84	\$3.50	\$3.5m
Oil Trader	\$84	\$84.50	50c	\$500k
End User	\$84.50	(Consumes oil)		-\$84.5m

Figure 3.4 An Example of How Money Will Flow in the Physical Market

But the producer also hedged and lost \$20 million, so where was this money transferred from?

The producer entered the derivative market to hedge and needed a buyer to satisfy the order.

Part of a market maker's service is to show a price throughout the trading day, so the market maker will likely be on the other side of this transaction. However, the market maker will not take this contract to

settlement, as this would be a considerable speculative risk. Instead, they will temporarily warehouse this risk and look for an outlet using the many contracts available in the oil derivatives market. Many traders might buy and sell this contract, but in the end, someone must take it to settlement.

The most idealistic scenario is that the market maker, or indeed the producer, sells to an end user, like an airline, that is also looking to hedge but protect its physical purchase rather than a sale. This way, both companies are willing to take the contract to settlement. But in reality, it is likely to be a combination of speculators and end-user hedging that takes the contract to settlement.

In the derivatives component of the transaction, the end user (the utility company) is transferred the money from the producer's loss. Figure 3.5 shows this cash flow.

	BUY PRICE	SELL PRICE	DIFFERENCE	NET MONEY TAKEN FROM MARKET
Producer		\$60	-\$20	-\$20m
Market Maker	\$60	\$60.05	\$0.05	\$50k
End User	\$60.05	\$80	\$24.95	\$19.95m

Figure 3.5 An Example of How Money Will Flow in the Paper Market

But to understand the cash flow overall, you must include both the physical and derivatives cash flows because the net money into the market comes from the end user, including the producer's extraction cost. But how much the end user pays depends on the notional price, while the producer also pays the market if the cost of sale is lower than the extraction cost. Further costs that decrease profit include costs to service providers in the derivatives market, including the exchange clearing and block costs, brokerage commissions and financing fees.

If the market goes up, end users pay a higher price than expected because they pay the price at which the market ultimately settles. If the end user won't buy at that price, the knock-on effect is that the market must decrease the price, depleting physical differentials. But the benchmark, as a financial contract, can stay at elevated levels irrationally. As John Maynard Keynes (or was it Gary Schilling?) said, 'The market can stay irrational longer than you can stay solvent'. If the producer hedges low and prices rise, they have effectively given the market money. This should flow to the end user, who hedged low and made money on the hedge. But this assumes the

end user hedged at the same time. More likely, an intermediary speculating in some form collects the money from timing the market right.

The price the end user pays is most often the spot price of oil, making hedging programmes useful for end-user companies like airlines. They can pass the cost onto the consumer if they lose money, or they can price tickets competitively or increase the size of their operations if they make money. On the other hand, gasoline can't be hedged by the end user, as the end user is retail. This means that the liquidity for the buy side needs to come from oil traders, who are the only willing buyers taking these contracts to settlement. The key concept here is that the oil market is only zero-sum when you include the physical and derivatives. Traders taking contracts to settlement is imperative to provide true end-to-end liquidity for hedgers. Despite the market maker not taking much to settlement as a risk-taking strategy, the reality is that these players, like Onyx, hold huge numbers of contracts against other contracts to settlement. This provides immediate liquidity in these contracts for traders in a given market. Yet if market makers neutralise the risk, you have to ask, 'Where are they getting the liquidity?' The answer is somewhere in the wider derivatives market, which is how the markets are ultimately linked.

Open Interest

Open interest refers to the total number of outstanding or unsettled futures contracts or options contracts in a particular market at a given time. It represents the total number of contracts entered into by buyers and sellers that have not yet been closed out through an offsetting transaction or delivery. Open interest is zero-sum in the sense that for every buyer, there must be a seller. If one side is a speculative trader, then this risk must be managed to the contract's expiry, meaning the position will be closed out at some point if the trader doesn't want exposure to the contract pricing. That position will be a function of the trader's risk tolerance, so if the contract moves too far in one direction, the trader may have to stop out. When a trader stops out, the price action tends to be aggressive and can have knock-on effects on the market, like high volatility and even panic. So it is important to gauge not just the net position but also the average entry prices of that speculative risk.

Open interest into expiries and settlements is also important as a gauge

of interest in convergence with derivatives and physical. The more open interest there is towards settlement, the more likely there is physical interest behind that position. On this basis, if open interest is high just before settlement, you can expect to see majors and trade houses in the PRA assessment periods with bids or offers for physical oil, which in turn will have a material influence on the final settled price.

It is unlikely that an oil swap contract will be reversed if used for a hedge. However, if the hedger is using a hedging strategy via exchange-traded futures contracts like Brent and WTI, which have an expiry period of one day, they will likely close this position to avoid the expiry. Closing out is similar to stopping out in that it means trading in the opposite direction but occurs less frequently in hedging than stopping out, and when it is done, it will probably have a different price action profile. Closing out a hedge does not cause the same concern for the market observer as stopping out. Closing out a hedge will likely be done over a three-to-five-day period to match with a physical loading period rather than abruptly getting out at all costs.

So it is important to dissect the open interest weighting and apply reasoning around what is speculative and what is hedging. Large amounts of hedging open interest are likely when the level reached on a given contract creates an opportunity for physical transactions on paper – for example, a geographical arbitrage open net of costs or attractive blending economics.

Open interest can be elevated on a contract when a market maker uses that contract as part of a deconstructed risk profile. A market maker will take down market flow on a given contract and use differentials of the same contract to close out the market risk. For example, a market maker might buy EBOB crack against selling the RBOB crack and then buy the RBOB vs EBOB contract. They are separate contracts that, in combination, net out the market risk; however, they need to be held on the balance sheet against each other to expiry, keeping the contracts live and open interest high until that point. Market makers can be heavily one side of the open interest of the contract for this reason, and discounting the side of the market maker can reveal the market's net position in a given contract. This strategy forms the basis of the Onyx COT data.

If the market's open interest is dominated by one-way speculative activity against passive positions (hedgers), speculative bubbles are likely to form. In speculative bubbles, price action is relatively predictable. Speculative bubbles get a lot of media attention, either deliberately from

traders and analysts talking up their views or because the price action is exciting and, therefore, newsworthy. This media attention can stoke the fire and does so regularly. So it is vital to understand the concentration of speculative risk versus the media narrative to predict future market behaviour.

Open Interest Data

Two data sources are available to dissect open interest in oil derivative markets: exchange data – published weekly on core futures contracts and accessible via the CFTC COT report – and the Onyx COT data – published daily on oil swap contracts and accessible through Onyx’s research service. The CFTC report breaks down the exchange-traded futures open interest into subcategories and shows how each category is made up of longs and shorts. The Onyx COT gives the overall market position net of Onyx for oil swaps and can be extrapolated for market makers.

In the context of the US CFTC, commercial and non-commercial positions are classifications used to differentiate between types of traders in the futures markets. These classifications help provide insight into the trading activities and market participation of market participants.

Commercial positions: Commercial positions refer to the positions held by entities engaged in the production, processing or use of the underlying commodity being traded in the futures market. These entities are considered to have a commercial interest or usage for the commodity. Commercial traders typically include producers, manufacturers, processors and other hedgers directly involved in the physical commodity. They use futures contracts to hedge against price fluctuations and manage business risks.

Non-commercial positions: Non-commercial positions, also known as speculative positions, are held by traders or investors not involved in the production, processing or use of the underlying commodity. These traders primarily participate in the futures markets for investment or speculative purposes, seeking to profit from price movements. Non-commercial traders include hedge funds, commodity pools, managed money and individual traders. They typically take positions based on market analysis, technical indicators or other trading strategies. The CFTC collects and publishes the COT report, which provides a breakdown of the positions held by commercial and non-commercial traders in various futures markets. The

report helps market participants analyse the trading behaviours of different groups and assess their potential impact on market dynamics. It can provide insights into market sentiment, trend reversals and potential price movements.

Differentiating between commercial and non-commercial positions allows regulators, market participants and analysts to monitor and understand the behaviours of different types of traders and assess overall market conditions. It helps promote transparency and effective risk management in the futures markets.

The CFTC reports the open interest and breakdown of positions in their core futures contracts by subcategory, releasing this information weekly. The data for the two days prior to the release date ($T - 3$) are generally released on a Friday. The subcategories are outlined below.

Producer, End User or Merchant

This subcategory reflects core hedging. Typically, producer hedging open interest overwhelms the end user, reflected in the long–short ratio range. The ratio is typically 0.4–1.2, showing that most of the time, selling outweighs buying, or in this case, producer hedging outweighs hedging by end users.

To get a true sense of end-user purchases in oil markets, you need to include buying in crude and product futures COT data, as there is material hedging from end users in these contracts that will respond to the same factors that influence the outright price of crude.

‘Merchant’ is the word used for an oil trade house like Vitol or Trafigura. There is a lot of hedging but also a lot of speculation in this subcategory.

Swap Dealer

The CFTC defines a swap dealer as a trader dealing primarily in OTC (bilateral) transactions for its own account or the accounts of others. Swap dealers are mostly banks and independent firms with a trade finance business model.

Other Reportables

This subcategory includes traders who hold reportable positions in futures and options markets but do not meet the reportable thresholds for the commercial or non-commercial categories. These traders may include small speculators, individual traders or small institutions.

Managed Money

This category includes traders who manage investment funds or other managed accounts on behalf of others, such as hedge funds, commodity trading advisors (CTAs) and commodity pool operators.

The managed money criterion is particularly important for understanding the macro-driven movement of flows. It is also an indicator of ‘fast money’, volume that will trade in and out in a reactive fashion to headline news and market momentum shifts.

Real World Example: COT report after the Oil and Petroleum Exporting Countries’ Cut in April 2023

On 2 April 2023, a Sunday, the media reported that OPEC had agreed a production cut of 1.16 Mbbl per day. On the open that evening (23.00 London time), the prompt Brent futures contract (June 2023) rallied from Friday night’s close of \$79.89 to \$85.35/bbl. The contract hovered around the same level for the rest of the week.

The market’s reaction to the news was understandable, as it was a cut to supply and, therefore, likely to support the physical market. A report released the following Friday gave insight into what happened at the open on Sunday night through to the following Wednesday.

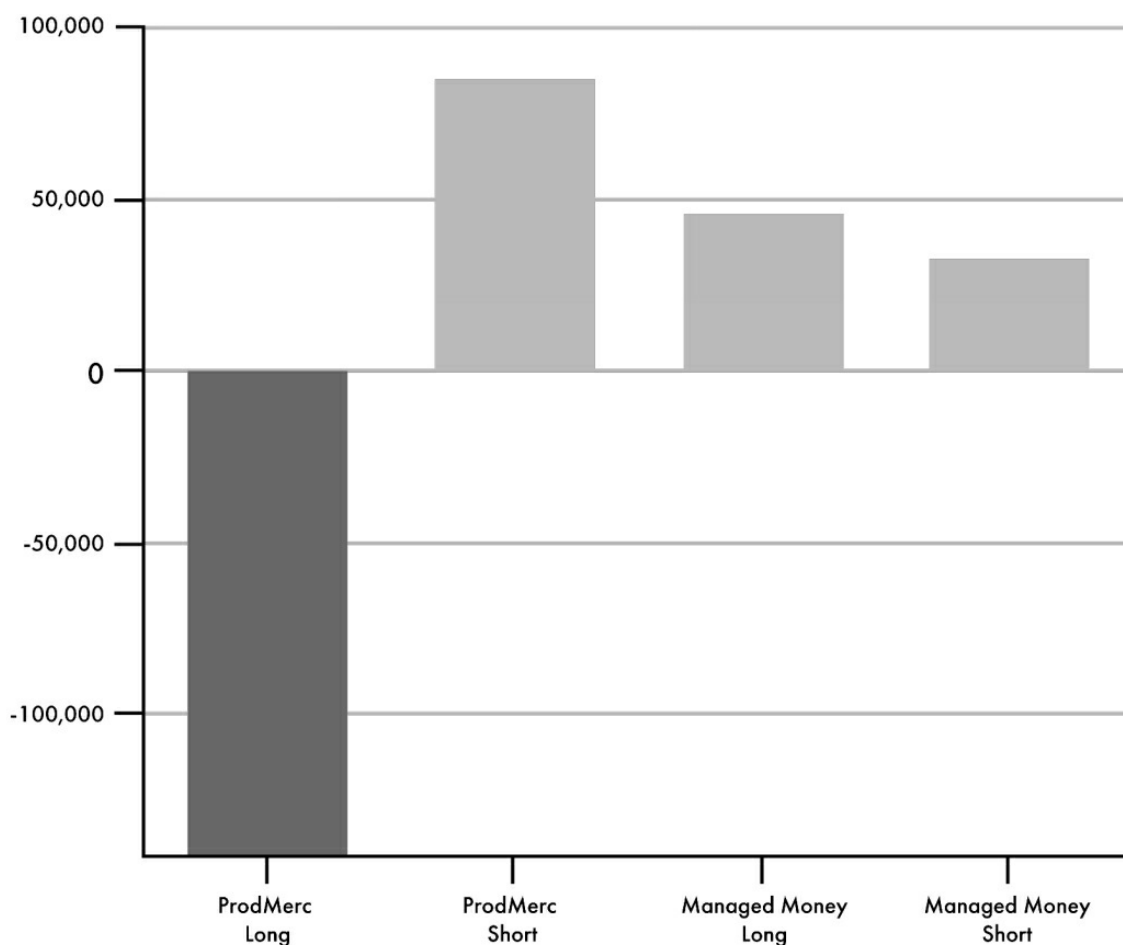


Figure 3.6 Buying and Selling Net Impact With Respect to the WoW Change To Open Interest (week to 4 Apr 2023) (Source: CFTC)

The categories traded proportionately in the following way:

Money managers added to their positions in Brent futures, with long positions increasing at the fastest rate since Nov 2020, adding over 46 Mbbl and becoming 19.5% of all long positions. Considerable buying also came from short-positioned players within this category closing out of positions at the fastest rate since 2016, closing over 33.8 Mbbl or 46.9% of all short positions.

But who provided the sell-side liquidity, and why?

Long positions decreased by 16% in the producer, end user and merchant category, or 150 Mbbl, while overall open interest in the contract decreased. As a result, new buyers got liquidity from this group, which was happy to close out its long positions.

Let's use this information to dive deeper into what went on in the market. Why would this category be selling? It could be the result of one of two things:

- Closing out an end-user hedge (end users buy to hedge)
- Taking profit on a speculative long position

We know there was a clear bullish headline, which should have materially impacted the supply and demand on the physical market. So it is unlikely that a buy-side hedger would take profit so soon after this event when the material impact was yet to transpire, given that the hedge is meant to protect against future rising fuel costs.

We also know it wasn't producer hedging, which would have resulted in new short positions. We know from the data that the existing longs closing out created the sell-side liquidity. So we can assume, with some degree of confidence, that this selling resulted from taking profit on a speculative position.

Why would this category be taking profit on a speculative position?

Even though the market did move higher, it was nothing compared to moves we had seen just a month before, and therefore the sellers must not have been convinced to hold this position long-term. We know that the merchants in this category hold very large speculative positions, usually holding them for months as they look to capitalise on their view over the year. We also know merchants had been announcing in the media up to this point and during International Energy Week that they expected Brent futures to head towards \$100/bbl. So it is likely that they were long. So why take profit at relatively low levels compared to this target in such a bullish event? One reason could be that they had substantial risk and knew that the result of this move would be a great chance to exit their position without moving the market against them. With such a binary bullish event, speculators will likely rush in at once and one way, making it very quickly a crowded trade and, consequently, highly volatile. When it came to exiting their position, they would be all too aware of the impact it could have on the market, first by the sheer volume of sell orders, but second, by the knock-on momentum that would lead to more selling and a fight for buy-side liquidity.

The market was also not very bullish in the underlying market until this point, and there may have been doubts about their view playing out. Getting a \$6/bbl move in seconds, having felt uneasy about the position the day before, may have made it an attractive time to exit.

Either way, looking at the COT report allows you to see what participants do rather than relying on what they say. It is your job to piece together what is happening using whatever means you have.

The Onyx Commitment of Traders Report

While the CFTC report serves as a staple for exchange-traded futures, there was no equivalent in the oil swaps market. The Onyx COT report has filled the void.

Onyx Advisory's COT raw data show the aggregated market position of Onyx Commodities Limited's total market share, presented by oil swaps contracts. The raw data are presented in the same format as the CFTC COT report, with the key differences outlined below.

Onyx's data focus on oil swaps open interest and volume. It is available daily (the CFTC COT report is available weekly). The data come from the Onyx Commodities trading floor (a subsidiary of Onyx Capital Group), widely known in the industry as the largest liquidity provider in oil swaps. The market-making business model is to take down the market's flow against other flows and profit in between each trade. Discounting Onyx's side of the trade, given that they are holding contracts against other contracts, can reveal the positions currently being held and the average entry prices of those positions. Additionally, as Onyx has such a large market share, these positions are highly reflective of the overall market position.

How can Onyx Commodities produce a more accurate aggregated market position than the exchange?

The ICE and NYMEX COT reports published by the CFTC focus on exchange-traded futures and are not designed for oil swaps. The reports are specific to the exchange that creates them and cannot distinguish between positions on different exchanges (and net them off). The Onyx COT report can reflect this net position.

Additionally, the netting of contracts from a market risk perspective requires data engineering, given variances between the contract arising from calendar differences and differing settlement periods.

Example: Over a few months, the trading arm of Onyx Capital Group, Onyx Commodities, entered a cumulative position in the EBOB crack contract for May 2022 pricing. The market flows varied on the EBOB crack, but over the few months, the net position held by Onyx in this contract was long 20,000 lots.

However, every time Onyx bought the EBOB crack, the book netted out the market risk from the EBOB crack contract using other available contracts. Given that the market risk was EBOB and Brent swaps, Onyx used

the available contracts to offset the market risk, creating an overall delta-neutral position. So Onyx bought the EBOB crack and sold the May 2022 Singapore RON 92 vs Brent crack against the EBOB crack in equal sizes. Onyx then had a market risk of Singapore RON 92 vs EBOB, otherwise known as the Gasoline E/W contract. By buying this E/W contract in equal volumes to the two crack contracts, Onyx had a delta-neutral position. However, despite the market risk being zero, Onyx was still holding the following on its balance sheet:

- Long 20,000 lots of May 2022 EBOB crack contracts
- Short 20,000 lots of May 2022 Singapore Ron 92 crack contracts
- Long 20,000 lots of May 2022 Gasoline E/W contracts

Onyx could then dig into the gasoline delta-neutral positions to extract what we called the COT raw data. With the knowledge that Onyx was delta neutral, the conclusion could be made that the market had the opposite position in these contracts. This made transparent the overall market's 'real' open interest in the contracts, excluding the market maker's risk, which was delta neutral.

Open interest data are publicly available as a regulatory requirement, and Onyx's data accurately reflect that open interest in oil swaps.

Real-world example: surprise Brent/Dubai reversal after the OPEC cut in Q4 2022

As Q3 drew to a close in 2022, the physical oil market was beginning to show signs of returning to balance after most of the year was incredibly tight. Rising inventories in the US and increased exports to Europe were pressuring the Western crude market, while OPEC+ was cutting the crude supply. The market remained relatively buoyant. Consequently, traders had positioned in the forward Brent/Dubai swap differentials net short on the basis that Middle Eastern crude would remain supported relative to the Western market, and this position was providing a strong return.

Early in Q4, OPEC+ announced a cut in crude oil production to arrest the flat price decline by reducing crude oil supply to global physical oil markets. The obvious impact was that the Brent/Dubai differential should move lower, as the OPEC+ cut would further strengthen the Dubai complex, being a benchmark for the Middle East region. However, unexpectedly, not only did the differential not move down, but it reversed aggressively.

The Onyx COT data showed that before the headline, the market showed signs of being overextended short. The total open interest sat at 53 Mbbl,

almost 60% above the five-year average, indicating a market risk far beyond normal. In the Onyx COT, the long-short weighting was clearly skewed to short, indicating that the high risk levels in the market were positioned short. So when the news came in, there were no new sellers in the market, as those who could sell already had. In other words, bearish information was already priced into the market, and the contract was now highly vulnerable to new buying flows. After an initial stagnant reaction, new buying emerged in the market. As the buying led to reduced short positions, the data heavily imply the new buying was short covering, possibly as the market panicked from the lack of reaction to the bearish news.

With no new sellers to absorb the new buying, the market needed to rally until sell-side liquidity was found. This resulted in a rally back up to the previous month's high, erasing the downward movement.

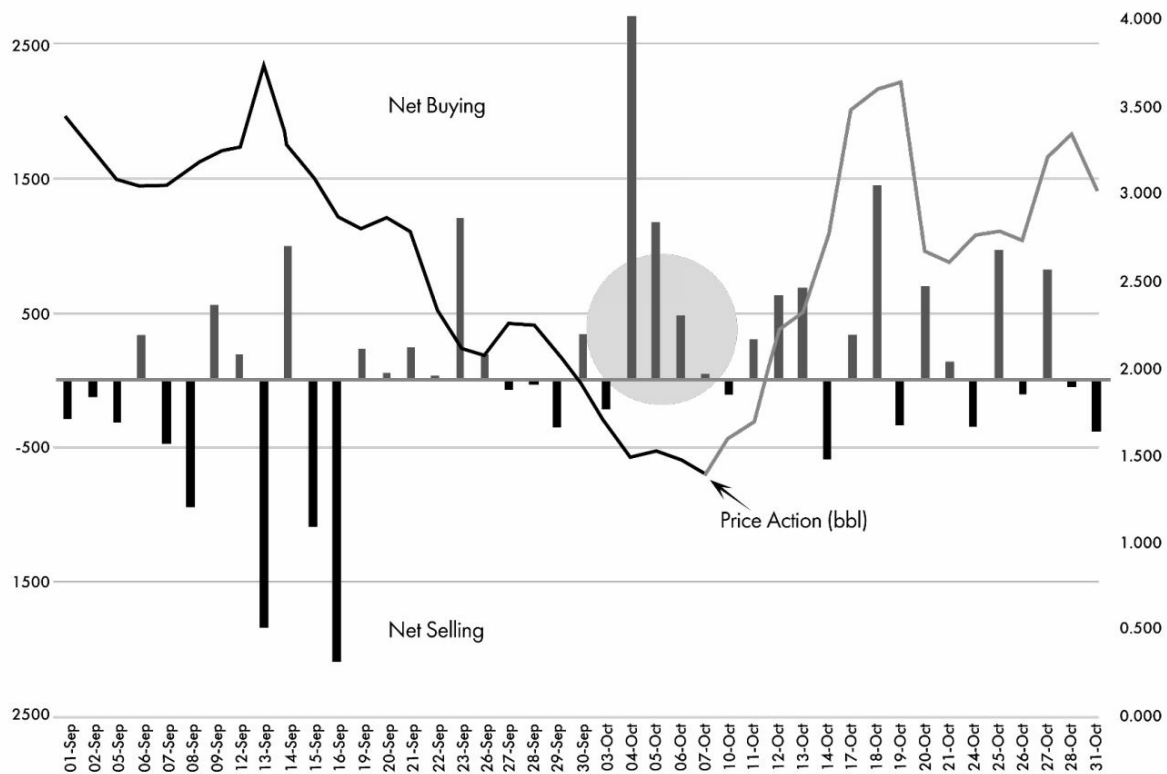


Figure 3.7 Net Market Positioning vs Price Action in September-October 2022 for November 2022 Brent/Dubai (Source: Onyx COT data)

Market Participants and Their Flows

A market **flow** is a group of transactions that have entered the market. You can learn to identify the types of participants by their characteristic flows. Here is an example of the daily buy and sell side flows in the Dated Brent market from Onyx's activity in the market:

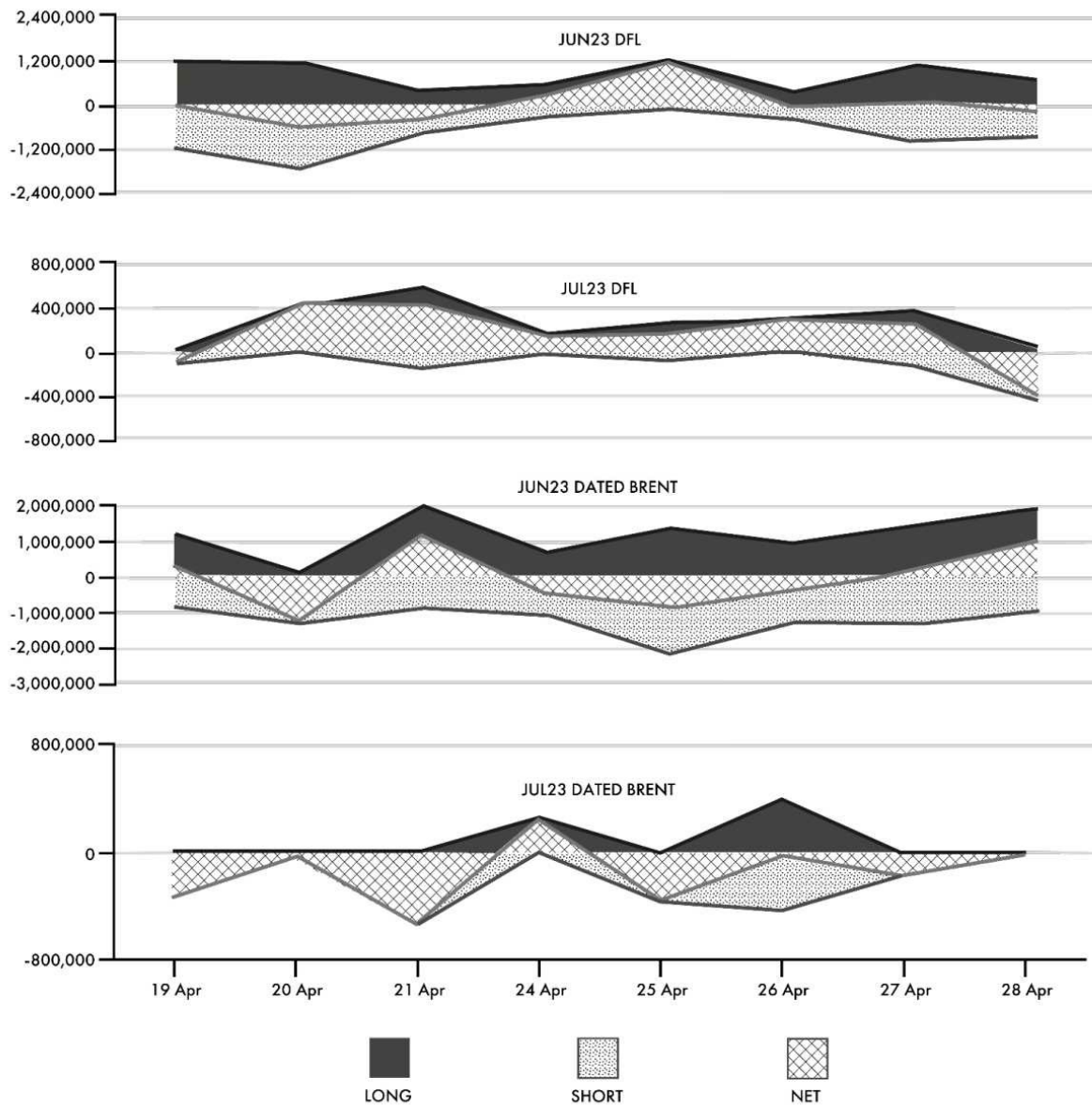


Figure 3.8 The Positioning Changes in Dated Contracts from 19 April to 28 Apr 2023 (Source: Onyx COT)

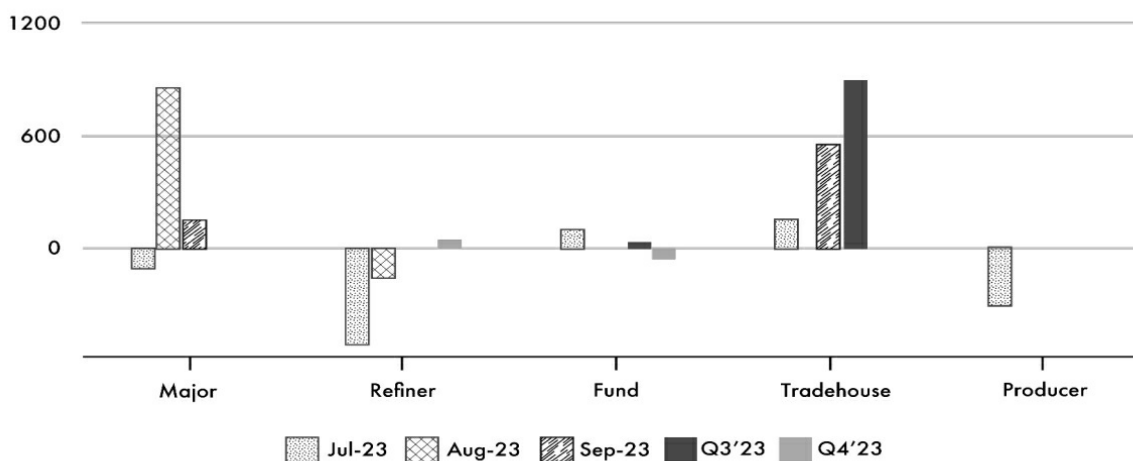


Figure 3.9 Snapshot of Positioning Changes by Counterparty Type in Brent/Dubai (Source: Onyx COT)

The flow might be a producer, for example, or a group of producers who are hedging their production for Q4 the following year and therefore have a fixed volume to sell of a given crude contract. All the transactions that comprise the producer's hedge would be considered part of the producer's flow. It might also be speculative positioning one way until a trader maximises their risk tolerance or an oil trader hedging until a price for the cargo volume is fully locked in. All traders will have a flow unique to their strategy, and each strategy has a maximum risk capacity, which is how flows and positioning come together.

Market flows will govern price action at any time, but working out for how long is based on a judgement of the participants' positioning or risk capacity. Market makers are different from the directional speculative trader or hedger, as their job is to take on the flow that the market needs to execute and de-risk this core flow against the flow of other contracts. In other words, a market maker provides liquidity for the trader entering a new position and has no maximum capacity as long as there are outlets for their risk. Market flows drive price action. Flows on a given contract can have knock-on effects on other contracts by the way price curves are linked via differentials. Information about the web of how curves are connected and flows absorbed gives the market maker the edge because they know that one flow must absorb another. It follows that significant distortions can form if there is an absence of flow in one way that can be used to offset the driving flow in another. Understanding flow is therefore necessary for context on recent price

action. The temptation is to believe that any move in financial oil markets is off the back of a shift in physical markets. When market observers see a flurry of volume pushing a contract or differential, they immediately look for a justification. The first assumption is that a headline must have sparked the move, and brokers and market participants will circulate snippets from news agencies or links to websites. Failing a material headline, market observers will assume a change to physical operations that has translated into a hedging move. Alternatively, market observers may assume that traders are speculating on a contract, with a view to taking the risk all the way to pricing against the physical product.

Flows can reveal information about the physical market, trader behaviour or a combination. Thinking of market prices and moves as a function of flows plus positioning will help you predict how far markets can move or the likelihood they will rebound objectively. The idea that moves that can't be explained by physical supply and demand 'do not make sense' or are a 'dislocation' is ingrained in the oil market. This can lead to bad trading decisions and bias, which can be hugely damaging to a trader over the long term. You can't think this way if you are going to be a complete financial oil trader who can stay active in all market conditions. When you trade in the derivatives market, you must think of what is possible in the context of a financial market, not what the market 'should' do, as if it were a perfect replica of the underlying supply and demand.

Types of Flows and How to Differentiate Them

Flows are either hedging or speculative.

How to identify a hedge flow?

Hedging flows have a few telltale signs. First, pure hedging will match the volume of a physical transaction. For example, oil hedges related to moving oil around the world will be in the form of cargo volumes. For crude, this would be an Aframax, Suezmax or very large crude carrier (VLCC), which have volumes of 600 KB, 950 KB and 2,000 KB, respectively. For products, there will be an equivalent; for example, a typical standard cargo transporting residual fuel oil will be 5 KT. Therefore, you see this volume trade frequently.

Aside from looking at volumes, you can look at the hedger themselves. A hedger will have a trading profile that can be mapped. If they enter a

hedge, they tend to be a market taker and will look to execute at market for their volume as soon as possible. This will be the case if the trader has low autonomy on the hedging execution and/or the physical transaction margin is relatively small and level dependent.

When entering a hedge, these traders tend to be less reactive than a speculative trader and stick to the levels they have in mind that make the economic work on the physical. This is also somewhat dependent on the market participant; for example, an airline might look to buy any dip below a given level in line with the budget it has set for the year but have a wide range that is acceptable and will only panic at the extremes of that range. An oil trader may have much more instantaneous reactions, given the tight margins in their business and the frequency of their transactions.

How to identify a speculative flow?

Speculative flows can be categorised in many ways, but some common ones are *day trading*, *short term*, *long term* or *pricing*.

Day trading can also be referred to as swing trading. The trader seeks to profit from momentum and patterns throughout the day. In exchange-traded futures, this tends to be driven by technical analysis, a lot of which is executed by the retail trader or automated traders using algorithms. Market makers dominate this type of trading in oil swaps (or block futures). The profile of this type of trading is higher frequency than other price action profiles, with traders looking to move in and out of risk.

Short-term speculative trading tends to be one to three weeks and is highly reactive to headlines and the market narrative. You can spot this type of trading by knowing the market participant and their trading style, and typically these traders make up the bulk of the 'herd' in oil trading. This trader will react in the morning of the trading day or the afternoon if trading in Singapore. The trader will decide on their position after reading the headlines from the night before and the morning commentary and after getting the consensus from the morning trader meeting (if they are at an oil trade house). Typically, the 'herd' will wait for the first significant flow of the day to make its move. If something unpredictable happens, these traders will look for justifications and be swayed by the narrative. So this group of traders is highly susceptible to rumours or sudden headlines, and if you are on the front foot, you can influence this group of traders, especially if trading OTC markets.

Long-term trading is normally dominated by large trade houses and

majors. These traders will find it difficult to disguise their position in OTC markets, as they will be one way for as long as it takes to enter into risk suitable for their revenue target. Some hedge funds trade this way in exchange-traded futures, and the only way to recognise this flow is by looking at the volume traded and the price action profile. If 10,000 lots of Brent or WTI futures are going through every minute for 10–15 minutes, this is likely a large player entering a position or stopping out. Stopping out has a much more violent price action profile than entering a position, which is typically steady and relentless. In oil swaps, this kind of execution is not as common as with exchange-traded futures, as it would be too easy for the market to move away from them. Instead, these traders tend to trade opportunistically at a given level and enter an initial position for the available size. If the size is not what they intended, the trader will typically wait until the market returns to the range they originally sought. Once long-term traders enter a position, they don't like to stand still, particularly in OTC markets. Traders within this profile are known to defend their position. Therefore, it is an advantage to know the level at which they entered (or at least to have a vague idea), as it is around this level they will add to their position and look to fight the momentum of the market that is going against them.

Pricing positions are almost exclusively held by trade houses and majors, who enter positions against the physical transactions they have already agreed on or in the knowledge they will need to transact physical barrels in the PRA's physical tendering window. The intention of a major or trade house to take a position in pricing will become evident by action in the PRA assessment windows at the beginning of the pricing period of a given contract, but it can also be opportunistic. If oil traders see a lack of interest in physical or a lack of offers, a pricing trade offers another route for speculative trading.

When analysing the oil derivatives market, it is important to understand every buyer's and seller's motives. Each trader with access to the exchange-traded oil derivatives market will have an influence, large or small, and the knock-on effects of flows can have a butterfly effect. Some participants will be executing and holding risk for purely hedging purposes, and others with the intention of trading on the physical markets that will be used to price the contract. There are many forms of speculators, investors and market makers, all of which have an impact. Therefore, it is important to know what risk is 'live'; in other words, what risk can be closed, reversed or supported in the

pricing.

Recognising Flows

Flows in the oil derivative market fall into the following categories:

- Oil traders
- Refiners
- Producers
- End users
- Market-makers
- Hedge funds
- Institutions
- Retail

Physical Oil Trading Flow

The physical oil trader's role is to move barrels around the world to satisfy demand. They are, in effect, market makers in the physical world, buying from the producer and selling to the refiner, then buying from the refiner and selling to the end user. Oil traders play a logistics game, sourcing relationships worldwide to become the most reliable supplier to achieve the greatest flow so they can make a small margin in each trade. Like market makers in financial markets, oil traders search for inefficiency. Inefficiency can come from a price dislocation between regions, blending components to make a finished product or varying prices over time set by the forward curve. So the primary trading activity of oil traders in the derivative market is hedging these inefficiencies. The result of this hedging activity is what we see playing out in the market.

Every product has a contract that reflects the economics of geographical arbitrage. For gasoline, the structural flow is from Europe to the US, so the RBOB vs EBOB contract exists, known as the 'transatlantic arb', with oil traders regularly on the sell side to lock in the voyage's economics. For naphtha, the shorts (demand outlets) are based in Asia, so oil traders interact with the E/W contract. For fuel oil, it is also the E/W contract, while the diesel market is typically about supplying Europe, so it is also the E/W contract but the reverse flow – buying the contract to hedge.

The flow direction can change, but you can expect these contracts to be dominated by oil traders. Flows and positioning can be localised in one region, and traders move to hold positions in that region, which can have

knock-on effects on how cross-regional differentials are priced. If prices for these contracts move to extremes, they can open up real physical arbitrage opportunities, which oil traders lock in by hedging the physical volumes they will move. Oil traders can also give themselves optionality by locking in this differential and then trading back out of it at a profit if it moves in the direction they traded it before the time to trade the physical product arrives. This allows them to play the ranges with more edge than the purely financial trader, who must accept the loss if it goes against them because an oil trader can commit the physical barrels against the differential and benefit from the move on the differential side in the physical world to offset the loss on the financial side.

Example: *A gasoline trader looks at the forward curve for US vs European gasoline by looking at the RBOB vs EBOB derivative swap contracts (also known as the gasoline transatlantic arb).*

The difference between the two contracts is the level at which they can generate a net profit after the costs involved in moving gasoline from Europe to the US. Therefore, they sell the differential, which acts as a hedge against the differential moving against the arbitrage economics. When it gets close to the time to move the physical barrels, close to when the contracts begin to price, they have the option of moving the physical and offsetting the gain or loss with the derivative gain or loss, or they can simply take profit on the hedge and cancel moving the barrels.

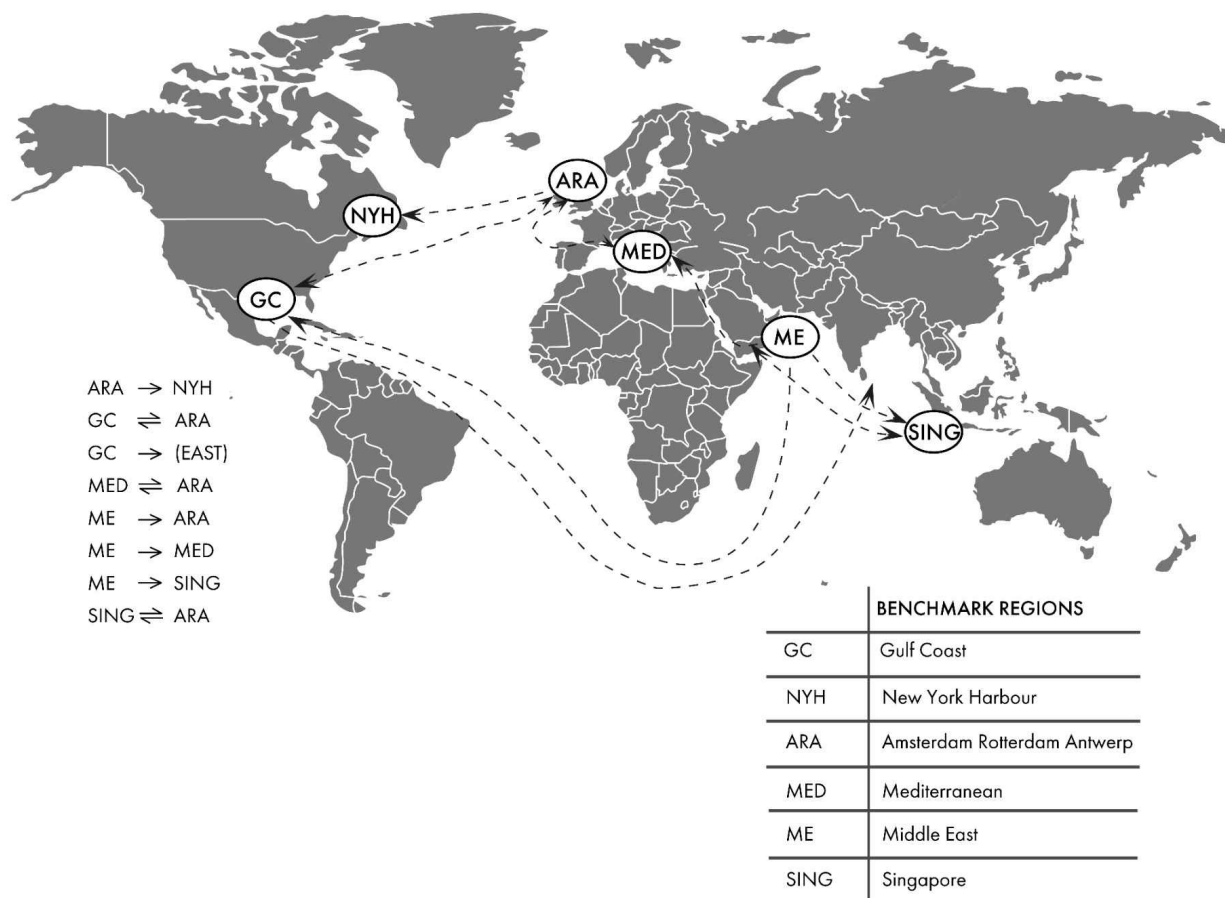


Figure 3.10 Key Benchmark Arbitrage Routes

You will also see oil traders active on time spreads, managing their exposure to oil storage. When they buy physical oil, they are long the outright price once they take delivery of that oil until they sell it in the market. Therefore, they hedge by selling a derivative contract with maturity at a later date. If they want to sell it later than the derivatives contract they hedged with initially, they will buy time spreads to roll that exposure into future months. Time spreads are also useful for managing the price risk of long voyages. Even though traders are buying and selling the product using the same benchmark as a reference price, the loading dates from lifting and delivering the oil can span two or more months.

But not all oil traders are the same. The large trade houses and majors have highly complex trading operations worldwide, with varying physical exposure. Some prefer to take a view of the overall market and gear all oil trading exposures to reflect that view. This could mean deliberately leaving forward purchases of crude (negotiated as a differential to a benchmark)

unhedged. Therefore, their market position is implied as short to the benchmark used to set the price. They could also agree to sell on one pricing formula and buy on another, for example, to sell Dated Brent priced crude in the future and buy crude to supply it at a differential to Brent futures. This leaves a net market risk of Dated Brent vs Brent futures, which is very close in price to a DFL, and the oil trader has this position without the wider market knowing about it, which can be useful to either exit positions losing money or enter positions without being obvious.

Traders will also be trading their overall internal pricing position to expiry or settlement of these contracts. What you see on the exchange (or any data derived from cleared volumes on the exchange) is the executed volume, in other words, the volume that one of these participants has chosen to trade vs their overall price exposure. Like the tip of an iceberg, the executed volume hides the net exposure of these companies. The large trade houses and majors who act as oil traders have huge balance sheets, which makes their speculative volumes disproportionately larger than the average oil trader and even the largest of hedge funds. As a result, the large trade houses (usually referred to as the 'Top 5') and the major producers positioning in and out of markets can have huge impacts on prices.

Refinery Flow

Refiners observe and monitor the oil derivatives market movements, looking at the basis risk for the crude they will be buying alongside the product prices, less the crude or feedstock prices. From a refinery margin perspective, they are agnostic on the movement of the outright price, as even in a high-price environment, they can still purchase the crude as long as they can profit from the processing. They also pay attention to the time spreads, as they will be holding crude in storage, the amounts of which will vary based on whether crude spreads are in contango – cheaper in the prompt period – or backwardated – more expensive.

The refinery management board will typically set a target for refinery margins for a given period, usually one year, relying heavily on the prevailing forward margin calculations. The target level will be considered against the company's view of long-term trends in the oil market and their costs.

These costs can fluctuate with power and gas prices, which can also be

hedged, since sophisticated financial markets exist in these contracts. In 2022, prices in the power and gas markets rose so much that despite some of the highest gross refinery margins ever, profitability was still under pressure, forcing European refiners to cut production towards the latter part of the year.

Once a target is agreed upon, the refinery may choose to forward hedge to lock in the margin if the forward curve is pricing at levels they are content with. For example, the board of a European major was reportedly infuriated that the business had to endure negative refinery margins in 2014, having left the margin to materialise in the spot market with little to no hedging for protection. As a result, the board ordered the immediate locking in of forward margins using derivatives for 2015 and into 2016. The major was highly active in the market for weeks, locking in these levels, and the volume was so large that it became a key flow driving the swaps market at the beginning of the year. However, the margins reversed in 2015, when product cracks surged in the spot market, and the hedges settled out of the money, leading to the abrupt end of this strategy the following year.

The expectations for ten or so years have been that European refinery margins will weaken and become uncompetitive in the global landscape of increasingly sophisticated refineries. There has certainly been volatility, but almost every year, the forward price action has offered opportunities to lock in what can be considered profitable and attractive margins.

The refinery margin is constantly rebalancing in tune with the relative strengths of light and heavy product markets, which can also be a function of the crude market. When one product crack, which forms a large part of the barrel, blows out, the overall margin can increase significantly. For example, in 2022, Gasoil cracks roofed while all other product markets were enjoying a bull run. The buying was relentless and focused on Gasoil, meaning all other cracks had to correct, or the overall refinery margin would be astronomically high, which it was for a short while. Astute refiners locked in margins for deferred months, which were dragged up by relentless buying. Gasoline corrected to accommodate the Gasoil crack prices and went briefly negative in European and Singaporean markets. It is common for refinery hedgers to limit the downside of waste products or secondary feedstock into the open market, as this can make or break the overall economics. The residual fuel oil market, in particular, can be highly volatile; cracks have been positive but also as low as $-\$30/\text{bbl}$ when the GOFO (Gasoil vs fuel oil) differential was targeted as a way to reflect the incoming changes in shipping fuel from IMO

2020.

The impact on derivatives from refiners' flows can be significant. Refiners execute much of their hedging through banks on a bilateral basis. Interestingly, they tend to enter the market at once, usually following a significant move in product markets the day before. You see the evidence of refinery activity in the OTC market when banks come into the market, looking at the DFL on the buy side and product cracks on the sell side. The flow is consistently one-way for a time, and the flow can distort differentials and cracks until the flow is absorbed. Given the buying in deferred DFLs, traders taking down the buy-side flow are usually called 'dated to lead' traders. Because of the mathematical link between DFLs and Brent spreads, these traders will sell DFLs against buying Brent spreads in associated tenors. Therefore, the knock-on effect of refinery hedging is Brent spreads rallying, which can also feed into the outright price and create strong momentum for the overall market.

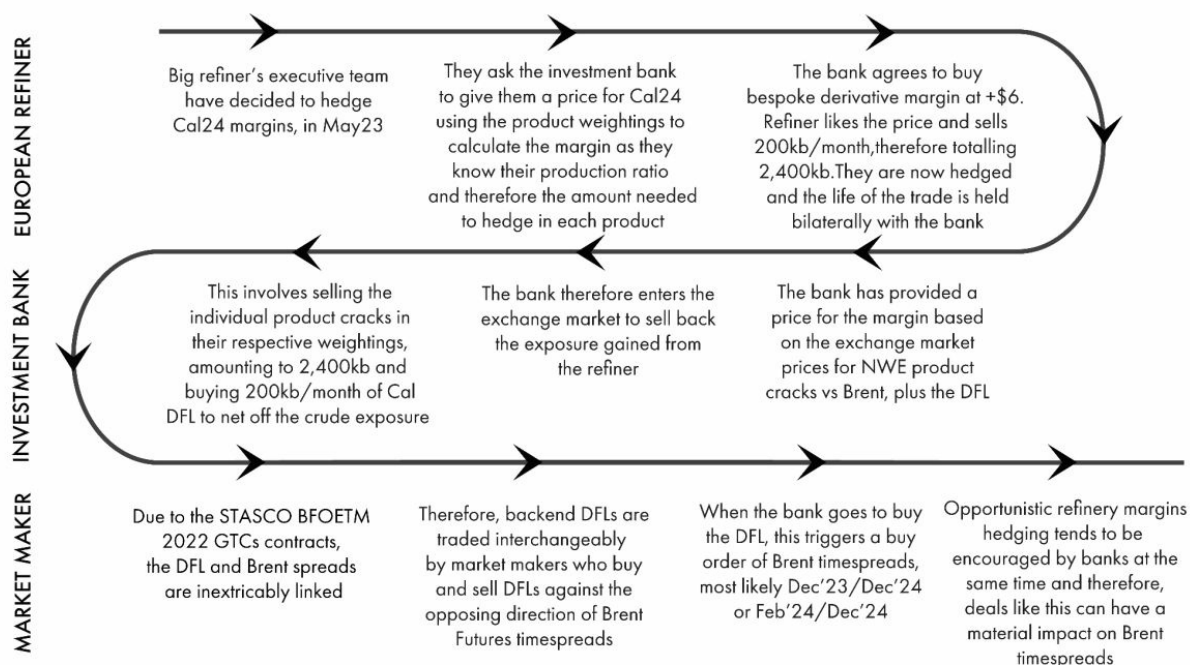


Figure 3.11 The Impact of Opportunistic Refinery Hedging

More nuanced impacts from refiners will come from storage hedging, as they buy time spreads to move the flat price hedge on their inventories to the following month. Dislocations between products are also hedged so they can shift the product yields of what they are refining. If the price difference between distillate and gasoline becomes large enough, refiners could hedge

that price difference and begin shifting their yields to produce more of one product than the other. This can lead to genuine operational changes to refinery units and the type of crude refiners buy, which can have knock-on effects on benchmarks.

For example, if naphtha were to blow out, naphtha-yielding crudes like CPC (a light, sweet oil with low sulphur) would be more in demand. If this crude in the Med was supported over Angolan crude coming from West Africa, then it may seem like there would be no shift in supply and demand. However, the naphtha-yielding crude could be replaced by Ekofisk in the North Sea, which an oil trader in the North Sea could recognise. This would alert the North Sea trader to specific tightness in BFOETM and, therefore, Brent-related contracts. So Brent spreads and DFLs could rally aggressively even though global supply and demand economics hadn't changed, just regional differences. This may seem abstract, but it really happens. This is why monitoring refinery margins and product crack differences is so important (for more on the Brent pricing mechanism, see the Appendix).

Producer Flow

Crude oil producers are relatively binary in their impact on financial oil markets. They are naturally long the crude oil prices of the region they market their crude oil into, so they look to hedge the outright price in the associated contracts. Producers typically trade Brent or WTI futures contracts, looking above all for liquidity. Some choose to roll this exposure into swaps contracts for exposure to a localised region, such as Dated Brent or Dubai swaps differentials contracts.

The clearest way to track producer trading and positioning is via the CFTC COT report, which has a separate section for producers and end users. You can be confident that producers will be short and end users long, given that this aligns with their natural hedging direction. Typical long–short ratios tend to be below one on Brent and WTI contracts. Generally, producers sell more crude futures contracts than end users buy. End users also use more closely related product exchange-traded futures or oil swap contracts. Typically, the range for long vs short in this section is 0.5–1.2. Therefore, looking at this ratio and overall open interest will give you a sense of how committed producers are. As this selling is for hedging, it is unlikely that the hedging will be reversed outside of extreme scenarios. Instead, high hedging

volumes can be taken as evidence of the selling flow being priced into the market, as producers have limited selling flows. In contrast, if the hedging commitment is low, it indicates that selling flows can still enter the market. When producers wait on the sidelines and observe markets without committing to a hedging programme, they are running a large, speculatively long position. So, when prices threaten to go lower, there tends to be a flurry of panic selling from producers who don't want to miss out on locking in higher prices. This can exacerbate directional moves and has the same price action profile as a trader stopping out, which in effect, they are.

End User Flow

Many end users, such as utilities, shipping companies and airlines, hedge similarly to producers in buying Brent and WTI futures contracts. However, they are increasingly moving to eradicate basis risk (final physical price vs final derivative settlement price) by hedging with contracts closer to the actual product they are buying. For example, shipping companies buy residual fuel at flat prices, and airlines buy jet fuel similarly. So tracking their positions is only partly possible via the CFTC report. It is important to monitor the Gasoil futures contracts end user section and Brent and WTI data.

In terms of impact on the market, end users tend to provide liquidity on sell-offs at the outright price. Typically, end users have an agreed budget for physical costs and may be free to lock in prices anywhere below that budget. This leads to the opportunistic buying flow on the flat price moving downward. When the flat price is perceived to be attractive, you must be careful when trading differentials linked to end-user buying.

For example, in 2014, the market was bearish across the board, and one trade house looked to position short on jet fuel. They chose to short jet fuel differentials against Gasoil to reflect the weak jet fuel market, but this turned out to be a bad choice because of end-user buying. Airlines rushed in to hedge as the flat price moved lower, and given the attractive levels, bought jet fuel swaps contracts in large volumes and were happy to pay whatever offer was available. This meant that the jet swap differential vs Gasoil was, and remained, propped up relative to Gasoil. This is quite common, so on Brent futures sell-offs, you need to assume that end-user buying will naturally elevate differentials and product cracks.

Hedge Funds, Institutions and Retail Flows

Retail flows come from financial investors looking to gain exposure to financial oil prices. These investments enter the market in the following ways:

Discretionary Hedge Fund

A discretionary hedge fund is a hedge fund in which the fund manager makes investment decisions based on their judgement and expertise rather than relying strictly on algorithmic or systematic strategies. The investment process in a discretionary hedge fund is qualitative, with decisions typically influenced by various macroeconomic indicators, company fundamentals, market trends, and other subjective factors.

Discretionary hedge funds enter the financial oil market in cycles, showing more interest in bull phases than in a bear regime. When oil gets attention, investors demand exposure to the commodity, and so hedge funds work to provide a solution. These types of hedge funds in oil were much more common in the 2000s than now, and very few who have survived or started in the last ten years have consistently done well. Perhaps the only prominent discretionary trading hedge fund left is Andurand Capital, run by Pierre Andurand, who has refreshed the business several times.

Risk Parity

Risk parity is an investment strategy that seeks to allocate risk, rather than capital, equally across the different components of a portfolio. The goal is to create a diversified portfolio where each asset contributes equally to the overall risk of the portfolio. 'Risk parity flow' refers to the flow of investments into and out of strategies that adopt this risk parity approach.

Risk parity might also refer to how the allocations within a risk parity portfolio change over time as different assets' risks rise and fall and as the portfolio is rebalanced to maintain its risk balance.

When market conditions change and volatility or risk in certain asset classes increases or decreases, the 'risk parity flow' would involve rebalancing the portfolio to ensure that risk remains evenly distributed across

the assets. This rebalancing might involve buying or selling assets to maintain equal risk contributions.

In a risk parity strategy, oil and other commodities would be allocated in the portfolio such that their risk contribution is in line with other assets in the portfolio. This means the allocation to commodities might not be based on their expected return or market value but rather on their volatility and correlation to other assets.

Commodity Trading Advisors

This is the term used to describe hedge funds that deploy systematic, trend-following systems. The flow from these funds will likely add weight to material moves in Brent and WTI.

All hedge funds prioritise liquidity when selecting a contract to trade in oil, so unsurprisingly, they are heavily focused on exchange-traded futures. Almost no discretionary hedge fund looks at oil swaps in any meaningful way, so the impact of hedge funds and managed money on oil swaps is currently low. But as the market becomes more democratised and transparent, this may change, and swaps liquidity may be seen as enough to present an attractive opportunity.

Exchange-traded Funds

Institutions and retail businesses access the oil derivative market via exchange-traded funds (ETFs), such as Wisdom Tree WTI crude and the United States Oil Fund. They are an increasingly popular route to the market.

Generally, ETFs are uni-directional – usually long – but some offer a short position against oil futures. They are leveraged, so the position reflects two or three times its notional value. In April 2020, when WTI went negative – to $-\$37.63$ – there was a lot of concern, as the share value of ETFs and exchange-traded products (ETPs) was based on oil having a maximum downside of \$0.

These ETFs include a combination of near prompt month contracts and roll the exposure forward using time spreads every month. So it is common that Brent and WTI time spreads have periods of selling volumes near known times when ETFs are rolling their positions, which is generally formulaic, with only some discretion in execution by the fund manager. Nevertheless, as

a market observer, you will see high volumes on time spreads going through on the exchange or a 'blocked' deal posted on the exchange trade blotter, which is usually simple to identify as fund rolling.

As the prompt time spread in Brent futures is linked to the Dated Brent and Dated Brent physical markets, the ETF rolls are very much part of the liquidity cycle until the Brent expiry. North Sea traders will either short the Brent spread ahead of expiry and use ETF rolls to take profits or cover their shorts, or they will take the other side of the time spread risk from the rolls to get the maximum length going into expiry settled by the Brent Index. Much of this fund rolling flow goes to futures brokers, who go directly to participants and execute directly on the exchange. A fair few participants will be happy to trade in the required volume, so the market isn't usually materially moved by this flow. This is why it is common to see Brent futures time spreads 'blocked' on the exchange during these times, indicating an OTC (person-to-person) transaction that must be reported to the exchange. These blocked transactions are in minimum sizes of 1,000 KB clips and sometimes as high as 5,000 KB, so they are relatively straightforward to identify as fund rolling when you consider the timing. The same can be said about all futures that need rolling, and the more the ETF market grows, the more volume will be traded in this way.

Retail Brokerages

Finally, retail brokerages, such as Interactive Brokers, which provide access for retail traders to trade oil derivatives. The trading through these platforms tends to be a day trading style, reacting to technical analysis indicators. Again, the focus for these traders is on exchange-traded futures, but I would argue the wider retail trading community simply hasn't been offered the oil swap market to trade and would, in fact, be very interested in a rapidly growing and interesting market to trade. This is certainly something that should, and most likely will, change in the near future.

Algorithmic/High-frequency Trading

The high-frequency trader (HFT) tries to exploit order flow into the exchange and is active in exchange-traded futures contracts. The HFT looks to gain an edge through technical innovations like fast access to live data

feeds, ever-faster connections to the exchange and software algorithms running on fast processors. They're described well by Michael Lewis in 'Flash Boys'[\[48\]](#).

The game for these traders is to buy ahead of buying flow and sell ahead of selling flow, hoping to exit seconds or even milliseconds later for a profit. They do this by riding the momentum of the flow, so the larger the volume entering the exchange, the more active these algorithms will be. These traders receive rebates from the exchange for adding liquidity to the market, which helps reduce the margin they need to make a profit. As futures execution becomes more sophisticated, so do HFTs. The systems HFTs use to trade, governed by algorithms, are constantly refined and replaced when they stop generating returns.

It is contentious for traders to see HFTs at play in the market. In other types of markets, this would be considered 'front running', which is an illegal practice. Even more contentious is how HFTs can access the order book breakdown, which exchanges sell as data feeds to these companies. In a conversation I had with one exchange, I learned that you can buy this data 'if you can show it is part of your business model'. In other words, only if the exchange wants you to see it.

Executing on exchange-traded futures comes with risks. For example, stop losses entered on the exchange are shown as orders once they are triggered, and HFTs will seek them out. When the market reaches the level where your stop is in place, your stop will be triggered. The stop will become a market order, which is then executed at the best price for the volume required, but which HFTs will know you are selling. The HFT can therefore front-run this order, and the result is that your fill is at a worse price than it would otherwise have been because the buy orders are taken out of latency that HFTs achieve through co-location or being closer to the exchange. This is particularly dangerous at illiquid times of the day, and, therefore, stop losses on exchange-traded futures overnight are a bad idea.

Traders frustrated with this have tried to 'game' the algorithms, but these attempts have led to investigations and fines. See the 'spoofing' example in the Compliance and Regulation section.

In any event, these high-frequency automated algorithms are active from exchange open to close and provide much of the liquidity we see in exchange-traded futures today. Without these algorithms (which you must remember are also taking risks), getting constant two-way markets going

from 01.00 to 23.00 would be a considerable struggle. In one case, a new HFT on an exchange launching an already popular benchmark on another exchange struggled to get the market going and ended up losing a considerable amount despite the rebates and the liquidity already in the market, so their profits are not guaranteed.

Market Maker Flows

Market maker flows are mainly present in the oil swaps market. These traders are taking down the market's flow and using flows in other contracts to offset most of the risk. Market makers are therefore left with a lot of cross-contract risk, which can build up if the flow is consistently one way. Therefore, you will sometimes see large volumes from market makers, which can confuse market participants, given that market makers traditionally have a lower risk profile than the trade houses and IOCs in the market.

As part of your trade cycle, you will undoubtedly encounter and trade with a liquidity provider, otherwise known as a market maker or 'curve stripper' in the oil swaps market. The demand for immediate pricing and depth in volume in oil swaps means there is a value-add from these firms that exclusively work to provide liquidity at all times of the trading day.

The role of the market maker is regarded negatively by some in the oil derivatives market and often presents a dilemma. You need liquidity, but you don't like how profitable some companies specialising in the field are. You feel you are paying over the odds by trading with them. You may not like their strategies, such as those of HFTs on exchange-traded futures, and feel you are getting turned over every time you execute. On the oil swaps side, where Onyx specialises, we've heard every complaint, from the understandable to the ludicrous. A broker working with a trade house once told me of a conversation he'd had with a trader, which went something like, 'These guys are just ruining the market; they just get in the way and add no value', and then a few seconds later, 'Anyway, can you get one of them to show me an offer on 3–7 Jan versus Cal Feb?'

In other markets, like bond trading, traders recognise the value of market makers. In those markets, market makers provide liquidity against the grain being traded, which traders acknowledge requires skills they do not necessarily have.

Traders also recognise that it is a role they do not want to fill. Perhaps

the main counter to the argument against liquidity providers is that it is incredibly difficult and risky, even for algorithms. There is so much information in the oil swaps market to manage, retain and assimilate for at least ten hours at a time from three different locations. To take down market orders in the sizes requested from the largest trade houses and majors requires holding a risk that can and does go against you considerably at times, leading to what are essentially leveraged losses. So, it is a highly stressful and complicated risk management job, which, if you do it well, will lead to an in-depth understanding of the market dynamics of derivatives that's hard to get in other roles. Like Ricky Bobby says in the comedy *Talladega Nights*^[49], 'If you ain't first, you're last', meaning that only the top market makers can generate meaningful profitability and maintain market share. If you are not the sharpest on price and accommodating volume, market participants and brokers will be less obliged to show you the flow, and you will rapidly lose the market pulse.

They only trade with you because they have to, as you offer the best price and are willing to go counter-market when no one else is, so if you're not doing these things, the market will have no use for you.

As the years go by, the training and development of market makers and the systems they use improve, making the barriers to entry increasingly high. At this point in futures and swaps, liquidity providers are a fixture in the market and are entirely necessary.

The voice-traded oil swaps market requires highly labour-intensive work by the traders themselves to market-make effectively. This style of market-making requires the trader to deconstruct incoming market flows and find other sources of market flow that can de-risk these contracts. To do this, the traders work with the concepts of how the market is pieced together, explained earlier in this book. Having a rigorous understanding of the net impact of flows on the forward curve and knowing the supported or vulnerable points are key to bringing the market's activity together and making a margin in between.

Market makers are highly influential in the pricing of swaps. Their most immediate impact is linking flows in the deferred parts of the forward curve into the prompt. Before market makers became significant influencers, sustained dislocations in the forward curve were common. These days, a significant flow in any market will have knock-on effects on all linked contracts, which can confuse market participants. For example, sudden heavy

buying in the Singapore 180 flat price will drive the Visco contract, the E/W and the Barge crack higher almost instantaneously, leaving European fuel oil traders scratching their heads as to the source of all the buying in their local market. In other scenarios, a differential might be forced to extremes by flows that market observers try to explain. If they understood the differential was being forced into its current price by concentrated flows and market makers using it to decrease their risk, market observers might draw different conclusions about what was driving the market at that time.

In my opinion, the biggest impact market makers have is their influence on pricing. Swap contracts are settled partly by physical prices in the Platts or Argus assessment windows, but they are also priced by trades and values in the forward curve. Oil traders whose flows influence the pricing windows are now coming up against the deferred market flow. Market makers use flows outside the window to match the aggressor *in* the window.

The Role of Interdealer Brokerages for Market Flow

Interdealer brokerages play a big part in the oil derivatives ecosystem, mostly in the voice-traded swaps market, but they also have a presence in exchange-traded futures. They deal mostly through cloud communication systems like Cloud-9 but also via e-brokerage software platforms. Like many others, you might wonder why we still need a voice broker between trades.

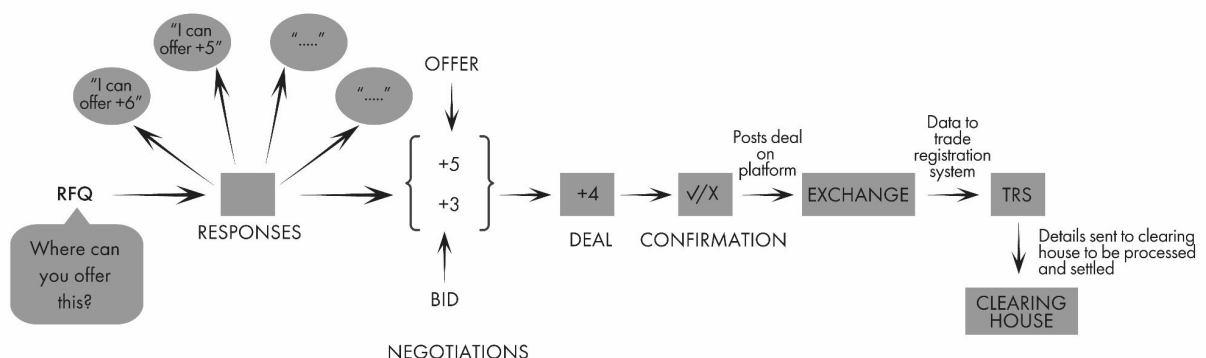


Figure 3.12 The Broking Trade Process

Brokerages have been under constant pressure over the last ten years from the monopolistic tendencies of the exchanges, which are trying to force

all trading onto one unified market platform. Despite this pressure, brokerages continue to operate and grow for the following reasons:

- They possess market information (flows, patterns and changes in the physical market).
- They receive a substantial volume of requests for quotations for non-standard deals.
- They can service a Request for Quotes (RFQ) function by sourcing markets on hundreds of contracts with thousands of possible differentials and time spreads linking those contracts.

They provide advice and guidance on various situations if they have regulatory permission to do so. A broker can help to determine the trade to be executed, enquire on the market value (especially if it's a 'non-vanilla' pricing structure) and agree on a strategy to execute a deal.

They have proven their ability to execute deals efficiently without alerting the market, especially when managing volumes. Large futures flows can move the market, but you pay brokerage fees for a brokerage firm to work on your behalf to find traders to take the other side of the deal without moving the market against you. This is common for hedge funds entering positions and macro funds rolling their position to the next tenor – like index funds and ETFs.

They build and sustain a legacy of trusted relationships. A broker can be helpful in an emergency; for example, if you are out of the office and need efficient execution you can trust, a broker would be helpful.

Predicting Market Behaviour (Market Psychology)

Trader Behaviour

The oil markets move by flows, but the extent to which they move depends on how the market is positioned or is positioning. There is a finite capacity for risk in the market, so price discovery is based on at what price level you are able to find a new buyer or seller with enough volume to satisfy an incoming market order. A trader's behaviour will drive them to suddenly sell when they are long or buy when they are short.

If a large buyer enters a market that is already short, and the open

interest of those shorts is close to the upper limit of the total shorting capacity, this buying activity will likely significantly impact the price. There won't be enough new sellers to absorb the flow effectively, as the players who can short the market, either for hedging or speculative purposes, have already done so. If the buyer is determined to continue buying at higher and higher levels until they gain enough liquidity for their desired position, the market will move to this new price discovery point.

How would you react in this scenario if you had shorted the market only to see a new and determined buyer enter and force the market against you? If you were a hedger, you should be indifferent to the price action, as hedging means fixing the floating price of a physical product using a derivatives contract. If you are a speculator, your trade is live and needs to be managed, which could prompt you to make trading decisions. The reaction of traders in situations like this is where behaviours become important.

Do you try to protect your position? Do you protest at how ridiculous the market is, how it 'makes no sense'? When do you stop out? Do you wait for the market to go back to where you entered? Do you have a stop in mind below this? Do you try to bargain with your risk manager to keep the trade going even though it is moving against you, arguing that the market move is illogical and will soon revert?

Identifying Psychological Responses

Traders are human and saddled with bias – even an algorithm is coded with human logic. So when a financially traded contract holds a lot of speculative open interest, the price action is highly susceptible to traders' reactions to a range of signals: the media, market gossip, liquidity changes, volatility, reaching meaningful market levels (entry levels or stop losses) and many more. The money being made or lost on a trade at a given time – sometimes called the profit and loss profile – heavily influences a trader's psychology and how they act in the market. Making money can lead to over-enthusiasm and greed, and losing money can lead to caution and fear. Extreme emotions are debilitating, but when managed, they can be beneficial when instinct and judgement calls are required in the face of high pressure. Theoretically, trading decisions are always clear and logical, but hesitation or a single wrong move has the potential for severe consequences, especially in a voice-traded market like oil swaps, where in extreme environments,

liquidity vanishes, and traders all move one way.

When a trader loses money, they will go through the psychological effects of loss like in any other scenario in their business or personal life. As humans, our response to loss is well-researched and fairly well-understood. *What I Learned Losing a Million Dollars*^[50] is a book on trading that applies the five stages of loss to trading and claims traders go through this process many times a day. I found this persuasive, giving me a more structured understanding of traders' reactions. I began to judge trader reactions based on what I knew about their positions and started to rationalise their behaviours in various situations.

I started using this approach in a niche contract where all the participants were known, but it has given me a sense of the wider market's behaviour over time.

Denial

Denial can be seen and heard in markets when the price action moves against a speculative position or, more widely, against the prevailing consensus.

As mentioned earlier, the market looks for a narrative which tends to be guided by the most relevant influencers. To identify denial, you can look for signs in media appearances. Phrases we've heard recently are the oil market being in a 'state of schizophrenia' and 'increasingly disconnected from signs of robust crude demand', as reported by *The Wall Street Journal*^[51]. Other examples are the 'oil market is broken'^[52] or 'the recent move is nothing to do with fundamentals'^[53].

Denial in traders is likely to be focused on a justification of the trading nuance. For example, I've heard traders reference a move against them as 'just short term, as there is a trader taking profit'.

Either way, denial can cause you not to react or to maintain a strategy or trade plan that needs to be changed. Unfortunately, denial is usually the emotion responsible for the biggest losses.

Bargaining

For trading managers, bargaining is less worrying than denial because at least you are going through the process. To identify bargaining, you need to

look for narratives that try to justify uncertainty, followed by a plan that allows you to avoid the feeling of loss. An example is a narrative that claims the market will play out as you said, 'as long as...' while making caveats that begin with 'unless...'. The follow-up plan will be something like, 'If it goes back up two dollars, then I'll exit and look to re-enter as...'.

At this stage, you begin feeling out of control and looking for ways to reclaim that control.

Anger

Logically, it follows that when you start to acknowledge you don't have control, you get angry at a third party. 'This stupid market' or 'the only reason this is happening is because they are so risk averse' are examples of what you might hear or say. Most of the time, the only way to deal with this is to get the trader out of the environment. This is not a good mental state to be in as a trader and can lead to some reckless decisions.

Depression

This is when you start to accept what has happened but become frustrated with yourself and self-defeatist. You might say, 'I'm too old', 'I can't do this anymore' or 'Maybe I'm just not cut out for this'. I've seen traders in this mental state look to make drastic decisions like leaving the desk, and I have had to remind people more than once (including myself) that needing a break is not the same as the right time to quit.

Acceptance

Ultimately, when you accept what has happened, you can effectively move on, and crucially, if you are a high performer, learn from what went wrong and adjust your process for next time. The evidence of acceptance is a return to conviction for traders who have moved on to the next trade. You cut losses and continue trading in new volumes that will hit the market and start the next trend.

Ideally, when applying this to your trading, you will self-identify these behaviours to accelerate you through these stages, limit any unnecessary losses and get back into it.

Sensing Fear in Markets

Denial can turn to fear when the market moves in seemingly irrational ways. Volatile spikes can be alarming, and profit or loss usually realised over a few weeks can materialise in a single trading session. When fear is present in markets, they become highly vulnerable. Fear can cause traders to stay out of markets, lowering liquidity and giving those with conviction the opportunity to drive the markets. Even a small volume traded with conviction in these scenarios can cause disruption, with a market stunned into inaction.

Greed

Uncontrolled greed is arguably the sole driver of interest in any financial market. It can blind you to reality as you chase short-term gains, leaving long-term effects forgotten, abusing leverage to amplify your risk with the hope of amplified reward. As a basic human instinct, greed is ever present in the market, but you can tell if a trader is being driven by this when they start to emanate a certain smugness. It is a quiet emotion, but you can spot it in someone else from a mile away.

In the oil community, greed taking hold of market sentiment presents itself in dismissive attitudes and phrases like ‘You don’t know what you’re talking about’. In markets, it feeds back into denial – but a willful denial as you make money.

Using Behaviour in Decision-making

Knowing the behaviours that drive trader flows is one thing, but using this knowledge will set you apart.

To include behavioural knowledge in your decision-making, you first need to model, as best you can, the participants in the contract you are trading. Getting to know the participants you are trading with is possible in OTC-driven markets but much more challenging in exchange-traded futures. However, the market has relatively few companies run by individuals who have disproportionate control over the volumes in the market. You need to study and analyse how these individuals react to markets and their general activity profile.

For example, there are significant financial volumes from hedge funds and asset managers within Brent futures contracts. CTAs move quickly to new trends and are, therefore, highly reactive to market shifts; discretionary money reacts to fundamental headlines, and portfolio managers in asset allocation tend to move positions slowly. Trade houses and majors generally take positions with a one-to-three-month view and don't have a stopping-out mentality like hedge funds do. They can take trades to pricing and trade around their derivatives positions with physical barrels. Producers and end users are sensitive to price levels.

Next, you need to understand the current state of the market, which some people refer to as the regime. For example, you could list the factors that make up the market state using the following parameters:

Risk capacity: Using open interest, you can gauge the current market state on a continuum from underinvested to overextended.

Entry price: Is the market approaching a key market level corresponding to entry prices or a level beyond an entry price which can trigger a stop loss?

Volatility: Are you currently in a high-, medium- or low-volatility environment?

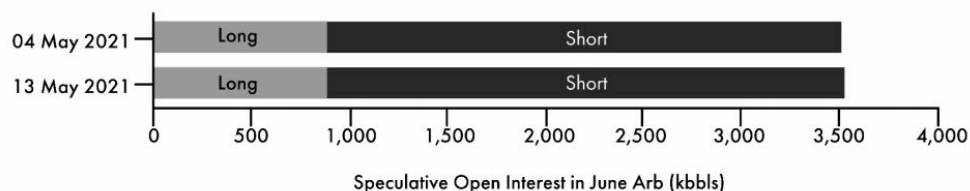
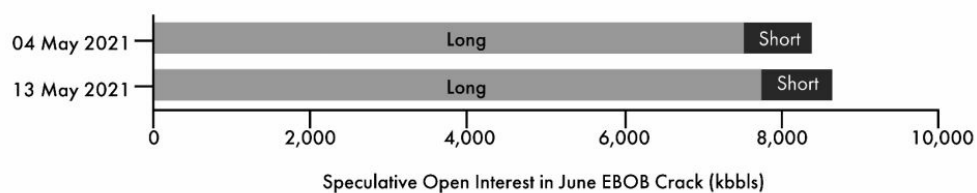
Liquidity: Are you in a high-, medium- or low-liquidity environment?

Next, you need to think about how you can combine what you know about the market state with what you know about the market participants and how they will react to upcoming market events or key moments.

Short story: EBOB Colonial Pipeline ransomware attack

On 7 May 2021, the US Colonial Pipeline, one of the US's largest pipeline systems for refined oil products (capacity of 3 Mbbl/day), suffered a ransomware cyber-attack that halted pipeline operations for five days.

During the Colonial Pipeline shutdown, the market didn't behave in a way that would be expected when a major pipeline goes offline on the back of market positioning. The open interest, speculative positions and long-short ratios of the EBOB crack showed a significantly long market in June and the third quarter, with both tenors highly saturated. The RBOB vs EBOB, on the other hand, was the opposite.



		OI – 5 YEAR OI AV ERAGE	SATURATION
04 May 2021	June EBOB Crack	31%	Saturated Long
	June Arb	68%	Saturated Short
13 May 2021	June EBOB Crack	27%	Saturated Long
	June Arb	65%	Saturated Short

Figure 3.13 Market Positioning and Open Interest for the for June 2021 EBOB Crack and June 2021 Arb in May 2021 (Source: Onyx COT)

Since the EBOB crack was already saturated long, not only did it not move up, but with selling pressure from arb and E/W buying, the crack drifted lower.

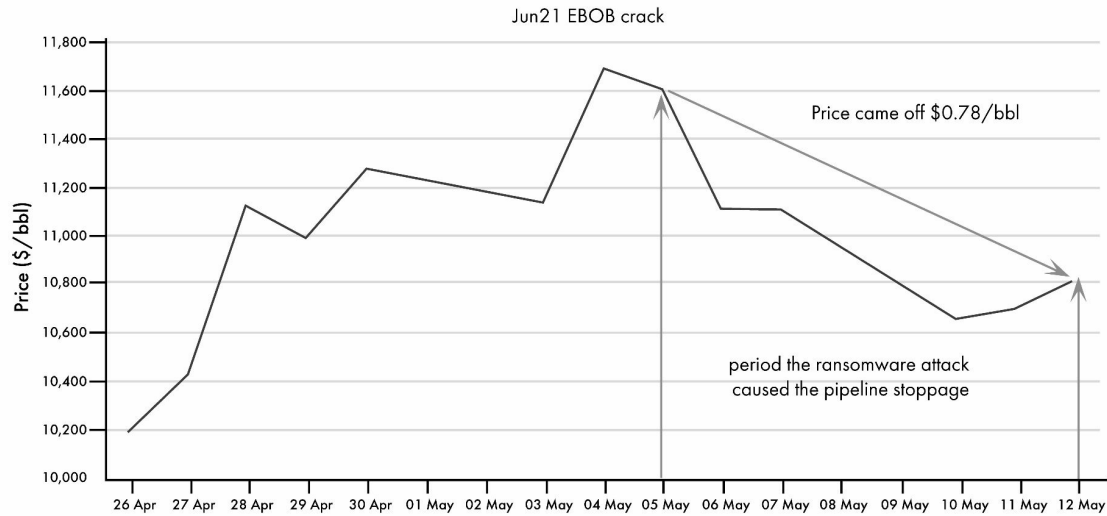


Figure 3.14 Price Action in the June 2021 EBOB Crack from 26 April to 12 May 2021

The arb, on the other hand, rallied, with short cover driving the differential upward.



Figure 3.15 Price Action in the Jun 21 Arb from 26 April to 12 May 2021

For policymakers, market behaviour is about whether a decision released to the market will be a surprise and have a material impact on traders' flows. If the decision is expected, people love to say, 'It's been priced in', but the definition of this is that those who can react in line with the sentiment of the announcement have already done so. For example, an interest rate hike is being talked up as having a potentially bearish impact on the markets. Crude futures contracts may already be volatile and heading down before the announcement. By the time the headline is released, it may materialise as an interest rate hike, but the market has already sold the volumes it would sell in reaction to this news. Policymakers also need to factor in whether there is confidence that the decision can make a difference. For a long time, a 'geopolitical premium' was the excuse used for Brent crude futures rallying in response to conflict like that of Syria. These days, unless it is set to significantly impact real spot physical crude or physical crude exports for a sustained period, it is completely ignored because the market no longer believes in the 'geopolitical premium' and doesn't have faith that the collective speculative trader positioning will buy into the news in this way.

If a policymaker is after control, then the control is only there for as long as traders value their decisions. If they continuously use the same bullet, the consequent impact diminishes. Sometimes it might be best to go counter to what traders expect to regain influence. Using the interest rate example, if a

rise was expected and the market reacted accordingly, lowering the rate would be a huge surprise, forcing traders to react and probably sending the market aggressively the other way.

Behaviour can also signal when you are too early. Perhaps you believe in a market hypothesis but haven't realised that you are in a contract with very little open interest and therefore haven't yet formed a consensus and started a trend. This stage is highly susceptible to the prompt or broader market. You can react to this by widening your stop and taking your time to enter the trade, or you can wait to see the open interest grow and react then.

Even if you are merely a market observer analysing the market, it is still important to examine market behaviours. We like to seek comfort in crowds, which is probably why so many people hold onto market narratives from the media or market gossip. Knowing this allows you to understand past and future market moves in the context of something that can be defined rather than believing there is some mystical market force that keeps supply and demand in perfect balance.

We should all be careful about market behaviours in times of hype. Speculative bubbles can form when a narrative has gained traction and is starting to work for many participants. This happens again and again in oil derivatives. Speculative bubbles are so common in financial markets that their general trend was captured on a graph created by Dr Jean-Paul Rodrique from Hofstra University in Long Island, US.

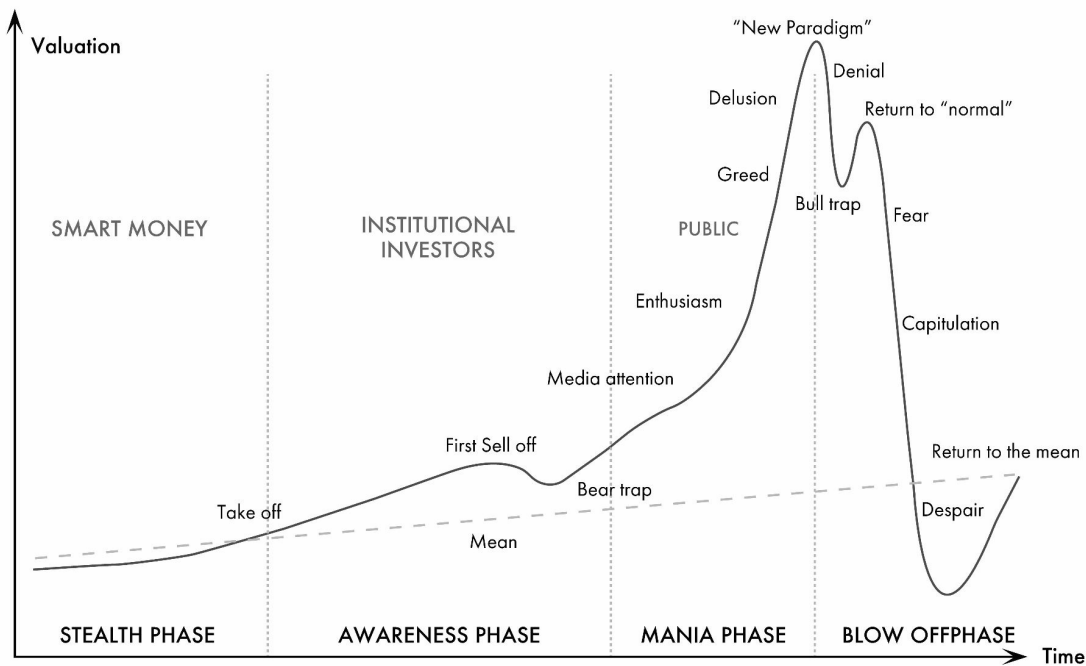


Figure 3.16 Typical Expected Price Action During a Bullish Speculative Bubble (Source: Dr Jean-Paul Rodrique)

A clear example of such a bubble in the oil derivative markets, which formed in 2019 and burst in early 2020, was the GOFO differential. Leading up to the IMO 2020 regulatory shift that would enforce low sulphur fuel consumption for the global shipping industry, the oil derivatives market sought direction on the impact in markets. The Gasoil swap vs 3.5% NWE fuel oil barges differential gained much traction as a reflection of the change, leading to excessive speculative positioning for trade houses and hedge funds.

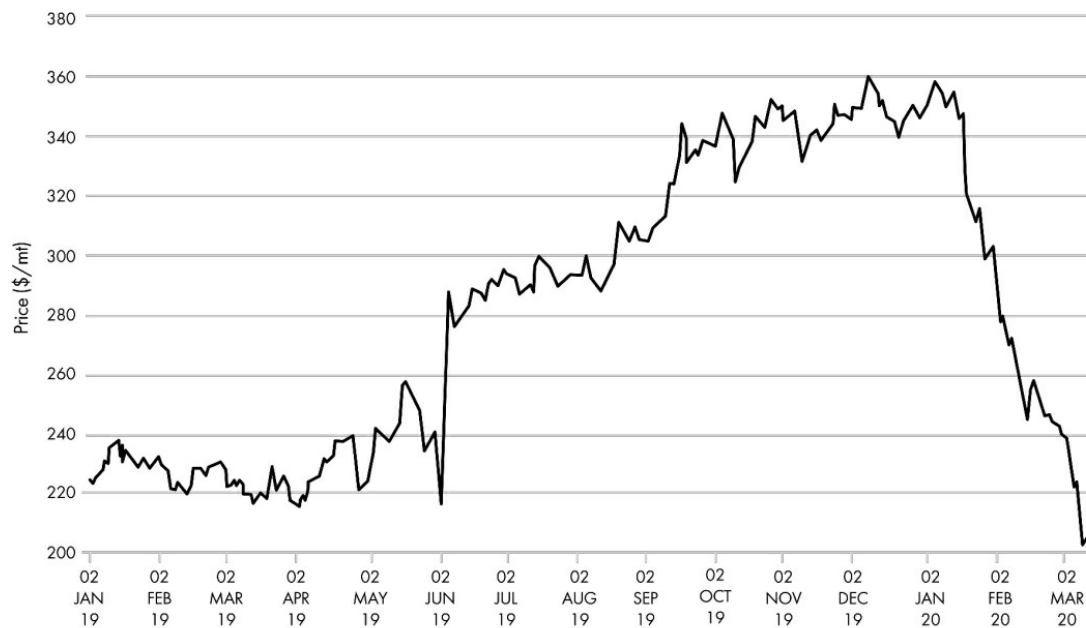


Figure 3.17 Prompt ICE Gasoil Swaps vs 3.5% Barges Fuel Oil

Blood on the Streets

You need confidence in your interpretation of market behaviours to make market sentiment work for you. A famous trading/investing saying is, 'Be greedy when the market is fearful, and fearful when the market is greedy'. The best traders can recognise when the market is overreacting and opportunistically benefit. In fact, when the market stops out, it can be the best time to enter.

Liquidity Pools

To understand distortions that can occur from market flows and positioning, you need to understand liquidity pooling. Each contract in financial oil contracts has a relatively fixed volume that can be traded, which steadily grows each year. So each contract has a finite capacity, a measurable maximum to how much open interest a given contract can take. When contracts are linked, unless the capacity to generate and absorb market flows is equal and simultaneous, distortions will inevitably occur.

The biggest liquidity pool is in the Brent and WTI flat price contracts. These contracts hold almost all the flat price risk traders execute in the financial oil market, and flat price risk makes up a significant open interest of the overall contracts traded in the oil derivative market. The next largest by open interest is the futures cracks, specifically Gasoil, heating oil and RBOB vs Brent or WTI. So outright contracts in product futures and swaps are priced by Brent and WTI – the largest liquidity pool.

If Brent goes up \$1/bbl, every outright contract will also increase by \$1/bbl unless the market widely adopts a delta to flat price or a flow counter to this movement occurs. With oil swaps, most of the contracts are implied from other, more liquid contracts, and therefore the disparity in contract capacities can sometimes cause dislocations.

Real-life Story: GasNap Breaking the Physical Economic Ceiling in 2022

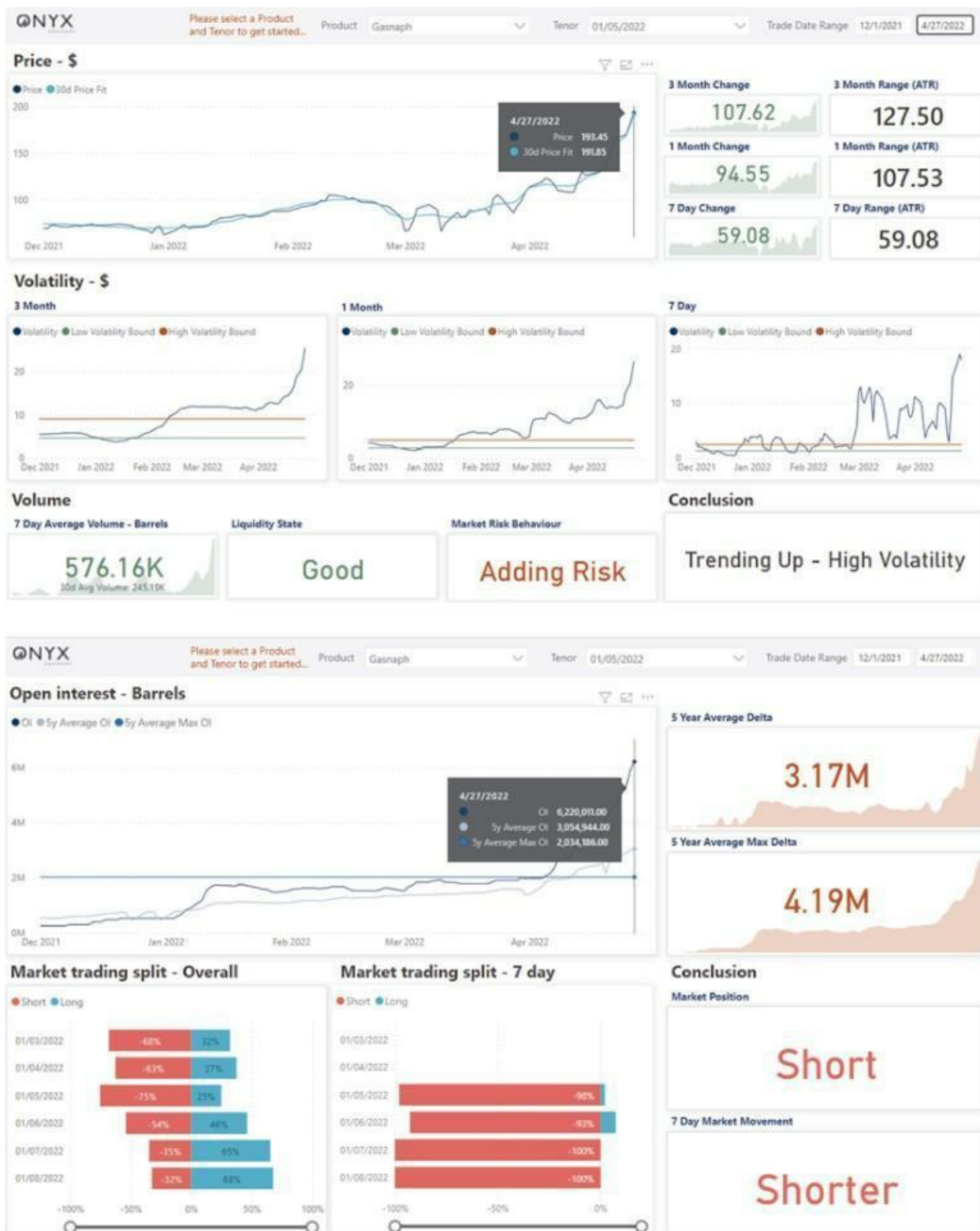


Figure 3.18 Snapshot of Open Interest and Market Trading Split for May 2022 GasNap. (Source: Onyx COT Dashboard)

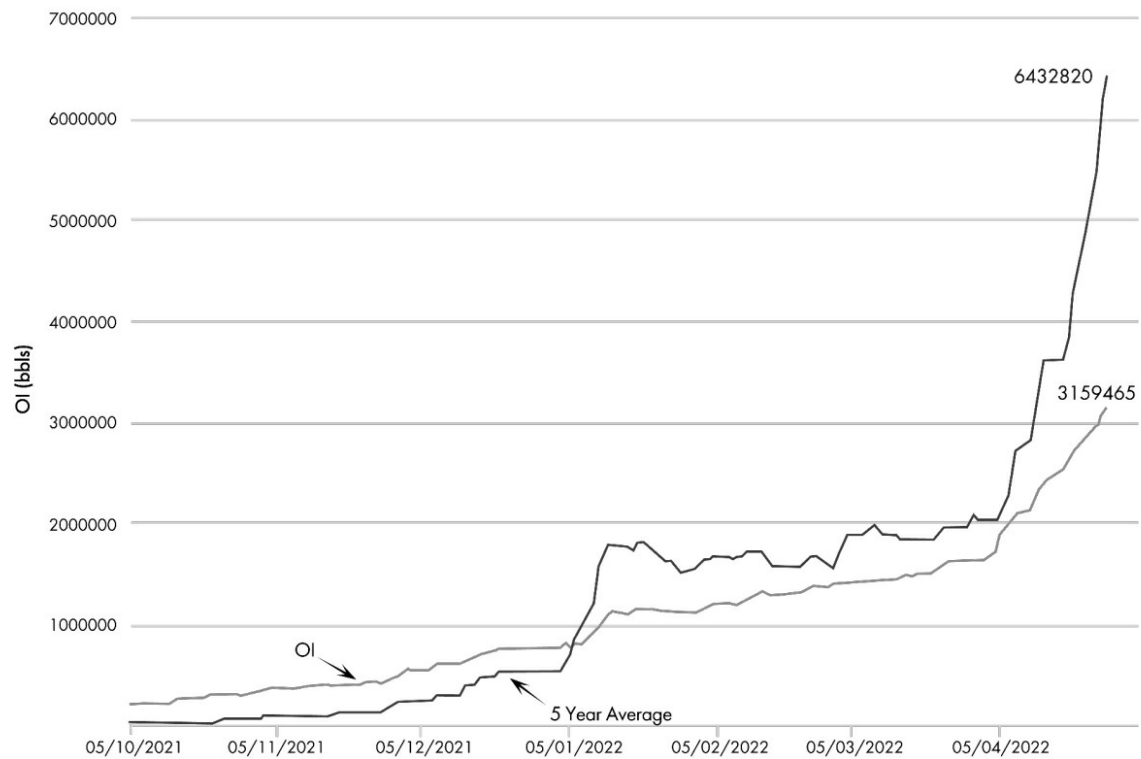


Figure 3.19 May 2022 GasNap Open Interest from 5 October 2021 to 5 May 2022

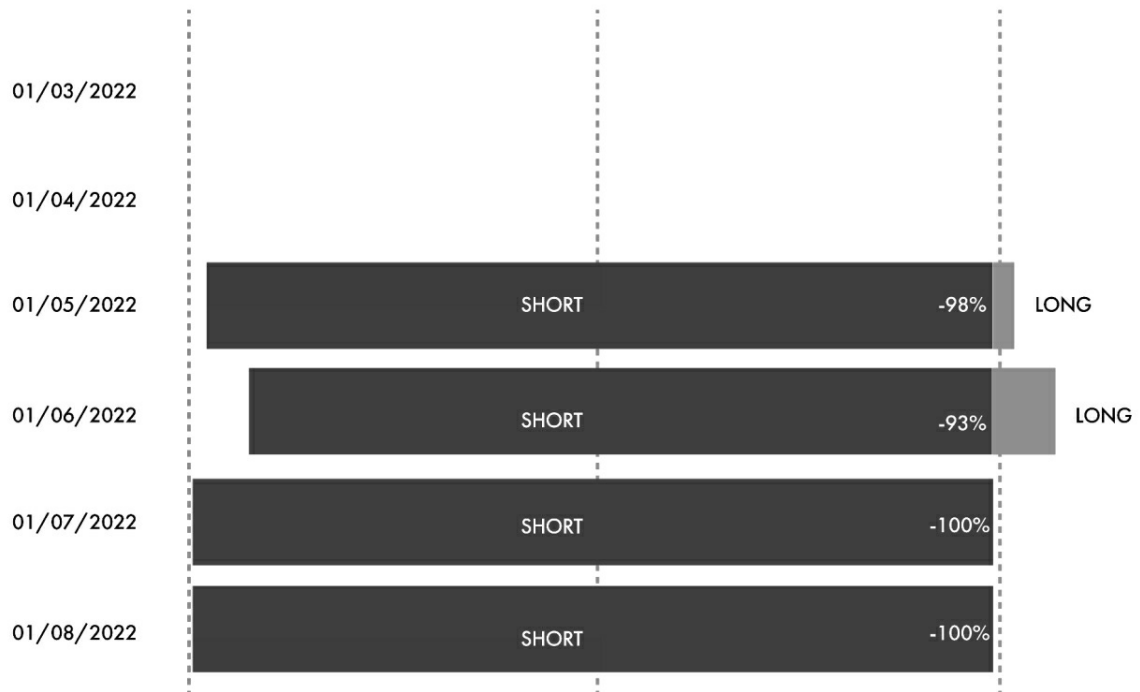


Figure 3.20 GasNap May 22 Market Trading Split from 5 October 2021 to 5 May 2022

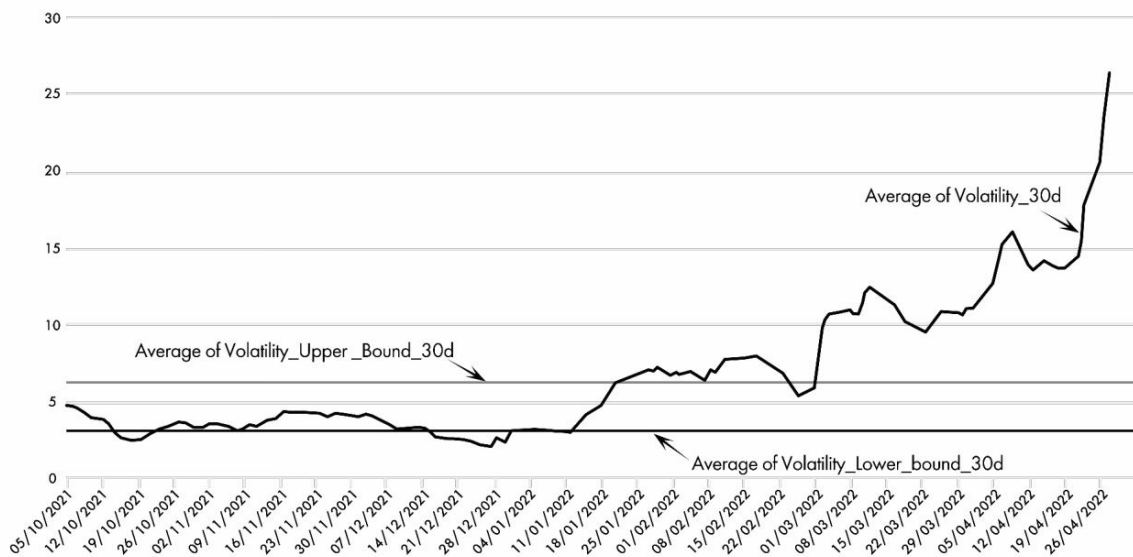


Figure 3.21 May 22 GasNap Volatility from 5 October 2021 to 5 May 2022

The May 2022 GasNap differential rallied significantly, reaching attractive levels for physical blenders. On the back of these levels, physical blending hedgers entered the GasNap differential short, and on the back of the same logic, many speculators also got short.

From mid-March to late April, the long–short ratio of the differential moved from around 50:50 to 30:70, while open interest escalated to more than 100% above the five-year average. Despite this, the GasNap maintained its rally and reached historical highs, surpassing \$200/mt for the first time. The GasNap’s unstoppable rise was further exacerbated by GasNap shorts reaching their pain tolerance and stopping out.

How could this happen when there was so much concentrated selling in this differential?

It comes down to the fact that there was an unusually large amount of interest in buying gasoline contracts, for which the main liquidity pool is RBOB, which in turn is linked to EBOB.

Traders hadn’t fully appreciated that even though the GasNap had real links to physical markets, the differential hasn’t priced yet and therefore to determine its value, you had to separate the differential into its constituent parts in terms of positioning and flow. Gasoline buying was overwhelming

any selling, while naphtha was more balanced in its flows. All the buying in naphtha from selling the GasNap differential was met with enough selling in naphtha cracks and flat price contracts to be absorbed. On the other hand, gasoline was struggling for sell-side liquidity; therefore, any differential linked to the contract would inevitably be supported. On top of this, gasoline volumes are much larger than naphtha, and any selling of gasoline from the GasNap was pitiful compared to the risk capacity in exchange-traded futures. In summary:

- *RBOB open interest capacity > EBOB open interest capacity > naphtha open interest capacity.*
- *Significant RBOB buying drags EBOB up via arb differential (RBOB vs EBOB) sell-side hedging.*
- *As a historically correlated contract, buying in RBOB also leads to outright buying in EBOB contracts.*
- *Gasoline buying is one way, whereas naphtha contract flows are balanced.*
- *The imbalance of flows leads to the GasNap (EBOB vs naphtha) rising, and it reaches the physical refinery economics incentivisation point for maximising naphtha blending into the gasoline pool, triggering selling in the differential from blending hedgers and speculators.*
- *GasNap open interest is much smaller than EBOB and naphtha contracts independently; therefore, selling is powerless if gasoline and naphtha market flows are being driven the opposite way independently.*
- *The net result is a continuation of a widening GasNap as long as the flows in the independent legs of the differential contract continue in the same way.*

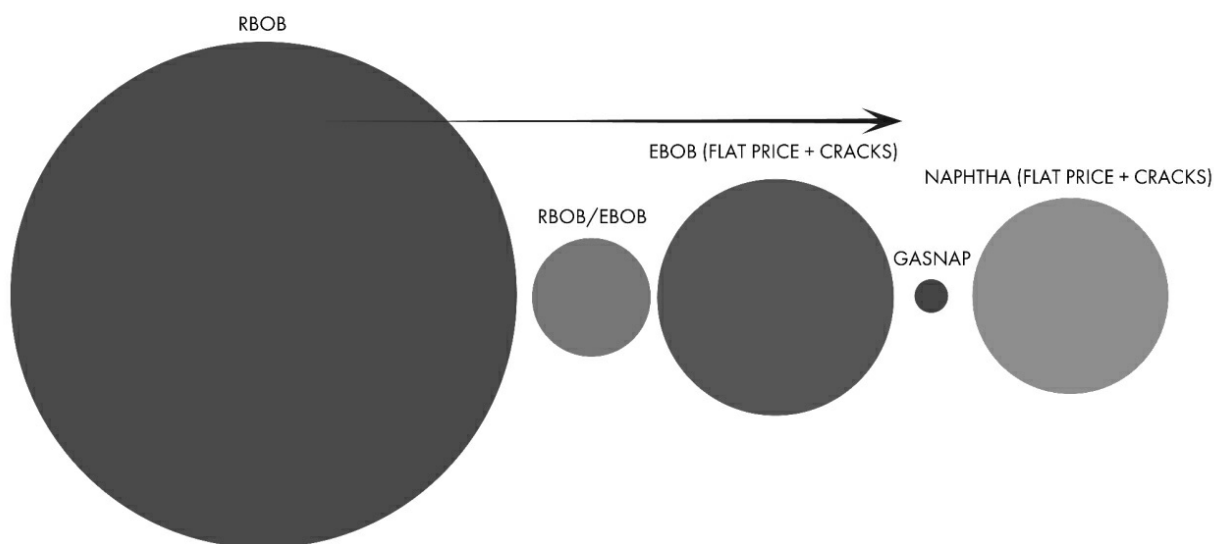


Figure 3.22 Relative Open Interest in Gasoline Differentials

Evaluating Market Conditions

The current market condition is a function of flows, positioning and behaviours and results from these factors playing out to a point in time. Volatility can be a function of a new flow entering the market, stopping out or a battle to prop up a position or force markets in a certain direction. Liquidity can be a function of the current risk tolerance in the market, certainty over the market direction, physical opportunity, balance sheets or even regulation. So market conditions reflect the market mentality and are crucial for analysing markets in combination with market position and flow. As a trader, you should define the market conditions before trading and incorporate this analysis into your strategy so that adjustments can be made to your risk management. This will reduce the risk of ruin and give you an advantage over those who don't do this analysis.

Volatility describes how much and over what period a contract's price moves. Market volatility is defined relative to historical contract price movements, shown in bands on the Onyx COT dashboard.

Market liquidity is assessed by how straightforward it is for a trader to make a transaction. Ideally, a trader can freely buy or sell contracts in the required volumes while expecting the transaction price to be as close as possible to fair value for the full volume traded. Quantifying liquidity is

trickier than volatility, but it can be measured historically by comparing transaction volumes and price moves within a range.

From a risk management perspective, gauging market conditions will allow you to structure your trade appropriately. If volatility is higher than normal, you must lower the position size so the current at-value risk stays consistent. If the liquidity is lower than normal, you must assume there will be slippage when exiting the trade, and you must reduce position sizes to factor this in.

You can gauge market behaviour at any time by observing market conditions. Fear and uncertainty in markets will usually lead to a period of high volatility as traders exit their positions, and some prices will be over-extended as a consequence. In addition, the open interest will be way down, indicating a period of quiet or continued volatility as hedging flows can't be executed because there are no interested parties to take the other side. So it is important to use market conditions in conjunction with positioning and flows as a proxy for market behaviour. It is useful for you to build your story to justify price action as long as it is tested and refined to build experience in observing markets.

Quantifying Market Conditions

Volatility

This is a measure of the degree of fluctuations or variation in the price of the market. In statistical terms, volatility and standard deviation from the mean are closely related, and sometimes standard deviation is used to measure the volatility of an asset.

You can use the ATR to assess volatility quantitatively. The ATR can be calculated by finding the average of the true range values, which represent the greatest value among the differences between the current high and low, the current high and previous close, and the current low and previous close over a specific period. ATR valuations have the advantage of incorporating intraday ranges and end-of-day ranges and are usually assessed over a three-day, seven-day or 30-day time frame.

Volatility assessments are done on a comparative basis, as one range that is considered high volatility in one contract would be considered low in another. The quantitative assessment is useful for forming stop losses. For example, multiples of the ATR can be considered to ensure the trade isn't

stopped out by market ‘noise’ in a long-term trending trade.

Volatility can be a function of a new flow entering the market or stopping out or a battle to prop up a position or force markets one way or another.

Liquidity

A market’s liquidity is assessed by how straightforward it is for a trader to make a transaction. Ideally, a trader will want to be able to freely buy or sell contracts in the volumes they need, expecting the transaction price to be as close as possible to fair value for the full volume traded. Quantifying liquidity is trickier than volatility, but usually, it can be measured historically by comparing transaction volumes and price moves within a range.

For example, you can compare the daily volume traded on a given contract with previous seven-day and thirty-day averages. If more volume is trading than the average, the market is in a ‘good’ liquidity state. If the volume is comparable with those averages, the market is in a ‘fair’ liquidity state. If the recent trading volume is lagging behind these averages, the market is in a ‘poor’ liquidity state.

Liquidity can be a function of current risk tolerance in the market, certainty over the market direction, physical opportunity, balance sheets or even regulation.

Technical Indicators

Technical indicators are useful tools to avoid subjectivity in drawing conclusions from volatility and liquidity assessments. The relative strength index (RSI) is a momentum oscillator that measures the speed and change of price movements, calculated using the average gains and losses of an asset over a specified time period. The RSI ranges from 0 to 100 and is commonly used to determine whether a contract is overbought or oversold.

Moving averages are useful for identifying trends. They signal changes to market momentum and act as critical indicators of support or resistance levels. Commonly used moving average indicators are exponential, simple and weighted moving averages. The moving average convergence divergence indicator uses exponential moving averages over different periods.

Bollinger bands contain a measure of volatility in the form of two

standard deviations away from the mean for the upper and lower bounds and a moving average between the bands. These three features can be thought of as the levels you expect the price to trade in over the next few candles.

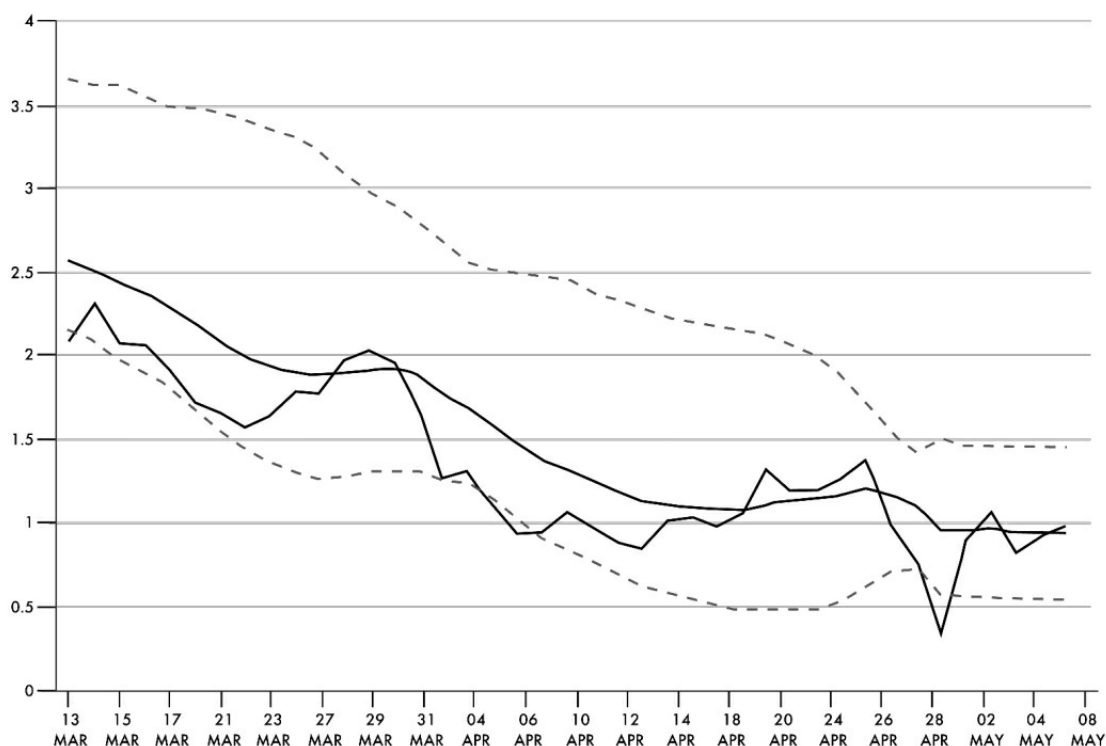


Figure 3.23 Price Action, Bollinger Bands and the Ten-day EMA on the June 2023 Brent Dubai Differential from 13 March to 8 May 2023

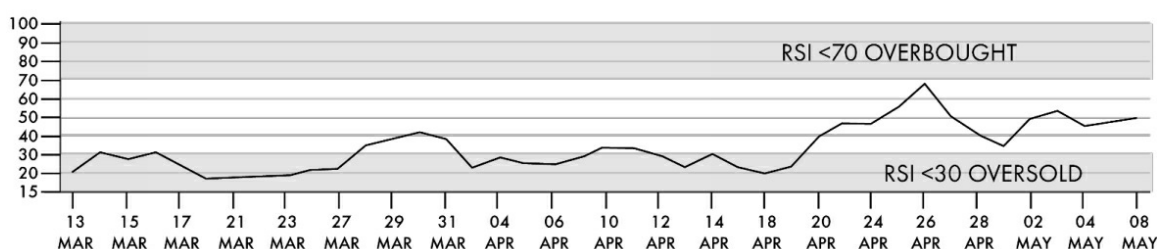


Figure 3.24 An Example of the Relative Strength Index Using the June 2023 Brent Dubai Differential from 13 March to 8 May 2023

The market conditions should be defined before trading so that adjustments can be made to your risk management, reducing the risk of ruin and potentially benefiting when others can't. If the volatility is higher than normal, you must lower your position size so the current at-value risk stays consistent. If the liquidity is lower than normal, it is similar to high volatility

– you must assume there will be slippage in exiting the trade; therefore, position sizes should be reduced to factor this in.

Fear and uncertainty in the markets will usually lead to a period of high volatility as traders exit their positions, resulting in some over-extended prices. The open interest will also be way down, which can either predict a period of quiet or continued volatility as hedging flows can't be executed, as there are no interested parties to take the other side. It is very important to use market conditions in conjunction with positioning and flows as a proxy for market behaviour. Building your own story to justify price action will be useful as long as this is tested and refined so you can build a proper experience of observing markets.

‘Regimes’ can be used to classify sustained market conditions:

High Volatility, Low Liquidity Market that is Trending

In this regime, you can expect the market to be in an extreme state, driven by a major supply shock or macro event that has led to a binary consensus. Greed is likely driving markets in this scenario, with positioning one way and speculative risk maxed out, drying out opposing market flows. The resulting market conditions could indicate a short or long squeeze and, therefore, the likelihood of at least one major player on the verge of ruin.

Low Volatility, High Liquidity that is Sideways

In this regime, the resulting market conditions are likely from a market being propped up, and this tends to predicate a major breakout. These regimes usually come at the end of a long-term, exhausted trend, with opposing flows balancing the market. It is not the case of the market becoming hands-off, but rather complacency building up and an echo-chamber environment. Despite the perceived inaction of this regime, it is one of the more dangerous trading environments. Complacency and liquidity lead to lower perceived risk and a build-up of volume on trading positions, leaving the trader highly vulnerable to a breakout.

Low Volatility, Low Liquidity that is Trending

In this regime, you can expect the market to be ‘drifting’ with seemingly no buy- or sell-side interest. This is more common in contracts derived from other contracts with much larger liquidity pools that have low risk at that time. These regimes tend to lead to fear-of-missing-out scenarios, with traders watching the move without being involved and suddenly rushing in.

Speculative Trading

What do Most Traders Do?

Most traders venture into the realm of speculation, believing they can utilise their market intuition and appetite for risk to generate substantial returns. This, however, can be a dangerous self-deception. For instance, in the oil market, it’s common for traders to align their strategies with larger trade houses and oil companies, believing these organisations’ significant profits come from successful market trend predictions. But if trade houses and oil majors make enormous trading profits, it has nothing to do with inspired forecasting of market trends. Despite what some might tell you, most oil traders aren’t playing that game. The nuances in the oil trading world – floating storage, operational tolerance and physical arbitrage – give trade houses’ or IOC’s directional trades a huge head start and provide highly valuable optionality – every trader’s dream. The trade houses’ and IOC’s derivatives trading also comes with orders from customers for which they can charge a high price, as well as in-house hedging from physical trading desks. It is not easy to identify where oil traders’ discretion comes into play if they are managing a portfolio of physical hedging and directional trading.

Traders often base their decisions on the same supply and demand reports and adopt a similar directional view for the entire year. Moreover, they are swayed by arbitrary exit targets, which could be identical to the next trader’s targets due to the lack of a distinct process. The market frequently punishes such herd behaviour, making this trade process risky.

What Should You Do Differently?

With the evolution of derivative analytics and risk modelling in the industry, there is no longer the same glory in swinging for the fences (to use a

baseball idiom) to have a great year, only to blow out spectacularly every other year with disastrous returns. Gone are the days when your reputation was protected by a market consensus that the futures market is ‘broken’ or ‘detached from fundamentals’, leading to market moves that couldn’t have been predicted to justify your loss. If you are working at a professional oil trading organisation, you will be expected to navigate the market as it presents itself. Trading managers and executives are much less forgiving than in the peak days of investment banks in the 2000s. Oil trading is capital-intensive, and financing a trading function is one of the most important drivers of this business. So spare capital for directional trading is expensive, especially when interest rates are rising. So the cash needed to fund the margin for exchange-traded positions is expected to provide a strong and consistent return.

Therefore, as opposed to following the herd, developing an independent approach is crucial. Successful trading is about cultivating a method that scales over time, utilising a predetermined amount of risk capital acceptable to your trading enterprise. It’s imperative not to be swayed by the perceived wisdom of other traders. If, for example, you find yourself seeking confirmation from traders at the top five trade houses or oil majors, it’s a sign that you’re falling into a potentially hazardous pattern. A robust trading plan focuses on self-learning and continuous improvement through objective analysis. Given the often irrational nature of markets, if your exit plan mirrors that of the majority, you’re more likely to incur losses.

What are the Benefits of Using this Approach?

A strong risk management framework coupled with an independent trading plan offers several benefits. This strategic approach shields your business from potential downsides and positions it for trading strategies that could yield a return ten times the investment. By maintaining focus and discipline, this strategy provides a competitive advantage in the current state of the oil derivatives market. Avoiding the distractions of herd trading and relying on objective analysis can offer protection against market irrationality, which is known to cause significant losses.

Forming a Hypothesis

The first step to a proprietary trading strategy is formulating a coherent and logical opinion of what will happen next in the market. Clarify your opinions and justify them with the appropriate objective detail. You can back-test a strategy using historical market data to simulate trades using a specific set of rules and then analyse the results to see how well the strategy performed over time. But you still need to justify your strategy to ensure you aren't susceptible to the classic correlational fallacy of drawing a conclusion solely from seeing apparent relationships between variables.

Typically, a trader will start with a view of the market.

I am bullish on NWE naphtha, as the physical differentials in the spot European naphtha market are being reported as strong, indicating a bullish physical market, while the forward demand outlook also looks strong according to supply and demand analysis. In addition, stockpiles in Europe for physical naphtha are at seasonal lows.

This is a view of the physical market, but how does it lead to the derivatives market? How will this view lead to buying flows? Which contract will be most affected and best reflect your view? Next, you can analyse what moves the derivatives market: the flows, positioning and behaviours discussed in previous chapters. How do you think this physical strength will drive buying flows? How is the market positioned to accommodate the new buying? How will the market react, given its current state?

Flows

Have there been consistent buying flows in any particular contract? Have the flows been speculative? Have there been hedging flows, and by which type of player? Could the flow(s) stop? Are there any threats to these flows? Why do you expect them to continue? What will encourage more flows?

Positioning

You should start asking questions about market capacity. Is the market positioned speculatively long, or is the hedging flow likely maxed out? How much open interest is in the market? Is there room for more long positioning, or have we reached the higher end for risk in the market? Is that risk skewed in one direction?

Market Behaviour

How do you expect the market to react to buying flows? If bearish information comes in, will there be a race to exit, given the market is already geared speculatively long? Are we close to long entry levels that, if we exceed, will lead to stop outs or, vice versa, short entry levels, which could push naphtha contracts higher?

Since derivatives contracts will ultimately partly price to physical transactions, there is the option of taking the contract to settlement. This is where the physical view comes into play, but this view needs to be challenged in the context of how it translates to derivative price action. In the example below, you predict that the physical demand will increase, but how *precisely* will that lead to a higher naphtha derivatives contract?

You may say the physical differential in the spot physical market is higher now than the forward time spread implies. Therefore, pricing the time spread will convert to these physical prices on the February leg and move it higher. But how much higher and why? Which of the participants who can trade the contract gains from this?

Ideally, your hypothesis should incorporate the following:

- Your market view
- An evaluation of flows
- An evaluation of positioning
- The expected market behaviour
- Physical pricing considerations (if taking to pricing)
- The specific financial contract selected
- Your determination of the expected fair value
- The period

Figure 3.25 shows how this could be used for a trader at Onyx.

MARKET VIEW	Bullish NWE Naphtha as the physical differentials in the spot European Naphtha market are being reported as strong, indicating a bullish physical market, whilst the forward demand outlook also looks strong according to supply and demand analysis. In addition, stockpiles in Europe for physical Naphtha are at seasonal lows.
EVALUATION OF FLOWS	According to Onyx Insight Trader Meeting Notes, selling flow in the NWE Naphtha market has been mostly refiners on the sell side of the February Naphtha cracks, coming in via banks most mornings, most likely as part of a refinery margin hedging programme. On timespreads it has been mostly market makers trading on the sell side of the February/March 2023 timespread. Buyers of the same contracts have been large trade houses that trade the physical NWE Naphtha complex, both on the cracks and time spread. There was buying at the beginning of the month, but since the market moved slightly higher there has been a slowdown, indicating a sensitivity to entry point.
EVALUATION OF POSITIONING	The open interest for February NWE Naphtha contracts in February is all around 20% lower than the seasonal average for this contract before expiry on the contract. Additionally, the market is relatively balanced in terms of positioning. According to Onyx Commitment of Traders data, the February 2023 NWE Naphtha crack is 60:40 long to short whilst the February/March 2023 Naphtha timespread is 55:45 long to short. The long positioning has averaged around -\$6.50/bbl in the February NWE Naphtha crack (now \$6.00/bbl) and \$1/mt on the February/March 2023 NWE Naphtha timespread (now \$2/mt)
EXPECTED MARKET BEHAVIOUR	Given the open interest is seasonally lower than normal, I expect that more risk will enter the market. Due to this, and the fact that the positioning is relatively balanced, there is capacity for either flow to come in and move the market. However, as the market is more long than short, and there are positive physical signals in the market, I expect any selling to be absorbed and the current market levels to be protected. I don't expect the same resistance on the short side, given the selling has mostly been reportedly hedging, and therefore I believe the market will move higher as open interest grows.
PHYSICAL PRICING CONSIDERATION (IF TAKEN INTO PRICING)	The physical Naphtha spot market is trading at \$8/mt versus M1 Naphtha. The February/March 2023 timespread will reflect the physical differential on pricing, and it is currently trading at \$2/mt right now. If the physical strength maintains into the physical that settles the February NWE Naphtha contract, then the timespread should move up towards \$8/mt also. As it is trade houses that are long on the timespread, I am confident that they will be buying physical Naphtha in the pricing window, with no clear signal that there are motivated sellers of the physical waiting in the wings.
SPECIFIC FINANCIAL CONTRACT SELECTED	As the timespread settles to physical transactions in settlement, I think it is the February/March timespread that is most undervalued. This is therefore the contract on which I will place my risk.
DETERMINATION OF EXPECTED FAIR VALUE	Given the physical market indications, I see fair value as at least \$5/mt to reflect a strong physical market and align with current spot physical market differentials.
TIME SPAN	Given my hypothesis is based on physical information, I will be prepared to take the contract into settlement if it has not reached the level I expect before that point. Therefore, holding the timespread is an appropriate contract choice, as its settlement methodology reflects my view on the physical market.

Figure 3.25 Examples of Factors Considered When Forming a Hypothesis

The view selected here still leaves a lot of work to be done in choosing the right contract and building a trade structure around it. The hypothesis, on the other hand, provides an expected scenario for prices that we can build around. The hypothesis clearly states that the view has come from physical demand, which will be evidenced by pricing according to the PRA's assessment methodology. Therefore, to structure a trade that reflects this view, we need to take that time spread to pricing. Pricing trades are difficult to exit, so we must leave the contract to price out in its entirety, which can be volatile, so an appropriately wide spot will be required.

We also have a condition that can signal a challenge to the hypothesis and, therefore, a reason to exit the trade: the long-short ratio moves below 55:45, and open interest goes up, indicating incoming selling flow and expected flows not coming in. Finally, we have indicated clearly that expected fair value on the time spread is at least \$5/mt, and so if the contract moves to that level before pricing, we have the choice of exiting the trade, or run the winner with a tight stop and exiting before pricing.

The idea is to use all available information to make your best-informed guess about the market's next move. Be clear that if the market moves in your favour, you should not feed your ego and think you were 'right'. It shouldn't be about 'right' or 'wrong' when it comes to trading. You ultimately never really know why there was a flurry of price action in a given direction; all you can do is hypothesise and give yourself the best chance of success by having a clear plan, entering efficiently and using sound risk management principles. You must own your decision with all factors considered and not trade someone else's position. But you must also recognise that there is always something to learn, and you can never know everything.

Determine Your Trade Classification

Your trade classification will either be **trending**, **contrarian** or **momentum**.

Trending trade classification: The trending trade classification refers to attempting to participate early in a new trend and exiting before the market becomes oversaturated. An example of this could be buying next year's August EBOB cracks as the contract begins to move up, being aware that the open interest is on the low end and, therefore, anticipating that traders will

bring further buying flows to the contract as bullish consensus grows. You would then look to exit before market buying flows reach their maximum capacity and have had their maximum impact on the direction of the August EBOB crack price.

In this case, your stop would be defined by the fluctuations in price around that trend to date, offering protection against market ‘noise’, often defined by volatility. The period of your trade is essential in quantifying this volatility – if the trade’s expected duration is short, say a week, then a one- to two-week period would be used to assess whether a trend is in play or you’re bucking the trend.

Contrarian trade classification: The contrarian trade classification involves trading against the herd in anticipation of a reversal of flows and a market position that cannot sustain itself in the current trend. Imagine a scenario where most traders are buying DFLs in anticipation of a North Sea supply shortage on the back of an OPEC+ cut. As a contrarian trader, you might sell instead, believing the market has overestimated the impact of the potential shortage and that positioning is already long and overextended. Your trade’s expected duration should align with your hypothesis about when the reversal of market sentiment will occur.

Momentum trade classification: The momentum trade classification strategy leverages the market’s momentum to enter and exit trades swiftly. Imagine a scenario in which there’s a sudden surge in Brent crude futures prices due to unexpected geopolitical tensions, such as the US and Iran tension in 2019. A momentum trader might buy oil futures on hearing the news, with the intention of being early to the move and riding the wave of increasing prices before selling quickly before the momentum subsides. As with other strategies, it’s crucial to define your trade’s expected duration and set your stop accordingly. You must remember that the future is unpredictable, and trends can reverse abruptly. Therefore, constructing a trade should aim to stack probabilities in your favour rather than striving for absolute correctness.

It is important to note that defining a trend isn’t straightforward, but technical analysis can provide some insights. For example, when a short-term moving average crosses a longer-term moving average, it may signal the beginning of a new trend.

Define Market Conditions

The final piece you need before defining your trade structure is an assessment of the market conditions for the contract you are choosing. To do this, you can use quantitative assessments of the conditions – described in the previous chapter. Market conditions include volatility and liquidity, but much like when judging a trend, an assessment of these two factors is relative.

Volatility

To define volatility, you must look back at the relevant time frame (which we defined for determining the market trend). Then, you can measure this against historical volatility to determine where the current volatility sits in the range.

The absolute volatility metric is important for quantifying your stop, but where it sits in the range is also vital, as it will allow you to prepare for transitioning market conditions. For example, you may be in a low-volatility environment, so you should expect that a material change in the contract's positioning could move it into the medium- or even high-volatility range, which could cause you to alter the stop to accommodate this change. You should also avoid being in a high-volatility environment if you do not understand why, as this could indicate something is going on of which you are unaware.

A spike in volatility, for example, could be the consequence of traders either rushing in through a fear of missing out or stopping out. On the other hand, it could mean that a new counterflow is coming in, which may force you to reevaluate your trade. You can correlate volatility spikes with price levels, open interest and the long–short ratio to ascertain whether it is a stop out, fear of missing out or a counterflow. If volatility is decreasing and the market range is narrowing, this can be a sign that the market is about to break out in a new direction.

On the Onyx COT dashboard, volatility is defined by the rolling standard deviation of price across seven, thirty and ninety days for short, medium and long periods, respectively. The volatility levels are given as high, medium or low based on quartiles of the historic rolling standard deviation. The dashboard also shows the ATR, which is the greatest of the following values: the current high minus the current low, the absolute value of the current high minus the previous close or the absolute value of the current low minus the previous close. The ATR can be adjusted depending on

the time frame being analysed and can be represented as an absolute value or a percentage of the asset's price. A higher ATR value indicates higher volatility and vice versa.

Liquidity

It is important to define current liquidity, since entry into an illiquid product will negatively impact the trade's risk-reward ratio. The liquidity can be defined, for example, using the Onyx COT dashboard price movement to volume ratio.

Protect your Downside with Stops

In proprietary trading, nothing is more important than your stop. It will protect your downside; without one, you will fall foul of every kind of bias and be on a pathway to ruin. You should first see your job as staying in the game, which is an achievement in itself.

By this stage, you should have your trade classification and expected period on the trade and understand the market conditions. Combining these will create the basis for your stop. But first, you need to decide on the type of trade you are making.

Trending Trades

For trending trades, the essential principle is to choose one stop and ensure that it is wide enough to ride out the market's expected noise. If your stop is triggered, it must be a quantitative signal that the trend has ended, and you should exit this trade. If you continue in the trade after this, you effectively change to a contrarian view, which has a very different profile of price action.

Contrarian Trades

For contrarian trades, you look to enter and exit as many times as is feasible to maximise your chances of finding the bottom or top of a trend. Fighting the market trend is like 'catching a falling knife', which indicates

how difficult it is, how quickly things can go wrong and how damaging it can be to your profit and loss. Therefore, you don't want to place a stop in a contrarian trade in the same way you would when joining a trend or short-term momentum.

Momentum Trades

For momentum trades, you are looking to profit from a continuation of a market move immediately and therefore want to be in and out of the market quickly, looking to catch a breakout move.

How do you Establish Your Stop?

Once you have identified the type of trade, you can establish your stop.

For a **trending trade** stop, you should:

- Use a wide stop that is outside the trending range
- Choose a number far enough outside the range that it protects against market noise but tight enough so that it would not be considered breaking the trend, thereby disproving your hypothesis

For a **contrarian trade** stop, you should:

- Split your capital at risk on the trade a minimum of four times to give yourself multiple entries
- Define your entry criteria
- Choose a tight stop and top out quickly if the trend continues
- Use a predefined move in your favour to signal that the trend has reversed
- Once the turn has occurred, scale in aggressively to take part in the new trend you were expecting

For a **momentum trade** stop, you should:

- Base the stop on price movement and flows
- Find a meaningful counter-flow to trigger an immediate stop out

When engaging in proprietary trading, nothing is more important than your stop. It will protect your downside, and without one, you will fall foul of every kind of bias and be on a pathway to ruin. You must first see your job as staying in the game, and that alone is an achievement.

Look for Attractive Risk–Reward Profiles

Once you have established the appropriate stop, you can weigh this against your expected return to evaluate the upside. This is where risk and reward come in. What are the seasonal and historical ranges? How much movement can we expect from this? Could this movement surpass historical ranges? If so, by how much?

Typically, the risk–reward profile should be at least 1:2; the expected reward on the trade should be at least double the movement from the entry price to your stop. This is the foundation of the risk–reward profile. It may seem obvious to say the risk versus reward needs to be in your favour, but when it comes to risk, it can be very difficult to think purely rationally. So many biases work against you, but the process of building a trade structure protects against those biases.

The first rule is that if the expected reward is less than the risk, the trade should not be placed, pure and simple. If you haven't evaluated the risk–reward profile, you should reflect on whether trading is the right career for you!

There are many opportunities in the market, and they're constantly presenting themselves, so don't let yourself think that you'll miss an opportunity by not making a trade with a poor risk profile. For this reason, many traders have looked back on their careers and wished they had traded less, not more.

The expected value against your stop should equal at least 1.5. And you should be looking for 3:1 or even 10:1 opportunities, but to achieve these kinds of returns, we must explore how you create leverage in your trading structure.

Position Sizing

Once you have done the work to calculate an appropriate stop loss, you can use the following process to determine your position size:

- Find the minimum volume you can trade in the contract you will be trading
- Use the stop loss you calculated to determine how much you will lose if you reach your stop on that minimum volume, i.e. minimum loss = minimum volume stop

- Decide the maximum loss you are willing to accept for this position
- Calculate the position size = maximum loss ÷ minimum loss × minimum volume

For example, let's say you were looking to buy Brent crude futures and were willing to risk \$100,000, and your stop was \$2/bbl.

The minimum volume you can trade on Brent futures is 1,000 barrels or 1 KB. The minimum amount that would be lost using the minimum size and the pre-determined loss would be:

$$\text{\$2/bbl} \times 1,000 \text{ barrels} = \text{\$2,000}$$

You are willing to risk \$100,000; therefore, your position size would be:

$$\text{\$100,000} \div \text{\$2,000} \times 1000 \text{ barrels}$$

$$= \textbf{50,000 barrels} \text{ (or 50 KB)}$$

Determining the Capital at Risk

This is another skill you need in purely financial directional trading, coupled with the scaling up of your strategy. Hedge funds typically use 1–5% of assets under management as the capital they are willing to lose. Once you scale, you can risk some of what you have made, allowing for increased capital at risk.

The amount you have to risk should be split into multiple trades, but how many times it is split will depend on your strategy. Large funds can take only a few positions a year, given they are long term. Medium-term traders will be willing to trade 20+ times a year. If you know your strategy and, therefore, how many times you are likely to trade in a year, you should have a good idea of how much to risk per trade.

Create Leverage in Your Trade Structure for High Returns

If you spot an opportunity that you think could bring a *significant* move in a financial contract you are trading, you can structure your trade to create leveraged returns without increasing your total capital at risk.

To achieve this, you must be confident that you see a strong, long-lasting trend. If you are early enough in this trend, you can scale into the trade once you hit your price level targets. You may be buying at a higher

price (or selling at a lower price), but you can use the returns you have already made, combined with the capital you were risking in the first place, to quickly run a leveraged position with considerable volume in the trade but no more at risk than you started with.

LONG SCALE EXAMPLE ATR = \$4

SIZE	ENTRY	SCALE IN	STOP	STOP SIZE	TOTAL SIZE	AVE. ENTRY	MAX. LOSS
100	100	N/a	96	4	100	100.000	\$-400,000.00
125	105	5	101	4	225	102.778	\$-400,000.00
75	110	5	106	4	300	104.583	\$-425,000.00
50	115	5	111	4	350	106.071	\$-1,725,000.00

EXIT PLAN

SIZE	EXIT	PROFIT SIZE	PROFIT
350	120	13,929	\$-4,875,000.00
		-106.071	\$ -
		-106.071	\$ -
		-106.071	\$ -

TOTAL SIZE	AVE. EXIT	AVE. PROFIT SIZE	TOTAL PROFIT	RR
350	120.000	13,929	\$-4,875,000.00	\$12.19

Figure 3.26 Example of How to Create a Leveraged Trading Position by Reinvesting Gains

In this example, we are anticipating a strong move in Brent futures. The initial risk is \$400,000, and given that the appropriate stop has been deemed to be \$4/bbl, the volume is 100 KB. The first target for the scale-in is \$105/bbl, which is 1.25 times greater than the initial risk and, therefore, already a strong move.

Maintaining the appropriate stop, defined as \$4/bbl, while accepting a net risk of -\$400,000 on the trade will mean that we can more than double the volume to 225 KB.

When it gets to the next scale-in, we have already had a significant move and a respectable return. We still believe in the long-term trend, but the closer we get to the target, the more the risk–reward ratio diminishes, so we will reduce the scale-in volume, maintain an appropriate stop on the full volume and be comfortable that we will be left with a minimum of a \$425,000 profit if we then stop out at this point.

By the final scale-in, we are well in the money, and even if stopped out,

we will still generate \$1.725 million in profit, which is a 4:1 risk–reward ratio. If the final target is reached at the expected \$120/bbl, we will have created a 12:1 return.

This is how you achieve leveraged returns in trades you really believe in. When you lose, you lose what you can manage, and you can always re-enter with a sound mind. When you win, you win spectacularly. When scaling into trades, however, you must factor in liquidity. First, the higher the volume you are holding, the greater your chance of slippage when you exit the trade, which could lead to a greater loss than planned.

Second, there could be issues on the scale-in levels because if you are trading in highly volatile conditions, you could miss your scale-in level and be forced to trade at a number much higher than the planned scale-in level to find liquidity. This could be followed by a violent reversal, which destroys the risk–reward profile.

In this case, it is better to wait for the market to calm down before scaling in. You can also do this with a contrarian trade that turns into a trending trade, as it is much the same principle.

Don't Overlook Your Exit Strategy

The power of tradecraft is that you are structuring your trade to account for how you believe the price action will play out, not just the direction.

Planning for New Information

Much of what could happen must be planned for as early as possible to avoid subjectivity, but sometimes force majeure situations exist. For example, if you are short WTI time spreads and the Cushing hub blows up, it doesn't make sense to maintain your short in such an obvious 'black swan' event.

Take this hypothesis, for example:

I am bullish on the January 2024 DFL swap contract, as Chinese refinery quotas for buying physical crude oil need to be fulfilled, and dirty tanker freight rates are cheaper than for fixtures last month, which will improve the economics of importing North Sea physical crude relating to the Dated Brent contract. There is also evidence of Chinese refiner buy-side hedging in the forward curve. I expect an upside of \$0.40/bbl in the next

month.

The specifics prompt clear signals for exiting. For instance, once the trade is live, you may see that Chinese selling comes in instead of the expected buying. As the whole justification is related to Chinese demand, this signal should justify an exit without bias. It could signal a real change to the physical crude market in the North Sea (the Chinese import a lot of North Sea crude), as a VLCC fixture (holding 2 Mbbl) may have been cancelled, or perhaps an alternative crude was bought that won't impact the Dated Brent market. By keeping the trade on despite this new information, you are now holding a long position without the original justification. So this trade becomes a gamble.

The Risks of Turning Winning Trades into Losing Ones

A winning trade idea can become a losing trade – and it happens *often*. Monetisation is difficult and often overlooked. Frequently, you hear in the market of traders who 'gave some back' to the market, which is trader-talk for losing money from the peak of the trade. With a lack of planning and discipline, your exit target can shift higher the more money you make. When the market retraces, you will be susceptible to bias, unwilling to accept that you're on a losing streak. Then, you start going through the five stages of loss. By the time you accept that you missed the high of the range, you may have lost all the money you made – and potentially more.

Using Trailing Stops

You can stack probabilities in your favour by using trailing stops. You are in the money and want to run your win, but you should also want to protect against 'giving it all back'. A trailing stop will move up as the market moves up but stay firm as the market retraces. This could be as simple as a fixed number with an allowance for weekly volatility or a defined percentage below the price.

In my second year of trading, I was short after an OPEC meeting that failed to yield the expected production cut, which, on balance, meant that there would still be a severely overhung physical market. I rode the contract

down from around \$55 into the 40s. My risk manager asked me about my target exit, which I had arbitrarily placed at \$40/bbl. It went all the way down to \$40.57 only to bounce back aggressively and go on a three-day rally. I ended up giving back a lot of gains, which was incredibly frustrating, and a trailing stop is all I would have needed to do well out of a trade that had moved very far in the money.

You can also set a **ratcheting trailing stop**, which trails the market price and moves up on a predefined criterion, like 10% below the previous night's market close. If this number is higher than the current trailing stop, the ratcheting stop is used and trails the market price from there.

Role of Technical Analysis in an Exit Strategy

You can use technical analysis to help with your exits, which can help in identifying a quantitative definition of an end to a trend or an overbought/sold scenario. However you choose to determine fair value, you will significantly improve your chances of long-term success by factoring in a prudent exit strategy.

Review in Detail

If you are going to become a successful trader, you must approach the entire process with humility. To really cement this, you should review each trade. You should reflect on your thought processes before, during and at the closing of the trade.

Did you deviate from the plan or use some form of discretion? Was this truly necessary, or were you acting out of fear and losing control of the process? What could be improved? If it was a winning trade, how could you have optimised it? Was there any luck involved? If so, did your structure allow you to benefit from this?

Make sure you scour each part of your process for improvements, documenting the learnings and, most importantly, have the courage to go again with the new learnings applied. If you don't follow this logic and repeat your mistakes, you will never improve or be a high performer.

Trading diaries are the ultimate way to hold yourself accountable. Your work ethic will vary between processes, and you will be lazy at times. The best way to stop this is to be as open as possible and get back to the

grindstone whenever you are led astray.

Track Performance

Tracking performance is vital for you to acknowledge where your returns are coming from and not let subjectivity get in the way. Look at your trading performance metrics: wins and losses, average return, Sharpe ratio and Sortino ratio.

Not only is it important to track your quantitative performance but also your behavioural performance. Did you exhibit any of the five stages of loss? Did you linger longer than you should have in one stage? Were you ill-equipped to deal with the markets because you were sick, tired or hungover?

In this respect, personal diaries or daily reports can be incredibly useful. Being open about your thought processes will make it easier for you or others to spot where your attitude could be refined to optimise your trading strategy.

Common Mistakes in Directional Trading

Not Sticking to the Trade Plan

The point of the plan is to remove subjectivity and accept that while it will never be perfect, forming a plan and sticking to it can be built on, but random discretion cannot.

I have seen many intelligent, cogent individuals transition to an irrational state of mind as risk increases. They will panic and close their position when a flurry of price action goes against them or sit frozen and unable to act when the market has moved well through their stop. It is well understood that in high-pressure scenarios like trading, your brain is like an athlete in a game or a soldier in battle. It is a classic fight-or-flight mechanism, and no matter how cool and calm you think you are, a thought-through, systematic process will always be more effective and repeatable than free-styling during a trade.

One of my irritants has become traders who say they trade off gut feel. I strongly believe in instinct, and it is very powerful, particularly for market makers who must react first. But when it comes to directional trading, you have the time to break down your gut feeling. Why exactly did you feel that

way? Usually, it is one or two things you saw, and your subconscious has historical data stored to signal to you in a split second that something isn't right or there is an opportunity. But if you break this down when it happens and document it, you can start to see patterns and bring your decision-making to the forefront of your conscious mind. You will increasingly make decisions you can explain, which can be converted into a rule book or heuristically derived pattern.

Trying to Finesse the Market

Beginner traders often try to be too clever; some never grow out of it. They typically don't appreciate the level of randomness in the market and think they immediately know a way to enter a trade that is smarter than what the market is doing or at least offers more protection.

An illustration: A young fuel oil trading team in the very early days at Onyx was given some risk capital to enter the market. They wanted to buy fuel oil cracks in the prompt, and their game plan was to buy the cracks at the beginning of the window and ride them out during the day with a ratcheting stop loss. At the end of the Singapore window, I asked how their position was, and I could see the cracks were higher. They told me they had thought that buying in the window was too aggressive, so instead of buying, they *sold* the cracks, looking to buy them back cheaper and going long.

Taking a step back, I'm sure they could see this made no sense, but this is where theory and practice are so different. They were essentially overwhelmed, and their actions were borne out of fear and a form of arrogance. Who were they to say that the buying was too strong? Hadn't they hypothesised that the market was going to be well bid? Did they not think others would be thinking the same thing?

Once you have done the work on the trade, there is a lot to be said for conviction. It might seem so simple, but those who are already in a trading situation can attest to the difficulty.

Attaching Too Much Meaning to Market Numbers

If you have done the market analysis and judged the flows and

positioning as well as possible, you cannot obsess about the number itself. Reluctance to buy a 'high' number (or sell a 'low' number), like in the previous example, is overthinking, and your biases creep in.

The more you attach meaning to numbers, the more you fall victim to crowd trading. For example, setting long-term targets for your trade at round numbers is what most traders do. So, in a given trend, there will be volatility spikes and liquidity issues around numbers that mean nothing other than the meaning given to them by the market.

There are many examples where there is an incentive to change physical operations at certain levels. For example, a crude futures price above breakeven for crude producers will incentivise new crude production and spur sell-side hedging. Or an E/W level on a fuel oil swap will open up a genuinely profitable geographical arbitrage opportunity if the level is locked in on the derivatives contract. If a physical opportunity can be monetised ahead of time using the derivatives market, only hedging at these levels will result in a material change that the derivatives market will see. Even if these flows do occur, only a few traders will be able to react to them, and the financial market is larger than the physical, so this physical-linked derivatives flow can still be absorbed, and the market will be materially unchanged by it.

The point is that numbers are just numbers, and in the derivatives market, you have to ensure they don't govern your decision-making. They are only a guide for what flows are to come.

Blurring Lines with Portfolio Risk

Pure directional traders don't have the luxury of a business model to fall back on, like physical arbitrage or market making. They can't hide. This is how you should treat your directional trading. Even if it is alongside another strategy, define a position as directional with the appropriate trade structure and trade plans, processes and reviews.

You have to clarify where you entered directionally, *with no implied advantage*. For example, if you wish to enter a trade on a directional basis, the entry point should be the current market bid if selling and offer if buying, *not* the value you have in the trade off the back of market making. To take the fair value and not consider execution costs gives your trade a head start that you would not be afforded if your entire strategy was directional financial oil trading.

Having to be Right

You must be strict with yourself and not tell yourself a story about what happened. It can be helpful to ask a trading manager or colleague to review your trade. You shouldn't judge the outcome. Making a 'right' call can feel great but is irrelevant. How you monetise the trade with a repeatable process is the only thing you should be concerned about. Conversely, obsessing about calling the market 'wrong' is also pointless, as a losing position can still be traded well with sound risk management. You should take pride in limiting loss as much as in maximising gain.

By protecting your internal narrative, you put huge pressure on yourself that you don't need. Formulating a process that you can demonstrate has improved over time is much more impressive than the huge one-off win you may have had. Ego is for the trader seeking guru status; no one is looking for this anymore.

Hedging Strategies in Oil Derivatives

Given the complexity of the oil derivatives market, it's easy to forget that the original purpose was to provide a futures contract to hedge price fluctuations on physical oil to help budgeting by smoothing cash flows. This is the first thing to remember for those looking at hedging for the first time. Every contract available is there to mimic the physical market ahead of its settlement.

How Do You Use Hedging?

Step 1: Define Your Hedging Strategy

The first principle is to try to replicate what you do in the physical market as accurately as possible in the derivatives market. No matter how many contracts exist with strong liquidity, there is no such thing as the perfect hedge. Instead, skilled hedgers understand their jobs are about narrowing the difference in price fluctuations between the physical market they are trading and the derivatives contract, known as *basis risk*. They can do this by choosing the right contract, executing under opportune market

conditions and optimising the execution of the trade.

As soon as you use hedging to generate a return on the derivatives side, you become a speculative trader, which defeats the purpose. Irish airline Ryanair sticks to a systematic approach which seems to work very well. It maintains at least some form of hedging every year, no matter the market volatility, and focuses on securing cash flows. Other airline treasury departments see themselves as hedge funds and attempt to trade purely for profit. For example, an article on SouthWest Airlines in the *Financial Times* featured a company treasurer boasting about his hedging profits. The article discussed his strategy, using bull versus bear analysis, trading WTI and Brent contracts rather than jet fuel and, evidently, using the airline's hedging book as a speculative tool. The treasurer was clearly of the opinion that he is equipped to challenge the best traders in speculative oil trading, having said in the article, 'Send me your most arrogant person on the Street and have them come work for us in the fuel hedge, and we will humble them in about 10 minutes'^[54]. This is not hedging. To look at hedging for investment returns is not the point. You shouldn't think that you could have made more profit or wouldn't have lost so much if you hadn't hedged or had hedged at a different level. You should be entirely dispassionate about the performance of the hedge, and all the work should be on minimising the cost of execution, financing the transactions and ensuring the administrative processes are as precise and secure as possible. Hedging provides certainty in cash flows, boosting confidence in the financing operation. Sudden nosedives in prices have destroyed local oil-producing economies in the US. Huge spikes have crippled airlines. If you have locked in a price level for your fuel, you have removed one of the variables which affect your bottom line, freeing you to concentrate on optimising other variables in the business.

Even with this principle well understood, traders have often used hedging to disguise rampant speculation, resulting in plenty of scandals. Even whether to hedge or not in the first place can be contentious.

Before you build a hedging plan, it is important to clarify what you are trying to achieve, specifically how far ahead you are looking for protection and how precise you want that hedge to be. But first, you need to recognise at what point you consider there is exposure to the forward oil prices.

If the need to buy or sell physical oil in the future is an integral part of a business, then it must manage its live and continuing price exposure. For example, an end user like an airline or shipping company needs fuel for

everyday business operations. For as long as oil is used as a fuel, they will have exposure to the oil price that is playing out in the derivatives market.

Exactly when they deem it necessary to hedge and in what amount will depend on the business managers. And it's not only oil, of course. If the business has foreign exchange exposure, they should be hedging this, too. Our business at Onyx requires that we trade and receive cash in USD, but we need to pay costs in GBP for our London operation. As long as we trade in London and in USD-cleared contracts, we have long exposure to USD/GBP. Do we convert all the USD as we receive them to GBP, or do we actively manage this exposure ahead of time? How much should we hedge? For how long should we look to guarantee the rate? These are management decisions we make all the time.

As an illustration, look at this chart from a *Reuters* article from 11 March 2022 summarising the hedging strategies of some major airlines^[55] (see Figure 3.27).

AIRLINE	HEDGING STRATEGY
Air France KLM	72% 1st quarter, 63% second quarter and 'similar' in second half of year
Easy Jet	60% for financial year
British Airways	60% for 2022 and 2 years forward
Lufthansa	63% hedged for 2022
Qantas	90% hedged for 6 months to June 30 and 50% for Q3
Singapore Airlines	30% - 40% hedged for 2022

Figure 3.27 Examples of Hedging Strategies for some Major Airlines

The business will have to form a view on the costs versus benefits of hedging in the first place. Shareholders have been known to resist hedging^[56], arguing about the expense and the losses which may have accrued on the derivatives trade. This is why getting an oil trader or encouraging an oil trading mentality to optimise the hedging process for a producer or end user is dangerous. It is a risk management job, not a speculative trading job, and these are very different skills. If you are looking to beat the market with your hedge, you are not hedging.

Step 2: Assess the Physical Price Exposure that

Needs to be Hedged

In every physical oil deal, there will be a pricing formula in your contract that will refer to a benchmark index, and the negotiation on the price will be based on a differential to this benchmark.

The pricing formula will be based on the following:

- A **benchmark index** settled by a reputable PRA
- A defined **pricing period** of that benchmark index
- A negotiated **physical differential**

Figure 3.28 provides an example of this.

PHYSICAL OIL TRANSACTION, FOB TERMS

TO: [COUNTERPARTY]

FROM: [NAME OF COMPANY]

Dear Sir/Madam:

[COMPANY] hereby confirms the transaction as follows:

TRADE DATE

[DATE]

SELLER

[NAME]

BUYER

[NAME]

Seller and Buyer each shall be a "Party" and taken together shall be hereinafter referred to as the "Parties".

PRODUCT: [PRODUCT DESCRIPTION]

QUANTITY: ([MT/Barrels])

DELIVERY:

Delivery shall be given and taken FOB [DELIVERY LOCATION], in the period commencing [DELIVERY DATES] (all dates inclusive) in one full cargo lot on vessel [INSERT VESSEL NAME]

DETERMINATION OF QUANTITY AND QUALITY

Product quantity shall be based on bill lading figures. Product quality shall be determined by independent inspectors as per standard practice at loadport based on measurement and analysis of product in loadport shore tanks.

Any charges of the inspector for performing duties not specified in the Agreement will be for the account of the Party so instructing the inspector.

PRICE:

[AMEND PRICING MECHANISM FOR THE TRANSACTION]

The price shall be the average of "[PRICE QUOTE]" meaning that the price for a Pricing Date (as defined below) will be that day's average of the high and low prices per [metric ton/barrel] of [PRODUCTION], stated in US Dollars, published under the heading "[HEADING]" in the issue of [PUBLICATION] that reports prices effective on that Pricing Date from and including [PRICING DATE] plus [PREMIUM] USD per [Metric Ton/Barrel] based upon bill of lading (or equivalent documentary) quantity.

The Pricing Date shall be defined as the date [on which the vessel completes loading at the loadport/date of the bill of lading/OTHER].

Figure 3.28 An Example of a Physical Oil Contract

Step 3: Select the Right Derivative Contract(s)

Once you are aware of the benchmark used for the physical price, you need to select a derivative contract or contracts which as closely align to that benchmark price on settlement as possible, weighed against the cost of the hedge.

To do this, you need to assess the following:

- The correlation of that derivative contract with the physical pricing benchmark
- The liquidity of the derivative contract
- How to align the pricing period of the physical contract with the settlement of the derivative
- The execution of holding that contract
- The risk management process required to manage the hedge life cycle
- The financing costs

Correlation

The benchmark index will typically be a derivative contract, covering most of the exposure to the final physical price. However, the more generic the benchmark index for the delivered physical product, the more variance there will be in the physical differential.

For example, a physical naphtha cargo may be loading in Genoa, Italy. If, for some reason, the supplier wanted to sell using Brent futures as a benchmark index, hedging the final physical price with this contract barely covers half the exposure. Much greater accuracy will be found using the Northwest Europe benchmark and the more niche Mediterranean contract. If the pricing formula is set using a Northwest Europe benchmark, slightly more accuracy can still be found using the Mediterranean contract.

BENCHMARK DERIVATIVE CONTRACT	BRENT FUTURES	RBOB FUTURES	PLATTS NORTHWEST EUROPE NAPHTHA CONTRACT	PLATTS NAPHTHA FOB MEDITERRANEAN CARGOS
Example correlation with physical Naphtha prices delivered into Goa, Italy	50%	60%	98.5%	99%

Figure 3.29 Examples of Correlation Between Physical Naphtha and Various Derivatives

Derivative contracts are commonly available for most oil across the

barrel in Northwest Europe, the Mediterranean, Singapore, the US Gulf Coast and New York Harbour regions. There are some less liquid derivatives contracts for the Middle East (usually using UAE as the base region) and ports in China. If you are in a region not reflected in the benchmark, it is a matter of finding the closest one.

To ensure accuracy, you could even use a combination of contracts in a basket, weighted based on correlations and regression analysis. This combination could also include forward freight agreements to value the difference in price for the same product and specifications in a different region. For example, if the physical is being delivered to India, the common contract is Singapore. But to ensure accuracy, you could include a freight derivative that accommodates the difference created by the journey from Singapore to India; the TC12 contract is a derivative that tracks the West Coast India to Japan route shipping prices.

But this type of hedging is complex, and you can quickly run into issues with the other factors, so searching for accuracy is only one step in the process.

If you work for a corporation consuming oil (end user), you may be delivered oil based on a benchmark that is opaque in its price discovery process and low in liquidity for the derivatives contract that prices to the benchmark index. You can be confident that the oil trader delivering the oil will calculate the price using a similar price to the one above. They will take the nearest, most liquid contract and calculate the value based on that benchmark's price plus the known costs of getting oil delivered to the new region.

For example, a fuel oil supplier may agree to deliver physical oil to an end user at a differential to the Platts Fuel Oil FOB Mediterranean daily assessments, using an average of five days around the bill of lading (details of delivery including the loading date). As previous chapters have demonstrated, all oil derivatives contracts are linked, with less liquid contracts deriving prices from more liquid ones. So in this example, the Platts Fuel Oil FOB Mediterranean price will be derived from the Platts Fuel Oil 3.5% FOB Rotterdam Barges contract, which is a lot more liquid. So the oil trader will likely hedge the transaction made with the end user using the more liquid contract and decide what to do with the Mediterranean-based risk against the Northwest Europe risk. They could hedge that differential, but it is illiquid, and a market maker will give a wide bid/offer. They might be

better off holding the risk and seeing it through to pricing.

If there is enough of this risk (implied short Mediterranean vs NWE), and they believe the market is weak, they could enter the Platts assessment window used to calculate the settlement price for the contract being used in the negotiations and offer the fuel oil publicly. If the oil is lifted at an attractive price, they may choose to buy more of that physical, as they now have a good gauge of the demand. If they cannot find a buyer, they can continue to offer the oil at lower prices as a signal to the PRA that the price needs to be settled lower, and they will benefit from the implied short derivatives position. The more opaque the benchmark, the fewer players there are in the market who can participate in PRA assessments to aid price discovery and the greater the risk of distortions to more robust benchmarks.

The overwhelmingly best strategy for corporates in the position of buying oil in locations that do not have liquid benchmarks associated with them is to renegotiate the physical pricing formula to a robust benchmark as a derivative. This will result in more variability in the physical differential, as there are costs in taking oil from the benchmark region to the delivery region. But it would be worth it so that the majority of the price risk (the benchmark) can be hedged effectively. You will also be able to track the prices much more effectively, as there are many solutions for fair value in oil swaps that become increasingly reliable the greater the liquidity on the contract.

Liquidity

Liquidity assesses the ease with which you can access a market bid or offer close to value in sufficient volumes. If the market is too illiquid, even though the contract may refer to an index that is aligned with the physical deal negotiated, it may still be better to trade a more liquid contract that is strongly correlated with the pricing formula benchmark.

The cost of a hedge can be hidden in the execution. A given contract will have a bid and an ask (or bid/offer), and the executable volume on those two numbers will be limited. You will get progressively worse prices as the volumes you want to trade in increase. The greater the price variance away from fair value, the greater the hedge cost, known as 'slippage'.

For instance, using the same naphtha example, the bid/offer on the highly correlated Mediterranean contract may be wider than the slightly less correlated NWE naphtha contract. The variance from fair value to traded

price may be so much more that it doesn't provide better accuracy as a hedge in practice.

	BID	OFFER	VARIANCE FROM FAIR VALUE (MID - POINT)	COST OF HEDGE FROM BID/OFFER	CORRELATION TO PHYSICAL PRICE
NWE Naphtha	\$600/mt	\$602/mt	\$1/mt	0.16% of volume	98.5%
Med Naphtha	\$598/mt	\$606/mt	\$4/mt	0.66%	99%

Figure 3.30 Example of the Typical Bid/Offer in Naphtha Derivative Contracts and the Correlation to the Physical Price

If the cost of executing the hedge is greater than the accuracy, the more liquid contract is the better choice. The cost of hedging a standard 5,000 metric tonne cargo (5 KT) would be $5,000 \times \$4/\text{mt} = \$20,000$.

The correlation using the Mediterranean contract is 0.05% greater than the NWE contract, which on a \$600/mt valued cargo is \$3/mt. Therefore, the cost of hedging is lower than the accuracy improvement of the contract.

If you were trading this contract once and taking it to settlement, then it is worth doing based on these numbers. But if you wanted to trade out of the contract before settlement, you are adding a further \$4/mt to the cost of the hedge, which would be costlier than the maximum saving from using the contract in the first place. So you have to consider if you value the option to exit the trade.

Pricing Period

Most physical cargoes are loaded in a three-day period (defined by the bill of lading), and therefore the physical price can be set using the average of those three days plus the physical differential.

Ideally, the pricing formula of the contract will align with how the derivatives price, and this does happen. For example, a supplier could agree to sell a cargo of residual fuel oil at the arithmetic average of the Platts 3.5% Fuel Oil Barges FOB Rotterdam assessments in March 2023 plus the physical differential but still commit to a 1–3 March 2023 loading period. This makes the hedge formation much easier, as the fuel oil contracts trade in monthly tenors, and even though there are daily settlements, there are no daily contracts. The only derivative contract with a liquid daily contract cleared on the exchange is in the Dated Brent market for crude oil.

If, however, the pricing formula is clear in that the price will be

determined during the bill of lading, then unless it is a contract that can be hedged with Dated Brent daily contracts, you will have a tricky hedge to manage.

Let's look at an example:

On 20 January 2023, a shipping company agrees to buy residual fuel oil with a sulphur content of 3.5% from a known supplier with which the shipping company has a formal working relationship in place.

The supplier agrees to deliver the oil under a pricing formula:

Volume: 100 KT

Benchmark: Platts Fuel Oil 3.5% FOB Barges NWE Loading Period: 1–3 April 2023

Pricing formula: the final price payable will be the average of the Platts assessment for 3.5% NWE Fuel Oil FOB Barges over the 1st, 2nd and 3rd of April, plus a differential of $-\$2.00/\text{mt}$.

Let's say the buyer of this fuel oil is looking to hedge as soon as this deal has been agreed on January 20th. There is a liquid oil swaps contract for 3.5% Fuel Oil FOB Barges NWE that can be cleared on the ICE or CME exchange. However, the fuel oil contracts are traded in monthly tenors and settle based on an average of the daily assessments from the PRA Platts, which are reported every working day as defined by Platts' calendar.

PRICING DAYS (PLATTS CALENDAR)	PLATTS FUEL OIL 3.5% FOB BARGES NEW APRIL 2023	3.5% RESIDUAL FUEL OIL PHYSICAL (LOADING PERIOD)	PRA ASSESSMENT PRICE (\$/MT)
April 1	5KT	33.33KT	650
April 2	5KT	33.33KT	625
April 3	5KT	33.33KT	630
April 4	5KT		640
...	5KT		
April 19	5KT		660
April 20	5KT		650
Total	100KT	100KT	645

Figure 3.31 Example of a Fuel Oil Swap Pricing Calendar and Breakdown of Daily Positioning

There are 20 trading days in this month, so a total of 600 KB of this fuel oil swap will have 30 KB per day rolling off to a price determined by the PRA. On the other hand, the physical has exposure to the PRA assessments only during 1–3 April, and therefore 33.33 KT per day. If you hold 100 KT as an April contract to match the physical volume, then only 15 KT ($5 \text{ KT} \times 3$) is actually pricing to the three-day loading period, and after this point, you have pure derivatives risk. Ideally, you would have derivatives volume that rolls off 33.33 KT per day on 1–3 April in line with the loading period, but as the fuel oil swaps only trade by months, that would require a much larger position: 667 KT. This would mean taking 667 KT risk on derivatives against a volume on the physical of 100 KT, so not a hedge at all. This strategy would create a lot of exposure to the flat price of fuel oil barges, which can be very volatile. Using the above, the average of the PRA assessments during the loading period is \$635/mt ($[(650 + 625 + 630) \div 3]$), whereas the monthly average is \$645/mt. This variance is large, and in reality, it can be a lot more.

You have several options:

Buy 100 KT of the April 2023 fuel oil swap as a hedge for your physical purchase and sell out of the contract on 4 April.

Buy 100 KT of a contract that can price from 1–3 April 2023, which would be an exchange-traded futures swap (Brent or Gasoil in this case, as they are closest to fuel oil) or a 1–3 Dated Brent swap.

Buy 100 KT of the May 2023 fuel oil swap and sell out 33.33 KT per day during 1–3 April.

In the first option, 85 KT will have to be exited immediately after the third day, which involves trading a partially settled contract, known as 'balance of the month' or *balmo*. Given the usual pricing volatility, getting a price for a balmo contract becomes more difficult the more the contract settles. The bid/offer on this contract, particularly if the settlement prices are heading in a direction opposite to your risk, can be wide and ruin the accuracy of the hedge. The volume that actually prices to the same dates as the physical pricing formula is only 5 KT per day (15 KT) compared to the 100 KT required. It is better than nothing but not ideal.

In the second option, you will have the same volume exposure on the derivative contract on 1–3 April as you do on the physical side, but there will be basis risk between the price of whatever derivative contract you choose (Brent swap, Gasoil swap or Dated Brent swap) and your fuel oil swap assessments. The differential between these contracts can vary significantly and ruin the performance of the hedge. In addition, the Dated Brent and Gasoil swaps will price off at the same time as the fuel oil (16.30 London time), but the Brent swaps will roll off at 19.30 London time, meaning there is a mismatched exposure to the outright price if you choose this contract, which can again be highly variable and ruin the hedge. To get around the temporary mismatching, you could trade a minute marker (16.30) for the exchange-traded future the futures swap is priced to and then buy back the future using TAS (19.30), but this increases execution time and will leave you with the basis risk between the futures swap and fuel oil assessments.

In the third option, you are exposed short to the pricing month time spread of the fuel oil swap, April vs May 2023. This spread can move significantly and is one of the more dangerous contracts to hold, but it is only for three days, and the liquidity to exit the May leg will be strong and execution straightforward.

Some will offer you a 1–3 April vs May price for fuel oil bilaterally (not cleared on the exchange), but you will pay a wide bid/offer for this, and effectively all the price provider will be doing is setting the price wide enough so that they have a degree of confidence in the 1–3 vs full month price variance. Given the price you are paying, you may decide it's better to take that risk yourself.

Either way, in this scenario, hedging is a challenge. Again, the overwhelmingly best thing you can do is push for the pricing formula of the physical agreement with your supplier to have the same assessment period as

the swap – a monthly average.

Execution and Timing of the Hedge

Liquidity is one thing, but the accessibility of that liquidity is also important. Is this contract available to be traded at all times of the trading day? If you want to hedge when it is convenient for you, you will need a market participant to give you a price at that time, which might not be straightforward. The Mediterranean contract is a niche contract, and only a few participants may be willing to give you a price at any one time. There will be times when you can't find a price at all.

When assessing the timing of a hedge, there is a psychology to it that is similar to a trader and investor, which means many of the same pitfalls are present. Producers tend to hedge the lows and not hedge the highs. Refiners do the same, and end users do the reverse.

How you will execute should come under your hedging strategy. Ideally, when an agreement is made to purchase or sell the physical product, you should enter into the hedge, but this is not always the smartest move. There are times of the day when liquidity is strong; for example, in the European fuel oil example, this would be during the 16:00–16.30 Platts window. Especially if your position is large, you don't want a lack of market liquidity to work against you.

What is alluring about exchange-traded futures for the average hedger is that you can trade at almost any time – the exchange is open every working day, bar Easter, Christmas and New Year's Day. The value that a highly liquid and available contract gives you is the freedom to react instantaneously to a change of circumstance. But of course, there can be enormous deltas to the actual physical price, and swap market makers are around very early in the morning to late at night. The swap market will continue to grow in sophistication.

A slightly more nuanced factor is the payment of the bid/offer. It always seems at the time that it is a smart move to negotiate to get an extra \$0.01 or \$0.02 in your favour, but this can be a very dangerous mentality. The bid–offer is there for a reason, and it is the broker's job to gather interest and get the best markets to give you the best price. By over-negotiating, you risk missing the trade, and you can lose a lot more by not trading than you would have by paying the bid/offer.

The best approach is to be systematic. Sometimes the timing will work in your favour; other times, it won't, but all you can do is control the controllable. Within the hedging process, you will want to ensure you obtain sufficient liquidity. Hedging on the buy side when everyone wants to sell may sound counterproductive, but remember, you are not looking to call the market; you are simply looking for efficient execution. This makes market opens and settlement periods a favoured time to hedge, as there is a lot of trading activity around these times.

Risk Management

There is also the work to manage the hedging trading cycle to consider. Some hedges are admin-heavy – for example, those with a lot of what is known as 'roll-off', which is a changing exposure as the contract starts to expire. Expiry times vary: European contracts roll off at 16.30 London time, but exchange-traded futures Brent, WTI and their derived swaps roll off at 19.30 London time. If you are holding multiple contracts at a time to create a hedge, this can be particularly dangerous if not managed diligently. One contract may decrease in exposure while another contract doesn't, creating exposure that wasn't intended.

The more complicated the hedge – the more admin work to manage it – the greater the risk of failure. This can be supported by a strong risk system geared towards hedging in oil derivatives.

Financing

As discussed in the Foundation section of the book, to trade oil derivatives to hedge, you need to access the market by establishing a bilateral line with a bank or trade finance operation or gain access to the exchange via a clearing member. Unless you can afford to self-fund a transaction on the exchange using your own capital, as a hedger, you will need access to financing to access the market either way.

To receive financing, the financier will need to examine the assets and liabilities of the corporation, using either the cash reserves or assets as collateral for the transaction. Financing hedges is, therefore, a pain point for every corporation looking to manage its oil derivative price exposure, as it strains the balance sheet.

Every hedger whose core business is not derivatives would prefer not to need to hedge, preferring to deploy cash to leverage business growth, especially with the costs of capital going up. In bilateral trades, an investment bank (or company acting as a principal) will take the other side of your hedge without the need for upfront cash that would be needed to trade on the exchange. This relieves the pain point of retaining the company's cash reserves, which is why bilateral trades remain big business.

Companies offering this service, referred to as swap dealers in the CFTC report, look to understand your physical position, verify that the financing of the physical assets is secured and assess the credit risk of the hedging company. Only then can a strategy be discussed. These companies play on the risk aversion of other companies by steering customers into hedge structures with low upfront cash requirements. But there is no such thing as a free lunch, as the saying goes, and the hedger will typically pay handsomely to avoid the cash strain on the business.

A company's decision whether to trade on the exchange or bilaterally should be based on an assessment of the costs. Derivatives on the exchange require up-front initial margin and ongoing variation payments throughout the life of the trade. The price of bilateral trades will be determined by the following:

Fair market value + exchange market bid/offer + discretionary bid/offer

For example, a refiner might be looking to sell a December 2023 Brent/Dubai crude swap differential and goes to an investment bank for a price.

Fair market value = \$3.50/bbl bid/offer = \$3.47/\$3.53.

Therefore, the bank knows that if it buys the differential from the refiner for less than \$3.47, it can sell the market bid and generate revenue. The bank determines the client's credit risk to be low, being an established refiner with a track record of paying losses on hedges promptly. Therefore, the bank determines an appropriate spread to account for the credit risk as \$0.05.

Bid to the refiner = \$3.47 - \$0.05 = \$3.42/bbl

The bank's profit is the credit spread and anything extra the trader can obtain on the market. Instead of immediately backing out of the trade at the market bid/offer, they may choose to run the risk for a time and get a better price. This is how these companies try to optimise this business.

The informed hedger will be aware of fair market value and the market bid/offer and can determine how much extra the bank or trade finance firm

has put into the price. The more bilateral lines the company has in place, the stronger its bargaining position and the more competitive the process will be.

A further benefit of a bilateral line is that, usually, the financing terms will allow the customer to have a loss on a hedging position without a requirement to immediately fund that loss with cash, as you would need if trading on the exchange. But the unfunded loss amount is also usually capped, and beyond this amount, the bank will insist on daily margining of the position, just like an exchange. Therefore, the advantages of trading bilaterally over trading on the exchange only extend up to a point.

With knowledge of the true cost of the hedge, this can be compared with the cost of trading on the exchange. If the saving on the bid/offer is greater than the cash required on the exchange, this weighs heavily in favour of gaining access to the exchange.

Zero-cost Collars

Commonly known offerings from banks (and other financing companies) that have appealed to corporate hedgers looking to be efficient with capital are the zero-cost collar and zero-margining deals. A zero-cost collar is a financial strategy used to limit the potential losses of an investment position without incurring any upfront cost. For a hedger looking for protection against the upside (e.g. an end user), this strategy involves purchasing a put option while simultaneously selling a call option. It is the reverse for a hedger looking for downside protection (e.g. a producer).

The limitation of this strategy is that the pay-off on a zero-cost collar is capped, so if the price moves out of the range set by the options, there is no protection. This option is, therefore, only viable if you believe the market has low volatility.

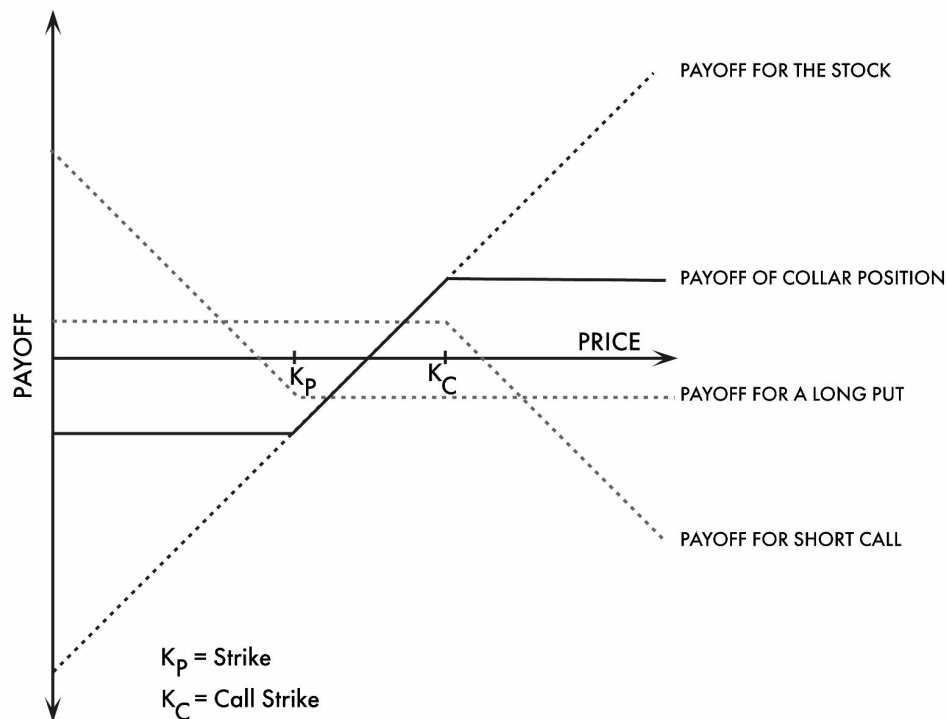


Figure 3.32 The Price vs Payoff for a Zero Cost Collar

The company may be happy with the pay-off structure and want to do it anyway, but they should still be aware of the cost. The price provider will transact with the company and not require it to pay any initial margin, but the provider will set the strike prices of those options. In effect, it is the same logic as the fair market value + market bid/offer + credit bid/offer, but it is easier to disguise. The provider will choose two numbers that net to zero but will trade out of it in the open market via two legs at a much better price.

Zero-margining Deals

In a zero-margining deal, on the other hand, there will be no demand for margin payments throughout the life of the trade, no matter how the market moves. This is the best deal from a credit perspective but requires the hedge provider to take a considerable credit risk on the company. Not many hedge providers are willing to offer this kind of financing arrangement. So providers who offer this arrangement are in a strong bargaining position and can charge huge amounts on the discretionary element of the bid/offer

explained above.

	PHYSICAL TRADE	PHYSICAL OIL PRICE EXPOSURE	WHEN	EXAMPLE	EXAMPLE HEDGE
PRODUCER	Sell crude oil	Long physical crude oil price	From when production is planned and given go - ahead	A shale oil producer who has a number of oil rigs operational with production estimates agreed over the period that the company's cash flow is available to support the operation	Sell crude oil futures or swaps with pricing periods that align with the known loading periods of the physical oil. OR Buy put options
REFINER	Buy crude, sell refined oil product	Short physical crude oil and long refined product	Short physical crude up to delivery into refiner and long refined product until sold into end user market	A refinery in Rotterdam knows it needs to buy crude oil in March, and will process into refined product by April, has a short March crude vs long April refined products position	Buy crude oil futures or swaps in the month the crude is loading, and sell refined product futures or swaps in the month they are refined, in volumes that align with the production of each product yielded
END USER	Buy refined oil product	Short physical refined oil	From when business operations depending on refined oil has been confirmed	An airline with ticket sales up to six months forward knows it has exposure to the jet fuel that is needed to fly planes for the passengers that have bought those tickets	Buy refined oil product futures, swaps of options
PHYSICAL GEOGRAPHIC ARBITRAGE	Buy crude or product in one region and deliver to end user for a product	Flat until one side of the deal is priced out	Initially short the timespread once a deal has been made to move oil from one region to the other. Position becomes long once oil is bought, and flat once oil is sold.	Buying 3.5% Barge residual fuel oil in ARA, booking a ship and sending the fuel to Singapore as the local fuel oil, Singapore 380	Sell (fuel oil) E/W contract
PHYSICAL BLENDER	Buy crude or product(s) and blend together to create a finished product of greater than the sum of its parts	Long the differential between finished product and blending components	From when blending components are purchased	Buy Naphtha at attractive levels relative to the final delivered gasoline product	Sell Gasoline v Naphtha differential
STORAGE	Buy the physical crude or product and store in tanks or ships	Long physical until sold in open market	Once the price for oil is settled by PRA	Buy crude oil in cash market with forward curve in contango	Sell outright contract of crude or product in month 1, and buy timespread every month until you are ready to sell

Figure 3.33 Why and How Different Players Hedge

Some products don't have a liquid benchmark, for example, Vacuum Gasoil. Traders must be inventive to hedge this product. Some have begun to use Dated Brent but lagged by one month, as this has in the past proven a decent enough correlation.

Airlines in the Pandemic

Jet fuel is vital for airline business operations. A rise in the price of jet fuel can be devastating to margins. Hedging can help smooth the price of jet fuel, allowing airlines to provide price stability in ticket prices.

Hedging to lock in the fuel price makes sense when an airline sells enough tickets for a flight that it is certain to fly. When considering hedging for flights a year in advance or even longer, the situation becomes a little more complex. This is not an easy position to manage.

A hedging strategy can benefit an airline when fuel prices rise because locking in a lower fuel price will give it an edge over competitors that have not hedged the fuel price and need to take a loss or pass on rising fuel prices to their customers through increased ticket prices. Of course, the reverse is also true. An airline's hedging strategy could lock in the fuel price before the price collapses. An even worse scenario occurs when a hedging strategy is losing money and there is a sudden lack of passengers because the falling price of jet fuel can't be used to lower flight costs.

This is what happened during the COVID-19 pandemic in 2020, and it was devastating to the industry. Live hedges served no purpose, as there was no demand for jet fuel. To make matters worse, the hedges were out of the money (live risk that is losing money), and with no physical demand, these losses couldn't be offset with cheaper jet fuel purchases. So many companies had to close out of their losing hedges, locking in a loss. It was reported that European airlines lost as much as \$4.66 billion on hedges, with many treasurers and risk management teams losing their jobs as a result.

When the industry started to recover, many had lost the appetite for hedging but lost out again when the jet fuel price rose more than 1,000% in just two years.

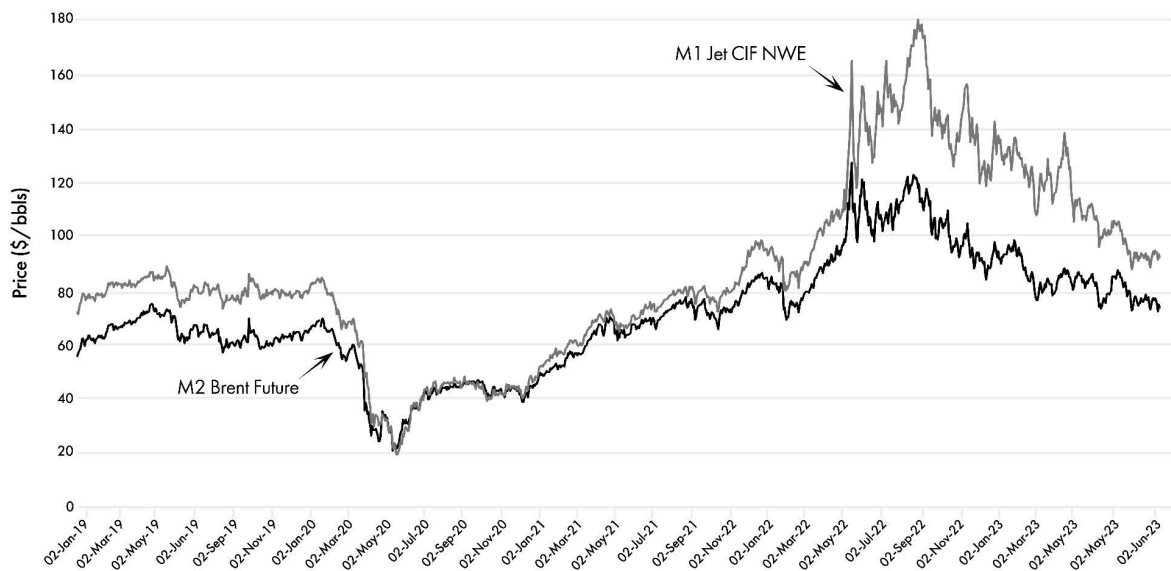


Figure 3.34 The Price of Month 1 NWE Jet Fuel from 2 January 2019 to 2 January 2023

Malpractice in Hedging

There are many examples of hedging operations falling foul of the law. For example, OW Bunker – said to be the largest bunker fuel trader with an estimated 7%^[57] of global market share in bunkering fuel oil (storing for clients) – went from an initial public offering (IPO) of over \$1 billion to collapse in November 2014 after unscrupulous risk management, which was blamed on two traders whom the company subsequently sued for fraud.

The company was financing bunker fuel physical deals and managing the hedges of consumers off the back of the deals, very much like investment banks would do. Because it was such a large and complicated operation, the traders at OW Bunker could muddy the waters between hedging activity and speculation.

For example, according to market participants, the traders were selling puts on Brent futures as the outright prices of crude futures markets were plummeting during the sell-off in 2014 when prices went from \$114/bbl to the low \$40s/bbl.

Selling put options ‘naked’, or unhedged, is notoriously risky in any market, because this tends to be a highly leveraged activity.

In early November 2014, OW Bunker announced that it had discovered fraud at its Singapore subsidiary involving losses of at least \$125 million^[58].

Rather than reporting the losses to the company's management when they occurred, the executives responsible for the trades allegedly engaged in a series of cover-up measures, such as creating fictitious contracts and misrepresenting the subsidiary's financial position.

The cover-up measures were ultimately unsuccessful, and the losses led to a liquidity crisis at the subsidiary, which could not meet its financial obligations. OW Bunker filed for bankruptcy on 7 November 2014, and its assets were sold. The fraud scandal had significant repercussions for the marine fuel industry and led to increased scrutiny of risk management practices and financial transparency in the sector.

Another major fraud scandal that made headlines in 2020 was Hin Leong Trading, a Singapore-based oil trading company. The fraud involved billions of dollars in losses, leading to the company's bankruptcy.

In April 2020, Hin Leong Trading disclosed that it had suffered significant losses on oil trades. The company (which had cancelled its IPO in 2014) bet long on China recovering quickly after the pandemic broke out, only to find that prices plummeted. Faced with loan repayments to twenty-three banks of billions of dollars, the company filed for bankruptcy protection in 2020.

However, it later emerged that the company had been concealing losses for years, amounting to \$800 million^[59], using fraudulent accounting practices to overstate its assets and profits.

According to reports, the founder of Hin Leong, Lim Oon Kuin, had engaged in a series of unauthorised trades using the company's assets as collateral without the knowledge or approval of the management or board of directors. Hin Leong had a long exposure to derivative prices via its oil inventory. The valuation of that inventory plummeted with the price rout during the pricing war in 2020, and the position was not hedged.

The unauthorised trading had been concealed through a complex web of offshore companies and accounting manipulation. The fraud was eventually uncovered in early 2020 when banks began to demand payment for unpaid debts. The revelation of the fraud led to the company's bankruptcy and a major investigation by the Singapore authorities, which resulted in charges against Lim Oon Kuin and other executives.

The core issue in both examples is how companies differentiate between

hedging and speculation. The two must be separated and accounted for; otherwise, you run the risk of malpractice. Additionally, the lesson from such malpractice in the past is to keep things simple. If the proposed trades can't be explained at the board level, the actions traders are suggesting should not be taken. Hedging is a simple process; as a manager or treasurer, you should never be made to feel you can't understand.

Determining Fair Value

Fair value for the oil derivatives market is a theoretical midpoint between the best bid and offer of a contract. This can be applied to every contract's maturities, time spreads and differentials to other contracts.

Traders rely on fair value to price contracts where it is not immediately obvious where the bid/offer will come in. This is the case for exchange-traded futures contract months that will settle years into the future and for the majority of the oil swaps market.

The oil swaps market is voice-traded, has much lower daily volumes than exchange-traded futures and has many more contracts. Therefore, knowing the best bid/offer outside the most liquid contracts will only come from engaging a broker or a market maker directly, which takes time.

Not knowing fair value can leave you susceptible to mispricing the contract you wish to trade and leaving money on the table. Therefore, you need a good sense of market value to monitor the market sufficiently.

There are providers of fair value through APIs and software platforms, but it still helps to be aware of how these values are determined, allowing you to appreciate how contracts are linked.

The most important thing to recognise when determining fair value is that WTI and Brent futures are the most important contracts.

The oil derivative market always prioritises the contract that has the most liquidity and is the most accessible, as this is the most reliable contract to trade to get exposure to the oil price. There are currently two highly liquid contracts that dominate: WTI futures for the US crude market and Brent futures for the global crude market.

The reliability of these contracts means that rather than creating new markets in every contract, oil derivatives contracts can be derived from these two contracts. Every contract in the financial oil market is somehow linked to either the outright price of these two contracts or their time spreads.

Determining the value of the prompt month Brent and WTI futures is straightforward, as the exchange has a reliable live price: either the last traded price or the value between the bid and the offer. Given how frequently the contracts trade and how tight the bid/ask spread is, the value is always clear when the exchange is open for trading.

The time spreads of Brent and WTI are very liquid and more stable than the outright price. The prompt time spread values are clear and well-defined from when the exchange opens to when it closes. The contract prices for Brent and WTI beyond the prompt month are derived from the prompt flat price minus the time spread. For example, the contract price for February Brent futures equals January minus the January–February time spread. There is direct trading in these months, but mostly the liquidity on these deferred contracts comes from automated trading by algorithms trading against the prompt month and the time spreads. Even though the bid/ask is wider the further away from the prompt you get, the Brent and WTI futures market is by far the most liquid.

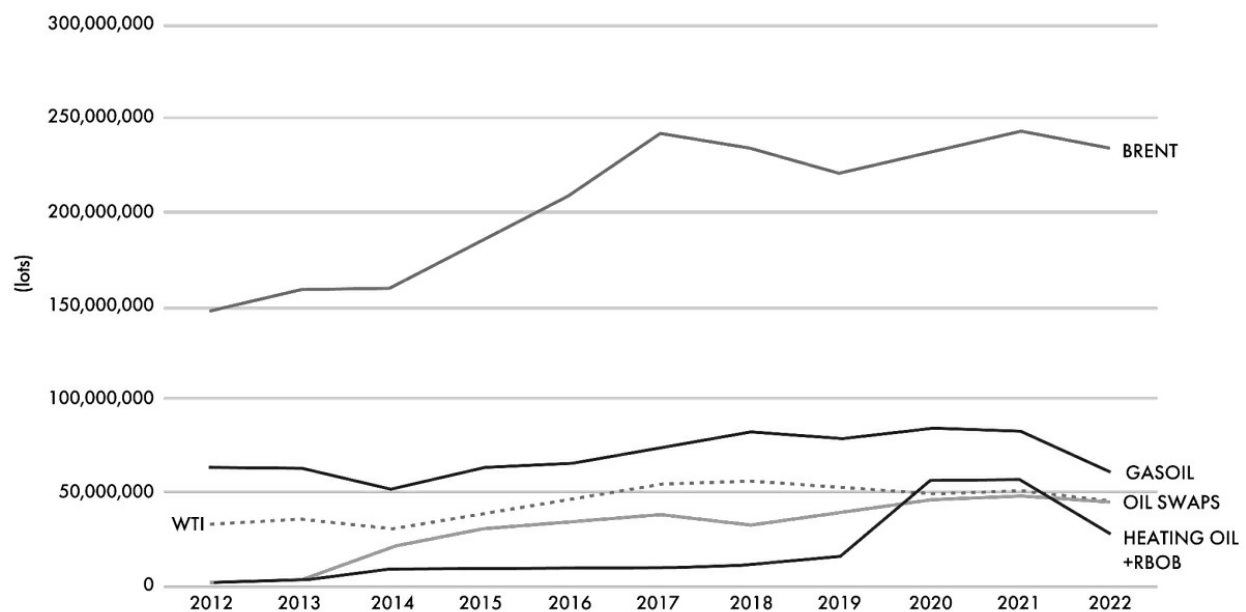


Figure 3.35 Comparing Open Interest on the Intercontinental Exchange from 2012-2022 (Source: ICE)

This chapter will illustrate how the oil swaps market is valued and how it is intimately linked with the Brent futures market for swaps out of the US.

First, to determine fair value effectively, you will need a tool in the form of a **pricing sheet**.

A pricing sheet is a document that provides current market prices for various grades of crude oil, such as Brent, WTI and Dubai, that you're interested in and prices for refined products like gasoline, diesel and jet fuel. It may also include market trends, supply and demand data, and news that may affect oil prices. You need to update it throughout the trading day as market conditions change.

Step 1: Start with a Futures Feed from the Exchange

The first step to determining fair value is a reliable feed of live crude futures prices. This is only practical with a programmed software link via an application program interface.

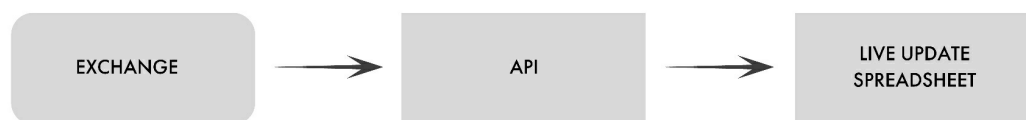


Figure 3.36 The Process of Getting Prices from the Exchange

For example, let's say we are at the beginning of January 2023. Figure 3.37 illustrates how futures might look.

	BRENT FUTURES	BRENT FUTURES TIMESPREADS
Jan 23		
Feb 23		
Mar 23	81.02	0.47
Apr 23	80.55	0.33
May 23	80.22	0.25
Jun 23	...	...

Figure 3.37 An Example of Futures Pricing in a Pricing Sheet Format

Step 2: Create Futures Swaps Prices

Remember: exchange-traded futures are settled on just one day at expiry, and the tenor reflects the loading period of the physical oil it is indexed to, not the day it expires. On the other hand, swaps settle daily, and

the contract tenor reflects the period in which the contract will settle.

Therefore, to enter the swaps market, we first derive an exchange-traded futures swap.



Figure 3.38 Transforming Exchange Traded Futures into Their Equivalent Swaps

For example, January 2023 Brent futures swaps (1st line futures) = $((\text{Brent futures March 2023} \times \text{pricing days in January} - 1) + (\text{Brent futures April 23} \times 1)) \div \text{pricing days in January}$

February onwards follows the same logic until you have a forward curve.

The time spreads are simply the January minus February values and so on.

	BRENT FUTURES	BRENT FUTURES TIMESPREADS	BRENT FUTURES SWAP	BRENT FUTURES SWAP TIMESPREADS
Jan 23(Pricing month)			81.00	0.47
Feb 23			80.53	Feb – March
Mar 23	81.02	0.47	(May x pricing days – 1) + 1 day June/pricing days in March	March – April
Apr 23	80.55	0.33	(June x pricing days – 1) + 1 day July/pricing days in March	April – May
May 23	80.22	0.25	(July x pricing days – 1) + 1 day Aug/pricing days in May	May – June

Figure 3.39 Creating Future Swaps Pricing Formulas in a Pricing Sheet

Step 3: Create a Column for the Oil Swap Contract You are Looking at

For this example, let's consider NWE naphtha. You can create a column for the differential that links Brent futures swaps to this contract.

M1 Brent futures swaps → manually marked differential → oil swaps inferred price from Brent swaps

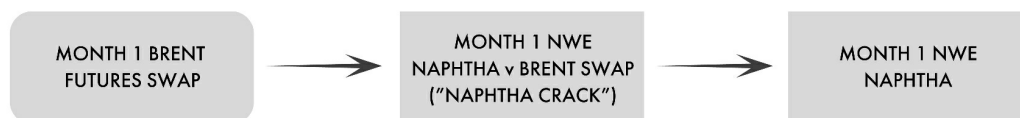


Figure 3.40 Linking Brent Futures Swaps to the Preferred Swap

The Brent swaps prices will be in \$/bbl, and the naphtha crack is also traded in \$/bbl.

However, the market for NWE naphtha is in \$/mt. The industry standard for converting the number of barrels to naphtha metric tonnes is multiplying by 8.9.

Therefore, your spreadsheet will have to use the following formula:

$$\text{NWE naphtha} = (\text{Brent swap} + \text{naphtha crack}) \times 8.9$$

The differential you pick to link futures to the swaps market is very

important, as this will price the other contracts. It should be the most stable and liquid differential in that market. In the naphtha example, this is the naphtha crack, but for gasoline, it is RBOB vs EBOB – or the arb contract – as it is much more stable than the EBOB crack (EBOB vs Brent swaps).

Step 4: Link the New Contract Formed with Contracts Reflecting Other Regions or Specifications

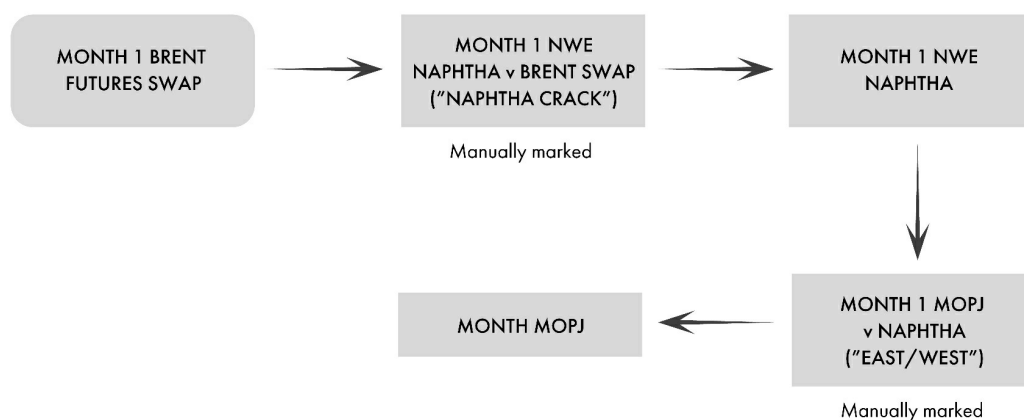


Figure 3.41 Linking a Contract with Contracts in Other Regions

The link between new contracts of different specifications or locations will be another manually marked differential each time.

TENOR	BRENT FUTURE	BRENT SWAP	NAPHTHA CRACK	NWE NAPHTHA	EAST/WEST DIFFERENTIAL	MOPJ
Jan 23 (Pricing Month)	-	81.00				
Feb 23	-	80.53	-8.50	641.00	16	657.00
Mar 23	81.02	...	...	...	...	...

*Shaded regions are manually marked

Figure 3.42 A Basic Version of a Naphtha Pricing Sheet

Now you have a flowing prompt market that will move up and down

with the Brent futures price. The notion of value for these is vital to keep your market in line.

Step 5: Build out the Forward Curve of your Swaps Through Manually Marked Time Spreads

The time spread is simply the value of one month against another. It is always chronological; in other words, always the nearest month minus the next.

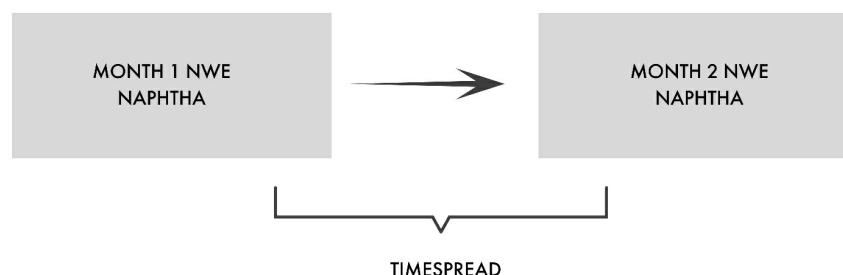


Figure 3.43 A NWE Naphtha Time Spread

The time spread values should be manually marked for the outright swaps contracts:

Implied Feb NWE naphtha = Jan NWE + Jan/Feb NWE time spread (manually marked)

Then the time spreads for differentials that use the sum of two outright contracts will be implied.

Implied Feb naphtha East vs West = Jan East vs West (manually marked) – (Feb MOPJ – Feb NWE naphtha).

	NWE NAPHTHA	NWE TIMESPREAD	EAST WEST	E/W TIMESPREAD	MOPJ	MOPJ TIMESPREAD
Jan 23 (Pricing Month)	= Jan/Feb NWE Naphtha + Feb Nwe Naphtha 641 + 6.50 = 647.50	+6.50	= Jan MOPJ – Jan NWE Naphtha 647.50 – 665.50 = 18	= Jan East/West – Feb East/West = 18 – 16 = +2	= Jan/Feb MOPJ + Feb MOPJ 657 + 8.50 = 665.50	+8.50
Feb 23	= (Feb Brent swap + Feb NWE Naphtha crack) * 8.9 647.50 – 6.50 = 641	+5.50	16	= Feb East/West – Mar East/West = 16 – 16 = +2	= Jan MOPJ – Jan/ Feb MOPJ timespread 665.50 – 8.50 = 657	+6.50
Mar 23	= Feb NWE – Feb/ Mar NWE timespread 641 – 5.50 = 635.50	...	= Feb MOPJ – Feb NWE Naphtha 651.50 – 635.50 = 15			...

* Shaded regions are manually marked

Figure 3.44 Marking Spreads and Key Differentials in an Example Naphtha Pricing Sheet

Hopefully, you can start to see how all the derivative contracts are linked for a given product. This should emphasise how the Brent futures price and Brent futures time spreads have a very real link to swap markets.

Note that time spreads of differentials (e.g. E/W) are known as *boxes*.

TENOR	BRENT SWAP	NAPHTHA CRACK	NWE NAPHTHA	NWE SPREADS	NAPHTHA E/W	E.W BOX	MOPJ	MOPJ SPREADS
Jan 23(Pricing Month)	81.0	-8.25	647.5	6.5	18.0	2.0	665.5	8.5
Feb 23	80.5	-8.48	641.0	5.5	16.0	1.0	657.0	6.5
Mar 23	79.5	-8.10	635.5	4.5	15.0	1.0	650.5	5.5

* Shaded regions are manually marked

Figure 3.45 An Example Naphtha Pricing Sheet With the Manually

Marked Cells

Step 6: Create a Space for the Quarterly and Calendar Year Values for Contracts at Any Time

Quarterly prices are simply the average of the three months, e.g. the sum of Jan, Feb and March prices divided by 3 is the first quarter price.

TENOR	NWE NAPHTHA	MOPJ
Jan 23	647.50	666.50
Feb 23	641.00	657.00
Mar 23	636.50	651.50
TENOR	NWE NAPHTHA	MOPJ
Q1 23	641.3	657.7

Figure 3.46 Adding the Monthly and Quarterly Values to a Naphtha Pricing Sheet

Calendar prices are the same but average for the full calendar year.

You can then ensure that quarter versus quarter time spreads (boxes) are implied, calendar versus calendar and so on.

Pricing Month

When the contract enters the pricing month, it is still possible to trade it, but more options become available. For the days that haven't settled, the contract becomes what is known as the *balmo* in the trading vernacular. For example, on day two, there may have been one day that has already priced. The balmo contract will be trading live as normal and will start settling from the end of that day onwards.

There is also the balmo starting the next day, called the 'balmo-ND'. A contract can be cleared on the exchange with any day as the starting day. For example, a broker may say they are interested in buying 'balmo pricing from the 13th', indicating the contract will price from the 13th through to the end of the month. If the contract is pricing and has had one or more pricing days, but the trader still wants to trade the contract as a whole month, this is known

as trading the 'full month'.

To calculate the final price of a swaps contract, you need to sum the settlement prices for each day and divide it by the total number of pricing days to find the average. If the contract is only partially priced out, you need to take the balmo price for the days that haven't settled, add the settlements for the days that have priced out and find the average.

Average = settlement price on day one + (remaining pricing days × balmo price) ÷ total pricing days in the contract month

		FULL MONTH	BALMO	BALMO - ND	BALMO PRICING FROM 4TH
SETTLED	Jan 1	72.45	-	-	-
LIVE PRICE	Jan 2	72.50	72.50	-	-
(YET TO SETTLE)	Jan 3	72.50	72.50	72.50	-
	Jan 4	72.50	72.50	72.50	72.50
					
	Jan 20	72.50	72.50	72.50	72.50
	Jan 21	72.50	72.50	72.50	72.50
	Live Price	$72.45 + (72.50 \times 20) / 21$	72.50	72.50	72.50

Figure 3.47 Calculating the Balance of the Month Contracts in a Pricing Sheet

Creating a Hierarchy of Contracts

As I hope this book has made clear, contracts are organised based on liquidity, and less liquid contracts derive their price from the more liquid contracts to form the network of oil derivative contracts.

When I first looked at the gasoline market in my first year of trading, it was as a junior to help the senior naphtha trader. He wanted to incorporate gasoline contracts to understand how they influenced naphtha and branch into trading those contracts over time. The naphtha market was well understood, as it had similarities to the fuel oil market. The market liquidity was centred around the NWE naphtha crack, so MOPJ flat price and NWE naphtha flat price values were derived from the crack plus the Brent futures swaps (as in the example above).

If the crack had the most liquidity, this was the best and most reliable definition of value we were going to get at any time in the naphtha market. The rest of the market could be derived from this, with diminishing degrees of certainty. So it was a matter of deriving the NWE naphtha contract from the liquid crack (naphtha vs Brent) by adding back the highly liquid Brent leg.

The E/W was relatively static, so it was straightforward to link the East

price (MOPJ) via the now implied NWE naphtha price and then add the E/W differential. It wouldn't be a problem to mark the prompt E/W as it didn't move that quickly.

The problem in creating the pricing sheet came with linking gasoline with naphtha. The GasNap, or EBOB vs NWE naphtha, was the clear link to either market, but we weren't sure exactly how to derive it. We knew we had the highly liquid exchange-traded contract for US gasoline, RBBR, and the liquid naphtha crack. But there was also a contract for EBOB vs Brent (European gasoline crack), so we thought we could manually mark this in the same way as naphtha and then derive the GasNap by adding the two.

As we discovered, the issue was that the EBOB crack wasn't trading frequently enough to rely on it as an independently marked contract. It was driven by the volatile exchange-traded RBBR contract. This is how we knew that the RBOB vs EBOB contract (the arb) was crucial in determining live, fair value in the gasoline market. The arb reflected arbitrage economics from Europe to the US and, therefore, was liquid from physical arbitrage hedging. It was stable enough to be manually marked, and the EBOB vs Brent price could be derived from RBBR minus RBOB vs EBOB. Now we had a reliable EBOB-derived price, the value of which was being driven by the correct driver, so we could link the derived EBOB vs Brent against the NWE naphtha vs Brent and derive an EBOB vs naphtha price, which was the GasNap.

This was very important from a value perspective, but understanding the hierarchy of contracts and how they should be linked via a spreadsheet opened insight into the market. We could then understand the link between flows in gasoline to naphtha. In other words, if the EBOB crack moved up in tandem with the RBOB crack, we understood the GasNap value that would drive how gasoline would filter into naphtha. For example, it made sense for the naphtha crack to move with the EBOB crack, so if a spurt of buying occurred in the EBOB and the GasNap remained stable, the naphtha crack would have to move in tandem with the EBOB crack. So seemingly independent drivers of one contract would have a very real impact on another.

This process is the essence of how we took the data from Onyx's market-making and formed a data solution to analyse markets. The physical trader will expect an open opportunity to hedge their physical arbitrage in the derivative market as a signal that the differential implying the arbitrage has

reached a ceiling in its value. However, if there happen to be larger, independent buying or selling flows in other contracts linked to how the differential is valued, they will drive that differential value over the physical hedging.

The overpowering flows from a derived contract can be frustrating for a trader looking at the derivative market as a reflection of the physical ahead of time. But as a modern- day oil derivative trader, you must accept that the curve is driven by flows only, and the only way the physical impacts the financial oil market (before settlement) is through the hedging flow. If other seemingly random flows overpower it, that is just the way it is.

4

THE WORLD OF OIL DERIVATIVES

The Impact of Financialisation

Financial oil contract volumes and open interest have been on a long-term growth trajectory since the first contracts gained traction on exchanges in the 1980s. As a result, every year, the derivatives market has grown in the form of contracts available to trade, depth of market volumes and rising open interest.

The derivative market has evolved, driven by worldwide data and technology advancements, and market dynamics have permanently changed from the days of the market being controlled by a select few. The key event that sparked this material shift was the exchanges listing oil swaps in 2012. From then, rapid adoption of exchange trading took place for almost all oil derivatives traded worldwide. As a result, soon after 2012, cleared financial volumes would not just supersede physical volumes but would dominate them.

Publicly available ICE volumes clearly show the growth trajectory of all oil derivatives contracts from 2012. Comparing ICE financial trading volumes with physical market volume estimates (using Statista and IEA)^[60]^[61] demonstrates how this impact has grown. In 2012, financial volumes on ICE were 401% higher than the underlying global physical volumes. In 2015, it was 503%, and in 2020, it was 757%.

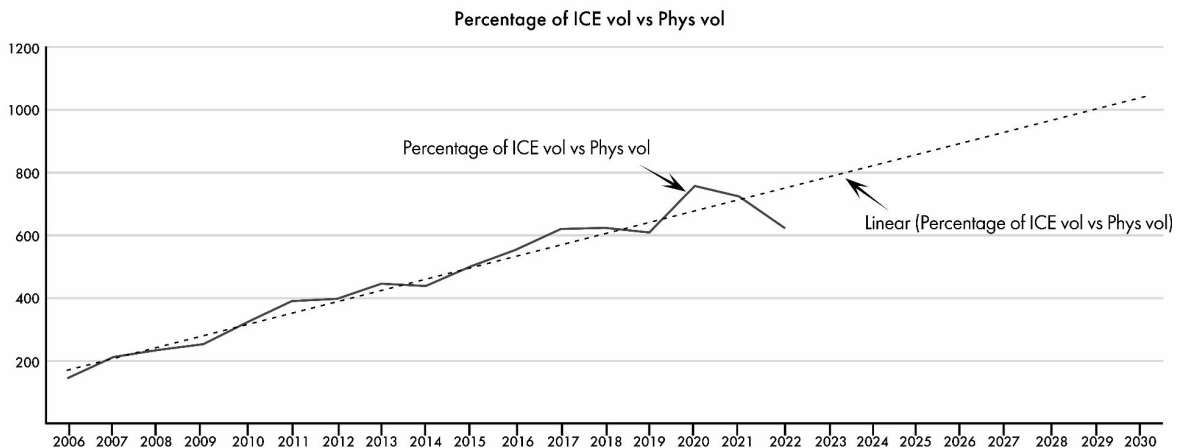


Figure 4.1 A Projection of the Percentage of Physical Volume that will Trade on the Intercontinental Exchange

After the turmoil of the last three years, financial volume and open interest slowed, but as markets settled in 2023, growth started picking up again. Extrapolating for the long term, by 2030, financial volumes on the ICE could be more than ten times the physical, especially with the IEA recently reporting the expectations of peak oil demand being reached this decade^[62].

This doesn't include the huge volumes on the CME, a major competitor to the ICE. Additionally, volumes in the East are growing rapidly, meaning financial contracts on the SGX, TOCOM and Shanghai Futures Exchange will also have a significant market share. You could therefore make a strong case for at least doubling the ICE numbers. Michael Greenberger, a former US government trading regulator, is quoted in the Guardian as predicting this value to be approximately 13 to 1^[63].

Purely financial volumes (non-commercial) are responsible for a lot of the growth in the last ten years. In exchange-traded futures, non-commercial position open interest has gone up from 2.8 million to 5.03 million from 1 July 2014 to 6 June 2023, according to CFTC data. As a share of the overall open interest, non-commercial positions on exchange-traded futures have gone from 53% of total open interest to 62% in the same time frame.

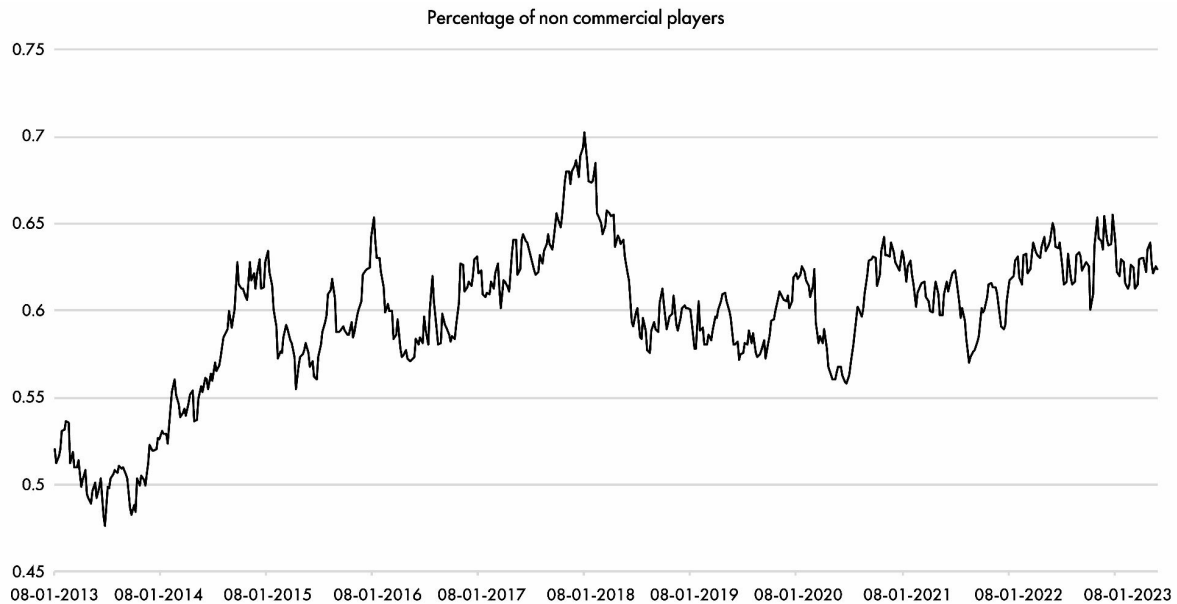


Figure 4.2 Percentage of Non-commercial Financial Volumes in Exchange-traded Futures from 8 January 2013 to 8 January 2023

What Has Been the Impact of This Financialisation?

As soon as the oil swaps market's contracts were listed on the exchange, visibility into the oil derivatives market was possible. Data that were previously only available to a very few companies were now posted digitally in real time. This much-improved visibility would set off a sequence of events over the following ten years that would cause a rapid increase in market participation, competition, technology and data enhancement, transforming the market. This evolution has resulted in a permanent change in how the market is accessed, analysed and traded.

The Growth of the Market Maker

The structural changes in the market provided the foundation, but what set the efficiency revolution in motion was the formation and proliferation of independent market makers in exchange-traded futures through algorithmic, high-frequency trading and the OTC market in oil swaps.

According to a Futures Industry Association report, algorithmic trading

accounted for around 46% of trading volumes in energy futures contracts (including oil futures) in 2019, up from around 25% in 2010^[64]. A *Financial Times* article suggested that Brent crude futures trading volumes were 80% automated transactions in 2019, up from 65% in 2013^[65]. Policymakers and governments must understand the impact of these volumes entering. But the changing market dynamics have considerably improved the liquidity of exchange-traded futures.

Oil swaps now trading on the exchange was not only about transparency but democratisation. Access to the market, previously only possible through the ISDAs discussed in the Foundation chapter, is now possible for traders with small seed capital. The first to take advantage of this was Muwaffaq Salti, Alf, who had a long trading career at Morgan Stanley, JP Morgan and J Aron (Goldman Sachs). Alf developed the market-making business model in fuel oil swaps while working at these banks and set up Mandara Capital to benefit from the opening of the oil swap market.

Alf's model was to hire graduates and begin their training in the model on the fuel oil desk before branching out into other, mostly nascent, swap markets. In 2013, around ten successful traders came through a bespoke programme designed by performance expert James King, including myself and co-founder of Onyx, Omar Kayaam. Together we took on new markets: Omar led gasoline swaps in Asia and Europe, and I led crude (Dated Brent and Dubai). Another equity partner at Onyx, Luke Mcdermott, built out distillates, US crude grades, propane and markets outside oil like ethanol. The Head of Trading at the time, Sean Cotter, built out naphtha. We all relished the challenge of being a newcomer in a growing market, but none of us could have expected what later materialised. Within three years, we were trading huge volumes daily, and the markets we were in grew enormously. How we went about this, starting as novices, is another whole book, which maybe one day I will write. But by 2016, there were many competitors in the space, and this is when Omar and I branched out to start our own market-making business, Onyx. We could see clearly that this market would keep growing for some time, and we knew there was a change coming that the market hadn't yet appreciated.

To put the growth into context, in 2012, Mandara was the only established independent market maker, trading fuel oil swaps with a market share of barely 5% of daily traded volumes. Today, there are at least fifteen established independent market makers in oil swaps, with a market share that

likely exceeds 50% of all daily swap volumes in the oil swaps market. Onyx Commodities (the trading arm of Onyx Capital Group) is now the largest, trading in excess of 20% of the entire oil swaps market volumes, with annual volumes above 25 billion barrels equivalent.

This growth in market-making activity in cleared swaps trading demonstrates just how significant the inefficiencies were in the oil derivatives market. But it also speaks to the growth of the market itself. The ten years after oil swaps were listed on the exchange saw a boom in trading activity driven by a virtuous liquidity cycle, further enhancing transparency and confidence in new markets, which inspired confidence in niche contracts, encouraging liquidity from hedgers. The much-improved reliability of core oil swaps markets led to the expansion and adoption of many more contracts, which are still growing today and increasingly mapping the wider physical oil market in financial form.

Growing liquidity and trading in oil swaps have undoubtedly improved trading volumes on exchange-traded futures.

Growing Sophistication of Financial Data

The increasing volumes and participation in the market have led to the formation and increasing sophistication of financial oil data, which are now critical for managing oil derivatives markets. Market price data have become more efficient, with exchange chat systems, WhatsApp and media platform chat systems processing data into pricing solutions, replacing Yahoo! Messenger as the primary source of market price data. This efficient communication has led to multiple offerings for end-of-day and live fair value assessments, streamlining the trading cycle and allowing more time for trade execution.

Settlement data have also improved with the advent of 'e-Windows' and software platforms allowing PRAs to take trading activity from the window and use it to settle each contract. This has improved the quality of PRA assessments and increased transparency in physical prices worldwide, leading to more efficient trading in physical markets.

Volatility calculations have also improved, enabling companies to better assess trader positions and reduce the risk of mismanaged losses. The availability of liquidity data has enabled assessments of the vulnerability of trader positions to slippage, providing an extra layer of sophistication in risk

management assessments. Moreover, liquidity data can be used to determine the real cost of doing business bilaterally, reflecting the difference between the bilateral bid/offer and market bid/offer, thereby aiding regulatory efforts.

Positioning data are regularly reported through the CFTC and ICE COT data in exchange-traded futures and recently through Onyx's COT data for oil swaps. This has led to increasingly sophisticated trading strategies in derivatives driven less by binary market calls from supply and demand models and more by information on what drives markets.

The increasing sophistication of financial market data has enabled more efficient pricing solutions, improved transparency in physical prices, reduced risk of mismanaged losses and increasingly sophisticated trading strategies.

The Power of Technology

The world of financial oil markets has seen a significant improvement in the speed and accuracy of receiving information thanks to technological advancements. The use of trading venues and market indication platforms has provided traders with multiple digital platforms to execute their trades, while the use of market indication platforms has further enhanced visibility and streamlined communication in the OTC market. These platforms have improved market data and streamlined the execution process, reducing time spent on price discovery and benefitting brokerages and market makers who help start the market daily.

Risk managers in oil trading companies now have increasingly sophisticated technology to evaluate risk assessments such as VaR or exchange margin requirements. The much-improved data feeds have enabled the software to scale these risk solutions, materially improving internal processes. These systems allow for live checking of risk, reducing rogue trader scenarios and raising awareness of large losses, which is particularly important for larger trading companies with many traders in multiple locations worldwide.

Execution software has further improved efficiency in the trade cycle. This software uses constantly evolving algorithmic techniques to enter orders into the exchange for the best possible fill. Additionally, tools that compare bids and offers in the market are written in text (chat systems) with a fair value assessment. This indicates to traders if there is an attractive number relative to what they see as fair value, providing trading opportunities.

While factors beyond supply and demand increasingly drive the market, the oil derivatives market is still based on a tangible commodity. Therefore, supply and demand data will always have their place. The improvements in delivering this data have been mainly due to technological advancements. While the data are publicly available, the challenge for traders is accessing this data in an easily digestible, condensed form. This requires heavy design work and technology platforms.

Infographics have become increasingly popular for conveying information concisely. Today, software provides streamlined information on physical arbitrage like Sparta Commodities, Kpler and Vortexa for ship tracking and OilX, simplifying supply and demand models, all of which have further enhanced the delivery of physical information.

Change in the Market Dynamic

Prior to 2012, with the oil swap market mostly bilaterally traded^[66], it was very difficult for traders to compete with investment banks, IOCs and the top-five trade houses. Businesses with financial oil risk feared oil swaps markets, viewed them as opaque and resigned themselves to these participants having full control of them.

Hedge funds and purely speculative players would sit on the sidelines, sticking to exchange-traded futures, knowing they would struggle if they entered the swaps market. The IOCs and trade houses had deeper pockets, greater expertise and unrivalled access to physical information. They were owners or, at least, marketers of the physical barrels that would be used to set the price on settlement of the largest and most influential derivative contracts.

It was very common for traders in niche contracts like Dated Brent to trade swaps deals of huge significance bilaterally and, therefore, be able to keep quiet about the transaction, as there was no obligation to report. If you had live risk and were not an influential player with a large balance sheet, you were in a highly vulnerable position in these scenarios. It was common to be 'run over', a trading term for executing a trade only for the market to move immediately and aggressively in the other direction. Markets could move on a whim if something had changed in the physical market that priced the oil contract, and, therefore, without a physical footprint, you were completely in the dark. You had to rely on gossip and relationships to be kept informed, but this information would be selective and third-hand.

PRAAs were also highly reliant on fair value and trades from voluntary trading participants and brokers, which was difficult to get right and could vary wildly from firm to firm. This made the contract settlements varied, opaque and susceptible to distortions.

The financial oil market today is vastly different from the pre-2012 market. Interpreting fair value is much more straightforward, and even traders with a minimal presence who would previously have been in the dark can be on the market immediately. Markets start more quickly at the beginning of the trading day, and trading activity is much more consistent than the stop-start dynamic previously. With such improvements to liquidity, swap markets can't be bullied by one dominant player, and distortions are a lot rarer and more explainable.

These improvements are the result of the following changes which occurred over the last ten years:

- The development of previously opaque markets
- Increased transparency in both derivative and physical markets
- Much improved efficiency in oil derivative markets as a whole
- Decreased barriers to entry for wider market participants in the oil derivatives market
- Democratised oil market data and information
- Democratised knowledge of markets and their complexities

Together, the changes have altered market dynamics permanently.

Future of the Oil Derivatives Market

Future of Data

The biggest threats to traditional oil traders today are the diminishing power of physical oil market information and the increasing influence of financial market flows. Data and technology advancements have democratised derivatives and physical markets and their information. As discussed in the Supply and Demand Modelling chapter, ship tracking of cargos, global refinery outputs, and storage and oil production are mapped and modelled by research agencies that rival the IOCs, who previously kept such information largely to themselves.

With so many improvements in data sourcing, accuracy and assimilation, physical market information is increasingly available to the

average trader, making forward modelling of supply and demand increasingly similar – traders who reflect their view of the physical market in the derivatives market using this methodology are moving in herds. This can be exacerbated by the media, conferences and market meet-ups that drive consensus rather than differing views.

There is also a huge amount of competition in this space, with the value evident in the revenues generated and the mergers and acquisitions activity. But where does all the available information from multiple sources go? This monopolisation of democratised information has been seen in macro analyses. With so many banks, wealth managers and independent research agencies analysing the same things, this research eventually had no edge.

Derivatives participants are increasingly financially motivated to hedge, and as financialisation grows, its impact will increasingly drive markets.

But financialisation also impacts contract settlement – as this book has hopefully demonstrated. All oil derivatives contracts are inextricably linked, with knock-on effects from one market reverberating into another. Liquidity in pricing windows comes from this link; therefore, the forward curve flows directly impact the final settled price.

The change in market dynamics may be frustrating to legacy oil traders, as increasing financial volumes increasingly negate the disadvantages of not having physical assets, especially if increasingly sophisticated physical information can be sourced through subscriptions to research houses.

This increasing financialisation will likely be permanent, and by 2030, we could be in a very different space. Refusing to acknowledge this evolution will leave legacy oil traders by the wayside, posing a problem for existing traders.

What is the Solution?

Oil market participants must now include financial data and analytics into the current well-trodden supply and demand analysis path. Financial data and analytics should now include the following.

Physical Opportunities Presented in the Forward Curve (and Hedging Flows)

Hedging flows cleared on the exchange reveal the details of forward physical transactions, meaning an increasing amount of information on forward physical markets can be found through the visibility of market flows. By centralising trade flow information from all time zones and evaluating this against physical trading economics, the motives of physical companies can be known before the evidence is shown in the movement of the physical oil.

The forward curve can present opportunities in the physical market if those numbers are hedged. For example, an attractive GasNap differential may incentivise more use of naphtha in a blending pool than could have been predicted based on the forward modelling of gasoline. If this is locked in on the derivatives side, the implications on physical trading flows are very real. The forward refinery margin and the evaluations of light and heavy margins are also very important to a macro view of oil markets, as they are highly indicative of how refiners will choose to operate. Just like in the GasNap example, they will have ramifications on physical trading flows worldwide that can highly disrupt any supply and demand model.

Increasingly, analyses will incorporate the two worlds to accommodate this. There are already subscriptions, like Sparta Commodities, which use live fair values on the forward curve and model the physical arbitrage opportunity that the curve represents. If the arbitrage is open on paper, it can be locked in and will likely drive physical flow in that direction.

Position Analysis

Exchanges offer data on open interest – the positions that are still open – which is the first important assessment of the risk currently in the market. In addition, the volumes traded indicate liquidity.

Beyond this, it is important to assess market positioning, which is possible through the CFTC and ICE COT reports for exchange-traded futures and Onyx's COT report for oil swaps. Using these data sources, you can assess how long or short the market is by category using the CFTC and ICE reports or overall using Onyx's COT report.

It is important to assess how long or short the market is and how saturated this positioning is because this will guide you to figure out if you are early into a trend, joining a herd or going against the market, each of which will create different expected price action. Finally, you should assess the entry level for this positioning, which will allow you to judge when the market is close to stopping out or taking profit and goes a long way in predicting how the market will behave.

Market Flow Analysis

It isn't just hedging that will move markets but any market flow. Learning as much as you can from objective sources about where flows are coming from will deeply enhance your analysis. Given how the oil derivatives market is interrelated, significant flows in one market can have very real knock-on effects on another. At the very least, you should be away from these flows to gauge whether something is coming from the physical market, speculative positioning or just a knock-on effect from another contract's flow. All this information will allow you to form a trading plan and may impact your timing or the contracts you hold.

Volatility and Liquidity Analysis

As seen in 2022, changes in market conditions have very real impacts on how markets move. A deeper analysis of volatility, market volumes, exchange margins and anything else that will impact market conditions will be increasingly necessary the more sophisticated markets become.

Market Behaviour Analysis

Traders, forecasters and policymakers need to focus on **behaviours**. A financialised market is a game of poker, with price action determined by when and how traders use their risk capacity in the form of buying or selling

flows. Without appreciating what governs the new market and sourcing the intelligence required, you are playing an old version of the game.

The die-hard, legacy belief that physical fundamentals drive derivatives prices and a long position in a strong market is safe (or short in a bearish market) is no longer sufficient. The wealth of information available makes it necessary to consider the sum of people's actions when assessing market movements, not just the underlying fundamentals.

Future Growth in Derivative Trading Volumes

The oil derivatives market has weathered some of the most extreme scenarios in recent years, proving its resilience in the face of unprecedented challenges. In 2020, the market reached the pinnacle of its growth trajectory in terms of volume and open interest, only to be hit by extreme volatility and liquidity issues due to the Russia–Ukraine war. But the market stood strong through not one but two unprecedented scenarios in just three years.

First, the market was thrown into turmoil by the negative pricing scenario in April 2020, which introduced a new set of parameters that many believed were impossible. Then, shortly after, a dangerously one-sided bull market emerged against the backdrop of shockwaves in the physical market, posing an existential liquidity threat to the entire market.

Despite these challenges, the market infrastructure has proven to be antifragile. The exchanges resisted an LME-style cancelling of trades, and no serious scandals were reported in the face of extreme volatility. While there was a temporary slowdown due to liquidity issues, the long-term growth trajectory of the market remains intact.

If the market can withstand such extreme scenarios in quick succession, it is poised for continued future growth. The oil derivative market has proven it can weather any storm, making it a robust and reliable option for traders and investors alike.

Where Will Continued Growth Come From?

The trajectory to date has come from the benefits of democratisation. More market participants are taking more ownership of their price risk or entering into hedging activity for the first time. There is greater education on the oil derivatives markets and their relationship with the physical market,

which has encouraged more financial players. The physical market information does not hold the same value or influence in the derivatives market as it once did, creating more confidence in financial trading.

More Hedging Participants

In the professional world, hundreds of physical oil trading entities still have no derivatives capability. Meanwhile, the years ahead will be a busy period for the physical oil trading world. The lasting impact of the events of 2022 was a reshaping of the world's oil trading routes and urgency from governments to be energy secure. The new trading routes should almost guarantee more oil trading activity, which should then filter into price risk management in the oil derivatives market.

The growing sophistication of the oil derivatives market will encourage more sophisticated hedging and the use of previously opaque contracts that will have robust liquidity in the future. For example, the petrochemical derivatives market is still relatively nascent, just as propane markets were in 2015. Since then, the propane market has become heavily financialised and a sizeable, independent market with liquid contracts in all major global hubs. The propane market was almost entirely the Mont Belvieu contract in the US until the NWE propane on the CME increased in volume by 339% from 2015 to 2019 – to 21,688 lots traded. This volume is dwarfed by the volume of contracts referencing the basis in the Middle East and Asia.

Just like with propane, with a more robust market available, hedgers of niche physical products will turn their current hedging strategy from roughly correlated contracts to contracts of high correlation to the underlying physical they are hedging. We have seen this virtuous cycle repeatedly over the last ten years.

Growth of Integrated Oil Companies in the Oil Derivatives Market

The largest oil companies in the world are increasingly turning to oil trading to protect and enhance their bottom line by competing with the likes of BP, Shell and Total. For example, Exxon Mobil has made its intentions to grow in this area clear and seems to be motivated by the trading performance

of its direct competitors. It is prepared to invest in a proper build-out of the operation, and because Exxon is the largest oil company in the US, it will be managing an enormous amount of price risk in the oil derivatives market. Saudi Aramco, the world's largest oil company, is rapidly increasing its sophistication in the oil trading space and is already an influential derivatives market player. ADNOC, from the United Arab Emirates, is on a similar trajectory. It can't be overstated how much more volume these enormous companies will bring by building in this area. Previously, there were only the big three majors with sophisticated operations as majors, and the impact of one of these entering or exiting the market was immense. Now, even bigger companies on the physical side are joining in, and we expect many years of growth in physical oil trading and derivatives volumes on the back of this.

Increasing Sophistication of Trade Houses

The large trade houses are generating more income than ever and will want to retain their dominant presence in oil trading. This will likely come with more diversified strategies in the oil derivatives market, as finding an edge moves beyond the binary long or short punt on the back of oil fundamentals. These companies are investing heavily in their data feeds and assimilating data. Complex, hedge-fund-style trading techniques will become commonplace in these companies. A significant drive for systematic trading styles is underpinned by increasingly advanced risk monitoring. This will likely lead to expanded derivative operations and more volume and open interest in this market.

The monumental shift of physical oil trading flows in 2022 led to inefficiencies worldwide that will take years to iron out. This is the core of the oil trading strategy, and any growth in this area will also translate into greater volumes in the derivatives market.

Liquidity Growth in Existing Contracts

Many oil derivative contracts are still early in their growth trajectory and have plenty of room to develop. Exchange-traded futures still dominate daily traded volumes, but interestingly, they are increasingly comparable to oil swaps contracts in terms of open interest.



Figure 4.3 Total Swaps and Futures Open Interest listed on ICE and CME from 2014 to 2023

Therefore, there is actual risk in oil swaps markets, but there is plenty of room for daily traded volumes from day traders or high-frequency trading. For these participants to be in this space will require another boost of liquidity, which is highly likely.

Institutional Investors

Large corporates in the enormous institutional investing space still use relatively blunt instruments to express their desired exposure to oil markets. For example, most portfolio managers trade outright Brent or WTI crude futures contracts to hedge against inflation or express a view on oil. They could start to look increasingly towards product futures instead of crude, as these can serve as better reflections of inflation.

Heating oil and diesel cracks performed very well throughout the rebound from COVID-19 and 2022, when the physical oil market became very tight, but crude futures were much more unpredictable.

There are also time spreads, differentials of exchange-traded futures and

hundreds of oil swaps contracts that offer a multitude of de-correlated price movements and more precise links to localised physical markets than exchange-traded futures. These offer opportunities to hedge equity portfolios with more accuracy; for example, long exposure to a plastics business could be hedged against Asian petrochemical derivatives prices using contracts listed on the SGX or the correlated product naphtha on other exchanges. Alternatively, these contracts outside of Brent and WTI can provide exposure that may reflect an economic view of a specific market that can be used in a basket of other asset classes.

Investors or portfolio managers must have confidence in the markets and ascertain an edge to enter the oil derivatives market outside the common exchange-traded futures. With increasing data availability, this should be possible and could be an interesting development over the next few years.

Retail Traders

Retail investors are currently limited in their offerings from retail brokerages. However, there is a growing demand for retail trading, and recent business models have shown that retail customers are seeking access to new investment opportunities. The oil derivative industry is ripe for opening up to these traders.

This logic mirrors that of institutional investors, where many contracts are de-correlated and offer exposure to different views. As demonstrated in the chapter on forming a speculative trading strategy, buying or selling is based on specific information and hypotheses. Swaps markets can offer a purer form of these hypotheses, such as trading the E/W contracts during the Ever Given crisis, which blocked the Suez Canal and disrupted key import and export channels.

Much of the growth from institutions and retail is about access. Just like exchange trading opened the door for independent market makers, so can offerings from retail brokerages and ETF providers do the same for institutional and retail.

Real End-user Hedging

In the long run, I believe democratisation could extend to everyday users of oil products, just like Revolut did for the foreign exchange market. Before

Revolut, there was no way for individuals to take ownership of their foreign exchange exposure when travelling abroad. Cash providers and banks charged worse rates than the market, similar to the bid/offer spread in the oil market. This issue was even worse at airport currency exchanges, where providers exploited travellers' anxiety to charge inflated rates.

To determine how much you are being charged if you do this, check the market rate online for the relevant currency pair, such as USD to GBP or EUR to CHF, and compare it with the airport rate, ensuring the bid and offer are accounted for. If you are converting GBP to USD, for example, you are selling GBP to buy USD; therefore, you want the GBP to USD rate, and you want to look at the buying price (bid). If the rate is shown in USD to GBP, then you must invert the calculation; in other words, you are paying the offer to buy USD with GBP. The foreign exchange market is so liquid that a trader can expect a bid/offer of 0.1% to 0.5%, but airport providers might charge as much as 5% below fair value, causing a considerable loss of cash with every transaction.

This issue extends to using debit or credit cards when paying for things abroad. Shops and restaurants may offer to convert the price into your home currency, but this will not provide the real market rate. Banks may offer poor exchange rates and add administrative fees when transferring money between accounts in different currencies. I have experienced this personally, having chosen to effect a transfer in dollars between accounts to avoid the poor exchange rate on offer, only for my bank to automatically make the exchange at their rate without discussion, which resulted in the bank profiting from the transaction in microseconds.

To avoid this loss of value, Revolut provides market exchange rates, so you can convert cash in your Revolut account at fair value. You can then transact in a country or transfer funds as normal without worrying about this exchange rate bid/offer.

In the same way, in the oil market, it should be possible for the end user of oil products to lock in a fixed rate for their fuel and avoid volatility. However, there isn't always transparency between the market price and the duties and commissions the provider charges. Strong competition or a major disrupter like Revolut would be necessary to avoid forward fixing being costly from a bid/offer perspective.

The impetus would likely need to come from the petrol station itself to implement this concept. For example, Texaco could offer an account where

customers can convert cash to fuel credits at a given rate, locking in the price and avoiding fluctuations. However, this would require strong competition among petrol stations to avoid being costly and ensure pricing transparency.

Threat to Growth in Oil Derivatives

Renewable Energy Market

Over the next ten years, as investments in clean fuel and renewable energy increase, there is potential for an impact on oil derivative trading. This could lead to decreased physical oil volumes and reduced hedging requirements. However, this assumption relies on the uncertain projection that physical oil demand will peak in the next decade. Additionally, the oil trading model involves the global movement of oil, not just its production and consumption. The growth of Eastern demand for oil is expected to continue for several decades, which means oil traders are likely to stay active for the foreseeable future.

Furthermore, oil derivatives are priced by low volumes of physical oil and can continue trading without obstruction, even if oil production and usage decrease. Since derivatives trading is part of the financial system, it can prosper regardless of physical commodity trends, similar to the precious metals market. Nonetheless, there could be societal pressures on trading these financial contracts, which may cause trading companies to avoid them. Eastern markets may be less affected by such societal pressures, as seen in the growth of the Shanghai Futures Exchange.

Increasing Capital Requirements for Exchange Trading

In recent years, margin requirements for trading derivatives on exchanges have risen, which is a valid concern. In 2012, former US President Obama tried to tackle speculation by increasing margin requirements. A future leader of the US or the UK, where the most dominant exchange trading occurs, could implement similar measures.

Investment banks are poised to return to bilateral trading, and we could see a significant increase in this type of trading. This happened in 2022, and

our current rising interest rate environment will prompt hedgers to protect their balance sheets and trade in this way. However, the work that has gone into regulating the market may resist this transition. As witnessed over the last decade, the push for markets to be on the exchange ensures transparency and a fairer market for all. The volumes that go bilateral eventually end up on the exchange, as that is where the best markets are and where banks back out the risk from these transactions.

Future of Service Providing

With so much advancement in the oil derivative industry, service provision has had to evolve, albeit more slowly than in other markets. An interdealer broker's value going back ten years was determined by their ability to provide market numbers, which is now a given for all brokerages, with the efficiency of fair value determination and the widespread coverage of market makers.

At a minimum, a competent broker should be able to assist you through the entire trading cycle, from price discovery to execution. They must always be aware of fair market value and are expected to be liberal with that information on request.

But there is increasing demand for brokers to help you navigate markets similar to how a stockbroker would invest in equities, which eventually evolved into what we now call wealth managers. The pricing of complex derivative structures, assisting in forming a hedge or even helping form a speculative trading plan is all in the remit of the broker of tomorrow. These changes require more regulation and qualifications, like an advisory licence from the FCA in the UK, SEC in the US or MAS in Singapore, to operate in this function effectively.

Undoubtedly, the improved confidence of the average corporate hedger due to the improved transparency in markets has had a knock-on effect on market education. The novice can now hold brokers and investment bank types to account for their prices or at least to provide transparency in where their costs truly lie. They can be much more aware of market drivers than ever before and are increasingly aware of market nuances. This raises the bar for training providers, and what was useful training even five years ago is not as relevant today. As with the theme of this book, education needs to increasingly focus on financial markets, with so much of the physical world

covered. The strategy and execution of hedges can save a lot of money, and this is where the focus should be for training providers to add value.

For the future, all service providers need to become multifaceted in their offers and transparent in their expertise to benefit new market entrants. The common belief among current service providers is that this mentality is too open and will hurt a business, but I have not found this to be true. Market participants are more than happy to pay for services, provided they understand the true nature of their costs. Trust is hugely important, and it is no longer acceptable for brokerages and trade finance offerings to pull the wool over the eyes of those with little upfront knowledge. It is not helpful to offer training services that hide elements of the process or avoid simplification to retain the perceived value of the training provider. Customers should demand clarity and practical outcomes; however, they should also not expect a silver bullet for predicting markets or best execution – it is an iterative optimisation process.

We all know that customers will be loyal if you create a trustworthy relationship. If you create empowered customers in oil derivatives, their growth will benefit you by greater volumes, commissions and money in the market. They will want to retain the trustworthy partner who helped them to flourish. Everyone in the oil derivatives market should want this, and the more this happens, the more service providers will evolve their service via data, technology and education. From what I have seen, the protectionism in the oil derivatives markets has stagnated innovation and fair market practice.

But whether service providers believe this or not, if you fail to adapt, you will eventually be picked off, which is not a particularly profound thing to say.

The Rise of Direct Trading as a Service

If it is just about getting a good price with efficient execution, there is the option to go directly to the market makers to trade, rather than via the brokers, with the immediate advantage of not having to pay brokerage commissions. In this scenario, market makers will provide the same price they would have provided in the market and therefore get the same benefit of lower costs.

This direct trading as a service has grown significantly in the oil derivatives market as independent market makers have increased their

volumes. Direct trading works much the same way as a brokerage but only trading with one counterpart – the market maker.

I expect this area will continue to grow, as it makes the most sense if you are just looking for execution. Advisory brokerage and direct trading may merge in the long run, supported by software, and this will look a lot like other financial markets, for example, foreign exchange.

Trade Finance

Banks and other providers offering credit for hedgers to trade bilaterally continue to be profitable but have had to adapt over the last ten years and will continue to adapt in the future.

Pre-2012, there were a huge number of commodity desks at different investment banks, all fighting to maximise volume and doing well. As the margins dissipated and competition increased, the trade finance model became increasingly about offering complex derivative structures, or ‘non-vanilla’ trades, such as the zero-cost collar. More recently, this practice has fallen out of favour, and trade finance is about the financing itself. Being aggressive and winning business is all about being competitive on financing, such as the no margining until settlement discussed in the chapter on forming a hedging strategy.

It is unlikely the bilateral trading business will ever disappear, but transparency in deal-making will improve. Purchasable data sets, such as market bid/offer data, can be used to compare the bilateral bid/offer and determine how much commission the financier is charging.

I expect a transparent and predefined ‘credit spread’ will be used more in the future. The trade financier will define the cost of financing as the credit spread, which will give a monetary value when multiplied by the volume. The end user will need to accept this cost of doing business, but at least they will know how much they are being charged.

Influencing the World’s Oil Price – Lessons from 2022

These days, derivative volumes have become so large that it makes the most sense to price physical transactions with reference to the financial

markets rather than the other way around. This means that derivatives set the physical price, so any distortions genuinely impact the physical world and the underlying consumer. It is a case of the tail wagging the dog. Given that financial markets are setting most of the prices used in physical negotiations, volatility in these markets has become a concern and the subject of government agencies' and policymakers' discussions of late. The prevailing question seems to be, is the market more volatile because of financialisation, or is it less volatile than it would have been?

The year 2022 was a defining year for the interaction between the physical and financial oil markets and the overwhelming impact of financialisation on prices. Leading up to 2022, the outright Brent futures price had been on a long-term bullish trajectory since the lows of April 2020. The market was being stirred up by a narrative that had taken hold and justified the continuation of this rally, which focused on the lack of investment in the oil industry and the lack of spare capacity in major producers.

Despite the rising Brent Futures prices, OPEC+ signalled to the market that it would maintain its coordinated production cut first announced in mid-2020 and the schedule of a steady release of output every month that would last until September 2022. The physical market was becoming short of oil with this oil held back and the rapid growth in demand, with stocks depleting and forward demand indicators on the forward curve (physical differentials and time spreads) signalling an expectation of a very tight physical market.

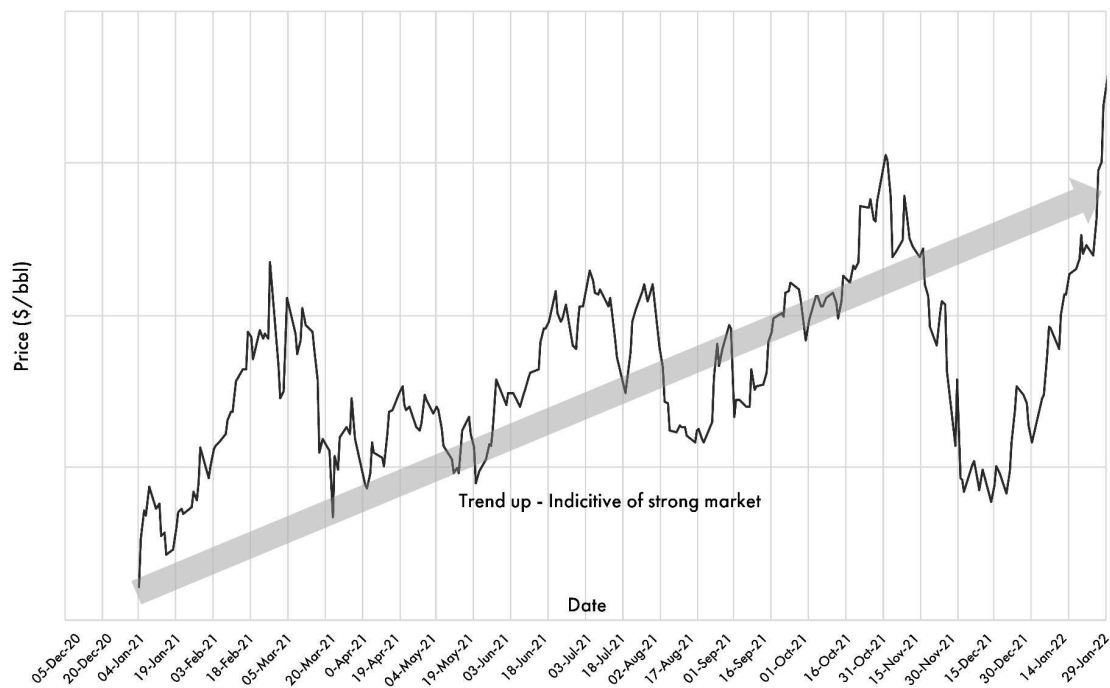


Figure 4.4 The Brent Rolling 6 Month Calendar Spread Before the Russian Invasion

Industry experts and analysts across various platforms consistently advocated for an increase in oil prices, causing many producers to hold off on entering the derivatives market for hedging purposes. This mentality was captured well in a *Reuters* article in which a shale oil executive expressed their reluctance to hedge, stating, ‘With every bank saying that oil will be at \$90–\$100, no one is going to put hedges on right now’^[67].

On 14 March, *Bloomberg* reported on the new mentality from shale oil producers, with one analyst quoted as saying, ‘Management teams have greater [fear of missing out] being hedged in a runaway market’. At this time, it was public knowledge that Pioneer Natural Resources Co., the biggest oil producer in the Permian Basin, had closed out almost all of its hedges for 2022, while shale producer Antero Resources Corp. was the ‘least hedged’ in the company’s history, according to its finance officer^[68]. Devon Energy Corporation was also only about 20% hedged, compared to around 50% normally, and it signalled it had plans to stay that way with all the bullish rhetoric in the market.

The positioning was, therefore, exaggerated beyond the direct speculative positioning in the derivatives market. Withholding hedging that the market has become accustomed to is, in effect, a long position and starves

the market of much-needed sell-side liquidity.

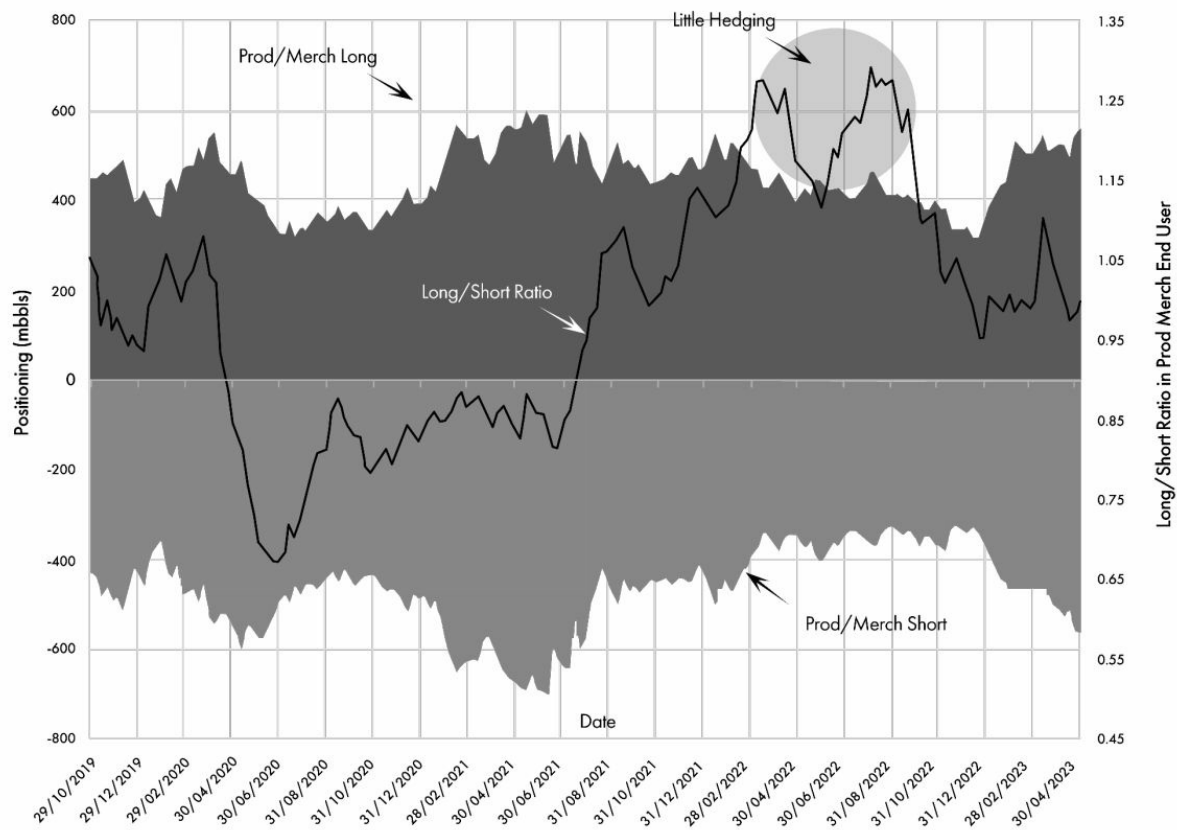


Figure 4.5 The Long/Short Ratio of Producer/Merchant/End User Open Interest in West Texas Intermediate-related Contracts During the Move-up in Flat Price from 2021 to Mid 2022 (Source: CFTC)

At this point, the market was very one-sided, and on 26 November 2021, the market was warned of the dangers of forming a speculative bubble. In just one day, there was a \$10/bbl sell-off in Brent futures, the largest one-day sell-off since the peak of the pricing war in 2020.

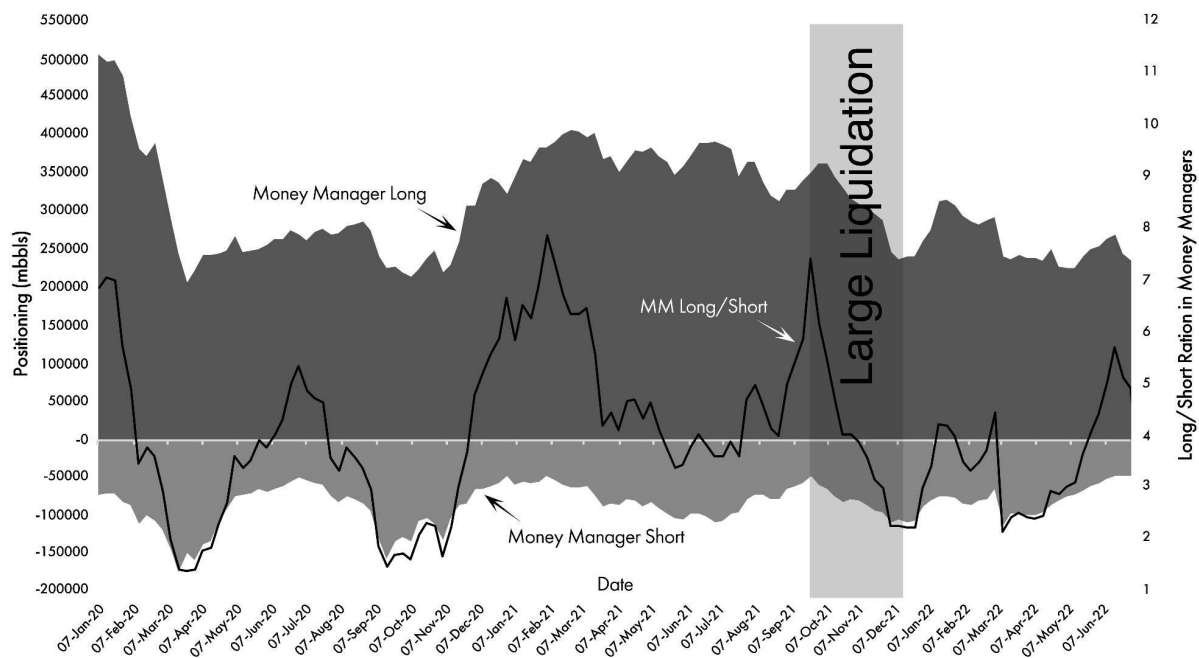


Figure 4.6 The Period of Liquidation of Money Managers Positions which Resulted in Black Friday Sell-off Last Year (Source: CFTC)

This resulted from a long-positioned market looking to exit its positions all at once, which had been made clear by the CFTC and ICE COT data. It wasn't just that traders unwound length but also new selling flows from producers who didn't want to miss out on the high-price environment that was slipping away with the downward price momentum. In this instance, the market's behaviour was also influenced by being towards the end of the year. Most trading years are assessed as a calendar year, and bonuses are paid the following spring in line with the prior year's performance. The new selling flow that entered the market could have been initiated simply as a derisking trade towards the end of the year to close out the books ready for the Christmas holidays and the new year. Whatever it was, the extreme sell-off showed the state of vulnerability the market was in.

But the market was ultimately not deterred, and by early 2022, the market was back in full swing, brushing off the sell-off and showing warning signs of getting into crowded trade. Reports in the media returned to push the narrative that there was a lack of investment in the industry and insufficient spare capacity from OPEC+, causing the group to miss its production targets. Traders were united in the belief that the markets could only go one way: Russia–Ukraine tensions were mounting, with the real prospect of a war on

Europe's doorstep. Once Russia invaded Ukraine, this perfect storm pushed the market to extremes.

Brent crude and WTI futures contracts were already struggling for sell-side liquidity after rampant speculation leading up to this point. Once the news broke, the reaction from the market put the financial system under serious strain.

Brent crude and WTI futures contracts trade on the exchange in a closed system, so the market can only react to buyers and sellers who enter into these contracts on the ICE and NYMEX (CME). While they are different exchanges, WTI and Brent contracts on these exchanges are connected by algorithms trading identically settled contracts, which instantly close any arbitrage opportunity that might arise, thus keeping the movements of these contracts aligned. The problem in such extreme scenarios is the very real prospect of insufficient selling volume in the market to satisfy buy-side liquidity. No available sell orders mean buyers must keep raising their prices until sell-side liquidity is found or enters the market.

The algorithms responding to the one-sided order book can move to find this liquidity instantaneously, and they did so. The market consequently moved up aggressively until the buy orders were filled, which was a very large move. But once the orders were filled, in a rush of mania, the crude futures prices gapped all the way back down. The price had gone up enough to invite new selling flows, but there wasn't enough liquidity to immediately satisfy this new selling flow, shifting the order book to be weighted towards sellers. This new selling could have been short-term traders who were long and waiting to see how far the market could go deciding it was the right time to take a profit. Or it could have been hedgers choosing to enter orders at around these levels, which were attractive for producers, and the algorithms had to fight for liquidity the other way. Or it could have been opportunistic short-selling.

Whichever of these was true, buying proved to be temporary and panic driven. This extreme volatility was clear evidence that the market's risk capacity had been stretched. Another spike like this followed as tensions and media hype built up once again.

By April, the flat price market was struggling to get back to the heights from the beginning of the invasion, but the industry was still doing its best to talk the market up. Even in the ETPs, which can offer liquidity to crude futures via crack differential trading, there was significantly less interest from

money managers in shorting. Oil products such as diesel and gasoline, as reflected in the CFTC, had almost no short positioning. The futures time spreads and oil swap differentials, more closely linked to physical markets, were surging to all-time highs across the board. Refinery margins completely blew out, and the oil derivatives market had reflected a bullish market in almost every way possible, with no end in sight.

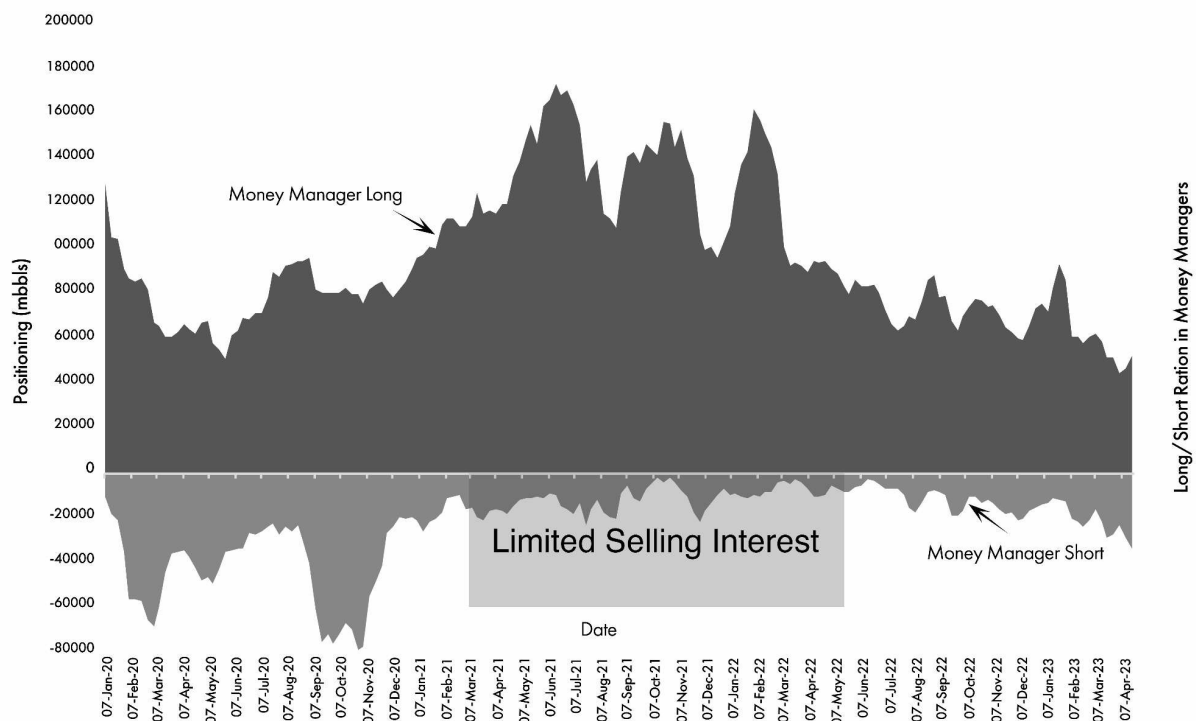


Figure 4.7 The Limited Sell-side Interest in Gasoil Futures Contracts From Money Managers in January 2020 to April 2023 (Source: CFTC)

Despite all this, prompt Brent crude futures contracts not only couldn't return to the highs set at the peak of the crisis but began to sell off aggressively back to below \$100/bbl by June. Throughout the rest of the year, there was a struggle to stay within range of this psychological level, and before the end of the year, it gave way and fell into the \$70s/bbl. What happened?

Exchanges and their clearing members faced a scenario they couldn't model, and it wasn't just oil under pressure. The global markets had just witnessed unprecedented events in nickel markets on the LME, where prices had gone up 250% in 24 hours, and margin calls threatened to devastate the industry in much the same way as prices had risen to their peak for the year in the height of the Ukraine invasion. It turns out that this was the consequence of a short squeeze on an enormous position held by Xiang Guangda of Tsingshan Holding Group Co., and there were simply not enough sellers to avoid the price spike. This price spike was so aggressive in nickel that the LME felt prices were 'becoming disconnected' from 'physical reality' and decided to cancel all trades that had occurred in that session. This was unprecedented and would lead to severe backlash from investors that has

continued until today and threatens contracts and the exchange itself.

Oil exchanges would have seen this. They would have realised that the oil market is much bigger than the nickel market, and the ramifications of this type of event in the oil derivatives market would be an overnight existential crisis for the business and threaten the exchanges' standing as hubs for oil derivatives trading.

ICE's clearing house decided that it needed a proactive capital injection to steady the waters. So it aggressively ramped up the initial margin – by as much as 90% for some contracts. The exchange did this in January, and it needed not one but two increases during March before it could finally stabilise the volatility to some extent. As a result, the market consolidated and began a more stable rally back up beyond \$120, while Gasoil and heating oil prices were closer to \$200/bbl.

However, the sustained impact of the heightened capital requirements would end up making a difference. It wasn't just that margins were high; the oil price was also very high, which meant the financing of physical transactions was being squeezed. Banks financing access to the exchange, bilateral trades and physical transactions saw the credit situation as alarming and raised customers' capital requirements beyond what the exchange's clearing houses asked.

The capital requirements considerably strained even the largest trade houses' and IOCs' balance sheets. All companies had to be selective about how they deployed cash and forced trading companies to reevaluate their positioning in outright futures. It was a particularly hard time for risk managers. Speculative traders of the crude futures Brent and WTI were participating in the rally from early 2022, generating a good income. They would have wanted to put their foot on the gas as the Russia–Ukraine conflict worsened. However, once volatility became astronomical, risk managers had to adjust companies' exposure to the outright price, with the risk being a function of the expected move (volatility) multiplied by the position size. They were therefore forced to be more restrictive on the traders and cut the long positions to align with their assessment of volatility, even if traders' positions were well in the money. There was also pressure from above, with executive management focusing on capital allocation in the overall business. The physical trading business in commodities is highly capital intensive, and while there were considerable opportunities with the physical flows reshaping around the world in real time, operations still needed to be financed to

participate.

There is no doubt that management was looking at the derivative function and thinking it was not worth the return to be holding almost double the capital on the exchange in return for holding a very volatile position in unprecedented times, especially when recent moves had put many positions under pressure after gyrations from the peak of \$138/bbl on Brent futures.

The trimming of positions seemingly happened all at once across the industry. But instead of a risk reduction, it became a significant liquidation event, just like ‘Black Friday’ for oil markets on 26 November the year before. Traders all fought for buy-side liquidity to unravel their positions, and all the potential sellers sitting on the sidelines waiting for a higher price to materialise suddenly rushed in to avoid missing out on locking in such high prices. The momentum from a downward trend in highly financialised markets can be incredibly difficult to stop.

Sure enough, with Brent futures on a long-term downward trend, research agencies and commentators trying to talk the market up could not stop the market move. JP Morgan’s well-publicised \$380/bbl call^[69] was followed days later by a \$10 sell-off over a few days. Even another OPEC+ cut later in the year wasn’t enough to stem the bearish sentiment taking hold. The damage had been done to investors who had rushed into the space to hedge against inflation or gain oil exposure and had been badly burned. If the market couldn’t sustainably perform in such a binary bullish event, why would it be a good trade after a \$50/bbl sell-off?

The market during 2022 demonstrated the pitfalls of a one-sided market but was also a watershed moment for the power of sentiment on the entire oil market and its ability to overwhelm physical signals.

How can a financial market mechanism be so devastating to the physical world? What happened to supply and demand driving oil prices? Analysts who make their name and vie for recognition from their market calls using supply and demand would blame liquidity constraints once derivatives experts explained what had happened. Traders called the oil futures market ‘broken’. It is no longer good enough to lament a market that isn’t moving how you expect: analysts, policymakers and traders are demanding to know more.

So, what does this mean for those who seek to use their influence to regulate markets in the modern oil market? It seems that the concern at this stage is that so much of this market is traded – and therefore influenced – by

algorithms. It is very difficult to see how you can control computers. Algorithms are the buzzword whenever financial markets are mentioned and are the generic explanation of trickier markets in recent times.

Andy Hall, a prominent oil trader, has been critical of the increasing use of algorithms in oil trading. He has expressed concerns that algorithms exacerbate market movements and lead to unintended consequences, as they often react to market movements automatically and indiscriminately. In an interview with *Bloomberg* in 2017, in which he discussed the closing of his hedge fund, he stated that ‘Algorithmic trading systems have increasingly come to dominate’ and that ‘Under current market conditions using an approach based primarily on fundamentals has therefore become increasingly challenging’^[70]. He has also criticised the use of algorithms in creating false liquidity and reducing the number of market participants, which can ultimately harm market efficiency. This view contradicts what many other industry participants believe: nothing should be changed, and more trading activity provides more liquidity and greater market functioning.

In my belief, it comes back to the need to model the behaviours of trading participants. Automated trading strategies are based on human coding, and it is not a given that they all make huge sums consistently. In fact, CTAs (who usually use trend-following algorithms) have not done well at all with the volatility in recent years. So their trading can only be classified as adding liquidity, as they traded much volume without benefit.

Market sentiment is complex, with motivations coming from human emotions like the fear of missing out, denial and greed. Consensus forms because we tend to seek comfort in crowds, as shown in the 2022 example. Surely, if anything, an objective machine learning algorithm with no bias would identify the dangers of herd trading and do anything but trade with a herd mentality that leads to these exacerbated moves.

My opinion is that if participants, policymakers and governments want lower volatility in markets, they should focus on seeking to calm market participants. Volatility is a measure of exuberance or uncertainty: in other words, emotions. The media seeks headlines because that is what draws attention. If you feed this, it filters into traders’ positioning and grows out of control, as shown in 2022.

Why didn’t the 2023 OPEC cut on 2 April have a sustained bullish impact on markets? There was a considerable liquidation of majors’ and trade houses’ length as the rest of the market bought into the news that there would

be less crude oil in the export market. They didn't believe in their position anymore and saw this as an opportunity to liquidate it, which is challenging to do without moving the market, given the size of their holding. The wider market, seeing the lack of reaction to the news (in fact, the opposite after a few weeks), got spooked and started selling out their new length.

You can read all the news that followed this volatility – about stagflation or demand worries – but this is sentiment playing out in the context of the rest of the year and the year before. The entry point at which the trade houses and majors entered Brent crude futures would have been lower, given where prices were at the end of the previous year and when positions were entering the market. So it was not only a liquidity opportunity but also a profit-taking opportunity.

Is the news a surprise to the market? Does anyone have the energy to react anymore with such volatility in the past, or would they rather sit on the sidelines? You can gauge the market's feelings at any time, much like governments gauge the population's feelings to determine how a new policy will go down. This is what testing is about for those marketing products to consumers.

Influencers have to ask themselves, what is the collective mindset right now? If you want action, you have to rile people up, playing on their fear of missing out if positioning is low, and drive them into action. Or can you play on their fear of loss if markets are too invested one way and stimulate a flight to safety? If you want the market to calm down and stop acting irrationally, you need to create confidence that things will remain stable and that there isn't another market shock around the corner. If the result is many investors exiting positions that, in turn, causes an adverse price reaction, then this is simply a necessity so the market can consolidate. From there, you can rationally influence market participants in the long run, but not before.

Lessons from Other Markets

There have been cautionary tales from the wider financial world in which markets are ahead in the growth of financial volumes.

GameStop

The GameStop saga was a monumental moment in financial history that

caught the world's attention. It was a story that seemed almost too incredible to believe: a group of retail investors banding together on Reddit's WallStreetBets to stick it to the Wall Street elite, particularly the hedge funds which had shorted GameStop stock. The story began with the discovery that certain hedge funds had overextended themselves in their short positions on GameStop, a struggling video game retailer.

What followed was a frenzy of buying from small investors that was fuelled by social media and led to a short squeeze. As the buying pressure continued, the stock price of GameStop skyrocketed, reaching levels completely disconnected from the company's fundamentals. The hedge funds that had shorted the stock were forced to buy back their shares at a huge loss, further driving up the stock price.

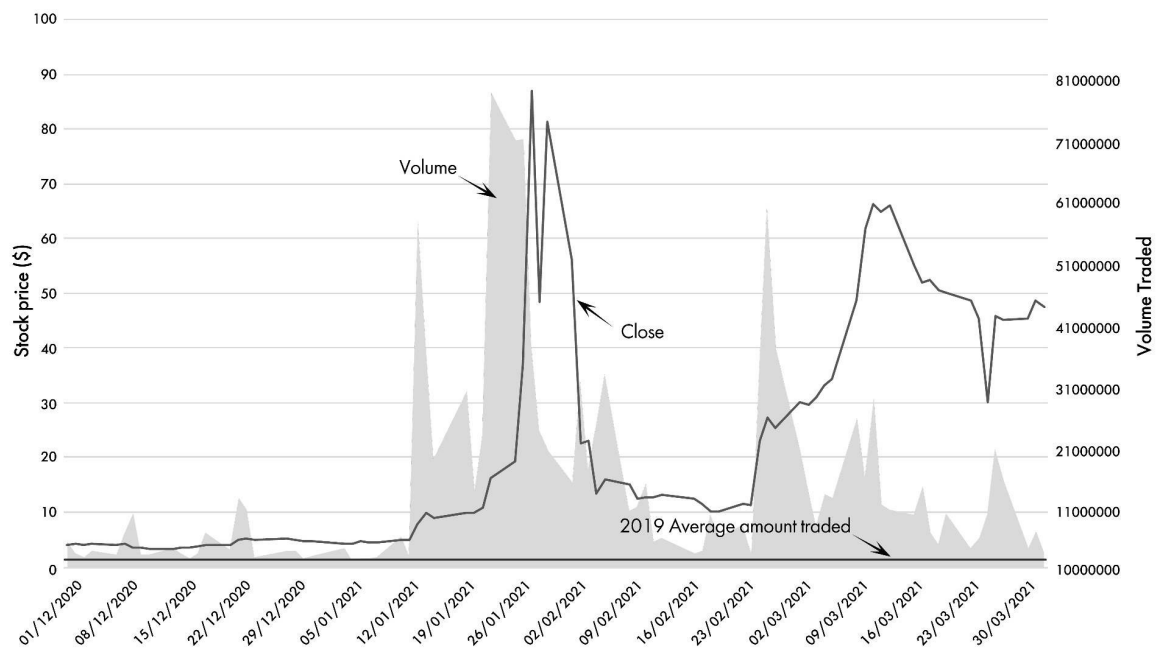


Figure 4.9 Gamestop (GME) Stock Price and Volume Traded December 2020 to March 2021(Source: Yahoo! Finance)

The short squeeze quickly became a target for the public, and it became the ‘in’ thing to join in the buying and force the hedge funds to stop out. Over time, the movement became about ‘sticking it’ to the hedge fund community, and a diverse range of participants got involved, fuelling the hysteria. The situation was so unprecedented that it led to dedicated documentaries on Netflix, lawsuits and endless blogs and articles.

However, despite the hype and excitement surrounding the GameStop saga, some experts and prominent figures in the industry, like Michael Burry, were cautious about the whole event. Burry, who was initially long the stock, made his misgivings about the GameStop event very clear with a Tweet that said the event was ‘unnatural, insane and dangerous’. He must have profited from the trade but most likely took his long position off way before the dizzying heights reached during the peak mania, with 2,500% gains at one point. During the height of the GameStop frenzy in January 2021, members of Congress in the US began to take notice of the situation and the potential risks it posed to the financial system. On 18 February, the House Financial Services Committee held a hearing to discuss the events surrounding GameStop, which featured testimony from various key players involved in the saga.

Among the witnesses who testified at the hearing were Vlad Tenev,

CEO of Robinhood, the popular trading app that played a significant role in the GameStop frenzy, and Keith Gill, the individual known as Roaring Kitty on social media, who became a prominent figure in the movement to buy GameStop stock. The hearing also included representatives from the hedge funds caught on the wrong side of the short squeeze, such as Melvin Capital.

During the hearing, members of Congress questioned the witnesses about various aspects of the GameStop saga, including the role of social media in driving the buying frenzy, the potential risks posed to retail investors and the actions of Robinhood in restricting trading of GameStop and other heavily shorted stocks.

The GameStop saga brought the concept of a short squeeze into the minds of the retail investor and showed just how detached a financial market can be from its fundamentals. The event also highlighted the potential risks and dangers when social media and the stock market collide. It highlighted the power of social media in financial markets and the challenges that traditional finance faces when attempting to regulate it. The GameStop saga ultimately changed the game, and it will be interesting to see how it impacts the financial industry moving forward. In the months following the hearing, there have been ongoing discussions among lawmakers and regulators about potential changes to the financial system to address some of the issues raised by the GameStop saga. Some proposed solutions include increased transparency, regulation of short selling and efforts to ensure retail investors are not put at undue risk by the actions of hedge funds and other market participants.

Cryptocurrency

Many lessons can be learned from the cryptocurrency market, but in the context of learnings for the oil derivatives market, the role of market hype in driving price action is most relevant.

Cryptocurrencies are often subject to extreme price fluctuations based on rumours, news headlines and speculation. At one point, Elon Musk's Tweets would send the dogecoin market flying, with no real fundamental stories to attach to the movements. Die-hard fans of the concept can be persuasive but also dangerously in denial at times.

The knock-on effect of early movers making millions and benefitting from the hype has led to a global fear of missing out on trades, leading to

trade participants probably in the hundreds of millions. It's one thing for professional traders to take losses, but we have all heard of the horror stories of everyday people who have bought into the hype and lost their life savings and beyond.

Just like in oil markets, bulls in cryptocurrency will insist on the most elaborate reasons why the market should or is moving up, leading to a one-way consensus that can be so dangerous in market reversals. I'm sure a lot of you will know very well the fanatic attitude some have had about cryptocurrency in the past, and this is also the best way to describe mentalities in the oil market in the past.

There is a deep human desire to be part of a club, and anyone outside of that club who questions its principles or ideas can be shunned or ridiculed. When these things unravel, as cryptocurrency markets sometimes have, there doesn't seem to be an 'I told you so' attitude. Rather, it's like watching a group of people awake from a trance and then go about their normal business. So what about us gets us into these frenzies of overexuberance in which we completely disregard reason for long periods? Whatever it is, capturing this is the essence of winning the narrative, and going against this (even if rational) is rarely a good strategy. Instead, you should be looking for ways to profit from the move without becoming a fanatic yourself.

Ed Seykota said it well in his interview with Jack D. Schwager for his book *Market Wizards*, 'Trying to understand the markets is futile. I don't think it makes any more sense trying to understand the market than it does trying to understand music. A lot of people would rather understand the market than make money'[\[71\]](#).

Another lesson from cryptocurrency is the lack of regulation and oversight that has made fraud and manipulation run rife. Unlike traditional financial markets, few safeguards are in place to protect investors. This is understandable with the market gaining so much traction globally so quickly, and its whole principle is based on being decentralised and away from government control. But for the trader, there are also worries about execution, especially in times when markets are aggressively selling off. Cryptocurrency exchanges will suspend trading or block access to their platform altogether, which can trap investors looking to exit. High-frequency trading is already dominant in the market, and trading in and out efficiently is not always possible. So even if you have a strong, proven strategy, you still have a lot of risk in executing the strategy if the infrastructure you are trading on is shaky.

Cryptocurrency is increasingly moving to established exchanges as futures. The market should welcome this to improve the trade cycle and create more certainty in its conditions, which is a lesson for renewable markets that hope to start in the near future.

The recent Future Exchange (FTX) scandal highlights the failures of the cryptocurrency market coming together. The lack of oversight is one thing, but the willingness to believe in the story led to many of their problems. The founder, Sam Bankman-Fried, now a convicted fraudster, played on the market's hype, and many paid-for influencers who backed the exchange have now been left blushing.

All in all, the cryptocurrency market has shown that it is important to remain level-headed and not get caught up in market hysteria. The fear of missing out can drive traders to make irrational decisions and chase short-term gains. Instead, traders should focus on developing and sticking to a sound trading strategy, even when the market is experiencing extreme volatility.

Inflated Technology Stock Valuations

As inflation took hold on the back of commodity prices skyrocketing and other macro factors, the big loser for global markets was not the equity market but the incredibly high-valued growth stocks, in the NASDAQ, in particular, throughout 2021 and much of 2022.

Like cryptocurrency, a wide-eyed frenzy of growth stocks went on a relentless rally that defied all classic valuation models. As monetary tightening came in with rising interest rates, the flow of money out of these stocks was aggressive, moving into value stocks – stocks deemed undervalued as the sector is currently out of favour (like oil companies!) – and into cash.

But it wasn't just stocks selling off; it was a change in attitude. With so much of the investing world no longer excited about high levels of growth and willing to buy into speculative forecasts from the next best thing, a more down-to-earth mentality has been evident ever since. In the investor world, private equity and public investors alike, there is now a demand for businesses to be at least cash flow positive, while discounted cash flow models are coming under scrutiny.

One particular stock that suffered was Peleton, which came down more

than 90% after being valued at around \$44 billion in market capitalisation at its peak in January 2021, despite operating with large losses. The hype in the stock from the swathes of subscribers during COVID-19 was an exciting story, but the business, even today, is unsure whether it can achieve cash flow positivity. *The Wall Street Journal* reported that Chief Executive Barry McCarthy was giving the unprofitable company about another six months to significantly turn itself around and ‘if that fails, Peloton likely isn’t viable as a stand-alone company’^[72].

Just like in cryptocurrency, as long as the bubble is working for the consensus, it is very difficult to break the cycle. If you are looking to be contrarian, timing is incredibly tricky, and given how common these bubbles are, it doesn’t help to be righteous about it. As a trader, your job is to navigate markets however you find them, and there is no point being derisory about everyone around you, as it is your job to figure out how to use it to your advantage.

Title Transfer Facility Price Cap

While oil derivatives were volatile, it was nothing compared to natural gas markets. The TTF contract, used as a benchmark for European natural gas, in particular, went on an incredibly wild ride, and the ramifications on the market were plentiful.

The margin calls on short hedging positions exacerbated the moves higher as these hedges were being pressured to be bought back, acting like a short squeeze even though the hedges were against physical longs. At one point, Equinor made headlines by estimating that margin calls could be as much as \$1.5 trillion, and fears of a Lehman-style collapse gripped European nations. European governments responded by lending billions to utility companies to plug the gap while alternative natural gas flows were sought to ease the tightness in the physical market.

The governments had to intervene. On the one hand, governments dealt very well with the situation, immediately seeking to replace the physical flows and preparing for the consequence of the energy war through plans to ration. The Liquid Natural Gas (LNG) market, sourced from the US and Qatar as a seaborne transported commodity, proved instrumental to replacing flows.

On the other hand, the government’s involvement in the financial gas

market moved from a securing cash flow to direct intervention and a price cap on the TTF benchmark. This was agreed at the end of 2022, with 180 EUR MWh set from 15 February 2023 to last one year and applied to the first three available contract months of the TTF contract^[73].

The problem with the price cap on the TTF futures contract is that if the physical market goes above the price cap, the TTF futures relationship with other derivatives contracts that make up the financial natural gas market breaks down. Like Brent crude futures, TTF futures are incredibly important for how the wider financial market and the forward curve determine fair value and how the contracts move together. Market makers and general traders of the LNG contract JKM trade TTF as a hedge like Brent crude futures and oil swaps contracts. There are also TTF swaps, and it is common for traders in the market to have huge TTF futures versus swaps positions that require managing as the swaps expire. This was covered at length in the benchmark chapter.

When trading out futures as the swaps expire, traders use the TAS contract; however, in the scenario of the price being above the price cap, those with short TTF futures against long TTF swaps will be unable to buy back the futures leg because the contract is physically deliverable, and why would anyone sell the futures contract if they can sell the physical product much higher in the open market? So in this situation, the market maker or financial trader will be stuck with short and physically deliverable TTF futures contracts, obligating them to deliver real physical gas on contract expiry. This is an unacceptable level of risk for the market maker. Therefore, they would be highly cautious about trading the contract in the first place, which would undo the liquidity improvements of the last decade.

The price cap on that TTF benchmark didn't get a huge amount of publicity, most likely because the levels decided now seem far off, with the prompt contract now trading in the 35–39 EUR MWh range. However, there is always the risk of returning to the high-price environment. The market was very unhappy about the cap when it was announced, but it wasn't just the traders: the ICE exchange, which owns the TTF contract, responded that the consequent margin calls arising from this change would reach \$33 billion, putting the industry under even more strain than it was at the peak of 2022. The exchange went even further, with a plan to move the contract to the UK to sit outside this price cap.

The price cap on a benchmark can only be considered necessary for

governments to cap their exposure to TTF prices after guaranteeing utility hedging, and it is a worrying precedent. As the ICE exchange is most likely all too aware, these kinds of actions lead to severe mistrust and are potentially fatal for the long-term viability of the contracts, just like what we now see after the LME nickel scandal.

Renewable Market Derivatives

The renewable energy market is poised for significant growth and enjoys strong political support to facilitate its expansion. As an oil trader, one can draw upon valuable knowledge and experience to help develop this industry. Despite one's personal feelings about the oil sector, it is undeniable that highly skilled individuals have played a vital role in shaping the market's system over the past four decades, and some of these experts will likely be instrumental in guiding the transition.

One essential lesson that can be applied to renewable energy is the importance of a well-functioning derivatives market. In oil, the derivatives market has facilitated the emergence of entirely new supply flows, such as the recent US shale oil boom. Independent oil producers, also known as 'wildcatters', were able to secure financing based on their reserves thanks to the derivatives market. As a result, they were able to scale up their operations by hedging forward their production on the WTI futures contract, enabling them to survive the 2014 crude oil futures crash.

Similarly, in the renewable energy sector, developing a robust forward curve for renewable-linked markets will help ensure cash flows, providing long-term security for new projects in the process of setting up. Such financial stability is crucial for the growth and success of the renewable energy industry, and lessons from the oil market can be valuable in achieving this goal.

Development of the Power Market

The power market is the most important to get right as early as possible. It's clear that the power market will be vital to the renewable market. There is already huge interest in its derivatives market that will likely grow in volume by multiples within the next five years.

Plenty of contracts exist; however, liquidity is nowhere near where it

needs to be, and liquidity considerations exist on the physical market.

The current difficulties in the power derivatives market are the constant volatility and uncertainty in the physical market that filters into financial trading. Therefore, the physical power market must evolve to behave more reliably and efficiently. This comes down to storage and arbitrage flows.

Physical Storage

Much confidence in the oil markets comes down to its global storage infrastructure. Oil is ultimately a stable liquid: any time it is not needed for immediate consumption, it can be placed in a storage facility or tanker. For example, Venezuela has had sanctions for years; however, the oil that would have been exported can stay in storage facilities and tankers. The situation is similar in Iran. When the situation changes politically, or a new buyer emerges who can navigate the sanctions, the oil can flow again with minimal product loss.

In situations of global market oversupply, storage provides an opportunity for a ‘cash-and-carry’ trade, an arbitrage strategy in which a trader simultaneously buys a physical asset and sells a related futures contract. This strategy aims to profit from differences in the prices of the asset and the futures contract.

The trader purchases the underlying asset at its cash price and simultaneously sells a futures contract to deliver the asset at a later date. The futures contract is priced based on the expected future price of the asset, considering factors such as interest rates, storage costs and supply and demand expectations. If the futures price is higher than the cash price, the investor can lock in a profit by buying the asset at the lower cash price and selling it at the higher futures price when the contract expires. This strategy can be particularly useful in markets where the cost of storing the asset is low and the futures contract is priced higher due to the cost of carry, including storage costs and financing charges.

So storage economics lead to real financial flows entering the market and are part of the market dynamic. Traders know to expect land storage, certain levels of time spreads and floating storage at deeper contango levels. This helps set a range and a floor to crude futures time spreads, which are critical for forming the forward curve for all oil derivatives markets.

But electricity storage is based on batteries and has a long way to go.

With solar and wind assets, energy production is continuous and, with the lack of storage, can exceed demand, leading to negative marginal costs. In such situations, producers must pay someone to take their excess energy production. Storage asset owners can be paid to take excess power and sell it at higher prices when energy prices turn positive, capturing the difference as revenue. This scenario briefly existed in 2020 in the oil market in the WTI contract and a few localised physical markets in North America.

The power, or electricity market, needs better storage capabilities. More storage means more buyers at negative prices can profit from this cash-and-carry trade, resulting in less volatile prices, greater energy security and more predictable returns for renewable energy assets. More predictable returns encourage greater investment and an accelerated energy transition. These returns then encourage the same confidence in time spread levels as in the oil market and increase liquidity from storage traders when price levels are reached that incentivise physical storage.

The investment is there, and according to research from BloombergNEF, the global installed capacity of grid-scale batteries reached around 56 GWh in 2021 and is expected to increase 15-fold to 1,194 GWh by 2030^[74]. Half of this increase will be in the US and China, according to the report, which bodes well for financial power market development in the US.

Physical Arbitrage

Another advantage of oil markets is the logistics infrastructure that improves efficiency in the physical market, which filters into the derivatives market. For example, if one region is pricing differently from another, oil traders can physically move barrels to profit from that discrepancy, thereby improving efficiency.

The derivatives market benefits from this infrastructure, as physical oil traders add liquidity via hedges to lock in these arbitrage levels forward. The volumes can be significant and sometimes come on the back of huge supply deals that need to be risk-managed in the financial oil market. Traders are aware of these levels and know to expect new hedging flows at certain levels, which speculators can react to, thereby creating a market and boosting market volumes.

Another benefit is the causal link between two derivatives contracts of different regions. Given the arbitrage opportunities that would arise if they

went out of whack, the two forward curves' movements are correlated, providing certainty for market makers or speculators and aiding price discovery.

With power, physical arbitrage is possible by sending electricity through transmission lines from one region to another. These lines can reach 1,000 miles and can be significant in interconnected power markets and flexible generation or storage assets.

However, this doesn't compare to the oil market, where shipping voyages and the pipeline infrastructure inland link far more of the world's key demand regions for oil than the power infrastructure. The gas market has also benefited from the addition of LNG cargoes as a means to reroute supply away from rigid pipeline infrastructure and satisfy demand imbalances around the world. In fact, this was crucial during the peak of the energy war in 2022 when LNG barrels from Qatar and other countries were secured to replace Russian gas coming via a pipeline.

The growth of renewable energy sources, such as wind and solar, is creating new opportunities for power arbitrage, as the production of these sources can be highly variable and location dependent. This variability creates price differences between regions and markets, which can be exploited through power arbitrage.

But to benefit, the power market must become less fragmented, which is likely to be highly political. Oil can move in ships and pipelines to refiners commonly located far from towns and cities. The refiners then transport refined products via road freight to where the products are needed.

Considerably ramping up transmission line infrastructure will likely come down to land disputes and jurisdiction awkwardness that arises when building and forming agreements between countries.

The commitment from the various parties to ensure the widespread expansion of power markets indicates the importance of this market in the future.

New Markets

Other nascent markets also need to gain critical mass. First, to create a new, complex and well-functioning derivatives market, we need a strong set of new indexes independently assessed by PRAs.

There are lessons to be learned from the oil market's journey over the

years with new contracts, and we can learn as much from the ones that failed as the ones that succeeded. The most important lesson is that the majority of these need to be cash-settled to encourage participation from the maximum number of counterparts. This may seem counterintuitive, but new, physically settled contracts are constantly shunned. To dip your toe into a market, you need to do an initial risk assessment, and the first thing you want to know is that the losses are comprehensible. You also want to know the ramifications of holding the contract. If it is cash-settled, you don't also have to worry about a miscalculation that will leave you with the physical product. Once you are confident that it is only a financial exposure, you want to start trading small volumes to get a sense of the market dynamic. You can build out a risk framework from there.

For example, there have already been early attempts at hydrogen benchmarks, but the market is only interested if there is sufficient trading volume or liquidity. Nevertheless, we have to start somewhere, and if the foundations are laid to accommodate the widest range of participants possible, this will give the best possible chance of success for new markets to gain traction. The market-maker role – investment banks in the early years of the financial oil market and the independent market maker over the last ten years – has proved critical to oil derivatives expansion. The market maker will only look to trade with vigour in a cash-settled contract, but trade houses and majors will only look to trade if they are at market numbers.

Hydrogen

The viability of hydrogen as a scalable fuel has come under scrutiny in recent years, and it will likely remain debated. But the physical oil industry is backing it, with hydrogen transport by ammonia a heavily favoured route. The similarities between transporting ammonia and oil are plentiful: the hope is that this can be a relatively smooth transition. Undoubtedly, there will need to be engineering work to overcome the properties of hydrogen, such as leakage and corrosiveness to steel, but with the investment and will of participants, it is likely hydrogen will have a role in the renewable energy pool.

The market should not be too difficult to start from the perspective of derivatives, given the correlation between power and gas. The early indexes that were touted take gas and power prices and reverse engineer them to get a 'grey hydrogen' and a 'blue hydrogen' price by adding voluntary carbon credit prices on top. The real goal for the renewable sector would be to get a 'green hydrogen' price, which would account for hydrogen created through electrolysis, which is considerably different from grey and blue hydrogen, making it more difficult to create an independent price. But price discovery is a process that commodity traders are more than used to, and it will figure itself out over time.

Expanding Metals Markets

Also important to the transition are the metals markets, and similar to power, there is growing interest in the metal derivatives markets.

One standout development that is likely is the lithium contract, which, while it exists, is very low in open interest and volume. Undoubtedly there will be more benchmarks for lithium created in different regions to help develop this market at scale.

The generalist view of rare earth metals is that they are predominantly owned by China, and there is a push for the US, in particular, to start mining and to develop those markets, which a derivatives market can help.

Metals don't have the same issues as the power market and should flourish as interest grows. Metals can be stored and transported worldwide in cargoes known as dry bulk by rail, trucks and even air freight for highly valued processed rare earth metals.

However, the metals derivatives industry is still reeling from the Nickel debacle, and trader confidence in currently listed contracts on the LME exchange is at a low point, let alone confidence in starting entire new markets.

Carbon and Renewable Credit Markets

The market for carbon credits has been established for some time, but it is a very different market to the rest of the commodities and energy derivatives.

The key challenge I can see is establishing a global benchmark. There are many different carbon systems, varying by their specific regulations, market structures and the scope of emissions covered. Ultimately they are all government-mandated markets, which makes the use of different contracts political.

It is predictable, therefore, that each major economy will want other economies to follow the rules that their domestic benchmark follows. If there were some standardisation, it would boost trade, reducing the variability between the contracts and allowing the global market to expand.

APPENDIX

The Brent Index

Since the web of contracts in the global oil swap markets returns to Brent crude futures time spreads, understanding the mechanisms that price crude futures is essential for competence in oil derivatives trading.

The basis for the Brent Index is that BFOETM = Brent futures.

What is BFOETM? This is known as the ‘cash’ contract or a bilaterally tradable instrument that gives the buyer physical cargo from the Brent, Forties, Ekofisk, Oseberg, Troll or WTI Midland oil fields. It is traded by month, corresponding to the Brent futures forward curve and can be traded interchangeably with the Brent futures contract via the EFP contract.

$M1\ EFP = M1\ BFOETM - M1\ \text{Brent futures}$

$\text{Month 2 (M2) EFP} = M2\ BFOETM - M2\ \text{Brent futures}$

A combination of BFOETM and Brent futures prices is used, with a consistent methodology, to settle the Brent futures price on settlement.

Brent futures expire to the Brent Index, assessed by the PRA ICIS. The index is settled using five 30-minute windows throughout the day: 10.00 to 10.30, 12.00 to 12.30, 14.00 to 14.30, 16.00 to 16.30 and 19.00 to 19.30 (London time).

The ICE created minute markers to align with the intra-day assessments, which take the volume-weighted average of the last minute of trading during each window for the next available future (second month) to the expiring contract. This price gives the exchange a reference point for the outright price, against which physical differentials can be assessed, and it is added to the second month outright price to determine a price for the expiring contract.

The physical differential used to bring the outright derivatives price to a physically aligned outright price is the EFP contract and the ‘cash spread’, which is the Brent futures time spread + the EFP time spread (called the EFP roll).

Therefore, the cash spread (physical BFOETM spread) = $(M1\ EFP + M1\ \text{Brent futures}) - (M2\ EFP + M2\ \text{Brent futures})$

The outright price of BFOETM can also be used in the assessment, but the pricing of the outright price by the North Sea oil traders is the same, using the $M2\ \text{Brent future} + M2\ EFP$ for $M2\ BFOETM$ and adding the $M1/M2$

cash spread to this to give a price for the first month BFOETM.

From the ICE website^[75], the exchange uses these components and calculates the final price by calculating the following:

- The volume-weighted minute marker for the second month ICE Brent futures contract at the sampling time plus the weighted average value of full cargo second month EFP trades plus the weighted average of full cargo spread trades (between the first and second months) in the Reference Quality Crude Oil market in the thirty-minute period concluding at the sampling point in question
- That same volume-weighted minute marker plus the sum of the straight averages of the independent assessment(s) specified in the index methodology for the second month EFP value plus the spread between the first and second month Reference Quality Crude Oil assessments at the sampling point
- A weighted average of full cargo first month Reference Quality Crude Oil trades (if any) in the five-minute period concluding at the sampling point in question

The ICE uses the PRA ICIS and other reporting agencies. The inter-dealer broker market is critical for assessing the value of EFPs and the cash spread, as well as reporting trades to ICIS. Importantly, the ICE makes it clear that it has the last word on setting the final price if there is any ambiguity.

Brent futures held to expiry are cash-settled and can be held by any derivatives trader to pricing. However, the ICE and CME impose strict position limits on the volume held to expiry. If the volume goes above this for hedging purposes, the trader must apply for an exemption ahead of time and be prepared to state their case clearly. An exemption could be granted to financial traders because they might be holding both ICE Brent futures and CME Brent futures, which makes the actual risk zero if they are long on ICE and short on CME (or vice versa).

ON EXPIRY OF THE BRENT FUTURES CONTRACT:

$$\text{MONTH 1 BFOETM} = \text{MONTH 1 BRENT FUTURES}$$

The tradeable instruments used to arrive at a settlement price for Month 1 BFOETM are:

1. MONTH 2 EFP 2. MONTH 2 BRENT FUTURES 3. M1 v M2 BFOETM TIMESPREAD 4. MONTH 1 BFOETM

$$1. \text{ MONTH 2 EFP} = \text{MONTH 2 BFOETM} - \text{MONTH 2 BRENT FUTURES}$$

$$= \text{VWA OF MONTH 2 BRENT FUTURES MINUTE}$$

MARKERS (MM)

ICE makes Minute Market contracts available to trade for each of the 5 assessment windows

$$3. \text{ M1 v M2 BFOETM} = \text{MONTH 1 BFOETM} - \text{MONTH 2 BFOETM}$$

TIMESPREAD

This is known as the 'cash spread' which ICE refers to as the 'full cargo spread'. To calculate fair value of the cash spread before the Brent Index, you use the EEPs and Brent spreads.

EXAMPLE

$$\text{MONTH 1 EFP} = \text{MONTH 1 BFOETM} - \text{MONTH 1 BRENT FUTURES}$$

$$\text{MONTH 2 EFP} = \text{MONTH 2 BFOETM} - \text{MONTH 2 BRENT FUTURES}$$

then to determine BFOETM timespread you need to add the M1 v M2 Brent Futures timespread

$$(\text{MONTH 1 BFOETM} - \text{MONTH 1 BRENT FUTURES}) + \text{MONTH 1 BRENT FUTURES}$$

$$(\text{MONTH 2 BFOETM} - \text{MONTH 2 BRENT FUTURES}) + \text{MONTH 2 BRENT FUTURES}$$

$$= \text{M1 v M2 BFOETM}$$

ICE USE THE TRADES, AND FAIR VALUES, OF THESE TRADEABLE INSTRUMENTS AS ASSESSED BY THE PRICE REPORTING AGENCY ICIS

There are 5 assessment periods throughout the day for which a price is determined:

10:00 - 10:30 | 12:00 - 12:30 | 14:00 - 14:30 | 16:00 - 16:30 | 19:00 - 19:30

The average of these 5 settlements is used for the final settlement price of the expiring Brent Futures contract:

During each window, ICE used 3 different methods to determine the M1 BFOETM

METHOD 1	METHOD 2	METHOD 3
1 AVERAGE TRADE VALUES OF CASH SPREAD Within 30 minute period concluding at the sampling point in question + 2 FAIR VALUE OF EFP Within a 30 minute period, the weighted average value of full- cargo second month EFP trades + 3 M2 BRENT FUTURES MINUTE MARKER ----- Calculated by the Exchange by calculating the volume weighted average price of trades executed in the ICE Brent Futures second month contract during the relevant sampling period	1 FAIR VALUE OF CASH SPREAD Spread between the first and second month Reference Quality Crude Oil assessments at the sampling point eg. 10:30 + 2 FAIR VALUE OF EFP Sum of the straight averages of the independent assessment(s) specified in the Index methodology for the second month EFP value + 3 M2 BRENT FUTURES MINUTE MARKER ----- Same volume weighted minute marker as method 1	1 AVERAGE OF M1 BFOETM TRADES A weighted average of full-cargo first month Reference Quality Crude Oil trades(if any) in the 5 minute period concluding at the sampling point in question

The value of the sampling time is determined by calculating the weighted average:
 $(\text{METHOD 1} + \text{METHOD 2} + \text{METHOD 3}) / 3$

Should there be no trades in either component market in METHOD 1 (1 or 2), the value is derived by the corresponding value in METHOD 2 (either 1 or 2) that is independently assessed by ICIS.

Should no trades occur in METHOD 3, the value for the sampling point is calculated using the mean of METHOD 1 and METHOD 2

Figure 5.1 The Brent Index Calculation Methodology

Creating a Financial Benchmark Index

The contracts in the financial oil market are in the many hundreds and growing, and these contracts need a reliable index published by a third party to create the settlement prices. In this section, I summarise how a PRA goes

from forming an index, to it becoming a benchmark and a tradable contract in the oil derivatives market.

i) Identify the physical oil that will be used to calculate the index

In picking the region, the PRA will need to consider price discovery, which becomes easier as the number of and variation in the participants on the producer and buyer or consumer sides increases. The following issues need to be considered: outright physical volumes and how they compare with regions around the world, the trading activity in the spot market of that oil and how the specification of that oil compares with the wider market.

ii) Engage the physical market for market monitoring

Once a PRA has identified the physical market representative of their new contract, they will need to reliably and efficiently source transaction data in this market to make a fair and balanced daily assessment. They may do this by engaging the industry for feedback. They would include a suggested methodology for how they will use the transactions to arrive at a daily price and state the settlement time and when it will be published. The PRA will want to encourage the proactive reporting of these physical transactions, and expect the industry participants will do this if they believe the index has value and that the transaction will be used to price a financial benchmark in the future.

PRAs have a team of price reporting analysts to discover what deals have been done by engaging with the market, much like journalists. The preference will always be what is visibly trading, but the PRAs acknowledge that capturing these trades in this tendering window every day is unlikely. So they look to a reliable network to ensure the best possible coverage of the physical markets.

The most efficient way to assimilate the data retrieved is via an automated process, and PRAs work on models to execute this. For further transparency, they reveal a breakdown of the calculation to the market.

iii) Create a methodology for the initial index

The index methodology will need to ensure maximum robustness, which, before it is a financial benchmark, will be about the timing of reporting and the reporting itself.

To widen the net of the physical prices included, multiple sources of oil and their physical transactions can be used, and rebased to be reflective of the region the PRA is looking to reflect in their Index. This is known as a 'netback' in PRA calculations. You take the delivered price in each region as

the price, including cost, insurance and freight, then take away the value of the journey from that region.

iv) Engage an auditor to ensure the Index calculation adheres to the International Organization of Securities Commissions (IOSCO) principles for financial benchmarks

For financial benchmarks, compliance with the IOSCO principles is the minimum standard for an index to be used by the industry. For a benchmark to comply, the creators must argue the benefits of accuracy to the underlying physical product and why it is superior to what is currently being used. They will need to ensure that the calculation methodology is objective and data collection methods are robust. Once the PRA is comfortable and ready to launch, an accounting firm should audit the benchmark to show IOSCO principles are being met. This stamp of approval is necessary for an exchange to start listing the contract.

v) Publish the index

Once we have a robust and verified methodology, the PRA needs to publish the data securely. Before the audit, it may have been fine to indicate the index prices daily, but once the audit is complete, the reporting itself is also important. There will need to be a secure website with login credentials required for access from which the data will be downloadable. In addition, there would be a database and an application program interface to deliver the data to registered consumers.

If the index proves useful and is well adopted, the PRA will want to push for deals on the back of the index. This could start as simple physical negotiations as differentials to the index. To verify the index validity, the PRA could model historical prices against other market assessments to prove the assessment is more in line with the physical oil they are attempting to represent.

If deals are made well into the future, participants will want to hedge as much of their exposure as possible. But to trade the contract, they will need this index listed as a financial contract as participants will need a bilateral transaction if it is not listed as a contract. Bilateral deals can be made with a bank, and the bank can give you a price for the index to hedge into the future as long as the deal is on its terms. This will be expensive initially, as the bank will need to manage its price and credit risks and, therefore, will charge a lot on the bid/offer.

vi) Lobby exchanges to list the index as a financial contract

Once we have evidence that the industry uses the index and there is demand for trading it, we can approach an exchange and negotiate a deal to list it as a contract. Once this is launched, financial oil participants can trade the new contract, which will settle to our methodology.

vii) Maximise liquidity to become a robust benchmark

For the contract to be properly adopted, it must be accepted as a functional contract, for which liquidity will trump all else. Tight bids/offers, trackable values throughout the day and market depth will be critical. The exchange will need to incentivise traders to use it, which they will only do if it is worthwhile, so the deal made with the exchange is also important. A new market will need intermediaries to warehouse risk and make markets, which banks and market makers are willing to do. Traders also need to believe they can make money trading these contracts, otherwise there will be no incentive.

Once the index becomes a financial contract, there are pricing concerns. Most likely, the PRA will need to update the methodology to include forward transactions, which will greatly improve price discovery, for example, if there is an absence of physical trades on a given day.

GLOSSARY

Argus

Argus Media is a global provider of news, prices and analysis for global energy and commodity markets. It offers services, including consultancy and conference events, and its price indexes are often used as benchmarks in physical contracts and the derivatives market. Argus covers a broad range of commodities, including crude oil, coal, natural gas, emissions, bioenergy, fertiliser, petrochemicals, metals and transportation.

Assessment window

The assessment window is a specific period during which market participants can submit bids, offers and trades for a particular product at a specific location. These data are combined with other sources to arrive at the prevailing market price for that product at that location. The assessment window typically lasts from thirty minutes to a full trading day and is a crucial component of the price discovery process.

Axe

Having an 'axe' is having a directional bias when trading. You may be willing to trade both sides of the market but will have a particular interest on one side of a given contract and will therefore be able to be more competitive on price when quoting this side. Brokers may ask you which way you are 'axed' to gauge interest in the markets they have to show you so as not to waste your time.

Backwardation

Backwardation is a situation in the futures markets where the future spot price is expected to be higher than the current futures price, i.e., futures prices are lower than the expected spot price at contract maturity. This typically occurs when the underlying commodity has a cost of carry or when market participants expect the spot price to rise. Backwardation can signal an expectation of a shortage or higher demand for the commodity in the future.

Balmo

Balmo is the short form of 'balance of the month' contract. The balmo is the pricing month contract that is settled every working day. For a September contract in swaps, this contract will become balmo once we are in September. The balmo will refer to the exposure that hasn't yet been priced. For example,

if it is 4 September, the balmo contract will expose the trader to 4–30 September and not include the first three days' settlement prices.

Basis Risk

Basis risk refers to the variance in the final physical price that settles a physical oil deal and the final derivative price on the contract(s) that were used to hedge. This risk can be mitigated, but no hedge is perfect.

Bid

A bid is a market order to buy. Traders will also refer to the market as 'bid' if there are more buyers than available sellers at a given time. 'Well bid' is also a term used.

Bid above

Bid above is a market maker's term used when a bid is above perceived fair value according to the trader's own assessment.

BFOETM (Brent, Forties, Oseberg, Ekofisk, Troll and WTI Midland)

BFOETM is a basket of physical crude grades used to set the price for Dated Brent contracts. The following are the crude grades included in the basket: Brent, Forties, Oseberg, Ekofisk, Troll and WTI Midland

Bill of Lading

A bill of lading is a legal document issued by a carrier to a shipper detailing the type, quantity and destination of goods being shipped. Serving as a shipment receipt, it also establishes an agreement for the transportation of the goods. In international trade, the bill of lading is often required for customs clearance and can act as an ownership title, which can be traded, sold or bought while goods are in transit.

Block Future

Block Future is a term used by exchanges to define oil swaps. Oil swaps used to be traded bilaterally until they were listed on the exchanges, and as exchange-cleared products, they strictly need to be defined as futures. The exchange differentiates between exchange-traded futures by calling these Block Futures.

CFTC (Commodities Futures Trading Commission)

The CFTC is an independent US government agency that regulates futures and options markets. It was established in 1974 to promote competitive, efficient and transparent markets while protecting market participants from fraud, manipulation and abusive practices.

CME (Chicago Mercantile Exchange)

The CME is one of the largest futures and options exchanges in the world, offering derivatives contracts for a wide range of assets, including commodities, currencies, equities and interest rates.

COT (Commitment of Traders; CFTC)

The COT is a weekly report provided by the CFTC that aggregates the positions of large traders in the US futures market. The report is used by traders and analysts to understand the behaviours and sentiments of major market participants, offering insights into potential price movements in various asset classes.

COT (Commitment of Traders; Onyx)

The Onyx COT is a daily report provided by Onyx Capital Group that aggregates the positions of traders in the global oil swaps market. The report is used by traders and analysts to understand the behaviours and sentiments of major market participants, offering insights into potential price movements in various oil swap contracts.

Contango

Contango is a situation in futures markets where the future spot price is below the current futures price, i.e. futures prices are higher than the expected spot price at maturity. This typically occurs in markets where there are costs associated with storing the commodity until delivery, such as with oil or natural gas. Contango can signal an expectation of surplus supply or lower demand in the future.

CTA (Commodity Trading Advisor)

A CTA (commodity trading advisor) is a US financial regulatory term for an individual or organisation who is retained by a fund or individual client to provide advice and services related to trading in futures contracts, commodity options and/or swaps.

Crack

A crack is a shortened word for the value of a refined oil product against the underlying crude price. It is a reference to the refining cracking process to convert crude into a refined product. The better the crack price, the more a refiner will look to maximise the yield on that product. To calculate the crack, a given refined product price is converted from metric tonnes or gallons to a barrel equivalent price to provide an accurate reference against crude prices. Product cracks are used to assess the refinery margin, a crucial indicator for refiner demand. They are a useful gauge of product-specific health, as the correlation with outright crude prices is removed.

Dated to lead

The value of a DFL contract vs the associated Brent spreads. Given the 10–30 day loading methodology that Platts uses to settle Dated Brent swaps, M1 Dated to lead will be assessed as the DFL vs the next month forward Brent spread. For example, January Dated to lead is equal to January DFL – February vs March Brent spread. As Brent futures can be converted to BFOETM physical to align the Dated Brent swap prices via the STASCO BFOETM 2022 GTCs at Brent futures expiry, the value of this switch can be valued by assessing the difference in value between the two. Dated to lead is therefore used as an indicator to assess the cost of keeping physical barrels from the Brent Index or Brent futures expiry.

Differentials (Diffs)

A differential refers to any contract priced against another, with appropriate volume conversions, so the value of the diff is decorrelated from the outright prices of oil. Differentials are not cracks, which are also

differentials but specific to a product value against the underlying crude price. Differentials are useful for oil traders to interpret variations, such as the same product in different geographical locations, different specifications within a core product or a link from crude futures to more physically related contracts.

Exchange-traded future

Exchange-traded future is a term used to distinguish Block Futures (oil swaps) and the core benchmark futures Brent, WTI, RBOB, Gasoil and Heating Oil.

Flow

A market flow is a way of labelling a group of transactions that has entered the market from a consistent source. The 'flow' might be a producer hedging their production for Q4 next year with a fixed amount of volume to sell of a given crude contract. All the transactions that comprise the producer's hedge would be considered part of the producer's flow. It could also be speculative positioning one way until a trader maximises their risk tolerance. Market makers will take on a flow that needs to be executed and derisk this core flow against the flow of other contracts.

Forward curve

A graph of what a commodity is currently worth based on a possible buy or sell in the future. A forward curve is built using the current day's price values to exchange oil in the future, and its value will change over time. A forward curve is not a price forecast like the formal weekly, monthly or annual predictions that analysts produce based on fundamentals data. It is an indicator based on the bids and offers in the market for that day.

GasNap

The GasNap is the difference in price between gasoline and naphtha in a region. It is used to interpret the value of placing naphtha as a blending component into the delivered gasoline price. An increase in the price incentivises blenders to use more.

Hitting

Hitting is the colloquial term used when selling a bid in the market, whether a derivative or physical transaction.

ICE (Intercontinental Exchange)

The ICE is an American company that operates global exchanges and clearing houses and provides data and listing services, including owning the New York Stock Exchange. Founded in 2000, ICE manages regulated

exchanges, marketplaces and clearing houses across multiple countries, facilitating financial transactions and trade.

ICIS (Independent Commodity Intelligence Services)

ICIS is a global source of independent commodity intelligence services. It provides data on over 180 commodities across the global petrochemical, energy and fertiliser markets and is known for its benchmark price assessments, in-depth market coverage and extensive data analytics.

IOCs (Integrated Oil Companies)

IOCs are large private corporations based in developed countries involved in all aspects of the oil and gas industry, including exploration, extraction, refining, distribution and marketing. They operate across national boundaries, often have their own production capabilities and hold significant influence in global oil and gas markets. Examples include ExxonMobil, Royal Dutch Shell and BP.

Lifting

Lifting is the colloquial term used when paying an offer in the market, whether a derivative or physical transaction.

Minute Markers

Minute markers are similar to TAS but for various times in the day and are calculated over one minute rather than two. Minute markers are provided at important times in the market, usually aligning with roll-off periods in PRA assessment windows or the Brent Index. For example, there is a minute marker at 16.30 Singapore time for Eastern contracts and at 16.30 London time for European-based contracts.

Liquidity

A market's liquidity is assessed by how straightforward it is for a trader to make a transaction. Ideally, a trader will want to be able to freely buy or sell contracts in the volume they need and will expect the transaction price to be as close as possible to fair value and for the full volume traded. Liquidity is therefore talked about by the volume available to trade at a given moment (depth) and the tightness of the bid/offer.

Offered

The market is said to be 'offered' when there are more sellers than buyers at a given time. 'Well offered' is also used.

Offered through

Offered through is a market maker's term for when an offer is below perceived fair value according to the trader's assessment.

Oil Future

An oil futures contract is a financial agreement that obligates the buyer to purchase and the seller to deliver a certain quantity of oil at a predetermined price on a specified future date. However, while some contracts involve the physical delivery of oil upon expiration, others are cash-settled, where the difference between the contract price and the market price at expiration is exchanged instead of the actual commodity.

Oil Inventory

Oil inventory refers to the stored reserves of crude oil and refined petroleum products at any given time. These inventories can be held by oil producers, refiners, governments and other entities involved in the oil supply chain. Monitoring changes in oil inventories, such as the weekly data reported by the US Energy Information Administration, is crucial for market participants, as it provides insight into supply and demand dynamics and can influence oil prices.

Oil Swap

An oil swaps contract is a financial agreement that obligates the contracted parties to exchange the cash difference between the contract price and the market price of oil at contract expiration. Oil swaps differ from oil futures in oil derivatives in that contracts are all cash settled, and the index is settled and published daily to a PRA assessment. Oil swaps are mostly voice-brokered rather than traded directly on the exchange, which is another difference from an oil future. Oil swaps are also sometimes called Block Futures.

Open Interest

The number of contracts that have not yet been closed out and are considered 'open'. The significance of an open contract is that it will either be held to expiry or closed out at some point. The closing of a position will lead to incoming flow that could have a material impact on the market's direction, especially if open interest is high.

OTC

This term classically refers to bilateral trading between counterparts. Counterparts extend each other credit, making an assessment of the counterpart's default risk internally. This type of trading can be facilitated in theory by any trader; however, it is commonly an investment bank that provides this facility to give traders access to the market. Confusingly, you will also hear the term 'OTC market' to define the oil swaps market, as the

trading is still mostly voice-executed via brokers who then pair two traders together. Historically the swap market was only accessible by trading bilaterally, and the terminology has stuck.

Physical Differential (or Premium)

The number at which physical transactions are taking place for a given crude oil or oil product relative to a benchmark. Physical trades are negotiated and dealt using a benchmark derivative contract as a reference price. The value of the physical differential is added to the derivative price to determine the full physical transaction value. It is done this way as outright prices of derivative contracts are highly volatile, and 90% or more of the price is correlated with outright prices anyway, making the differential key in assessing material differences in the specification, product type or location.

Platts

Platts is a provider of energy and commodities information and a source of benchmark price assessments in the physical energy markets. Owned by S&P Global, Platts covers a wide range of sectors, including oil, natural gas, metals, agriculture and petrochemicals. Its price assessments are widely used as benchmarks in the global energy sector.

PRA (Price Reporting Agency)

A PRA is a company that collects, assesses and publishes information regarding commodity prices. PRAs provide price transparency in markets by reporting on transactions, bids and offers, thus enabling buyers and sellers to make informed trading, pricing and risk management decisions. Key commodities covered by PRAs include energy products, metals and agricultural goods.

Priced In

This is when a given driver of markets is considered already represented in the current price. It occurs when all those capable and willing to react to a given market event or headline have already done so, and therefore you can no longer expect more news of the same situation to lead to more of the same flows.

Scaling In

Scaling in refers to a trader adding more risk to an already open position.

Slippage

Slippage refers to the difference between fair value at the time of a trade and the final price received once the trader has executed all the desired

volume. Slippage is more likely with larger volumes or when trading in the same direction as the rest of the market.

Sharpe Ratio

The Sharpe ratio is calculated by subtracting the risk-free rate of return from the expected return of the investment and dividing the result by the standard deviation of the investment returns. The higher the Sharpe ratio, the better the risk-adjusted performance of the investment.

Sortino Ratio

The Sortino ratio is calculated by subtracting the minimum acceptable return from the expected return of the investment and dividing the result by the downside deviation of the investment returns. The higher the Sortino ratio, the better the risk-adjusted performance of the investment in terms of avoiding downside risk.

Stopping Out

Stopping out refers to the closing of a trader's position that is out of the money. Stopping out will occur when the trader has reached their maximum risk tolerance on the trade, either from a pre-set level or direction from management. Stopping out usually causes a spike in volatility, reflecting the urgency of traders exiting a position.

Tenor

In oil swaps, tenor is used to reflect the pricing period of the contract. In oil futures, the term is used to indicate the loading period of the oil and can settle up to two months ahead of time.

Time spreads

A time spread, or 'calendar spread', in oil derivatives, is a strategy where a trader simultaneously buys and sells two futures or options contracts for the same oil quantity but with different expiration dates. This strategy aims to profit from the change in the price difference, or spread, between these two contracts over time.

Trade House

A trade house, or trading house, refers to a business that specialises in the international trade in commodities. They engage in various activities, such as procurement, distribution, arbitrage and risk management. Trade houses can deal in a range of commodities, including oil, gas, metals, grains and more. They often have a global presence, with operations in various countries, and play a critical role in the supply chain, bridging gaps between producers and end-users. Examples include Glencore, Trafigura and Vitol.

TAS (Trade at Settlement)

TAS is a contract that becomes a live exchange-traded future but only at

the point at which the exchange-traded future settles and at that settlement price determined by the exchange. This allows a trader to execute a trade at the settlement price of a futures contract rather than at the current market price.

Volatility

Volatility refers to by how much and over what period the prices of a given contract move. How volatile a market is will be defined relative to the historical movement of that contract.

ABOUT THE AUTHOR



Greg Newman is the Chief Executive of Onyx Capital Group - the world's largest liquidity provider in oil derivatives. Greg started his trading career at Mandara Capital aged twenty-two where he created the crude oil desk. Within three years it became the largest independent market-maker in Dated Brent and Dubai swap markets. In 2016 Greg left Mandara with his colleague Omar Kayaam to launch Onyx. Greg has been the driving force behind the development of Onyx on the trading floor and beyond. He established the group's highly successful training programmes, its unique research offering, data services, brokerage, news and analysis. He has featured in the book "40 Classic Crude Oil Trades : Real-Life Examples of Innovative Trading" and the crude desks he formed at Mandara and Onyx are mentioned in "Brent Crude Oil: Genesis and Development of the World's Most Important Oil Benchmark". Greg holds a BEng in Biochemical Engineering from

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